



Naval Facilities Engineering Command Mid-Atlantic
Norfolk, Virginia

Draft

**Tier II Sampling and Analysis Plan, (Field Sampling
Plan and Quality Assurance Project Plan) Long-Term
Groundwater Monitoring**

Site 17 – Former Building 32, Gould Island

Naval Station Newport
Jamestown, Rhode Island

July 2020



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Title and Approval Page

(UFP-QAPP Manual Section 2.1)

Document Title: Tier II Sampling and Analysis Plan (SAP), (Field Sampling Plan and Quality Assurance Project Plan), Long-Term Groundwater Monitoring, Former Building 32, Gould Island (Site 17), Operable Unit 06, NAVSTA Newport, Jamestown, Rhode Island

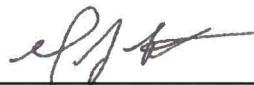
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EXECUTIVE SUMMARY

This Sampling and Analysis Plan (SAP) was prepared to guide personnel in fulfilling the requirements of a Long-term Monitoring/Management (LTM) Plan for groundwater set forth in the Record of Decision (ROD) for Site 17, Former Building 32, Gould Island, Operable Unit 06 at Naval Station (NAVSTA) Newport, in Jamestown, Rhode Island. Included in this SAP are requirements for monitoring of groundwater.

This SAP describes long-term groundwater monitoring at 21 existing monitoring wells. As described in the ROD, there are three purposes for performing groundwater monitoring at Site 17: (1) Monitoring natural attenuation (MNA) of pentachlorophenol (PCP), tetrachloroethene (PCE), and manganese concentrations in groundwater; (2) confirming that the source control measures for semi-volatile organic compounds (SVOCs), polychlorinated biphenyls (PCBs), benzene, trichloroethylene (TCE), PCE, arsenic, and manganese were effective; and (3) confirming that contaminants present in soil at concentrations exceeding industrial and leachability criteria before source control measures (including SVOCs, PCBs, arsenic, manganese, cadmium, and lead) are not impacting groundwater to the extent where groundwater remedial goals are being exceeded. Risk associated with subsurface soils at Site 17 has been mitigated by implementation of land use controls (LUCs).

The Site 17 ROD, which was signed on June 27, 2014, presents and describes the selected remedy for soil, sediment, and groundwater contamination at Operable Unit 06. The remedy includes soil and debris removal (completed in 2016), sediment removal, confirmation sampling of “shallow groundwater,” implementation of LUCs, MNA, long-term groundwater monitoring, limited sediment monitoring, annual inspections, and reporting. The scope of work described in this SAP describes activities associated with confirmation sampling of “shallow groundwater”, MNA, and long-term groundwater monitoring as described in the ROD.

This SAP is meant to be used as a long-term document, with modifications to be made on an annual basis as necessary when optimizations to sampling location, frequency, or parameters are deemed appropriate. Modifications will be recommended and included as part of each Long-Term Monitoring Report.

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FIGURES

- 4-1 Site Location
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APPENDICES

- A Tetra Tech SOPs
- B Laboratory Certifications

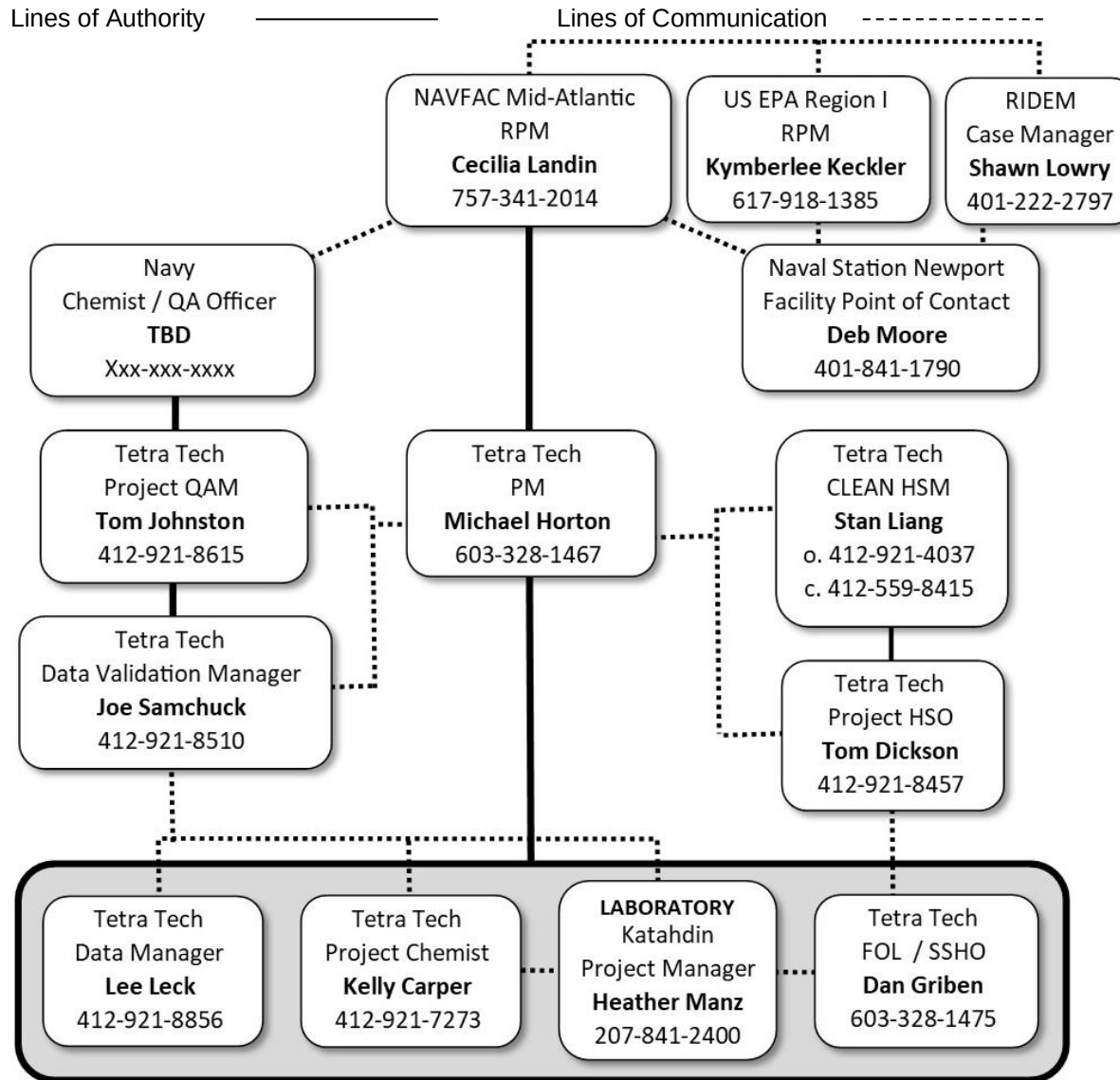
ACRONYMS AND ABBREVIATIONS

AES	Atomic Emission Spectroscopy
CAS	Chemical Abstract Service
COC	Contaminant of concern
CoC	Chain of Custody
CSM	Conceptual site model
DCE	Dichloroethene
DI H ₂ O	Deionized water
DL	Detection limit
DO	Dissolved oxygen
DoD	Department of Defense
DQO	Data Quality Objective
DV	Data validation
DVM	Data Validation Manager
EB	Equipment blank
ECD	Electron capture detector
EE/CA	Engineering Estimate/Cost Evaluation
ELAP	Environmental Laboratory Accreditation Program
EPA	United States Environmental Protection Agency
ERA	Ecological risk assessment
ERP	Environmental Restoration Program
FID	Flame ionization detector
FOL	Field Operations Leader
FS	Feasibility Study
FTMR	Field Task Modification Request
GC	Gas chromatography
GW	Groundwater
HCl	Hydrochloric acid
Helios/TT JV	Helios, Tetra Tech Joint Venture
HNO ₃	Nitric acid
HSM	Health and Safety Manager
ICAL	Internal calibration
ICP	Inductively coupled plasma
IDW	Investigation derived waste
IS	Internal standard
LCS	Laboratory control spike <i>or</i> Laboratory control sample
LCSD	Laboratory control spike duplicate
LOD	Limit of detection
LOQ	Limit of quantitation
LTM	Long-term monitoring
LUC	Land use control
MCL	Maximum Contaminant Level
mL	Milliliter
MNA	Monitored natural attenuation
MPI	Measurement performance indicator
MS	Mass spectroscopy
MS/MSD	Matrix Spike/Matrix Spike Duplicate
NA	Not applicable
NAVFAC	Naval Facilities Engineering Command

NEDD	Navy electronic data deliverable
NELAP	National Environmental Laboratory Accreditation Program
NIRIS	Naval Installation Restoration Information Solution
NUWC	Naval Undersea Warfare Center
ORP	Oxidation reduction potential
PAH	Polycyclic aromatic hydrocarbons
PAL	Project action limit
PCB	Polychlorinated biphenyls
PCE	Tetrachloroethene
PCP	Pentachlorophenol
PDF	Portable document format
PM	Project Manager
POC	Point of contact
PQO	Project quality objective
QA	Quality assurance
QAM	Quality Assurance Manager
QAO	Quality Assurance Officer
QC	Quality Control
QSM	Quality Systems Manual
RG	Remedial goal
RI	Remedial investigation
RIDEM	Rhode Island Department of Environmental Management
ROD	Record of Decision
RPD	Relative percent difference
RPM	Remedial Project Manager
RSL	EPA Regional Screening Value
SAP	Sampling and Analysis Plan
SDG	Sample delivery group
SIM	Selected ion monitoring
SOP	Standard operating procedure
SSHO	Site Safety and Health Officer
SVOC	Semi-volatile organic compound
TBD	To be determined
TCE	Trichloroethene
VC	Vinyl chloride
VOA	Volatile Organic Analysis
VOC	Volatile organic compound
%R	Percent recovery

1.0 Project Organizational Chart

(UFP-QAPP Manual Section 2.4.1 – Worksheet #5)



TBD – To be determined;
 POC – Point of Contact;
 RPM – Remedial Project Manager;
 QA – Quality Assurance
 QAO – QA Officer
 QAM – QA; Manager
 PM – Project Manager;
 SSHO – Site Safety and Health Officer;
 HSM – Health and Safety Manager
 DVM – Data Validation Manager;
 FOL – Field Operations Leader;
 IDW – Investigation-Derived Waste

Note: All Helios/Tetra Tech Joint Venture (Helios/TT JV) staff have communication access to the HSM and QAM at all times, especially if health and safety or quality incidents adverse to personnel safety or work performance are identified.

2.0 Communication Pathways

(UFP-QAPP Manual Section 2.4.2 – Worksheet #6)

Communication Drivers	Responsible Entity	Name	Phone Number	Procedure (Timing, Pathway To/From, etc.)
Draft Sampling and Analysis Plan (SAP) Review/Revisions	Helios/TT JV PM	Michael Horton	603-328-1463	Within 2 days of completing the draft SAP, the SAP will be submitted to NIRIS by the Helios/TT JV PM. NAVFAC QAO/Chemist comments will be resolved to the satisfaction of the QAO/Chemist and the SAP will be uploaded to NIRIS for QAO/Chemist signature.
	Naval Facilities Engineering Command (NAVFAC) RPM	Cecilia Landin	757-341-2014	
	NAVFAC QAO/Chemist	TBD	TBD	
Regulatory Agency Interface	NAVFAC RPM	Cecilia Landin	757-341-2014	When due according to the project schedule the draft SAP will be submitted to the regulators by the NAVFAC RPM or designee. NAVFAC RPM will interface directly, via phone or via email with regulators as needed.
Request for Utility Clearance/Permit	Helios/TT JV FOL	Dan Griben	603-328-1475	The FOL will submit a request for a utility clearance/permit with appropriate supporting documentation to Dig Safe and NAVSTA Newport a minimum of 14 days before intrusive activities are planned to be conducted.
Preparation for Sampling	Helios/TT JV PM Laboratory PM	Michael Horton Heather Manz	603-328-1463 207-87-2400	At a time that ensures sample bottles will be available on site early enough to collect samples, the Helios/TT JV PM will contact the laboratory PM via email or teleconference to order bottles. The Helios/TT JV PM or project chemist will maintain weekly contact with the laboratory throughout the project to ensure project goals, including changes to laboratory-specific limits, are acceptable prior to and during the analyses.
Field Progress Reports	Helios/TT JV FOL Helios/TT JV PM	Dan Griben Michael Horton	603-328-1475 603-328-1463	The Helios/TT JV FOL will provide email or verbal reports to the Helios/TT JV PM on a schedule arranged by the PM.
Stop Work Due to Safety Issues	Helios/TT JV FOL Helios/TT JV PM NAVFAC RPM Site POC Subcontractor PM	Dan Griben Michael Horton Cecilia Landin Deb Moore Marine Transport (TBD)	603-328-1475 603-328-1463 757-341-2014 401-841-1790 TBD	If Helios/TT JV is the responsible party for a stop work command, the Helios/TT JV FOL will inform on-site personnel, subcontractor(s), the NAVFAC RPM and Site POC, and the identified Project Team members within 1 hour (verbally or by e-mail). If a subcontractor is the responsible party, the subcontractor PM must inform the Helios/TT JV FOL within 15 minutes, and the Helios/TT JV FOL will then follow the procedure described above.

Communication Drivers	Responsible Entity	Name	Phone Number	Procedure (Timing, Pathway To/From, etc.)
SAP Changes Prior to Field/ Laboratory Work	Helios/TT JV FOL Helios/TT JV PM NAVFAC RPM Site POC NAVFAC QAO	Dan Griben Michael Horton Cecilia Landin Deb Moore TBD	603-328-1475 603-328-1463 757-341-2014 401-841-1790 TBD	The Helios/TT JV PM will document the required change via a Field Task Modification Request (FTMR) form within 5 days and send it to the NAVFAC PRM and Site POC. If necessary, the PM will submit a concurrence letter on a schedule agreeable to the NAVFAC RPM. SAP amendments will be submitted by the Helios/TT JV PM to the NAVFAC RPM and Site POC for review and approval. At the RPM's discretion, SAP changes will be submitted to the NAVFAC QAO for approval (e.g., when significant Data Quality Objective [DQO]-related changes occur, or the laboratory is changed). The Helios/TT JV PM will communicate scope changes to the Project Team via e-mail within 1 business day.
SAP Changes in the Field	Helios/TT JV FOL Helios/TT JV PM NAVFAC RPM Site POC	Dan Griben Michael Horton Cecilia Landin Deb Moore	603-328-1475 603-328-1463 757-341-2014 401-841-1790	The Helios/TT JV FOL will verbally inform the Helios/TT JV PM within 24 hours of realizing the need for a change. The Helios/TT JV PM will document the change via a FTMR form within 5 days and send it to the NAVFAC RPM and Site POC. If necessary, a concurrence letter will be submitted within 7 days of identifying the need for change. The NAVFAC RPM will establish the required level of review. At the RPM's discretion, SAP changes will be submitted to the NAVFAC QAO for approval (e.g., when significant changes are made to DQOs).
Field Corrective Actions in Response to Unusual Field Conditions or Conditions Otherwise Adverse to Quality (e.g., sample representativeness, comparability or other quality concerns)	Helios/TT JV FOL Helios/TT JV PM NAVFAC RPM Site POC NAVFAC QAO/Chemist	Dan Griben Michael Horton Cecilia Landin Deb Moore TBD	603-328-1475 603-328-1463 757-341-2014 401-841-1790 TBD	The FOL will alert the Helios/TT JV PM verbally of the condition adverse to quality as soon as it is safe to do so. The Helios/TT JV PM and FOL will immediately develop appropriate corrective actions to preserve quality. This may require consult with other Helios/TT JV technical staff (such as the Project Chemist) or subcontractor PM. If an appropriate action cannot be developed quickly or upon developing an appropriate action, the Helios/TT JV PM will notify the NAVFAC RPM verbally or via email and, at the RPM's discretion, other NAVFAC personnel such as the Site POC or NAVFAC QAO/Chemist. Upon agreement between the Helios/TT JV PM and NAVFAC RPM regarding appropriate corrective actions, significant deviations from this SAP (if any) will be documented in the field logbook and on an FTMR form.

Communication Drivers	Responsible Entity	Name	Phone Number	Procedure (Timing, Pathway To/From, etc.)
Sample Receipt Variances	Laboratory Sample Custodian Helios/TT JV FOL Helios/TT JV Project Chemist Helios/TT JV PM NAVFAC RPM	Various Dan Griben Kelly Carper Michael Horton Cecilia Landin	Various 603-328-1475 412-921-7273 603-328-1463 757-341-2014	The Laboratory Sample Custodian will report variances to the Helios/TT JV FOL, Helios/TT JV Project Chemist, or PM verbally or via email within 24 hours of identifying a variance and those individuals will attempt to resolve the variance with the laboratory. If the variance cannot be resolved, the Helios/TT JV PM will notify the NAVFAC RPM verbally or via email within 1 business day of the variances being brought to his attention. If necessary, the NAVFAC RPM will take corrective action commensurate with the deficiency.
Laboratory Quality Variances	Laboratory PM Helios/TT JV Project Chemist Helios/TT JV PM Data Validators NAVFAC RPM NAVFAC QAO/Chemist	Heather Manz Kelly Carper Michael Horton Various Cecilia Landin TBD	207-874-2400 412-921-7273 603-328-1463 Various 757-341-2014 TBD	The Laboratory PM will notify (verbally or via e-mail) the Helios/TT JV Project Chemist within 1 business day of when an issue related to laboratory data is discovered. The Helios/TT JV Project Chemist will notify (verbally or via e-mail) the data validation (DV) staff and Helios/TT JV PM within 1 business day. The Laboratory PM will document all quality variances in the Case Narrative of the Analytical Laboratory Report. Corrective actions may include a consult with the RPM and, at the RPM's discretion, with the NAVFAC QAO/Chemist.
Analytical Corrective Actions	Laboratory PM Helios/TT JV FOL Project Chemist Helios/TT JV PM NAVFAC RPM NAVFAC QAO/Chemist	Heather Manz Dan Griben Kelly Carper Michael Horton Cecilia Landin TBD	207-874-2400 603-328-1475 412-921-7273 603-328-1463 757-341-2014 TBD	If the impact of an identified deficiency is limited to this project, it will be resolved between the laboratory PM and Helios/TT JV PM and support staff and will be documented in the project report. If the deficiency is systemic and potentially affects other projects, the Helios/TT JV PM will verbally advise the NAVFAC RPM within 24 hours of notification from the Project Chemist or DVM. The NAVFAC RPM will take corrective action appropriate for the identified deficiency. Corrective actions may include a consult with the NAVFAC QAO/Chemist and coordination with the laboratory.
Reporting DV Issues	Helios/TT JV DV Chemist Helios/TT JV Project Chemist Helios/TT JV PM NAVFAC RPM	Various Kelly Carper Michael Horton Cecilia Landin	Various 412-921-7273 603-328-1463 757-341-2014	Helios/TT JV DV Chemist will document data qualifications in the DV report and database. For serious deficiencies the DV Chemist will notify Helios/TT JV Project Chemist or PM verbally or via e-mail within 48 hours of recognizing that a significant laboratory quality deficiency has been detected that could affect this project and/or other projects. The Helios/TT JV PM will relay information regarding serious data concerns or laboratory concerns to the NAVFAC RPM.

Communication Drivers	Responsible Entity	Name	Phone Number	Procedure (Timing, Pathway To/From, etc.)
DV Corrective Actions	Helios/TT JV DV Chemist Helios/TT JV Project Chemist Helios/TT JV PM NAVFAC RPM NAVFAC QAO/Chemist	Various Kelly Carper Michael Horton Cecilia Landin TBD	Various 412-921-7273 603-328-1463 757-341-2014 TBD	If a DV issue cannot be resolved between the Helios/TT JV DV Chemist and the laboratory or if the issue appears to be systemic, the Helios/TT JV validation chemist will alert, verbally or via email, the Helios/TT JV PM who will verbally advise the NAVFAC RPM within 24 hours of notification. The NAVFAC RPM will take corrective action that is appropriate for the identified deficiency. This may include a consultation with the NAVFAC QAO/Chemist.
Project Report Review	Helios/TT JV PM Helios/TT JV support staff NAVFAC RPM	Michael Horton Various Cecilia Landin	603-328-1463 Various 757-341-2014	Internal reviews will be conducted by the Helios/TT JV PM and support staff. Comments will be resolved internally to the satisfaction of the Helios/TT JV PM. When satisfied with resolution of all comments, the report will be submitted by the Helios/TT JV PM or designee to the NAVFAC RPM in accordance with the project scope of work requirements. Upon resolution of NAVFAC comments, the report will be submitted to the NAVFAC RPM for transmittal to the regulators for review. Comments will be resolved to the satisfaction of the NAVFAC RPM, revisions will be made, and the report will be finalized for final distribution and submittal to NIRIS.
Identification of Systemic Quality Deficiencies on the Project	Helios/TT JV PM Helios/TT JV support staff Helios/TT JV QAM	Michael Horton Various Tom Johnston	603-328-1463 Various 412-921-8615	The person identifying the concern will communicate to the Helios/TT JV PM and QAM in a timeframe that supports correction of the concern as soon as practicable and sharing of "Lessons Learned" to prevent propagation of deficiencies.

3.0 Project Planning Session Participants Sheet

(UFP-QAPP Manual Section 2.5.1 – Worksheet #9)

Project Name: Groundwater LTM		Site Name: Site 17, Former Building 32, Gould Island			
Projected Date(s) of Sampling:	Bi-annually (March/April and September/October)	Site Location:	Jamestown, Rhode Island		
Project Manager:	Michael Horton				
Date of Session:	September/October 2019				
Scoping Session Purpose:	Optimization of Long-Term Groundwater Monitoring at Site 17				
Name	Title	Affiliation	Phone #	E-mail Address	Project Role
Cecilia Landin	RPM	NAVFAC Mid-Lant	757-341-2014	cecilia.landin@navy.mil	Navy Manager
Kymberlee Keckler	RPM	Environmental Protection Agency (EPA)	617-918-1385	kymberlee.keckler@epa.gov	EPA Manager
Greg Kemp	EPA Consultant	Mabbett	603-625-9116	kemp@mabbett.com	EPA Consultant
Shawn Lowry	RPM	Rhode Island Department of Environmental Management (RIDEM)	401-222-2797	shawn.lowry@ri.dem.gov	RIDEM Manager
Deborah Moore	NAVSTA Newport POC	NAVSTA Newport	401-841-1790	deborah.j.moore@navy.mil	Facility POC
Michael Horton	Project Manager	Helios/TT JV	603-328-1467	mike.horton@tetrattech.com	Helios/TT JV Manager
Stephen Parker	Activity Coordinator	Tetra Tech	603-328-1469	stephen.parker@tetrattech.com	Project Support
Laurel Boucher	Facilitator	The Management Edge	703-407-1030	lboucher@up.net	Meeting Moderator

Comments/Decisions:

After completing the initial eight sampling events the project team agreed to meet for the purpose of discussing optimization of the Site 17 groundwater LTM. The Navy initially proposed the following optimizations:

- Reduce the frequency of monitoring to two times per year (March and September).
- Eliminate analysis of volatile organic compounds (VOCs).
- Eliminate analysis of pentachlorophenol (PCP).

- Eliminate analysis of polycyclic aromatic hydrocarbons (PAHs) from all wells where PAH exceedances of clean-up goals have not been reported during the first eight events and analyze samples at other wells only for the PAHs that have exceeded clean-up goals during the first eight monitoring events.
- Eliminate analysis of polychlorinated biphenyls (PCBs) from all wells where PCBs have not been detected during the first eight events.
- Continue analyzing samples collected from all wells included in the monitoring program for total and dissolved metals, anions, and alkalinity.

EPA formally responded to these proposals with the following:

- PAHs – EPA recommends that all PAHs that can reasonably contribute to drinking water risk be included in the optimization plan and analyzed in all wells. Based on EPA Regional Screening Level (RSL) values, these PAHs include benzo(a)anthracene, benzo(a)pyrene, benzo(b)fluoranthene, benzo(k)fluoranthene, dibenzo(a,h)anthracene, indeno(1,2,3-CD)pyrene, and naphthalene.
- PCP – Since PCP contributes to be a drinking water risk and the detections were an order of magnitude greater than the RSL (the 10-6 RSL is 0.041 ug/L), PCP should remain an analyte for all wells.
- Aroclor PCBs - EPA recommends that sampling for PCBs continue at wells where PCBs have been detected. This includes the seven wells that the Navy identified in Worksheet 18 plus MW303B where PCBs were also detected. Although PCBs were not detected at MW300B, EPA recommends that it be retained because it is proximate to the former transformer vault 53 where PCB releases occurred. (The greatest PCB detection to date, 1.5 µg/L, occurred at MW406, proximate to former transformer vault 54.)
- VOCs – EPA recommends continued sampling for trichloroethylene and vinyl chloride at wells MW304SR, 401, and 405 where TCE has consistently been detected at concentrations near the EPA Drinking Water Maximum Contaminant Level (MCL).
- Metals – EPA concurs with sampling all wells for arsenic, cadmium, lead, and manganese as well as anions and alkalinity.
- Semi-annual monitoring as proposed by the Navy is appropriate after the completion of the Year 2 monitoring event.

As of the date of this SAP RIDEM has not provided input regarding optimization of the Site 17 Groundwater LTM plan.

Consensus Decisions:

For the purpose of continuing to perform long-term monitoring (LTM) activities at Site 17, the Navy has prepared this SAP to implement bi-annual groundwater monitoring. The analytical requirements for groundwater samples have not been updated based on the optimization proposals presented here because project team consensus has not been reached; the Navy intends to continue monitoring for the full analytical suite at all monitoring locations until the project team agrees that adjustments are appropriate. When/if adjustments are made to the analytical suite, updates to this SAP will be made with FTMRs.

4.0 Conceptual Site Model

[\(UFP-QAPP Manual Section 2.5.2 – Worksheet #10\)](#)

Overview

The NAVSTA Newport facility encompasses approximately 1,400 acres on the west shore of Aquidneck Island facing the East Passage of Narragansett Bay and is located primarily in the towns of Portsmouth, Middletown, Jamestown, and Newport, Rhode Island. Environmental Restoration Program (ERP) Site 17 is located on Gould Island, in Jamestown, Rhode Island. Gould Island is located in the East Passage of Narragansett Bay, approximately 1.5 miles west of Newport, Rhode Island, between Aquidneck and Conanicut (not shown on Figure 4-1) Islands. Former Building 32, located on the northeastern end of Gould Island, occupied approximately 6 acres of land, not including the firing pier and former Building 35 to the north. Site 17 encompasses this 6-acre area around former Building 32, including the soil and groundwater and approximately 1 acre of marine sediment adjacent to the northern shoreline where contaminants from the former Building 32 and its operation have come to reside. The location of Gould Island within the NAVSTA Newport facility is presented on Figure 4-1. The layout of Site 17 is included on Figure 4-2.

Gould Island is generally unoccupied, with the exception of Building 35. However, the surrounding waters are used for electronic equipment testing by Naval Undersea Warfare Center (NUWC), and the grounds of Site 17 are intended for similar use. The Site is occasionally accessed by trespassers via recreational boating. At the northern end of Gould Island where Site 17 is located, the island consists of a constructed shoreline that is a combination of filled land, man-made structures, and natural island formations. These include the Firing Pier, a now demolished rigging platform (a timber dock), a piling field originally constructed to create a partial breakwater feature and support a pier platform, and constructed shoreline (filled land behind bulkhead walls). The northeastern and northwestern intertidal shorelines of the Site are exposed and subject to wave action. The entire northeastern shoreline consists of a deteriorated sheet-piling bulkhead wall and a stony beach face. The entire northwestern shoreline is composed of riprap.

Two reference documents are available that best describe the operational history, physical characteristics, and other key elements of the conceptual site model (CSM): the Record of Decision (ROD) for Site 17 – Former Building 32 Gould Island, Operable Unit 06, Jamestown, Rhode Island, Naval Station Newport, Rhode Island (NAVFAC, 2014); and the Feasibility Study for Site 17 – Former Building 32, Gould Island, Naval Station Newport, Newport, Rhode Island (Tetra Tech, 2014). These two reference documents provide most of the background information presented in this Section.

Operational History and Current Use

Building 32 was a Navy torpedo overhaul shop from the 1940s until it ceased major operations in the 1950s. During that time, torpedoes were brought to the overhaul shop for dismantling, cleaning, and reassembly. Operations within Building 32 included degreasing, parts washing, electroplating, sandblasting, mechanical and electrical testing, etc. Peripheral to Building 32, but featuring in the environmental investigations were Building 33 (a steam plant); Building 34 (a small building for generating acetylene); Building 44 (a fuel pump house and associated underground fuel tanks); Building 41 (a storage shed); and a series of five small transformer buildings – Buildings 53, 54, 56, 58, and 60. All these former buildings are within the footprint of the Site 17 boundary. Minor structures that featured in the environmental releases were an acid storage shed and dust storage equipment associated with indoor sandblasting operations. Other storage sheds and

structures (administration building, guard shacks, etc.) were also present within the Site 17 boundary (all have been removed) and are not pertinent to the environmental conditions at the Site.

Groundwater at the Site is not currently being used for potable use, and there are no drinking water wells present at or within the immediate vicinity of the Site. Groundwater beneath Site 17 is classified as GA by the State of Rhode Island.

In 2016, the soil remedy (“source control measure”) as described in the ROD took place. The remedy consisted of removal and offsite disposal of sump debris from Former Building 32 foundation sumps, and excavation and offsite disposal of impacted vadose zone soils from identified impact areas. These actions not only mitigated risk from contact with contaminants of concern (COCs) in surface soils and sump debris, but also acted as a “source control measure” for shallow groundwater.

Summary of Topography, Geology, and Hydrogeology

During development of Gould Island in the 1940s, the Site was altered by cut-and-fill operations, cutting the natural soil from the hillside at the southern end of the Site and filling the sub-tidal waters to the north to create a level area large enough to construct Building 32 and its support systems. As a result, the northern portion of the Site contains fill comprised primarily of disturbed native soils from the area, consisting of a mix of silty sand and gravel. The thickness of the fill ranges from about 6 to 12 feet, with greater thicknesses at the northern end of the Site. Portions of the Site are underlain by glacial till consisting of silt, sand, and gravel. The till unit has not been encountered at the southern end of the Site but was encountered during drilling at thicknesses of up to 74 feet at the northern end of the Site beneath the fill. Bedrock underlying the Site is characterized as metamorphosed sedimentary rock, predominantly phyllite. This rock was found to be weathered in the upper 3 to 18 feet. As a result of the softness of the weathered zone, the transition from overburden to bedrock has not been clearly defined at the northwestern end of the Site where the depth to rock is greater than 30 feet. At the eastern shoreline of the island (and south of the Site), the overburden is very thin or nonexistent; the bedrock is exposed in places and is eroding due to wave action, forming a shingle-style beach face.

Although it has not been specifically assessed, local tidal cycles are expected to have an impact on local groundwater elevations and flows. In general, overburden groundwater flow at Site 17 radiates towards the north, east, and west, generally away from the island interior, following topography. Bedrock groundwater has been observed to flow towards the north and east, but a lack of bedrock wells along the western side of Site 17 likely skews the interpretation toward the north and east. Groundwater contours from the September 2019 LTM event are depicted on Figure 4-2. Bedrock groundwater has not been adequately characterized with respect to flow pattern, although it is expected to behave in a similar manner as overburden groundwater.

Conceptual Site Model Summary

Historical investigations and long-term groundwater monitoring performed by, or on behalf of the Navy, at Site 17 have indicated that presumed releases from various industrial processes related to the Site’s former torpedo overhaul operations have resulted in the presence of contaminants in soil and groundwater in specific areas of the Site and in nearby marine sediments, primarily in Stillwater Area sediments in the vicinity of the now demolished rigging platform. COCs were identified in the Remedial Investigation (RI) and

associated risk assessments. COCs for surface soil (0 to 2 feet below ground surface [bgs]) and subsurface soil (greater than 2 feet bgs or below a foundation) are: select semivolatile organic compounds (SVOCs), PCBs (total Aroclors), arsenic, cadmium, lead, and manganese. COCs for groundwater are: select PAHs, pentachlorophenol (PCP), benzene, trichloroethylene (TCE), tetrachloroethene (PCE), PCBs (total Aroclors), arsenic, and manganese. COCs for sediment are: total PAHs, total PCBs, arsenic, cadmium, chromium, copper, lead, nickel, and zinc.

PCBs were released at previous transformer areas, and although the bulk of contaminated soil and sump debris has been removed, residual concentrations of PCBs appear to have intermingled with other contaminants in marine sediment. PAHs were found in sump debris and soil within and around the former Building 32 footprint, where drains and/or overland runoff also carried them to nearby marine sediments. Sediments in limited areas at the northeast shoreline, generally near some of the outfall pipes, have been found to contain PCBs, PAHs, and metals at lower concentrations than those detected in Stillwater Area sediments. These chemicals at the Northeast Shoreline are most likely a result of discharges from pipes that drained waste from specific building operations when the Site was active.

The SVOC PCP was found in building sump debris as well as groundwater. The MCL for PCP was exceeded during RI investigations at one groundwater sample location, only. During the RI, PCE was also detected in groundwater at a concentration exceeding the MCL at only one location, near a former parts shop in former Building 32, indicating possible use and release of cleaning solvents. PCP was not detected at concentrations exceeding the MCL in groundwater samples collected during the first two years of groundwater LTM.

Metals used and metal contaminants produced through electroplating and sandblasting operations in the southwestern corner of the former Building 32 appear to have been released to nearby sediments via building and roadway drainage systems. Manganese, a naturally occurring element in soil and rock, is present at elevated concentrations (above the EPA drinking water health advisory) in the groundwater across the Site.

Contaminants such as PCBs and PAHs are highly persistent and when released to the environment and are likely to remain adsorbed to the soil or sediment matrix, bound to particulate matter. As a result, these contaminants tend to migrate from source areas via bulk movement processes (e.g., transported as particles in surface water runoff or in wind erosion of soil). If leaching from soil to groundwater occurs, any migration that may take place is usually over relatively short distances. For these compounds, biodegradation is possible but is likely to be slow.

Metals are also persistent in the environment, and many metals such as cadmium and lead tend to adsorb strongly to soil particles or colloids, especially when the soil is of high organic content, and become more soluble under reducing conditions. Arsenic and manganese, however, are examples of metals that are generally more mobile. Soluble metals may be leached from native or contaminated soil to groundwater by infiltration of precipitation and through the seasonal rise and fall of the water table. Once dissolved in groundwater, metals will travel with groundwater flow. As groundwater migrates, some of the metals will undergo transformation processes that result in their return to an insoluble state. Oxidation-reduction reactions, precipitation, and adsorption reactions can cause the dissolved phase ions to leave the aqueous phase. The required timeframe for manganese to attenuate to concentrations in groundwater below cleanup goals is currently estimated at between 17 and 51 years for overburden, and between 23 and 69 years for

bedrock; this is based on a predicted rate for three volumes of groundwater to fully flow through the saturated zone at the Site. The predicted rate was calculated with groundwater flow data collected from select LTM program wells during the September 2018 LTM event; details are presented in the Draft Year 2, LTM Report (Tetra Tech, 2020).

During the RI, PCE and PCP were present in water-table overburden groundwater near the shoreline and were not found in deep overburden or bedrock groundwater at the Site. Given their location and concentration, it is likely that these COCs would flow with groundwater to discharge into the ocean to the east. Surface water samples were not collected from Narragansett Bay as part of the investigations because the tidal nature of the bay and the extremely large mixing zone associated with tidal waters limits confidence that data would have been representative of site related contamination. As a conservative measure, the groundwater data were evaluated in the Baseline Ecological Risk Assessment (ERA) to determine potential risks to marine organisms after groundwater discharges to surface water. Groundwater results from the first two years of groundwater LTM indicate that natural attenuation of PCE and PCP to concentrations less than clean-up levels has already occurred because there have been no exceedances of PALs during the first eight LTM events.

A ROD which presents and describes the remedy for soil, sediment, and groundwater at Site 17 was signed on June 27, 2014. The selected remedies were removal of sediment, removal of soil and debris (completed 2016), implementation of land use controls (LUCs), monitored natural attenuation (MNA), long-term groundwater monitoring, limited sediment monitoring, and annual inspections and reporting.

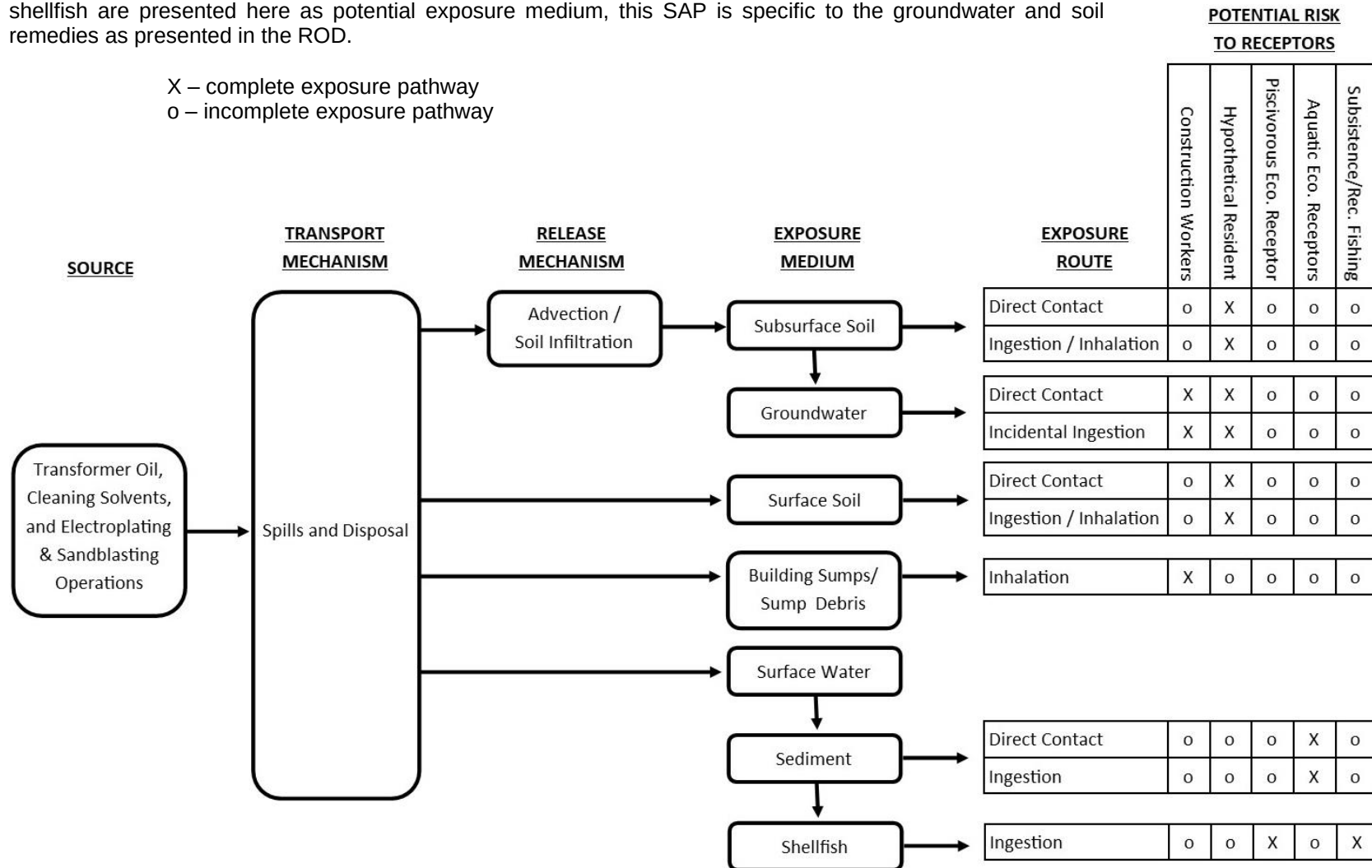
This SAP pertains only to the MNA and LTM of groundwater required by the ROD.

The following receptors and exposure pathways were identified in the ROD:

- Future construction workers by exposure from direct contact/incidental ingestion of shallow groundwater, and from inhalation of associated trench air, primarily due to concentrations of SVOCs, including PAHs and pentachlorophenol, and PCBs.
- Adults and children consuming shellfish at a subsistence or recreational level, primarily due to concentrations of PCBs and PAHs in shellfish (mussels and clams).
- Exceedances of regulatory criteria in the screening assessment indicated a potential risk to hypothetical future residents from exposure to soil and groundwater.
- Ecological receptors from exposure to arsenic, chromium, copper, lead, nickel, zinc, PAHs, and PCBs in sediment

CSM Wire Diagram – Based on risk identified during the RI and described in the ROD. Although sediment and shellfish are presented here as potential exposure medium, this SAP is specific to the groundwater and soil remedies as presented in the ROD.

X – complete exposure pathway
 o – incomplete exposure pathway



5.0 Data Quality Objectives/Systematic Planning Process Statements

[\(UFP-QAPP Manual Section 2.6.1 – Worksheet #11\)](#)

5.1 Problem Statement

This monitoring effort is necessary to comply with the groundwater remedy described in the ROD. The remedy includes monitoring concentrations of PCP, PCE, manganese, and natural attenuation parameters in groundwater to support evaluation of MNA; and monitoring concentrations of both soil and groundwater COCs in groundwater to document whether they are present at concentrations exceeding cleanup goals.

Based on the problem statement, the specific study goals of the planned monitoring are listed:

- Goal 1 – Evaluate concentrations of PCP, PCE, and manganese in groundwater and monitor natural attenuation parameters to monitor natural attenuation of these COCs and ensure that the MNA remedy as stated in the ROD is appropriate.
- Goal 2 – Evaluate concentrations of soil and groundwater COCs in groundwater to identify potential data trends. Monitoring for both the soil and groundwater COCs in groundwater will allow for comparisons to be made regarding chemicals detected in groundwater and whether they are related to Site operations.

5.2 Information Inputs

Information inputs required to resolve the problem statement are:

- Groundwater sample analytical results for soil and groundwater COCs.
- Filtered and unfiltered metals analytical results.
- Analytical results for MNA parameters (anions, alkalinity, dissolved oxygen (DO), oxidation-reduction potential (ORP), conductivity, and pH) to evaluate conditions for natural attenuation.
- Field parameter measurements (pH, DO, ORP, temperature, turbidity, and conductivity) to evaluate well stabilization conditions prior to sampling and natural attenuation.
- Static groundwater gauging data to establish groundwater elevations at the time of the field events.
- Project Action Levels (PALs) that are based on federal and state promulgated risk-based criteria and are presented in Section 9, which will be used to evaluate data through comparison with measured analyte concentrations.

To conduct comparisons of Site data to PALs, the selected laboratory should have Limits of Quantitation (LOQs) that are low enough to measure constituent concentrations that are less than the PALs. Analytical data reported by the laboratory must use the following reporting conventions: All results less than the Detection Limit (DL) will be considered non-detects; positive results reported at concentrations between the DL and LOQ will be reported with a "J" qualifier; and analytes not found (not detected) in a sample will be reported at the Limit of Detection (LOD) with a "U" qualifier. If the PAL is less than the DL or LOQ such results will be discussed in the uncertainty portion of the report. To remain consistent with previous LTM events EPA stage 2A data validation is needed on 100 percent of the data.

5.3 Boundaries of the Study

Site 17 encompasses the 6-acre area around former Building 32, including the soil and groundwater and approximately 1 acre of marine sediment adjacent to the northern shoreline where contaminants from the

former Building 32 and its operation have come to reside. The focus of this SAP is groundwater in bedrock and overburden. General spatial and temporal boundaries are described below.

Spatial Boundaries

The population of interest for this investigation is shallow groundwater potentially impacted by COCs within the 6-acre Site but also includes the groundwater upgradient of the Site where background monitoring wells are located.

Evaluation of LTM concentration trends to evaluate natural attenuation progress must be made for individual wells, but for estimating extent of contamination and other potential uses, the collection of data across a larger number of Site wells may be appropriate and will be dictated by context.

Temporal Boundaries

The LTM program is expected to extend for the foreseeable future; therefore this 2020 SAP is designed to support LTM for the foreseeable future with minor changes made with FTMRs. The types of modifications anticipated are adjustments to the sampling frequency, installation of new wells and COCs, or removal of existing wells and COCs from the program, etc. The modifications will be based on qualitative analysis of each year's data.

Biannual sampling and groundwater COC analyses are required for the foreseeable future. Synoptic water level gauging events (described in Section 8) must be performed within one hour to limit influence from tides.

5.4 Analytic Approach

The analytic approach for the planned monitoring is comprised of a series of "if... then...else..." statements. In each case, it is assumed that regulators will be involved in the decision making as part of the project team.

Goal 1 – Evaluate concentrations of PCP, PCE, and manganese in groundwater and monitor natural attenuation parameters to determine if natural attenuation of these COCs is an effective remedy as stated in the ROD and if natural attenuation conditions exist at the Site. Additionally, although daughter products of PCE (including vinyl chloride [VC] and 1,2-dichloroethene [1,2-DCE]) are not identified as COCs in the ROD, as part of monitoring natural attenuation, their concentrations must be monitored as part of the MNA process.

Decision Rule Number 1 (applied to each well for the COCs PCE, PCP, and manganese annually):

If the measured groundwater concentrations of PCE, PCP, and manganese have all decreased to less than PALs in any well for two full years of monitoring, and if visual inspection of data trends indicate that such concentrations are not increasing, recommend cessation of monitoring for these parameters and MNA parameters in that well; otherwise, convene the project team to evaluate whether MNA parameters and COC concentration trends support natural attenuation of these COCs and develop a path forward that is protective of human health and the environment.

Note: Indications of viable natural attenuation include decreasing COC concentration trends and MNA parameter concentrations that support biodegradation of these three COCs as described in the *Use of Monitored Natural Attenuation at Superfund, Resource Conservation and Recovery Act*

(RCRA) Corrective Action, and Underground Storage Tank Sites; US EPA Office of Solid Waste and Emergency Response Directive 9200.4-17P (EPA, 1999).

Goal 2 – Evaluate concentrations of soil and groundwater COCs (VOCs excluding PCE, SVOCs excluding PCP, PCBs, and metals excluding manganese) in groundwater to identify potential data trends. Soil sampling is not required to support this goal.

Decision Rule Number 2 (applied to each well for VOCs excluding PCE; SVOCs excluding PCP; PCBs; and metals excluding manganese after 8 rounds of sampling):

If the measured concentrations of any COC is less than its PAL in any well for two full years of monitoring, and visual inspection of the LTM data does not indicate that that COC concentration is increasing; or if any COC concentration exceeds its PAL but appears to be decreasing toward the PAL in the selected well, then evaluate the option to cease monitoring for that parameter in accordance with Navy optimization guidance (DoN, 2010); otherwise review project objectives against the observed spatial and temporal concentration patterns to develop a strategy that continues to be protective of human health and the environment with regard to groundwater contamination.

5.5 Performance Criteria

Potential Sources of Error

Potential sources of error in the investigation may be divided into two main categories: sampling errors and measurement errors. A sampling error can occur when the sampling design, planning, and implementation do not provide for a representative range of heterogeneity at the Site. A measurement error can occur when performance variance from laboratory instrumentation, analytical methods, and/or operator error occurs. EPA identifies the combination of these errors as a “total study error” (EPA, 2006). One objective of the planned investigation is to reduce the total study error so that decision-makers can be confident that the data collected accurately represent the chemical characteristics of the Site.

Managing Decision Error

The investigation will utilize decision-error minimization techniques in sampling design, sampling methodologies, and laboratory measurement of compounds of interest. Possible decision errors will be minimized during the field investigation by using the following methods:

- Use standard field sampling methodologies (as discussed in [Sections 8.2 and 8.3](#)). Perform sampling activities in accordance with the standard operating procedures (SOPs) referenced in this SAP. Tetra Tech SOPs are included in Appendix A.
- Use applicable analytical methods (discussed in [Section 10](#)) for sample analysis by a competent analytical laboratory certified by the DoD Environmental Laboratory Accreditation Program (ELAP) to reduce measurement errors.
- Confirm analytical data to identify and control potential laboratory error and sampling error by using spikes, blanks, and replicated samples.
- For analytical results that do not achieve desired project action limits (PALs), include a discussion of alternate information inputs as part of the assessment conclusions and uncertainty analysis. The PAL represents the limits of detection that analytical data must meet in order to be of sufficient

quality for use to support decisions regarding continued monitoring and remedy effectiveness. Decision errors associated with judgmental sampling are based on sample design and measurement errors. Assuming that the best possible professional judgment was used to develop the sampling plan (e.g., position sampling locations), the most important decision errors will be associated with field and laboratory techniques involved in the collection and analysis of the data.

Sampling Methods and Procedures

Possible decision errors generated by sampling errors will be minimized during the field investigation by applying standardized field sampling and field measurement methodologies (to be discussed in [Sections 6, 7, and 8](#)).

Field Data Collection Methods and Procedures

Possible decision errors generated by data collection errors will be minimized during the field investigation by applying standardized field measurement methodologies (to be discussed in [Sections 6, 7, and 8](#)).

Laboratory Measurement of Compounds of Interest

Possible decision errors generated by laboratory measurement errors will be minimized by using applicable analytical methods (to be discussed [Section 10](#)) for sample analysis by a competent analytical laboratory evaluated and certified by the DoD ELAP.

Managing Laboratory Sampling Error

Control of potential laboratory error and sampling error will be minimized by using spikes, blanks, and duplicates. Sampling error may be introduced when the laboratory chemist selects a single portion of the field sample for laboratory analysis.

5.6 Plan for Obtaining Data

The selected sampling design is a biased design that make use of existing groundwater monitoring wells spanning the areas of interest to ensure adequate spatial coverage (including horizontally across the site as well as vertically in both overburden and bedrock aquifers). The LTM groundwater monitoring well network was selected with input from the project team during development of the initial LTM SAP (Resolution, 2017). Additional details are provided in Section 8.0.

6.0 Field Quality Control Samples

(UFP-QAPP Manual Section 2.6.2 – Worksheet #12)

Measurement Performance Criteria Table – Field QC Samples

QC Sample	Analytical Group	Frequency	Data Quality Indicators (DQIs)	Measurement Performance Criteria
Field Duplicates	All Groups	1 per 10 field samples	Precision	If both the original and duplicate results are $\geq 2 \times$ LOQ Relative percent difference (RPD) must be $\leq 30\%$. If either the original or duplicate result is $< 2 \times$ LOQ, use professional judgment.
Trip Blank	VOCs only	1 per cooler containing aqueous VOC samples	Accuracy/Bias/Contamination	No analytes $\geq \frac{1}{2}$ LOQ, except common laboratory contaminants, which must be $< \text{LOQ}$.
Equipment Rinsate Blank	All analytical groups included in the associated samples	1 per 20 field samples per sampling equipment ⁽¹⁾	Accuracy/Bias/Contamination	No analytes $\geq \frac{1}{2}$ LOQ, except common laboratory contaminants, which must be $< \text{LOQ}$.
Cooler Temperature Indicator	All analytical groups	1 per cooler	Representativeness	Temperature must be between 0 and 6 degrees Celsius ($^{\circ}\text{C}$), but samples must not be frozen.
Matrix Spike (MS) ²	All analytical groups	1 per 20 discrete field samples collected	Accuracy/Bias	See Section 11.0.
Matrix Spike Duplicate (MSD) ²	All analytical groups (except metals)	1 per 20 discrete field samples collected	Accuracy/Bias and precision	See Section 11.0.

Notes:

1 The lot number of the source water (if applicable) will be recorded in the field notes.

2 MS/MSDs are prepared in the laboratory but additional sample volume must be collected to support these analysis.

7.0 Sampling Design and Rationale

[\(UFP-QAPP Manual Section 3.1.1 - Worksheet #17\)](#)

Groundwater

This section describes the sampling rationale and design for the collection of groundwater samples at Site 17.

The LTM groundwater monitoring well network includes 21 monitoring wells. Groundwater monitoring wells selected for sampling are intended to provide sufficient geographic coverage for evaluating MNA parameters, and confirming COC concentrations in groundwater.

LTM activities will be performed for the purpose of supporting the eventual regulatory closure of Site 17. Further details regarding the specific DQOs for the planned investigation are presented in [Section 5](#), and PALs are listed in [Section 8.3](#).

Groundwater Sampling

The LTM program will include biannual sampling of the groundwater LTM monitoring well network. In 2017 (the start of groundwater LTM), 22 monitoring wells were initially selected to be included in the groundwater LTM program monitoring well network; 12 wells were existing, 4 wells were proposed to replace destroyed wells, and six new wells were proposed for installation to provide adequate geographic coverage. During February and March of 2018, one of the groundwater monitoring wells initially scheduled to be included in LTM at Site 17 (G44S-MW207; presented on Figure 4-2) was destroyed when a series of powerful storms eroded the ground surrounding the well and subsequently filled it with sand after the PVC riser broke. The Navy opted to not replace this well because any well in the vicinity of this well is very likely to also be destroyed. This decision was documented in FMR007 dated March 27, 2018, and submitted to the project team in June 2018 with the March 2018 LTM event data memorandum. The LTM well network now consists of 21 monitoring wells and is presented on Figure 7-1.

To support Goals 1 and 2 (listed in [Section 5](#)), wells will be gauged to measure depth to groundwater, and groundwater samples will be collected from the monitoring wells listed in [Section 8.3](#) using a low-flow method. The specific methodology and procedures to be implemented for groundwater collection and sampling are included within the SOPs provided with this SAP ([Section 8.2](#)). The list of analytical parameters, which are based on objectives of this LTM program, is included in [Section 8.3](#) and [9.0](#). Additionally, QA/QC samples will be collected in accordance with [Section 8.3](#) of this SAP.

Analytical parameters include the Site COCs and MNA parameters (refer to [Section 8.3](#)). COCs are listed in [Section 9](#). All 21 groundwater monitoring wells will be sampled biannually. [Section 8.3](#) presents the specific analysis requested for each sample location.

8.0 Field Project Implementation (Field Project Instructions)

This section presents details as to how specific LTM tasks will be completed.

8.1 Field Project Tasks

[\(UFP-QAPP Manual Section 2.8.1 – Worksheet #14\)](#)

8.1.1 Synoptic Water Level Gauging

A synoptic water level gauging event will be performed at LTM program monitoring wells prior to purging and sampling to gather information to assess groundwater flow direction. Water levels will be measured in accordance with Tetra Tech SOP GH-2.5. An oil water interface probe will be used to complete this task so that if oil is present in any wells, it will be identified.

Because the Site is surrounded by Narragansett Bay which is tidal, the synoptic water level gauging event will be performed within one hour. Additionally, field personnel will record at which point in the tidal cycle the synoptic water level begins and ends.

8.1.2 Groundwater Quality Monitoring and Sample Collection

Each well will be gauged for depth to groundwater in accordance with SOP GH-1.2. Groundwater will then be purged and sampled in accordance with Tetra Tech SOP SA-1.1 and US EPA Region 1 Low-Flow Sampling Guidance GW 0001 (EPA, 2010). Groundwater will be purged using a peristaltic pump or bladder pump, and stabilization parameters will be monitored using a YSI multi-meter with a flow-through cell or similar equipment. Stabilization parameters are: temperature, pH, dissolved oxygen (DO), specific conductivity, oxidation reduction potential (ORP), and turbidity per low-flow sampling guidance. Additionally, field personnel will record at which point in the tidal cycle groundwater purging begins and ends at each well. Groundwater samples will be placed in laboratory-provided containers, preserved as required per analytical methods, stored on ice, and transported under chain-of-custody (CoC) to the analytical laboratory. Refer to [Sections 8.3 and 9](#) for analytical parameters, methods, and procedures.

8.1.3 Quality Assurance/Quality Control

The QA/QC sample collection frequency is presented in [Section 8.3](#).

8.1.4 Equipment Decontamination

To the maximum extent possible, Helios/TT JV field personnel will utilize dedicated and disposable sampling equipment to avoid the potential for cross-contamination of samples due to inadequate decontamination processes. Sampling equipment will include disposable non-Teflon or polyethylene tubing and bladders, disposable gloves, and laboratory supplied sample bottles.

Non-disposable or non-dedicated sampling equipment will be decontaminated prior to sampling and between sample locations. Equipment decontamination will be completed in accordance with Tetra Tech SOP SA-7.1, included in Appendix A. All decontamination water generated during decontamination of sampling equipment will be containerized as IDW and properly disposed of.

8.1.5 Investigation-Derived Waste Management

In accordance with RIDEM's Policy Memo 95-01, Guidelines for the Management of Investigation Derived Wastes (RIDEM, 2010a), IDW consisting of purge water and water generated during decontamination processes will be collected in properly labeled 55-gallon drums and temporarily stored on the Site (or at NAVSTA Newport waste storage warehouse¹) for subsequent off-site disposal as necessary. Subsequently, the containers will be characterized and properly disposed at a Navy approved disposal facility. Note that, if laboratory analytical results indicate purge and decontamination water are less than remedial goals, the water will be returned to the ground surface.

Waste removal from the Site will be documented by manifest or bill of lading. The Navy will be named as the generator of the waste and a representative of the Navy will sign waste profile forms and manifests. The waste disposal subcontractor will prepare disposal manifests or bills of lading and documentation for the Navy's use.

8.2 Laboratory Coordination, Data Management, and Validation

Helios/TT JV's Project Chemist will track the samples from collection through analysis. Data deliverables will be provided by the laboratory within 21 business days of sample receipt. Final data deliverables will be submitted in hard copy, pdf (bookmarked and searchable), and in electronic form. Electronic data results will be automatically downloaded into the Tetra Tech database in accordance with the proprietary Tetra Tech processes and will subsequently be converted by Helios/TT JV to the Navy NEDD data structure).

8.2.1 Analytical Tasks – Fixed-base laboratory chemical analyses for all analytes will be performed by Katahdin of Scarborough, Maine. This laboratory has been accredited to conduct the analyses required by this SAP by the DoD ELAP, and Rhode Island Department of Health through the National Environmental Laboratory Accreditation Program (NELAP). Copies of the applicable laboratory accreditations are located in Appendix B.

Analyses will be performed in accordance with the analytical methods specified in [Section 8.3](#). The laboratory will meet the PALs as shown in [Section 9](#). The laboratory will perform chemical analysis following laboratory-specific SOPs ([Section 10](#)) developed based on the analytical methods listed in [Section 8.3](#).

8.2.2 Data Handling and Management – The principal data generated for this project will be field records and laboratory analytical data. Field sampling log sheets will be prepared in hard copy, organized by date and medium, and stored in the project files. The field logbooks for this project will be used only for this Site and will also be categorized and maintained in the project files after the completion of the field program. Project personnel completing concurrent field sampling activities may maintain multiple field logbooks. When possible, logbooks will be segregated by sampling activity. The field logbooks will be titled based on date and activity.

The data handling procedures to be followed by the laboratory will meet the requirements of the technical specifications. Electronic data results will be automatically downloaded into the Tetra Tech database in accordance with proprietary Tetra Tech processes.

¹ IDW can only be stored at the NAVSTA Newport Waste Storage Warehouse if it has been characterized within the previous 1 year as Non-Hazardous, Non-DOD, Non-RCRA regulated waste. In accordance with NAVSTA Newport direction, IDW generated during support of the Site 17 LTM program is to be characterized once per year.

The Helios/TT JV PM (or designee) is responsible for the overall tracking and control of data generated for the project.

8.2.3 Data Tracking – Data are tracked from generation to archiving in the Helios/TT JV project-specific files. The Helios/TT JV Project Chemist (or designee) is responsible for tracking the samples collected and shipped to the laboratory. Upon receipt of the data packages from the laboratory, the Helios/TT JV Project Chemist will monitor the data validation effort, which includes verifying that the data packages are complete and results for all samples have been delivered by the laboratory.

8.2.4 Data and Records Storage, Archiving, and Retrieval – The data packages received from the laboratory are tracked in the data validation logbook. After the data are validated, the data packages are entered into the Tetra Tech Navy file system and archived in secure files. The field records including field log books, sample logs, chain-of-custody records, and field calibration logs will be submitted by the Helios/TT JV FOL to be entered into the Navy file system prior to archiving in secure project files. Project files are audited for accuracy and completeness. As documents are finalized, all relevant data are uploaded electronically to NIRIS and retained there indefinitely. Upon project completion, paper copies go to the Federal Record Center for long-term storage (50 years archival).

8.2.5 Data Security – Access to Helios/TT JV project files is restricted to designated personnel only. Records may only be borrowed temporarily from the project file using a sign-out system. The Helios/TT JV Data Manager maintains the electronic data files, and access to the data files is restricted to qualified personnel only. File and data backup procedures are routinely performed.

8.2.6 Electronic Data – All electronic data are validated, and qualifiers are manually added to the database. The manually entered qualifiers are verified by the DVM by direct QA, as well as an electronic verification step using proprietary Tetra Tech processes. Data will then be compiled into a NEDD and loaded into NIRIS in accordance with Navy procedures. This process includes a QA review of the data to ensure that the content and format of the data satisfy the requirements of NIRIS uploads. The NEDD is submitted through a data checker into NIRIS, which also ensures the format is acceptable.

8.2.7 Data Reporting – Data will be generated by the laboratory in a Contract Laboratory Program-like format.

8.2.8 Data Review – This review comprises data verification, validation, and usability assessment. The data verification and validation processes and requirements are described in [Section 12](#). The data usability assessment will, at a minimum, constitute evaluations to ensure that the amount, type, and quality of data are sufficient to achieve project objectives. The means of conducting these evaluations will vary depending on the nature of the data. For example, soil borings and well construction logs will generally be evaluated qualitatively or semi-quantitatively whereas precision, accuracy, and sensitivity of analytical data will generally be evaluated quantitatively and may be based on, or may supplement, data validation findings.

Analytical results from the final data deliverables will be reviewed according to the procedures in [Section 12](#). Only results from final data deliverables will be formally validated. Helios/TT JV will add data validation qualifiers to the electronic version of the final data. The validated data will subsequently be loaded into the

Tetra Tech and NIRIS databases. The level of validation to be performed, EPA Stage 2A, focuses evaluation of batch-specific and sample-specific measurement performance indicators (MPIs) and the MPIs which capture sampling error. The underlying assumption is that more rigorous validation is unnecessary because the laboratory maintains DoD ELAP accreditation and will follow the DoD Quality Systems Manual (QSM) in performing analyses and because the data trends are useful for identifying deviations from the trend that could be indicators of quality deficiencies. The types of review elements to be evaluated during the data validation are:

- Data Completeness
- Preservation (and Technical Holding Time)
- Blanks
- Surrogates
- MS/MSDs
- Laboratory Control Spike/Spike Duplicate (LCS/LCSD)
- Field Duplicates
- Compound Quantitation Limits (including percent solids)

These Stage 2A review elements are defined in the following EPA National and New England data validation guidelines.

- EPA New England Environmental Data Review Supplement For Regional Data Review Elements and Superfund Specific Guidance/Procedures (US EPA, 2013).
- USEPA National Functional Guidelines for Inorganic Superfund Methods Data Review (SOM02.4). EPA-540-R-2017-001. January (EPA, 2017a). <https://www.epa.gov/clp/contract-laboratory-program-national-functional-guidelines-data-review>.
- USEPA National Functional Guidelines for Organic Superfund Methods Data Review (SOM02.4). EPA-540-R-2017-002. January (EPA, 2017b). <https://www.epa.gov/clp/contract-laboratory-program-national-functional-guidelines-data-review>.

8.3 Report Preparation

All data will be reported in an LTM Report that is to be produced every 2 years. The LTM Report will evaluate monitoring data to determine if the goals of this SAP have been met. The report will be prepared using the following outline as guidance:

- Introduction
- Site Description
- Selected Remedy and Monitoring Program
- Long-Term Groundwater Monitoring Activities
- Data Summary and Evaluation
- Conclusions and Recommendations

In addition to the biannual LTM reports, data will be provided to the project team following each LTM event once it has been validated.

Field Project Implementation (Field Project Instructions), continued

8.4 Field SOPs Reference Table

(UFP-QAPP Manual Section 3.1.2 – Worksheet #21)

SOP Reference Number	Title/Author	Revision Date or Version Number	Location of SOP (if not included in the SAP)	Any planned deviation for Project Work	Comments
EPA SOP GW-001	Low Stress Purging and Sampling Procedure for the Collection of Groundwater from Monitoring Wells/ US EPA, Region 1, Quality Assurance Unit	Revision 4. September 19, 2017.	Various	No	None
GH-1.2	Evaluation of Existing Monitoring Wells and Water Level Measurement/Tetra Tech	Revision 5. April 30, 2020.	Oil Water Interface Probe/Water Level Indicator	No	None
GH-2.5	Groundwater Contour Maps and Flow Determinations/Tetra Tech	Revision 3. February 20, 2017.	None	No	None
SA-1.1	Groundwater Sample Acquisition and Onsite Water Quality Testing/Tetra Tech	Revision 9. July 15, 2016.	Various	No	None
SA-1.6	Natural Attenuation Parameter Collection/Tetra Tech	Revision 3. February 17, 2017.	Various	No	None
SA-6.1	Non-Radiological Sample Handling/Tetra Tech	Revision 6. April 30, 2020.	None	No	None
SA-6.3	Field Documentation/Tetra Tech	Revision 7. April 30, 2020.	None	No	None
SA-7.1	Decontamination of Field Equipment and Investigation-Derived Waste Management/Tetra Tech	Revision 9. April 30, 2020.	None	No	None
RIDEM Policy Memo 95-01	IDW Management/RIDEM	<i>No revision number or date referenced</i>	None	No	None

SOP Reference Number	Title/Author	Revision Date or Version Number	Location of SOP (if not included in the SAP)	Any planned deviation for Project Work	Comments
CT-04	Sample Nomenclature//Tetra Tech	Revision 6. April 30, 2020.	None	No	None

Notes:

If the field crew suspects that high concentrations of target analytes exist in any of the samples, the chain of custody form should be marked to identify to the laboratory those samples that potentially contain high concentrations. This will help the laboratory prevent cross-contamination of other samples and instrument down time that can be caused when high concentration samples are analyzed.

Field Project Implementation (Field Project Instructions), continued

8.5 Sample Details Table

(UFP-QAPP Manual Sections 3.1.1 and 3.5.2.3 – Worksheets #18, 19, 20 and 30)

CTO N4008520F4592, Site 17 Long-Term Monitoring Site 17, Operable Unit 06 Naval Station Newport, Jamestown, Rhode Island						Analysis Group	VOCs	Metals ⁴	PAHs/SVOCs	PCBs	Anions ⁶	Alkalinity ⁶
						Preparation and Analytical Method	SW-846 8260C	SW-846 3010A/60 20A	SW-846 3510C or 3520C, 8270D SIM	SW-846 8082A	US EPA 300.0	SM 2320B
						Analytical Laboratory/ Analytical SOP Reference	KAS / CA-202	KAS / CA-627	KAS / CA-213	KAS / CA-329	KAS / CA-742	KAS / CA-739
						Data Turnaround Time ¹	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days
Katahdin Analytical Services ⁵ 600 Technology Way, Scarborough, ME 04074 Project Manager: Heather Manz Ph. 207-874-2400, hmanz@katahdinlab.com						Container Type/Volume required (if different than container volume)	3 x 40mL VOA	500-mL plastic	2 x 1000-mL Amber Jar	2 x 1000-mL Amber Jar	250 mL plastic or glass	250-mL plastic
						Preservative	pH<2 with HCl	pH<2 with HNO ₃	Cool to ≤ 6 °C; store in the dark	Cool to ≤ 6 °C; store in the dark	Cool to ≤ 6 °C	Cool to ≤ 6 °C
						Holding Time (Preparation/ Analysis) ³	7 days to analysis	6 months to analysis	7 Days Extraction 40 Day Analysis	30 Days Extraction 40 Day Analysis	Nitrate 48 hours to analysis Sulfate and Chloride 28 days to analysis	14 days to analysis
Site	Matrix	Station ID	Sample ID ⁷	Coordinates ⁸		Sample Type						
				X	Y							
17	GW	300B	G32-MW300B	370706	165380	Normal	X	X	X	X	X	X
17	GW	300S	G32-MW300S	370707	165368	Normal	X	X	X	X	X	X
17	GW	301B	G32-MW301B	370580	164711	Normal	X	X	X	X	X	X
17	GW	301S	G32-MW301S	370586	164705	Normal	X	X	X	X	X	X
17	GW	302B	G32-MW302B	370554	165123	Normal	X	X	X	X	X	X
17	GW	302S	G32-MW302S	370560	165114	Normal	X	X	X	X	X	X
17	GW	303B	G32-MW303B	370851	165079	Normal	X	X	X	X	X	X

CTO N4008520F4592, Site 17 Long-Term Monitoring Site 17, Operable Unit 06 Naval Station Newport, Jamestown, Rhode Island						Analysis Group	VOCs	Metals ⁴	PAHs/SVOCs	PCBs	Anions ⁶	Alkalinity ⁶
						Preparation and Analytical Method	SW-846 8260C	SW-846 3010A/60 20A	SW-846 3510C or 3520C, 8270D SIM	SW-846 8082A	US EPA 300.0	SM 2320B
						Analytical Laboratory/ Analytical SOP Reference	KAS / CA-202	KAS / CA-627	KAS / CA-213	KAS / CA-329	KAS / CA-742	KAS / CA-739
						Data Turnaround Time ¹	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days
Katahdin Analytical Services ⁵ 600 Technology Way, Scarborough, ME 04074 Project Manager: Heather Manz Ph. 207-874-2400, hmanz@katahdinlab.com						Container Type/Volume required (if different than container volume)	3 x 40mL VOA	500-mL plastic	2 x 1000-mL Amber Jar	2 x 1000-mL Amber Jar	250 mL plastic or glass	250-mL plastic
						Preservative	pH<2 with HCl	pH<2 with HNO ₃	Cool to ≤ 6 °C; store in the dark	Cool to ≤ 6 °C; store in the dark	Cool to ≤ 6 °C	Cool to ≤ 6 °C
						Holding Time (Preparation/ Analysis) ³	7 days to analysis	6 months to analysis	7 Days Extraction 40 Day Analysis	30 Days Extraction 40 Day Analysis	Nitrate 48 hours to analysis Sulfate and Chloride 28 days to analysis	14 days to analysis
Site	Matrix	Station ID	Sample ID ⁷	Coordinates ⁸		Sample Type						
				X	Y							
17	GW	304S	G32-MW304SR	370372	165367	Normal	X	X	X	X	X	X
17	GW	305B	G32-MW305B	370507	165337	Normal	X	X	X	X	X	X
17	GW	305S	G32-MW305S	370506	165344	Normal	X	X	X	X	X	X
17	GW	306BR	G32-MW306BR	370549	165571	Normal	X	X	X	X	X	X
17	GW	306SR	G32-MW306SR	370560	165588	Normal	X	X	X	X	X	X
17	GW	209	G44S-MW-209	370448	165449	Normal	X	X	X	X	X	X
17	GW	202RR	G44S-MW-202RR	370497	165558	Normal	X	X	X	X	X	X
17	GW	400	GI-MW400	370373	165093	Normal	X	X	X	X	X	X
17	GW	401	GI-MW401	370729	164895	Normal	X	X	X	X	X	X

CTO N4008520F4592, Site 17 Long-Term Monitoring Site 17, Operable Unit 06 Naval Station Newport, Jamestown, Rhode Island						Analysis Group	VOCs	Metals ⁴	PAHs/SVOCs	PCBs	Anions ⁶	Alkalinity ⁶
						Preparation and Analytical Method	SW-846 8260C	SW-846 3010A/60 20A	SW-846 3510C or 3520C, 8270D SIM	SW-846 8082A	US EPA 300.0	SM 2320B
						Analytical Laboratory/ Analytical SOP Reference	KAS / CA-202	KAS / CA-627	KAS / CA-213	KAS / CA-329	KAS / CA-742	KAS / CA-739
						Data Turnaround Time ¹	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days
Katahdin Analytical Services ⁵ 600 Technology Way, Scarborough, ME 04074 Project Manager: Heather Manz Ph. 207-874-2400, hmanz@katahdinlab.com						Container Type/Volume required (if different than container volume)	3 x 40mL VOA	500-mL plastic	2 x 1000-mL Amber Jar	2 x 1000-mL Amber Jar	250 mL plastic or glass	250-mL plastic
						Preservative	pH<2 with HCl	pH<2 with HNO ₃	Cool to ≤ 6 °C; store in the dark	Cool to ≤ 6 °C; store in the dark	Cool to ≤ 6 °C	Cool to ≤ 6 °C
						Holding Time (Preparation/ Analysis) ³	7 days to analysis	6 months to analysis	7 Days Extraction 40 Day Analysis	30 Days Extraction 40 Day Analysis	Nitrate 48 hours to analysis Sulfate and Chloride 28 days to analysis	14 days to analysis
Site	Matrix	Station ID	Sample ID ⁷	Coordinates ⁸		Sample Type						
				X	Y							
17	GW	402	GI-MW402	370826	164872	Normal	X	X	X	X	X	X
17	GW	403	GI-MW403	370649	165116	Normal	X	X	X	X	X	X
17	GW	404	GI-MW404	370806	165184	Normal	X	X	X	X	X	X
17	GW	405	GI-MW405	370895	164979	Normal	X	X	X	X	X	X
17	GW	406	GI-MW406	370834	165113	Normal	X	X	X	X	X	X
17	GW	FD	GI-GW-DUP01	NA	NA	Duplicate ²	X	X	X	X	X	X
17	GW	FD	GI-GW-DUP02	NA	NA	Duplicate ²	X	X	X	X	X	X
17	GW	FD	GI-GW-DUP02	NA	NA	Duplicate ²	X	X	X	X	X	X
17	DI H ₂ O	EB	GI-GW-EB01	NA	NA	Equipment Blank ²	X	X	X	X	X	X

CTO N4008520F4592, Site 17 Long-Term Monitoring Site 17, Operable Unit 06 Naval Station Newport, Jamestown, Rhode Island						Analysis Group	VOCs	Metals ⁴	PAHs/SVOCs	PCBs	Anions ⁶	Alkalinity ⁶
						Preparation and Analytical Method	SW-846 8260C	SW-846 3010A/60 20A	SW-846 3510C or 3520C, 8270D SIM	SW-846 8082A	US EPA 300.0	SM 2320B
						Analytical Laboratory/ Analytical SOP Reference	KAS / CA-202	KAS / CA-627	KAS / CA-213	KAS / CA-329	KAS / CA-742	KAS / CA-739
						Data Turnaround Time ¹	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days	21 Calendar Days
Katahdin Analytical Services ⁵ 600 Technology Way, Scarborough, ME 04074 Project Manager: Heather Manz Ph. 207-874-2400, hmanz@katahdinlab.com						Container Type/Volume required (if different than container volume)	3 x 40mL VOA	500-mL plastic	2 x 1000-mL Amber Jar	2 x 1000-mL Amber Jar	250 mL plastic or glass	250-mL plastic
						Preservative	pH<2 with HCl	pH<2 with HNO ₃	Cool to ≤ 6 °C; store in the dark	Cool to ≤ 6 °C; store in the dark	Cool to ≤ 6 °C	Cool to ≤ 6 °C
						Holding Time (Preparation/ Analysis) ³	7 days to analysis	6 months to analysis	7 Days Extraction 40 Day Analysis	30 Days Extraction 40 Day Analysis	Nitrate 48 hours to analysis Sulfate and Chloride 28 days to analysis	14 days to analysis
Site	Matrix	Station ID	Sample ID ⁷	Coordinates ⁸		Sample Type						
				X	Y							
17	DI H ₂ O	EB	GI-GW-EB02	NA	NA	Equipment Blank ²	X	X	X	X	X	X
17	GW	Same as Parent Sample				MS ²	X	X	X	X	X	X
17	GW	Same as Parent Sample				MSD ²	X	X	X	X	X	X
17	GW	Same as Parent Sample				MS ²	X	X	X	X	X	X
17	GW	Same as Parent Sample				MSD ²	X	X	X	X	X	X

Notes:

- Turn-around times for data are based on the time of receipt at the laboratory. Courier service will be provided by the laboratory on a daily basis, Monday through Friday. Samples will arrive at the laboratory on the same day as courier pickup.
- Field QC samples will be collected with the following frequency: Field Duplicate - One per 10 field samples; MS/MSD - One pair per 20 field samples (including FD samples); Equipment Blank- One per day per type of sampling equipment used or one per 20 samples per matrix and type of equipment used, whichever is more frequent; Trip Blanks – One per cooler.
- Maximum holding time is calculated from the time the sample is collected to the time the sample is prepared/extracted.
- Note: Both field-filtered (dissolved) and unfiltered (total) metals samples will be collected during each sampling round.
- Katahdin Analytical Services will perform VOC, metals, PAHs/SVOCs, PCBs, Anions, and Alkalinity analysis.
- Natural Attenuation parameters for determining if favorable natural attenuation conditions exist.
- Sample IDs will consist of what is presented in this column with a date in the form of “-mmddyy” added to the end to represent the day that the sample was collected.
- Coordinates are referenced to NAD83, Rhode Island State Plane, feet.

mL - milliliter
VOA- Volatile Organic Analysis
HCl - Hydrochloric acid
HNO₃ – Nitric Acid
GW – Groundwater
DI H₂O- deionized water
EB – equipment blank
MS – matrix spike
MSD – matrix spike duplicate

8.6 Usability Assessment

(UFP-QAPP Manual Sections 2.6.2 and 5.2.3 – Worksheet #37)

After data validation, the data and data quality will be reviewed to determine whether sufficient data of acceptable quality are available for decision making. The Helios/TT JV PM and designees will be responsible for conducting this data usability assessment. Summary statistics for target analytes such as Sampling Completeness (Cs), Analytical Data Completeness (Ca), maximum concentration, minimum concentration, number of samples with non-detected results, number of samples with detected results, and the proportion of samples with detected and non-detected results will be compiled at the discretion of the Helios/TT JV PM. The Project Team will consider whether any missing or rejected data have reduced the Cs and Ca to less than the associated goals and will consider factors that affect sample integrity such as holding times, preservation, and storage conditions. The assessment findings will be presented to the NAVFAC RPM, NAVSTA Newport POC, EPA RPM, and RIDEM RPM.

The goal for Cs is 90% and for Ca it is 90%. These values will be calculated as follows:

$$\%Cs = \frac{\text{No. of Valid Samples}}{\text{No. of Planned Samples}} \times 100\%$$

$$\%Ca = \frac{\text{No. of Valid Analytical Results}}{\text{No. of Planned Analytical Results}} \times 100\%$$

Field and laboratory precision will be evaluated for each matrix, analytical fraction, and concentration level, as applicable, with the expectation that laboratory duplicate results will be no less precise than field duplicate results.

Precision will be computed in terms of RPD as follows:

$$RPD = \frac{200 * |\text{Result}_A - \text{Result}_B|}{(\text{Result}_A + \text{Result}_B)}$$

Accuracy will be computed in terms of percent recovery (%R) as follows for MS/MSD samples:

$$\%R = \frac{\text{Amount in Spiked Sample} - \text{Amount in Sample}}{\text{Known Amount Added}} \times 100 \%$$

The %R calculation for LCSs and surrogate spikes will be as follows:

$$\%R = \frac{\text{Experimental Concentration}}{\text{Certified or Known Concentration}} \times 100$$

Data comparability and representativeness will be evaluated by reviewing sample collection processes and associated documentation; and by comparing overall precision and bias among data sets for each matrix and analytical fraction. This will not require quantitative comparisons unless professional judgment of the Helios/TT JV Project Chemist indicates that such quantitative analysis is required. If comparability or representativeness deficiencies are identified, limitations on the data will be described in the project report.

If data gaps exist they will be identified and, if appropriate, the project team will take them into consideration if additional work is required to meet project objectives.

Data validators apply “X” qualifiers to results that are seriously compromised and potentially unusable from a quality perspective. The Project Team will consider whether any missing or “X”-flagged data have compromised the ability to make decisions or to make decisions with the desired level of confidence. In these cases “X”-flagged data will be qualified with an “R” qualifier if rejected. The data will be evaluated to determine whether available data can compensate for missing or rejected data.

The project report will identify and describe the data usability limitations and suggest resampling or other corrective actions, if necessary. If applicable, this will include a description of unacceptable levels of bias/contamination, non-representativeness of data, or poor data comparability that could affect the accuracy and usability of reported results. These discussions may be specific to a matrix, analytical fraction, or other logical grouping of results.

9.0 Reference Limits and Evaluation Tables

(UFP-QAPP Manual Section 2.8.1 – Worksheet # 15)

Matrix: Groundwater

Method Group/Analyte	CAS No.	PAL ⁽¹⁾	Basis	PQL Goal ⁽²⁾	Laboratory Specific Limits (see analyte group header for units)		
					LOQ	LOD	DL
VOCs (µg/L)							
Benzene	71-43-2	5	Drinking Water Criterion ⁽³⁾	1.67	1.0	0.75	0.38
1,2-dichloroethene	78-87-5	70	Drinking Water Criterion ⁽³⁾	23.3	2.0	1.0	0.5
PCE	127-18-4	5	Drinking Water Criterion ⁽³⁾	1.67	2.0	1.0	0.50
TCE	79-01-6	5	Drinking Water Criterion ⁽³⁾	1.67	1.0	0.75	0.38
Vinyl chloride	75-01-4	2	Drinking Water Criterion ⁽³⁾	0.67	2.0	1.0	0.5
Metals (Katahdin referencing 6020 limits; µg/L)							
Arsenic	7440-38-2	10	Drinking Water Criterion ⁽³⁾	3.33	8.0	5.0	2.5
Cadmium ⁽⁴⁾	7440-43-9	5	Drinking Water Criterion ⁽³⁾	1.67	5.0	3.0	1.5
Lead ⁽⁴⁾	7439-92-1	15	Drinking Water Criterion ⁽³⁾	5	5	4.0	2.0
Manganese	7439-96-5	300	EPA Health Advisory ⁽⁵⁾	100	5.0	4.0	2.0
PAHs (Katahdin referencing 8270 SIM limits; µg/L)							
2-Methylnaphthalene	91-57-6	350	Risk-Based Criterion ⁽⁶⁾	117	0.30	0.20	0.10
Benzo(a)anthracene	56-55-3	380	Risk-Based Criterion ⁽⁶⁾	127	0.20	0.10	0.05
Benzo(a)pyrene	50-32-8	0.2	Drinking Water Criterion ⁽³⁾	0.067	0.20	0.15	0.075
Benzo(b)fluoranthene	205-99-2	380	Risk-Based Criterion ⁽⁵⁾	127	0.30	0.20	0.10
Benzo(g,h,i)perylene ⁽⁴⁾	191-24-2	120	EPA RSL ⁽⁷⁾	40	0.20	0.15	0.075
Benzo(k)fluoranthene ⁽⁴⁾	207-08-9	2.5	EPA RSL ⁽⁷⁾	0.8	0.20	0.10	0.05
Chrysene	218-01-9	37,700	Risk-Based Criterion ⁽⁶⁾	12,600	0.20	0.10	0.05
Dibenzo(a,h)anthracene	53-70-3	38	Risk-Based Criterion ⁽⁶⁾	12.7	0.20	0.15	0.075

Method Group/Analyte	CAS No.	PAL ⁽¹⁾	Basis	PQL Goal ⁽²⁾	Laboratory Specific Limits (see analyte group header for units)		
					LOQ	LOD	DL
Fluoranthene	206-44-0	157,200	Risk-Based Criterion ⁽⁶⁾	52,400	0.20	0.15	0.075
Indeno(1,2,3-cd)pyrene	193-39-5	375	Risk-Based Criterion ⁽⁶⁾	125	0.20	0.15	0.075
Naphthalene	91-20-3	100	Drinking Water Criterion ⁽³⁾	33	0.20	0.15	0.075
Phenanthrene	85-01-8	118,000	RG per ROD	0.33	0.20	0.15	0.075
Pyrene ⁽⁸⁾	129-00-0	120	EPA RSL ⁽⁷⁾	40	0.30	0.20	0.1
SVOCs (Katahdin referencing 8270 SIM limits; µg/L)							
Pentachlorophenol (PCP)	87-86-5	1	Drinking Water Criterion ⁽³⁾	0.33	1.0	0.50	0.33
PCBs (µg/L)							
Aroclor-1016	12674-11-2	0.056 ⁽⁹⁾	1/9 Remedial Goal	0.019	0.50	0.40	0.20
Aroclor-1221	11104-28-2	0.056 ⁽⁹⁾	1/9 Remedial Goal	0.019	0.50	0.40	0.20
Aroclor-1232	11141-16-5	0.056 ⁽⁹⁾	1/9 Remedial Goal	0.019	0.50	0.25	0.12
Aroclor-1242	53469-21-9	0.056 ⁽⁹⁾	1/9 Remedial Goal	0.019	0.50	0.40	0.20
Aroclor-1248	12672-29-6	0.056 ⁽⁹⁾	1/9 Remedial Goal	0.019	0.50	0.40	0.20
Aroclor-1254	11097-69-1	0.056 ⁽⁹⁾	1/9 Remedial Goal	0.019	0.50	0.25	0.12
Aroclor-1260	11096-82-5	0.056 ⁽⁹⁾	1/9 Remedial Goal	0.019	0.50	0.40	0.20
Aroclor-1262	37324-23-5	0.056 ⁽⁹⁾	1/9 Remedial Goal	0.019	0.50	0.25	0.12
Aroclor-1268	11100-14-4	0.056 ⁽⁹⁾	1/9 Remedial Goal	0.019	0.50	0.25	0.12
Total Aroclors	NA	0.5	Drinking Water Criterion	NA	0.50	0.25	0.12
Alkalinity (Katahdin referencing SM-2320B limits; mg/L) ⁽⁸⁾							
Alkalinity	NA	NA	NA	NA	5.0	4.0	2.0
Anions (mg/L) ⁽⁸⁾							
Chloride	16887-00-6	NA	NA	NA	2.0	1.0	0.50

Method Group/Analyte	CAS No.	PAL ⁽¹⁾	Basis	PQL Goal ⁽²⁾	Laboratory Specific Limits (see analyte group header for units)		
					LOQ	LOD	DL
Nitrate	84145-82-4	NA	NA	NA	0.050	0.040	0.020
Sulfate	18785-72-3	NA	NA	NA	1.0	0.5	0.25

Notes:

CAS Chemical Abstracts Service
 DL Detection Limit
 LOD Limit of Detection
 LOQ Limit of Quantitation
 PAL- Project Action Limit
 PQL Project Quantitation Limit
 RG Remediation Goal
 ROD Record of Decision
 TBD To be determined
 NA Not Applicable
 µg/L Microgram per liter
 µg/kg Microgram per kilogram

- (1) The PAL is equivalent to the "Cleanup Level" as listed in the ROD with the exception of the PALs for 1,2-DCE (sum of cis and trans isomers) and vinyl chloride because they are not listed in the ROD as COCs. They are monitored because they are daughter products of PCE.
- (2) The PQL Goal is conservatively set to one-third of the PAL.
- (3) Drinking water criteria were used where available (the current EPA MCL or RIDEM GA groundwater objective).
- (4) Analyte not listed in ROD as a groundwater COC. It is a soil COC and is included because one remedial goal is to monitor concentrations of soil COCs in groundwater.
- (5) Per US EPA request the US EPA Drinking Water Health Advisory (lifetime) guidance value was used for manganese.
- (6) Risk-based criteria developed for construction worker exposure through dermal contact, incidental ingestion, and inhalation of vapors from shallow groundwater. Risk-based criteria are based on US EPA's target cancer risk of no greater than 1×10^{-4} , and non-cancer risk (HI) no greater than 1 (as described in the ROD).
- (7) The PALs are based on most current US EPA RSL for Tap Water.
- (8) These parameters are needed only to support an evaluation of natural attenuation conditions and not for evaluating potential adverse environmental impacts.
- (9) Individual PCB PALs are equal to 1/9th of the Total PCB PAL. The Total PCB PAL is what PCB results will be compared to.

10.0 Analytical SOP References Table

[\(UFP-QAPP Manual Section 3.2.1 – Worksheet #23\)](#)

Katahdin Analytical Services
600 Technology Way, Scarborough, Maine, 04074
Project Manager – Heather Manz
207-874-2400

Katahdin SOP Number	Title, Revision Date, and Number	Definitive or Screening Data	Matrix and Analytical Group	Instrument	Variance to QSM ¹	Modified for Project Work? (Y/N)
CA-202	Analysis of VOAs by Purge and Trap GC/MS: SW-846 Method 8260, 01/20, Revision 20.	Definitive	Water / VOCs	Gas Chromatography (GC)/Mass Spectroscopy (MS)	No Variance	N
CA-213	Analysis of Semivolatile Organic Compounds By: SW 846 Method 8270 – Modified for Selected Ion Monitoring (SIM), 01/19, Revision 15. (Reviewed 02/20)	Definitive	Water / SVOCs and PAHs	GC/MS	No Variance	N
CA-329	Analysis of PCBs as Total Aroclors By Gas Chromatography/Electron Capture Detector (GC/ECD): SW-846 Method 8082, 01/19, Revision 19.	Definitive	Water / PCBs	GC/ECD	No Variance	N
CA-502	Preparation Of Aqueous Samples for Extractable Semivolatile Analysis, 03/19, Revision 12. (Reviewed 01/20)	Definitive	Water / SVOCs and PAHs	Not applicable (extraction)	No Variance	N
CA-515	Preparation of Aqueous Samples for Pesticides/PCBs Analysis, 01/20, Revision 15.	Definitive	Water / PCBs	Not applicable (extraction)	No Variance	N
CA-604	Acid Digestion of Aqueous Samples by USEPA Method 3010 for Inductively coupled plasma Analysis of Total or Dissolved Metals, 01/19, Revision 9. (Reviewed 03/20)	Definitive	Water / Metals	Not applicable (digestion)	No Variance	N
CA-627	Trace Metals Analysis By ICP-MS Using USEPA Method 6020, 01/19, Revision 13.	Definitive	Water / Metals	Inductively Coupled Plasma (ICP) / Mass Spectroscopy (MS)	No Variance	N
CA-742	Anions by Ion Chromatography (IC) – Method 300.0, 12/17, Revision 11. (Reviewed 02/19)	Definitive	Water / Anions	Ion Chromatograph	No Variance	N

Katahdin SOP Number	Title, Revision Date, and Number	Definitive or Screening Data	Matrix and Analytical Group	Instrument	Variance to QSM ¹	Modified for Project Work? (Y/N)
CA-739	Titrimetric Determination of Total Alkalinity by EPA Method 310.1 and SM 2320 B using the Mettler DI25 Autotitrator, and Calculation of the Component Forms of Alkalinity by SM 4500-CO ₂ D, 07/18, Revision 13. (Reviewed 01/20)	Definitive	Water / Alkalinity	Autotitrator	No Variance	N

Notes:

Laboratory SOPs are available directly from the laboratory upon request.

11.0 Laboratory QC Samples Table

(UFP-QAPP Manual Section 3.4 – Worksheet #28)

Matrix	Groundwater					
Analytical Group	VOCs					
Analytical Method/ SOP Reference	SW-846 8260C/ CA-202					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
Method Blank	One per preparatory batch.	No analytes detected > ½ LOQ or > 1/10 the amount measured in any sample or 1/10 the regulatory limit, whichever is greater. Common contaminants must not be detected > LOQ. Results may not be reported without a valid MB.	Correct problem. If required, re-prepare and reanalyze MB and all QC samples and field samples processed with the contaminated blank. If reanalysis cannot be performed, data must be qualified and explained in the Case Narrative. Apply B-flag to all results for the specific analyte(s) in all samples in the associated preparatory batch. Flagging is only appropriate in cases where the samples cannot be reanalyzed.	Analyst, Laboratory Department Manager, and Data Validator	Bias/ Contamination	Same as Method/SOP QC Acceptance Limits.
Surrogate Spike	All field and QC samples. Alternative surrogates are recommended when there is obvious chromatographic interference.	Use QSM 5.3 Appendix C limits.	Correct problem, then re-prepare and reanalyze all failed samples for all surrogates in the associated preparatory batch, if sufficient sample material is available. If obvious chromatographic interference with surrogate is present, reanalysis may	Analyst, Laboratory Department Manager, and Data Validator	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	VOCs					
Analytical Method/ SOP Reference	SW-846 8260C/ CA-202					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
			not be necessary, but the client must be notified prior to reporting data and the failures must be discussed in the Case Narrative. Apply Q-flag to all associated analytes if acceptance criteria are not met and explain in the Case Narrative.			

Matrix	Groundwater					
Analytical Group	VOCs					
Analytical Method/ SOP Reference	SW-846 8260C/ CA-202					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
Laboratory Control Sample (LCS)	One per preparation batch of twenty or fewer samples of similar matrix.	Use QSM 5.3 Appendix C Limits for batch control.. but the client must be notified prior to reporting data and the failures must be discussed in the Case Narrative. Apply Q-flag to all associated analytes if acceptance criteria are not met and explain in the Case Narrative	Correct problem, then reprep and reanalyze the LCS and all samples in the associated preparatory batch for failed analytes, if sufficient sample material is available. If reanalysis cannot be performed, data must be qualified and explained in the Case Narrative. Apply Q-flag to specific analyte(s) in all samples in the associated preparatory batch. Flagging is only appropriate in cases where the samples cannot be reanalyzed.	Analyst, Laboratory Department Manager, and Data Validator	Accuracy/ Bias	Same as Method/SOP QC Acceptance Limits.
Matrix Spike (MS)	One per preparatory batch.	Use QSM 5.3 Appendix C Limits for batch control. Must contain all surrogates and all analytes to be reported.	Examine the project-specific requirements. Contact the client as to additional measures to be taken. For the specific analyte(s) in the parent sample, apply J-flag if acceptance criteria are not met and explain in the Case Narrative.	Analyst, Laboratory Department Manager, and Data Validator	Precision/ Accuracy/ Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	VOCs					
Analytical Method/ SOP Reference	SW-846 8260C/ CA-202					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
			For matrix evaluation only. If MS results are outside the limits, the data shall be evaluated to determine the source(s) of difference, i.e., matrix effect or analytical error.			
Matrix Spike Duplicate (MSD) or Matrix Duplicate (MD)	One per preparatory batch.	Use QSM 5.3 Appendix C Limits for batch control. MSD or MD: Relative Percent Difference (RPD) of all analytes $\leq 20\%$ (between MS and MSD or sample and MD). MSD: Must contain all surrogates and all analytes to be reported. For Sample/MD: RPD criteria only apply to analytes whose concentration in the sample is greater than or equal to the LOQ.	Examine the project-specific requirements. Contact the client as to additional measures to be taken. For the specific analyte(s) in the parent sample, apply J-flag if acceptance criteria are not met and explain in the Case Narrative. The data shall be evaluated to determine the source of difference.	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.
Internal Standard (IS)	Every field sample, standard, and QC sample.	Retention time within ± 10 seconds from retention time of the midpoint standard in	Inspect mass spectrometer and GC for malfunctions and correct problem. Reanalysis of samples	Analyst, Laboratory Department	Accuracy/ Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	VOCs					
Analytical Method/ SOP Reference	SW-846 8260C/ CA-202					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
		the ICAL; EICP area within - 50% to +100% of ICAL midpoint standard. On days when ICAL is not performed, the daily initial CCV can be used.	analyzed while system was malfunctioning is mandatory. If corrective action fails in field samples, data must be qualified and explained in the Case Narrative. Apply Q-flag to analytes associated with the noncompliant IS. Flagging is not appropriate for failed standards.	Manager, and Data Validator		

Matrix	Groundwater					
Analytical Group	SVOCs and PAHs					
Analytical Method/ SOP Reference	SW-846 8270D SIM/ CA-213					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
Deuterated Monitoring Compounds (DMCs) (surrogates)	All field and QC samples.	PAH analysis: DMCs required for PAH target analytes: fluoranthene-d10 and 2- methylnaphthalene- d10. Minimum Relative Response Factor (RRF) for PAH DMCs: 0.40. All DMCs: Requires 50-150% recovery until in- house limits can be established. For non-PAH target analytes, other DMCs with similar chemistry must be assigned. May use the same extract for full scan and SIM analysis if the SIM-specific DMCs are added prior to extraction.	Correct problem, and then re-prepare and reanalyze all samples with failing DMCs if sufficient sample material is available. If obvious chromatographic interference is present, reanalysis may not be necessary, but the client must be notified prior to reporting data and the failures must be discussed in the Case Narrative. Apply Q-flag to all associated samples and analytes if acceptance criteria are not met and explain in the Case Narrative.	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.
Method Blank	One per preparation batch of twenty or fewer samples of similar	No analytes detected > ½ LOQ or > 1/10 the amount measured	Conduct investigation to determine the source of the contamination and	Analyst, Laboratory Department	Bias/contamination	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	SVOCs and PAHs					
Analytical Method/ SOP Reference	SW-846 8270D SIM/ CA-213					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
	matrix, prior to analysis of any field samples.	in any sample or 1/10 the regulatory limit, whichever is greater. Results may not be reported without a valid method blank. Laboratories may use the same extract for full scan and SIM analysis provided the applicable DMCs and IS are spiked at the appropriate concentrations.	take appropriate corrective actions. Correct problem. If required, reprep and reanalyze MB and all samples processed with the contaminated blank. If reanalysis cannot be performed, data must be qualified and explained in the Case Narrative. Apply B-flag to all results for the specific analyte(s) in all samples in the associated analytical batch. Flagging is only appropriate in cases where the samples cannot be reanalyzed.	Manager and Data Validator		

Matrix	Groundwater					
Analytical Group	SVOCs and PAHs					
Analytical Method/ SOP Reference	SW-846 8270D SIM/ CA-213					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
LCS	One per preparation batch of twenty or fewer samples of similar matrix.	Laboratory must use the QSM 5.3 Appendix C Limits for batch control. Must contain all analytes to be reported. Results may not be reported without a valid LCS.	Correct problem, then reprep and reanalyze the LCS and all samples in the associated preparatory batch for failed analytes, if sufficient sample material is available. If reanalysis cannot be performed, data must be qualified and explained in the Case Narrative. Apply Q-flag to specific analyte(s) in all samples in the associated analytical batch. Flagging is only appropriate in cases where the samples cannot be reanalyzed.	Analyst, Laboratory Department Manager, and Data Validator	Accuracy / Bias	Same as Method/SOP QC Acceptance Limits.
MS	One per preparation batch.	Laboratory must use the QSM 5.3 Appendix C Limits for batch control. Must contain all analytes to be reported spiked at concentrations appropriate for SIM analysis.	Examine the project- specific requirements. Contact the client as to additional measures to be taken. For the specific analyte(s) in the parent sample, apply the J-flag if acceptance criteria are not met and explain in the Case Narrative. For matrix evaluation only. If MS results are outside	Analyst, Laboratory Department Manager, and Data Validator	Precision/Accuracy/ Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	SVOCs and PAHs					
Analytical Method/ SOP Reference	SW-846 8270D SIM/ CA-213					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
			the limits, the data shall be evaluated to determine the source(s) of difference (i.e., matrix effect or analytical error).			
MSD or MD	One per preparation batch.	Laboratory must use the QSM 5.3 Appendix C Limits for batch control . The MSD must contain all analytes to be reported spiked at concentrations appropriate for SIM analysis. MSD or MD: Relative Percent Difference (RPD) of all analytes $\leq 40\%$ (between MS and MSD or sample and MD). For Sample/MD: RPD criteria only apply to analytes whose concentration in the sample is greater than or equal to the LOQ.	Examine the project-specific requirements. Contact the client as to additional measures to be taken. For the specific analyte(s) in the parent sample, apply the J-flag if acceptance criteria are not met and explain in the Case Narrative. All data must be evaluated to determine the source of difference.	Analyst, Supervisor	Precision/Bias	Same as Method/SOP QC Acceptance Limits.
IS	Every field sample, Standards,	Retention time within ± 10 seconds from retention time of the	Inspect mass spectrometer or gas chromatograph for malfunctions. Mandatory	Analyst, Laboratory Department	Accuracy/ Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	SVOCs and PAHs					
Analytical Method/ SOP Reference	SW-846 8270D SIM/ CA-213					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
	blanks, and QC sample.	midpoint standard in the ICAL; EICP area within - 50% to +100% of ICAL midpoint standard. On days when ICAL is not performed, the initial CCV is used. Internal Standard is spiked no greater than 0.40 ng/μL concentration. (According to the EPA Contract Laboratory Program Statement of Work (CLP SOW), this is the concentration of IS specified for SIM analysis. The SOW indicates calibration standards range from 0.10 to 1.0 ng/μL, so 0.40 ng/μL is mid- range. 1, 4-dichlorobenzene-d4 is ignored for SIM.	reanalysis of samples analyzed while system was malfunctioning. corrective action fails in field samples, data must be qualified and explained in the Case Narrative. Apply Q-flag to analytes associated with the non-compliant IS. Flagging is not appropriate for failed standards.	Manager, and Data Validator		
Characteristic Ions for MS Confirmation	Minimum 3 ions. Need 3 structurally significant ions that are logical	The relative intensities of the characteristic ions of target analytes agree within 30% of the relative	No data can be reported without MS confirmation.	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	SVOCs and PAHs					
Analytical Method/ SOP Reference	SW-846 8270D SIM/ CA-213					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
	fragments – not isotopic clusters. Internal standard and DMC can use fewer than 3 ions.	intensities in the reference spectrum and the relative intensities must be > 0. Confirmation requires S/N ratio of ≥ 3 for each quant and confirmation ion.				

Matrix	Groundwater					
Analytical Group	PCBs					
Analytical Method/ SOP Reference	SW-846 8082A / CA-329					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
IS	If employed, every field sample, standard, and QC sample.	Retention time within ± 0.06 Relative Retention Time (RRT) units from retention time of the midpoint standard in the ICAL; Internal standard signal (area or height) within -50% to +100% of ICAL midpoint standard. On days when ICAL is not performed, the daily initial CCV can be used.	Inspect GC for malfunctions and correct problem. Reanalysis of samples analyzed while system was malfunctioning is mandatory. If corrective action fails in field samples, data must be qualified and explained in the Case Narrative. Apply Q-flag to analytes associated with the non-compliant IS. Flagging is not appropriate for failed standards.	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.
Method Blank	One per preparatory batch.	No analytes detected > 1/2 LOQ or > 1/10 the amount measured in any sample or 1/10 the regulatory limit, whichever is greater. Results may not be reported without a valid method blank.	Correct problem. If required, reprep and reanalyze MB and all samples processed with the contaminated blank. If reanalysis cannot be performed, data must be qualified and explained in the Case Narrative. Apply B-flag to all results for the specific analyte(s) in all samples in the associated preparatory batch. Flagging is only appropriate in cases where the samples cannot be reanalyzed.	Analyst, Laboratory Department Manager and Data Validator	Bias/contamination	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	PCBs					
Analytical Method/ SOP Reference	SW-846 8082A / CA-329					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
Surrogate Spike	All field and QC samples. Alternative surrogates are recommended when there is obvious chromatographic interference.	Use QSM 5.3 Appendix C limits for batch control.	Correct problem, then reprep and reanalyze all failed samples for all surrogates in the associated preparatory batch, if sufficient sample material is available. If obvious chromatographic interference with surrogate is present, reanalysis may not be necessary, but the client must be notified prior to reporting data, and the failures must be discussed in the Case Narrative. Apply Q-flag to all associated analytes if acceptance criteria are not met and explain in the Case Narrative.	Analyst, Laboratory Department Manager, and Data Validator	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.
LCS	One per preparatory batch.	Laboratory must use the QSM 5.3 Appendix C Limits for batch control. Results may not be reported without a valid LCS.	Correct problem, then reprep and reanalyze the LCS and all samples in the associated preparatory batch for failed analytes, if sufficient sample material is available. If reanalysis cannot be performed, data must be qualified and explained in the Case Narrative. Apply Q-flag to specific analyte(s) in all samples in the associated preparatory batch. Flagging is only appropriate in cases where the samples cannot be reanalyzed.	Analyst, Laboratory Department Manager, and Data Validator	Accuracy/ Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	PCBs					
Analytical Method/ SOP Reference	SW-846 8082A / CA-329					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
MS	One per preparatory batch.	Laboratory must use the QSM 5.3 Appendix C Limits for batch control .	Correct problem, then reprep and reanalyze the LCS and all samples in the associated preparatory batch for failed analytes, if sufficient sample material is available. Contact the client as to additional measures to be taken. For the specific analyte(s) in the parent sample, apply J-flag if acceptance criteria are not met and explain in the Case Narrative. For matrix evaluation only. If MS results are outside the limits, the data shall be evaluated to determine the source(s) of difference (i.e., matrix effect or analytical error).	Analyst, Laboratory Department Manager, and Data Validator	Precision/ Accuracy/ Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	PCBs					
Analytical Method/ SOP Reference	SW-846 8082A / CA-329					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
MSD or MD	One per preparatory batch.	Laboratory must use the QSM 5.3 Appendix C Limits for batch control RPD $\leq$ 30% (between MS and MSD or sample and MD). For Sample/MD: RPD criteria only apply to analytes whose concentration in the sample is greater than or equal to the LOQ.	Examine the project- specific requirements. Contact the client as to additional measures to be taken. For the specific analyte(s) in the parent sample, apply J-flag if acceptance criteria are not met and explain in the Case Narrative. The data shall be evaluated to determine the source of difference.	Analyst, Supervisor	Precision/ Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.
Second Column Confirmation	All positive results must be confirmed.	Calibration and QC criteria for second column are the same as for initial or primary column analysis. Results between primary and second column must be RPD $\leq$ 40%.	None. Apply "J" qualifier if RPD >40% and discuss in the case narrative. The higher of the two results will be reported unless matrix interference is apparent.	Analyst, Laboratory Department Manager, and Data Validator	Precision	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	Metals					
Analytical Method/ SOP Reference	SW-846 6020A / CA-627					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
IS	Every field sample, standard and QC sample.	IS intensity in the samples within 30% to 120% of intensity of the IS in the ICAL blank.	If recoveries are acceptable for QC samples, but not field samples, the field samples may be considered to suffer from a matrix effect. Reanalyze sample at 5-fold dilutions until criteria is met. For failed QC samples, correct problem and rerun all associated failed field samples. Flagging is not appropriate. Samples suffering from matrix effect should be diluted until criteria are met, or an alternate IS should be selected.	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	Metals					
Analytical Method/ SOP Reference	SW-846 6020A / CA-627					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
Method Blank	One per preparatory batch.	The absolute values of all analytes must be $< \frac{1}{2}$ LOQ or $< 1/10^{\text{th}}$ the amount measured in any sample or $1/10^{\text{th}}$ the regulatory limit, whichever is greater. Results may not be reported without a valid method blank. Non-detects associated with positive blank infractions may be reported. Sample results $> 10\text{X}$ the LOQ associated with negative blanks may be reported.	Correct problem. If required, re-prepare and reanalyze MB and all QC samples and field samples processed with the contaminated blank. If reanalysis cannot be performed, data must be qualified and explained in the Case Narrative. Apply B-flag to all results for the specific analyte(s) in all samples in the associated preparatory batch. Flagging is only appropriate in cases where the samples cannot be reanalyzed.	Analyst, Supervisor	Accuracy/Bias/ Contamination	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	Metals					
Analytical Method/ SOP Reference	SW-846 6020A / CA-627					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
LCS	One per preparatory batch.	Laboratory must use QSM 5.3 Appendix C Limits for batch. Must contain all reported analytes. Results may not be reported without a valid LCS.	Correct problem, then re- prepare and reanalyze the LCS and all samples in the associated preparatory batch for failed analytes if sufficient sample material is available. If reanalysis cannot be performed, data must be qualified and explained in the Case Narrative. Apply Q-flag to specific analyte(s) in all samples in the associated preparatory batch. Flagging is only appropriate in cases where the samples cannot be reanalyzed.	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.
MS	One per preparatory batch.	Laboratory must use QSM 5.3 Appendix C Limits for batch.	Examine the project- specific requirements. Contact the client as to additional measures to be taken. For the specific analyte(s) in the parent sample, apply J-flag if acceptance criteria are not met and explain in the Case Narrative. For matrix evaluation only. If MS results are outside the limits, the data shall be evaluated to determine the source(s) of difference (i.e., matrix effect or analytical error).	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	Metals					
Analytical Method/ SOP Reference	SW-846 6020A / CA-627					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
MSD or MD	One per preparatory batch.	Laboratory must use QSM 5.3 Appendix C Limits for batch. MSD or MD: RPD of all analytes $\leq 20\%$ (between MS and MSD or sample and MD). For Sample/MD: RPD criteria only apply to analytes whose concentration in the sample is greater than or equal to the LOQ.	Examine the project- specific requirements. Contact the client as to additional measures to be taken. For the specific analyte(s) in the parent sample, apply J-flag if acceptance criteria are not met and explain in the Case Narrative. The data shall be evaluated to determine the source of difference.	Analyst, Supervisor	Precision/ Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.
Dilution Test	One per preparatory batch if MS or MSD fails. Only applicable for samples with concentrations > 50 X LOQ (prior to dilution). Use along with MS/MSD or PDS data to confirm matrix effects.	Five-fold dilution must agree within $\pm 10\%$ of the original measurement.	No specific corrective action unless required by the project. For the specific analyte(s) in the parent sample, apply J-flag if acceptance criteria are not met and explain in the Case Narrative.	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	Metals					
Analytical Method/ SOP Reference	SW-846 6020A / CA-627					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
Post Digestion Spike (PDS) Addition	One per preparatory batch if MS or MSD fails (using the same sample as used for the MS/MSD if possible). Criteria apply for samples with concentrations < 50 X LOQ prior to dilution.	Recovery within 80- 120%.	No specific corrective action unless required by the project. For the specific analyte(s) in the parent sample, apply J-flag if acceptance criteria are not met and explain in the Case Narrative.	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.
Method of Standard Additions (MSA)	When dilution or post digestion spike fails and if the required by project. Document use of MSA in the Case Narrative.	NA.	NA.	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	Anions					
Analytical Method/ SOP Reference	EPA 300.0 / CA-742					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
Method Blank	One per preparatory batch.	No analytes detected > 1/2 LOQ or > 1/10th the amount measured in any sample or 1/10th the regulatory limit, whichever is greater. Results may not be reported without a valid method blank.	Correct problem. If required, re-prepare and reanalyze MB and all QC samples and field samples processed with the contaminated blank. If reanalysis cannot be performed, data must be qualified and explained in the Case Narrative. Apply B-flag to all results for the specific analyte(s) in all samples in the associated preparatory batch. Flagging is only appropriate in cases where the samples cannot be reanalyzed.	Analyst, Supervisor	Accuracy/Bias/ Contamination	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	Anions					
Analytical Method/ SOP Reference	EPA 300.0 / CA-742					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
LCS	One per preparatory batch. Must contain all reported analytes.	Laboratory must use QSM 5.3 Appendix C LCS Limits for batch control. Results may not be reported without a valid LCS. Must contain all reported analytes.	Correct problem, then re-prepare and reanalyze the LCS and all samples in the associated preparatory batch for all reported analytes if sufficient sample material is available. If reanalysis cannot be performed, data must be qualified and explained in the Case Narrative. Apply Q-flag to specific analyte(s) in all samples in the associated preparatory batch. Flagging is only appropriate in cases where the samples cannot be reanalyzed.	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.
Matrix Spike (MS)	One per preparatory batch.	Laboratory must use QSM 5.3 Appendix C LCS Limits for batch control. Must contain all reported analytes.	Follow project specific requirements. Contact the client as to additional measures to be taken. For the specific analyte(s) in the parent sample, apply J-flag if acceptance criteria are not met and explain in the Case Narrative. For matrix evaluation only. If MS results are outside the limits, the data shall be evaluated to determine the source(s) of difference (i.e., matrix effect or analytical error).	Analyst, Supervisor	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	Anions					
Analytical Method/ SOP Reference	EPA 300.0 / CA-742					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria
MSD or MD	One per preparatory batch.	Laboratory must use QSM 5.3 Appendix C LCS Limits for batch control. Must contain all reported analytes. MSD or MD: Relative Percent Difference (RPD) of all analytes $\leq 15\%$ (between MS and MSD or sample and MD). For Sample/MD: RPD criteria only apply to analytes whose concentration in the sample is greater than or equal to the LOQ.	Follow project specific requirements. Contact the client as to additional measures to be taken. For the specific analyte(s) in the parent sample, apply J-flag if acceptance criteria are not met and explain in the Case Narrative. The data shall be evaluated to determine the source of difference.	Analyst, Supervisor	Precision/ Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.

Matrix	Groundwater					
Analytical Group	Alkalinity					
Analytical Method/ SOP Reference	SM2320 B / CA-739					
QC Sample:	Frequency/ Number	Method/SOP QC Acceptance Limits	Corrective Action	Person(s) Responsible for Corrective Action	Data Quality Indicator (DQI)	Measurement Performance Criteria (MPC)
Method Blank	One per analytical batch of 20 or fewer samples.	No target analytes > LOQ or > 1/10 sample concentration or >1/10 regulatory limit, whichever is greater.	Correct the problem. Rerun MB prior to analysis of samples, if possible. Evaluate the samples and associated QC. Re-prepare/re-analyze blank and affected samples.	Analyst, Laboratory Department Manager and Data Validator	Bias/contamination	Same as Method/SOP QC Acceptance Limits.
LCS	One per analytical batch of 20 or fewer samples.	%R must be within 80-120	Investigate source of problem. If the LCS recovery is high but the sample results are <LOQ, request client approval to qualify and narrate. Otherwise, if sample volume available and <2x holding time, re-prepare and reanalyze.	Analyst, Laboratory Department Manager, and Data Validator	Accuracy/Bias/Contamination	Same as Method/SOP QC Acceptance Limits.
MS	One for every set 10 samples	%R must be within: 75-125	Evaluate the samples and associated QC: i.e. If the LCS results are acceptable, narrate. If both the LCS and MS are unacceptable reprep and reanalyze the samples and QC. Notate sample result in raw data if matrix interference suspected.	Analyst, Laboratory Department Manager, and Data Validator	Accuracy/Bias	Same as Method/SOP QC Acceptance Limits.
Laboratory Duplicate	One sample duplicate per 20 samples.	RPD $\leq$ 20 for samples.	Investigate problem and reanalyze sample in duplicate. If RPD still >20, sample should be reanalyzed. If still high, result is qualified.	Analyst, Laboratory Department Manager, and Data Validator	Precision	Same as Method/SOP QC Acceptance Limits.

12.0 Data Verification and Validation (Steps I and IIa/IIb) Process Table

(UFP-QAPP Manual Section 5.2.1, UFP-QAPP Manual Section 5.2.2, Figure 37 UFP-QAPP Manual, Table 9 UFP-QAPP Manual – Worksheets #34, 35, 36)

Data Review Input	Description	Responsible for Verification (name, organization)	Internal/ External
Sample Log Sheets, Chain of Custody Forms, SAP, and Laboratory Sample Login Documentation	Verify that samples were correctly identified, chain of custody records are legible, data will be traceable to the corresponding samples, all samples listed in Sections 6, 8, and 11 were collected from intended locations, and the correct sampling and analytical methods/SOPs were assigned to samples listed on the chain of custody record. The PM will verify that the sampling plan was implemented and carried out as written and will ensure that any significant deviations are documented in the project report.	FOL and PM, Helios/TT JV	Internal
Chain of Custody Forms	Verify that the chain-of-custody form is complete and accurate; and was signed and dated by the sampler relinquishing the samples and by the laboratory receiving the samples. Resolve discrepancies, if possible. Alert the Helios/TT JV PM verbally or via email if discrepancies are unresolvable.	FOL, Helios/TT JV	Internal
Chain of Custody Forms	Verify sample shipment completeness against the chain-of-custody record, verify proper sample preservation/integrity, sign to indicate receipt, note any discrepancies, and correct them as necessary. Notify the Helios/TT JV FOL or PM of any deviations from sample shipping requirements such as damaged sample containers, or inappropriate temperature or pH. Note uncorrectable discrepancies in the data package case narrative.	Laboratory Sample Custodian, Katahdin Analytical	External
Analytical Calibration Standards	Determine that standards are traceable and meet contract, method, and procedural requirements, as applicable, and include Certificates of Analysis in the laboratory data package to document the traceability. Verify that the certified values of each analyte of concern were correctly applied to create the calibration curve, that all analytes of concern were in the calibration mix, and contained both branched and linear isomers, if commercially available. If discrepancies in traceability are found, bring the discrepancies to the laboratory PM attention for correction.	Laboratory analyst, Katahdin Analytical	Internal
SAP, Analytical SOPs, and Analytical Data Packages	Verify that the correct analytical methods/SOPs were applied. Establish that all method QC samples were analyzed and in control as listed in the analytical SOPs. If	Laboratory PM, Katahdin Analytical	Internal

Data Review Input	Description	Responsible for Verification (name, organization)	Internal/ External
	method QC is not in control, the Laboratory PM will contact the Helios/TT JV Project Chemist or PM verbally or via e-mail for guidance prior to laboratory data package preparation.		
Laboratory Analytical Data Package	Verify all analytical data packages for completeness, including certificates of analysis for calibration and check standards. The Laboratory QAM will sign the case narrative for each data package.	Laboratory QAM, Katahdin Analytical	Internal
Laboratory Analytical Data Package	Review chain-of-custody records to ensure that the required analytical samples were collected, appropriate sample identifications were used, and correct analytical methods were applied to each sample. Verify the analytical data package for completeness and accuracy, including certificates of analysis for calibration and check standards. Obtain missing data package elements from the laboratory. Document unrecoverable elements, if any, in the DV report submitted to the Helios/TT JV PM and alert the Project Chemist or PM.	Data Validators, Helios/TT JV	External
EDDs/ Analytical Data Packages	Verify 100% of EDD results for accuracy and completeness against hard copy data package and chain of custody records at the start of validation. If required elements are missing, obtain missing elements from the laboratory before completing the validation. If any element cannot be obtained, document the omission in the DV report and identify the missing elements to the Helios/TT JV Project Chemist or PM as early as possible.	Data Validators, Helios/TT JV	External
Sample Shipment and Storage Conditions; and Holding Times for Representativeness	Verify that sample shipping and storage conditions satisfy Section 8 requirements. Document deviations from requirements in the DV report and notify the Helios/TT JV Project Chemist or PM if deviations from the SAP requirements are serious enough to warrant data rejection. Document findings in the DV report.	Data Validator, Helios/TT JV	External

Data Review Input	Description	Responsible for Verification (name, organization)	Internal/ External
QC Samples/MPC Compliance	Ensure that the scheduled laboratory and field QC samples were submitted for analysis and that the MPCs listed in SAP Sections 6, 8, 9, and 11 were met for all field samples and QC samples. Document deviations in the DV report. Evaluate sample results for laboratory contamination and qualify false detections using the laboratory method/preparation blank summaries. Qualify analyte concentrations between the DL and the LOQ as estimated ("J" qualifier). Replace laboratory qualifiers with validation qualifiers on validated data in accordance with the laboratory DV process described below and document findings in the DV report. Retain laboratory flags in the database and provide them in the DV reports to document data as received from the laboratory.	Data Validators, Helios/TT JV	External
Field and Laboratory Duplicate Analyses for Precision	Verify field sampling precision by checking RPDs for field duplicate samples. Verify laboratory precision by checking RPDs or %D values from calibrations, laboratory duplicates, MS/MSDs, and LCS/LCSDs. Ensure compliance with MPC accuracy and precision goals listed in Sections 6 and 11. Document findings in the DV report.	Data Validator, Helios/TT JV	External
SAP/Laboratory Data Packages/ EDDs	Conduct USEPA Stage 2A DV on 100% of the definitive laboratory data generated by the selected methods using QC criteria listed in Sections 6.0, 9.0, and 11.0, and QSM 5.3 . Apply qualifier flags in accordance with logic provided in DoD General Data Validation Guidelines (November 2019), and National Functional Guidelines (NFG) for Inorganic Superfund (USEPA, 2017a) or Organic Superfund (USEPA, 2017b) data review (including EPA Region-specific requirements, if applicable). If DoD guidelines and NFG conflict, defer to the logic in the DoD guidelines. Document findings in the DV report. If analysis-specific DoD guidelines become available prior to the start of validation, use those guidelines and the General DoD guidelines in place of the National Functional Guidelines.	Data Validator, Helios/TT JV	External
SAP/ Laboratory Data Packages/ EDDs	Check that all data and data qualifier flags have been transferred correctly and completely to the project Structured Query Language (SQL) database. Document deviations in the DV report.	Data Validators, Helios/TT JV	External

Data Review Input	Description	Responsible for Verification (name, organization)	Internal/ External
SAP/ Laboratory Data Packages/ EDDs	Describe in the validation report the impact on DLs that are elevated because of matrix interferences. Be especially cognizant of and evaluate the impact of sample dilutions on low-concentration analytes when the dilution was performed because of the high concentration of other target analytes. Inform the Helios/TT JV PM of compromised data usability. Review and add screening criteria to the laboratory EDDs. Flag samples and notify the Helios/TT JV PM of samples that exceed criteria listed in the SAP.	Data Validators, Helios/TT JV	External
SAP/ Laboratory Data Packages/ EDDs	Summarize deviations from methods, procedures, or contracts in the project report. Determine the impact of any deviation from sampling or analytical methods and SOPs requirements and matrix interferences effect on the analytical results. Determine if the data met the MPCs specified in this SAP and determine the impact of any deviations on the technical usability of the data. Document significant deviations from the SAP in the project report.	PM or designee, Project Chemist, Helios/TT JV	External

DoD, 2019. *General Data Validation Guidelines*, Environmental Data Quality Workgroup, Revision 1, November.

[EPA, 2017a.](#) *USEPA National Functional Guidelines for Inorganic Superfund Methods Data Review (SOM02.4)*. EPA-540-R-2017-001. January.

EPA, 2017b. *USEPA National Functional Guidelines for Organic Superfund Methods Data Review (SOM02.4)*. EPA-540-R-2017-002. January.
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NAVFAC, 2014. Record of Decision, Site 17 – Former Building 32, Gould Island, Jamestown, Rhode Island. Operable Unit 06, Naval Station Newport, Rhode Island. June 20.

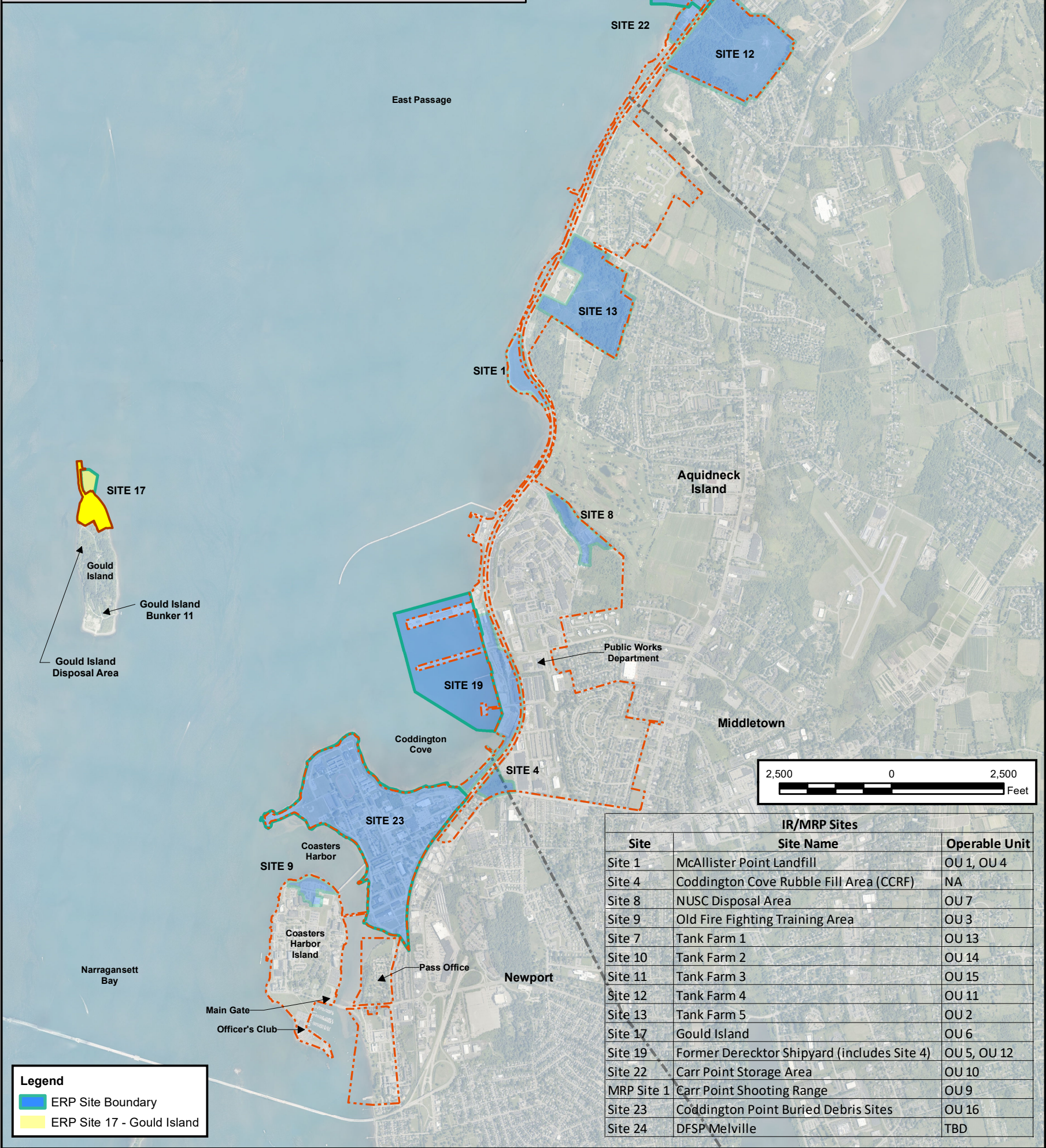
NAVFAC, 2019. NAVFAC Environmental Restoration Program Recordkeeping Manual, October, or most current version. Downloadable from the Internet at:
https://www.navfac.navy.mil/products_and_services/ev/products_and_services/env_restoration/RKM.html

Resolution, 2017. Tier II Sampling and Analysis Plan for Long-Term Groundwater and Sediment Monitoring. Former Building 32, Gould Island (Site 17), Operable Unit 6. Naval Station Newport, Jamestown, Rhode Island. June 5. (*Appendix A of the Long-Term Monitoring/Management Plan, Site 17 – Former Building 32, Gould Island*)

Tetra Tech, 2020, Draft Annual Long-Term Groundwater Monitoring Report, Years 1 and 2. Site 17 – Former Building 32, Gould Island. Naval Station Newport, Newport, Rhode Island. March.

FIGURES

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Legend

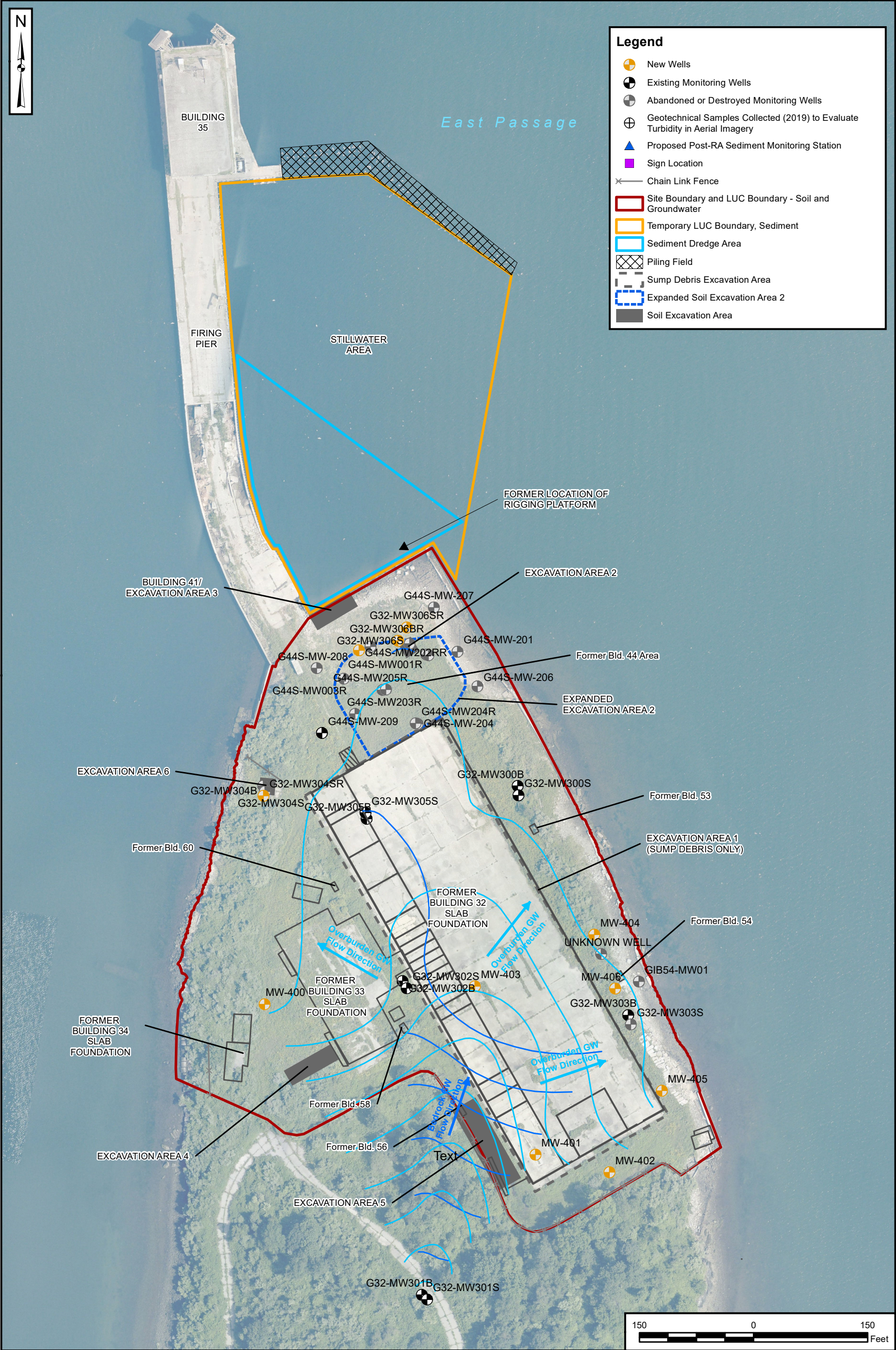
- ERP Site Boundary
- ERP Site 17 - Gould Island

IR/MRP Sites		
Site	Site Name	Operable Unit
Site 1	McAllister Point Landfill	OU 1, OU 4
Site 4	Coddington Cove Rubble Fill Area (CCRF)	NA
Site 8	NUSC Disposal Area	OU 7
Site 9	Old Fire Fighting Training Area	OU 3
Site 7	Tank Farm 1	OU 13
Site 10	Tank Farm 2	OU 14
Site 11	Tank Farm 3	OU 15
Site 12	Tank Farm 4	OU 11
Site 13	Tank Farm 5	OU 2
Site 17	Gould Island	OU 6
Site 19	Former Derecktor Shipyard (includes Site 4)	OU 5, OU 12
Site 22	Carr Point Storage Area	OU 10
MRP Site 1	Carr Point Shooting Range	OU 9
Site 23	Coddington Point Buried Debris Sites	OU 16
Site 24	DFSP Melville	TBD



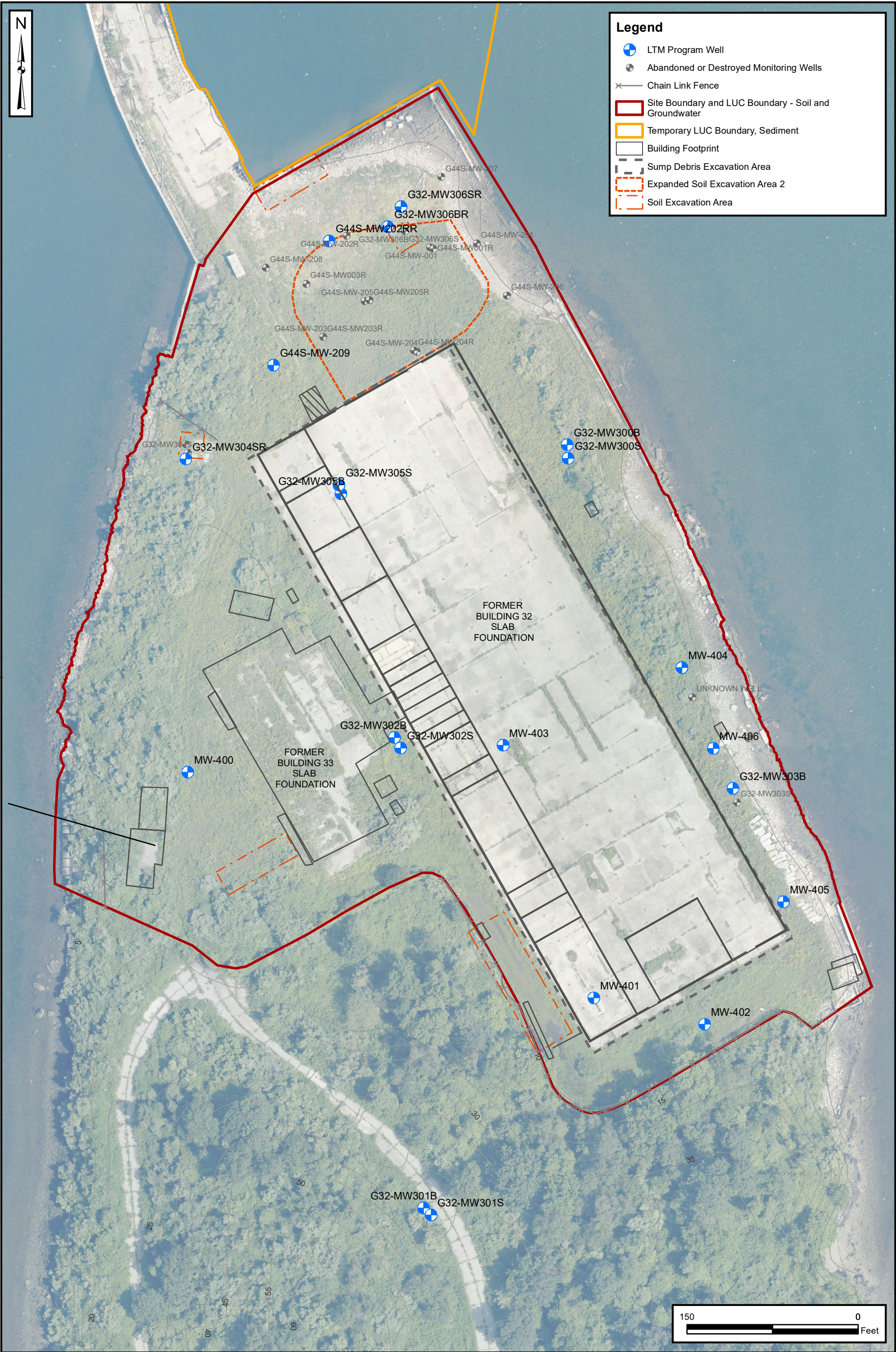
Groundwater LTM SAP
Site 17 - Gould Island
Site Location Map
Naval Station Newport
Jamestown, Rhode Island

CTO WE10	
DRAWN BY	DATE
M. HORTON	5/26/2020
CHECKED BY	DATE
M. HORTON	05/11/20
FIGURE NUMBER 4-1	



Groundwater LTM SAP
Site 17 - Gould Island
Site Detail Map
Naval Station Newport
Jamestown, Rhode Island

CTO WE10	
DRAWN BY	DATE
M. MASON	01/25/2018
CHECKED BY	DATE
M. HORTON	05/11/20
FIGURE NUMBER 4-2	



Groundwater LTM SAP
Site 17 - Gould Island

LTM Plan

Naval Station Newport
Jamestown, Rhode Island

CTO ML204592	
DRAWN BY	DATE
M. MASON	01/25/2018
CHECKED BY	DATE
M. HORTON	05/06/20
FIGURE NUMBER 7-1	

Appendix A

Tetra Tech Standard Operating Procedures

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TETRA TECH

STANDARD OPERATING PROCEDURE (GUIDANCE)

Number

CT-04

Page

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Effective Date

4/30/2020

Revision

6

Applicability

EGS Operating Unit

Prepared

Earth Sciences Department

Subject

SAMPLE NOMENCLATURE

Approved

T. Johnston

04/30/2020

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1.0 PURPOSE

This document specifies a consistent sample nomenclature system that facilitates sample and data management in a cost-effective manner. The sample nomenclature system has been devised such that the following objectives can be attained:

- Sorting of data by matrix
- Sorting of data by depth
- Maintenance of consistency (field, laboratory, and database sample numbers)
- Accommodation of all project-specific requirements
- Accommodation of laboratory sample number length constraints (maximum of 20 characters)

2.0 SCOPE AND APPLICABILITY

The methods described in this document are designed to be used consistently for all projects requiring electronic data. Other contract- or project-specific sample nomenclature requirements may also be applicable. If project-specific requirements dictate the use of a different system of nomenclature than what is described in this guidance, the objectives identified above must be borne in mind when developing the alternate nomenclature. Consultation with the EGS Operating Unit Information Solutions Group (database group) is advisable in those cases. The database group will help ensure that sample nomenclature is compatible with the affected databases.

3.0 GLOSSARY

None.

4.0 RESPONSIBILITIES AND PERSONNEL QUALIFICATIONS

Program Manager – The Program Managers (or their designees) are responsible for informing contract-specific Project Managers (PMs) of the existence and requirements of this document.

Project Manager – Project Managers (PMs) determine the applicability of this document based on: (1) program-specific requirements and (2) project size and objectives. PMs (or their designees) ensure that sample nomenclature requirements are thoroughly specified in the relevant project planning document (e.g., sampling and analysis plan) and are consistent with this document, if relevant. PMs ensure that the Field Operations Leader (FOL) is familiar with the sample nomenclature system. Program Managers appoint PMs based on their experience and past performance and qualifications..

Field Operations Leader – The FOL ensures that all field technicians or sampling personnel are thoroughly familiar with this document and the project-specific sample nomenclature system and that they implement the nomenclature system in accordance with this document.

Support Personnel – These personnel, whether in the field or office, are responsible for adhering to this procedure or with superseding project-specific requirements, if applicable.

5.0 PROCEDURES

5.1 INTRODUCTION

The sample identification (ID) system may consist of as few as eight but not more than 20 distinct alpha-numeric characters. The sample ID will be provided to the laboratory on the sample labels and chain-of-

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custody forms. The basic sample ID provided to the laboratory has three segments and shall be as follows, where "A" indicates "alpha," and "N" indicates "numeric":

A or N 3 or 4 Characters	AAA 2 or 3 Characters	A or N 3 to 6 Characters
Site Identifier	Sample Type	Sample Location

Additional segments may be added as needed. For example:

(1) Soil and sediment sample ID

A or N 3 or 4 Characters	AAA 2 or 3 Characters	A or N 3 to 6 Characters	NNNN 4 Characters
Site identifier	Sample type	Sample location	Sample depth

(2) Aqueous (groundwater or surface water) sample ID

A or N 3 or 4 Characters	AAA 2 or 3 Characters	A or N 3 to 6 Characters	NN 2 Characters	A 1 Character
Site identifier	Sample type	Sample location	Round number	Filtered sample only -F

NOTE 1: Sampling dates may be used in place of round numbers.

NOTE 2: The "-F" may be omitted from the sample ID if there is another way to definitively indicate on chain of custody forms and in the EGS Operating Unit database which portion of a samples is designated for total species analysis (unfiltered samples) and which is designated for dissolved species analysis (filtered samples).

(3) Biota sample ID

A or N 3 or 4 Characters	AAA 2 or 3 Characters	A or N 3 to 6 Characters	AA 2 Characters	NNN 3 Characters
Site identifier	Sample type	Sample location	Species identifier	Sample group number

5.2 SAMPLE IDENTIFICATION FIELD REQUIREMENTS

The various fields in the sample ID include but are not limited to the following:

- Site identifier
- Sample type
- Sample location
- Sample depth
- Sampling round number
- Filtered
- Species identifier
- Sample group number

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The site identifier must be a three- or four-character field (numeric characters, alpha characters, or a mixture of alpha and numeric characters may be used). A site number is necessary because many facilities/sites have multiple individual sites, Solid Waste Management Units (SWMUs), Operable Units (OUs), etc. Several examples are presented in Section 5.3.

The sample type must be a two- or three-character alpha field. Suggested codes are provided in Section 5.3.

The sample location must be at least a three-character field but may have up to six characters (alpha, numeric, or a mixture). The six characters may be useful in identifying a monitoring well to be sampled or describing a grid location.

The sample depth field is used to note the depth below ground surface (bgs) in units of feet at which a soil sample is collected; and the depth in units of inches bgs at which a sediment sample was collected. The first two numbers of the four-number code specify the top interval, and the third and fourth specify the bottom interval in feet bgs of the sample. If the sample depth is equal to or greater than 100, then only the top interval would be represented and the sampling depth would be truncated to three characters. The depths will be noted in whole numbers only; further detail, if needed, will be recorded on the sample log sheet or boring log, in the logbook, etc.

A two-digit round number will be used to track the number of aqueous samples collected from a particular aqueous sample location. The first sample collected from a location will be assigned the round identifier 01, the second 02, etc. This applies to both existing and proposed monitoring wells and surface water locations.

Aqueous samples that are field filtered (dissolved analysis) will be identified with an "-F" in the last field segment. No entry in this segment signifies an unfiltered (total) sample.

NOTE: If samples are filtered in the field, the results will be segregated in the EGS Operating Unit database according to whether they represent the unfiltered sample or filtered sample. The chain of custody form must be marked to indicate which portion of a sample is filtered. This may be accomplished by appending "-F" to the end of the sample number or by otherwise indicating on the chain of custody form which sample portion is filtered (for dissolved species analysis) as opposed to unfiltered (for total species analysis).

The species identifier must be a two-character alpha field. Several suggested codes are provided in Section 5.3.

The three-digit sample group number will be used to track the number of biota sample groups (a particular group size may be determined by sample technique, media type, the number of individuals caught, weight issues, time, etc.) by species and location. The first sample group of a particular species collected from a given location will be assigned the sample group number 001, and the second sample group of the same species collected from the same location will be assigned the sample group number 002.

5.3 EXAMPLE SITE CHARACTERIZATION SAMPLE FIELD DESIGNATIONS

Examples of each of the fields are as follows:

Site identifier - Examples of site numbers/designations are as follows:

- A01 - Area of Concern (AOC) 1
- 125 - SWMU 125
- 000 - Base- or facility-wide sample (e.g., upgradient well)
- BBG - Base background

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The examples cited are only suggestions. Each PM (or designee) must designate appropriate (and consistent) site designations for their individual project.

Sample type - Examples of sample types are as follows:

AH	-	Ash Sample
AS	-	Air Sample
BM	-	Building Material Sample
BSB	-	Biota Sample Full Body
BSF	-	Biota Sample Fillet
CP	-	Composite Sample
CS	-	Chip Sample
DS	-	Drum Sample
DU	-	Dust Sample
FP	-	Free Product
IDW	-	Investigation-Derived Waste Sample
LT	-	Leachate Sample
MW	-	Monitoring Well Groundwater Sample
OF	-	Outfall Sample
RW	-	Residential Well Sample
SB	-	Soil Boring Sample
SD	-	Sediment Sample
SC	-	Scrape Sample
SG	-	Soil Gas Sample
SL	-	Sludge Sample
SP	-	Seep Sample
SS	-	Surface Soil Sample
ST	-	Storm Sewer Water Sample
SW	-	Surface Water Sample
TP	-	Test Pit Sample
TW	-	Temporary Well Sample
WC	-	Well Construction Material Sample
WP	-	Wipe Sample
WS	-	Waste/Solid Sample
WW	-	Wastewater Sample

Sample location - Examples of the location field are as follows:

001	-	Monitoring Well 1
N32E92	-	Grid location 32 North and 92 East
D096	-	Investigation-derived waste drum number 96

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Species identifier - Examples of species identifier are as follows:

BC	-	Blue Crab
GB	-	Blue Gill
CO	-	Corn
SB	-	Soybean

5.4 EXAMPLES OF SITE CHARACTERIZATION SAMPLE NOMENCLATURE

A first round monitoring well groundwater sample collected from existing monitoring well 001 at SWMU 16 for a filtered sample would be designated as 016MW00101-F (or without the "-F" if otherwise indicated as to which portion of sample is filtered and which is unfiltered)..

The second round monitoring well groundwater sample collected from existing monitoring well C20P2 at Site 23 for an unfiltered sample would be designated as 023MWC20P202.

The second surface water sample collected from point 01 at SWMU 130 for an unfiltered sample would be designated as 130SW00102.

A surface soil sample collected from grid location 32 North and 92 East at Site 32 at the 0- to 2-foot interval would be designated as 032SSN32E920002.

A subsurface soil sample from soil boring 03 at SWMU 32 at an interval of 4 to 5 feet bgs would be designated as 032SB0030405.

A sediment sample collected at SWMU 19 from 0 to 6 inches at location 14 would be designated as 019SD0140006. The sample data sheet would reflect the precise depth at which this sample was collected.

During biota sampling for full-body analysis, the first time a minnow trap was checked at grid location A25 of SWMU 1415, three small blue gills were captured, collected, and designated with the sample ID of 1415BSBA25BG001. The second time blue gill were collected at the same location (grid location A25 at SWMU 1415), the sample ID would be 1415BSBA25BG002.

NOTE: No dash (-) or spacing is used between the segments with the exception of the filtered segment. The "F" used for a filtered aqueous sample is preceded by a dash (-F).

NOTE: It may be more useful for project staff to use feet bgs instead of inches bgs to indicate sediment sampling depths. Whichever units of measure are used for sampling of any matrix, the units must be clearly documented to ensure there is no confusion. Consistency with previous sample nomenclature also will reduce any chance of confusion.

5.5 FIELD QA/QC SAMPLE NOMENCLATURE

Field Quality Assurance (QA)/Quality Control (QC) samples are designated using a different coding system than the one used for site characterization samples. The QC code will consist of a three- to four-segment alpha-numeric code that identifies the sample QC type, the date the sample was collected, and the number of this type of QC sample collected on that date.

AA	NNNNNN	NN	-F
QC type	Date	Sequence number (per day)	Filtered (aqueous only, if needed)

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NOTE: The “-F” may be omitted from the sample ID if there is another way to definitively indicate on chain of custody forms which portion of a samples is designated for total species analysis (unfiltered samples) and which is designated for dissolved species analysis (filtered samples).

The QC types are identified as:

TB = Trip Blank

RB = Rinsate Blank (Equipment Blank)

FD = Field Duplicate

AB = Ambient Conditions Blank

WB = Source Water Blank

The sampling time recorded on the chain-of-custody form, labels, and tags for duplicate samples will be 0000 so that the samples are "blind" to the laboratory. Notes detailing the sample number, time, date, and type will be recorded on the routine sample log sheets and will document the location of the duplicate sample (sample log sheets are not provided to the laboratory). Documentation for all other QC types (TB, RB, AB, and WB) will be recorded on the QC Sample Log Sheet (see SOP SA-6.3, Field Documentation).

5.6 EXAMPLES OF FIELD QA/QC SAMPLE NOMENCLATURE

The first duplicate of the day for a filtered groundwater sample collected on June 3, 2016, would be designated as FD06031601-F (or without the “-F” if otherwise indicated on the chain of custody form as to which portion of sample is filtered and which is unfiltered).

The third duplicate of the day taken of a subsurface soil sample collected on November 17, 2016, would be designated as FD11171603.

The first trip blank associated with samples collected on October 12, 2016, would be designated as TB10121601.

The only rinsate blank collected on November 17, 2016, would be designated as RB11171601.

6.0 DEVIATIONS

Deviations from instructions specified in this document are allowed but sufficiently accurate and precise descriptions of sample nomenclature must be described in the site-specific planning documents to prevent confusion over sample identification.

U.S. ENVIRONMENTAL PROTECTION AGENCY REGION I

LOW STRESS (low flow) PURGING AND SAMPLING PROCEDURE FOR THE COLLECTION OF GROUNDWATER SAMPLES FROM MONITORING WELLS

Quality Assurance Unit
U.S. Environmental Protection Agency – Region 1
11 Technology Drive
North Chelmsford, MA 01863

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Prepared by: _____
(Robert Reinhart, Quality Assurance Unit) Date _____

Approved by: _____
(John Smaldone, Quality Assurance Unit) Date _____

Revision Page

Date	Rev #	Summary of changes	Sections
7/30/96	1	Finalized	
01/19/10	2	Updated	All sections
3/23/17	3	Updated	All sections
9/20/17	4	Updated	Section 7.0

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1.0 USE OF TERMS

Equipment blank: The equipment blank shall include the pump and the pump's tubing. If tubing is dedicated to the well, the equipment blank needs only to include the pump in subsequent sampling rounds. If the pump and tubing are dedicated to the well, the equipment blank is collected prior to its placement in the well. If the pump and tubing will be used to sample multiple wells, the equipment blank is normally collected after sampling from contaminated wells and not after background wells.

Field duplicates: Field duplicates are collected to determine precision of the sampling procedure. For this procedure, collect duplicate for each analyte group in consecutive order (VOC original, VOC duplicate, SVOC original, SVOC duplicate, etc.).

Indicator field parameters: This SOP uses field measurements of turbidity, dissolved oxygen, specific conductance, temperature, pH, and oxidation/reduction potential (ORP) as indicators of when purging operations are sufficient and sample collection may begin.

Matrix Spike/Matrix Spike Duplicates: Used by the laboratory in its quality assurance program. Consult the laboratory for the sample volume to be collected.

Potentiometric Surface: The level to which water rises in a tightly cased well constructed in a confined aquifer. In an unconfined aquifer, the potentiometric surface is the water table.

QAPP: Quality Assurance Project Plan

SAP: Sampling and Analysis Plan

SOP: Standard operating procedure

Stabilization: A condition that is achieved when all indicator field parameter measurements are sufficiently stable (as described in the "Monitoring Indicator Field Parameters" section) to allow sample collection to begin.

Temperature blank: A temperature blank is added to each sample cooler. The blank is measured upon receipt at the laboratory to assess whether the samples were properly cooled during transit.

Trip blank (VOCs): Trip blank is a sample of analyte-free water taken to the sampling site and returned to the laboratory. The trip blanks (one pair) are added to each sample cooler that contains VOC samples.

2.0 SCOPE & APPLICATION

The goal of this groundwater sampling procedure is to collect water samples that reflect the total mobile organic and inorganic loads (dissolved and colloidal sized fractions) transported through the subsurface under ambient flow conditions, with minimal physical and chemical alterations from sampling operations. This standard operating procedure (SOP) for collecting groundwater samples will help ensure that the project's data quality objectives (DQOs) are met under certain low-flow conditions.

The SOP emphasizes the need to minimize hydraulic stress at the well-aquifer interface by maintaining low water-level drawdowns, and by using low pumping rates during purging and sampling operations. Indicator field parameters (e.g., dissolved oxygen, pH, etc.) are monitored during purging in order to determine when sample collection may begin. Samples properly collected using this SOP are suitable for analysis of groundwater contaminants (volatile and semi-volatile organic analytes, dissolved gases, pesticides, PCBs, metals and other inorganics), or naturally occurring analytes. This SOP is based on Puls, and Barcelona (1996).

This procedure is designed for monitoring wells with an inside diameter (1.5-inches or greater) that can accommodate a positive lift pump with a screen length or open interval ten feet or less and with a water level above the top of the screen or open interval (Hereafter, the "screen or open interval" will be referred to only as "screen interval"). This SOP is not applicable to other well-sampling conditions.

While the use of dedicated sampling equipment is not mandatory, dedicated pumps and tubing can reduce sampling costs significantly by streamlining sampling activities and thereby reducing the overall field costs.

The goal of this procedure is to emphasize the need for consistency in deploying and operating equipment while purging and sampling monitoring wells during each sampling event. This will help to minimize sampling variability.

This procedure describes a general framework for groundwater sampling. Other site specific information (hydrogeological context, conceptual site model (CSM), DQOs, etc.) coupled with systematic planning must be added to the procedure in order to develop an appropriate site specific SAP/QAPP. In addition, the site specific SAP/QAPP must identify the specific equipment that will be used to collect the groundwater samples.

This procedure does not address the collection of water or free product samples from wells containing free phase LNAPLs and/or DNAPLs (light or dense non-aqueous phase

liquids). For this type of situation, the reader may wish to check: Cohen, and Mercer (1993) or other pertinent documents.

This SOP is to be used when collecting groundwater samples from monitoring wells at all Superfund, Federal Facility and RCRA sites in Region 1 under the conditions described herein. Request for modification of this SOP, in order to better address specific situations at individual wells, must include adequate technical justification for proposed changes. All changes and modifications must be approved and included in a revised SAP/QAPP before implementation in field.

3.0 BACKGROUND FOR IMPLEMENTATION

It is expected that the monitoring well screen has been properly located (both laterally and vertically) to intercept existing contaminant plume(s) or along flow paths of potential contaminant migration. Problems with inappropriate monitoring well placement or faulty/improper well installation cannot be overcome by even the best water sampling procedures. This SOP presumes that the analytes of interest are moving (or will potentially move) primarily through the more permeable zones intercepted by the screen interval.

Proper well construction, development, and operation and maintenance cannot be overemphasized. The use of installation techniques that are appropriate to the hydrogeologic setting of the site often prevent "problem well" situations from occurring. During well development, or redevelopment, tests should be conducted to determine the hydraulic characteristics of the monitoring well. The data can then be used to set the purging/sampling rate, and provide a baseline for evaluating changes in well performance and the potential need for well rehabilitation. Note: if this installation data or well history (construction and sampling) is not available or discoverable, for all wells to be sampled, efforts to build a sampling history should commence with the next sampling event.

The pump intake should be located within the screen interval and at a depth that will remain under water at all times. It is recommended that the intake depth and pumping rate remain the same for all sampling events. The mid-point or the lowest historical midpoint of the saturated screen length is often used as the location of the pump intake. For new wells, or for wells without pump intake depth information, the site's SAP/QAPP must provide clear reasons and instructions on how the pump intake depth(s) will be selected, and reason(s) for the depth(s) selected. If the depths to top and bottom of the well screen are not known, the SAP/QAPP will need to describe how the sampling depth will be determined and how the data can be used.

Stabilization of indicator field parameters is used to indicate that conditions are suitable for sampling to begin. Achievement of turbidity levels of less than 5 NTU, and stable drawdowns of less than 0.3 feet, while desirable, are not mandatory. Sample collection

may still take place provided the indicator field parameter criteria in this procedure are met. If after 2 hours of purging indicator field parameters have not stabilized, one of three optional courses of action may be taken: a) continue purging until stabilization is achieved, b) discontinue purging, do not collect any samples, and record in log book that stabilization could not be achieved (documentation must describe attempts to achieve stabilization), c) discontinue purging, collect samples and provide full explanation of attempts to achieve stabilization (note: there is a risk that the analytical data obtained, especially metals and strongly hydrophobic organic analytes, may reflect a sampling bias and therefore, the data may not meet the data quality objectives of the sampling event).

It is recommended that low-flow sampling be conducted when the air temperature is above 32°F (0°C). If the procedure is used below 32°F, special precautions will need to be taken to prevent the groundwater from freezing in the equipment. Because sampling during freezing temperatures may adversely impact the data quality objectives, the need for water sample collection during months when these conditions are likely to occur should be evaluated during site planning and special sampling measures may need to be developed. Ice formation in the flow-through-cell will cause the monitoring probes to act erratically. A transparent flow-through-cell needs to be used to observe if ice is forming in the cell. If ice starts to form on the other pieces of the sampling equipment, additional problems may occur.

4.0 HEALTH & SAFETY

When working on-site, comply with all applicable OSHA requirements and the site's health/safety procedures. All proper personal protection clothing and equipment are to be worn. Some samples may contain biological and chemical hazards. These samples should be handled with suitable protection to skin, eyes, etc.

5.0 CAUTIONS

The following cautions need to be considered when planning to collect groundwater samples when the below conditions occur.

If the groundwater degasses during purging of the monitoring well, dissolved gases and VOCs will be lost. When this happens, the groundwater data for dissolved gases (e.g., methane, ethene, ethane, dissolved oxygen, etc.) and VOCs will need to be qualified. Some conditions that can promote degassing are the use of a vacuum pump (e.g., peristaltic pumps), changes in aperture along the sampling tubing, and squeezing/pinching the pump's tubing which results in a pressure change.

When collecting the samples for dissolved gases and VOCs analyses, avoid aerating the groundwater in the pump's tubing. This can cause loss of the dissolved gases and VOCs in

the groundwater. Having the pump's tubing completely filled prior to sampling will avoid this problem when using a centrifugal pump or peristaltic pump.

Direct sun light and hot ambient air temperatures may cause the groundwater in the tubing and flow-through-cell to heat up. This may cause the groundwater to degas which will result in loss of VOCs and dissolved gases. When sampling under these conditions, the sampler will need to shade the equipment from the sunlight (e.g., umbrella, tent, etc.). If possible, sampling on hot days, or during the hottest time of the day, should be avoided. The tubing exiting the monitoring well should be kept as short as possible to avoid the sun light or ambient air from heating up the groundwater.

Thermal currents in the monitoring well may cause vertical mixing of water in the well bore. When the air temperature is colder than the groundwater temperature, it can cool the top of the water column. Colder water which is denser than warm water sinks to the bottom of the well and the warmer water at the bottom of the well rises, setting up a convection cell. "During low-flow sampling, the pumped water may be a mixture of convecting water from within the well casing and aquifer water moving inward through the screen. This mixing of water during low-flow sampling can substantially increase equilibration times, can cause false stabilization of indicator parameters, can give false indication of redox state, and can provide biological data that are not representative of the aquifer conditions" (Vroblesky 2007).

Failure to calibrate or perform proper maintenance on the sampling equipment and measurement instruments (e.g., dissolved oxygen meter, etc.) can result in faulty data being collected.

Interferences may result from using contaminated equipment, cleaning materials, sample containers, or uncontrolled ambient/surrounding air conditions (e.g., truck/vehicle exhaust nearby).

Cross contamination problems can be eliminated or minimized through the use of dedicated sampling equipment and/or proper planning to avoid ambient air interferences. Note that the use of dedicated sampling equipment can also significantly reduce the time needed to complete each sampling event, will promote consistency in the sampling, and may reduce sampling bias by having the pump's intake at a constant depth.

Clean and decontaminate all sampling equipment prior to use. All sampling equipment needs to be routinely checked to be free from contaminants and equipment blanks collected to ensure that the equipment is free of contaminants. Check the previous equipment blank data for the site (if they exist) to determine if the previous cleaning procedure removed the contaminants. If contaminants were detected and they are a concern, then a more vigorous cleaning procedure will be needed.

6.0 PERSONNEL QUALIFICATIONS

All field samplers working at sites containing hazardous waste must meet the requirements of the OSHA regulations. OSHA regulations may require the sampler to take the 40 hour OSHA health and safety training course and a refresher course prior to engaging in any field activities, depending upon the site and field conditions.

The field samplers must be trained prior to the use of the sampling equipment, field instruments, and procedures. Training is to be conducted by an experienced sampler before initiating any sampling procedure.

The entire sampling team needs to read, and be familiar with, the site Health and Safety Plan, all relevant SOPs, and SAP/QAPP (and the most recent amendments) before going onsite for the sampling event. It is recommended that the field sampling leader attest to the understanding of these site documents and that it is recorded.

7.0 EQUIPMENT AND SUPPLIES

A. Informational materials for sampling event

A copy of the current Health and Safety Plan, SAP/QAPP, monitoring well construction data, location map(s), field data from last sampling event, manuals for sampling, and the monitoring instruments' operation, maintenance, and calibration manuals should be brought to the site.

B. Well keys.

C. Extraction device

Adjustable rate, submersible pumps (e.g., centrifugal, bladder, etc.) which are constructed of stainless steel or polytetrafluoroethylene (PTFE, i.e. Teflon®) are preferred. PTFE, however, should not be used when sampling for per- and polyfluoroalkyl substances (PFAS) as it is likely to contain these substances.

Note: If extraction devices constructed of other materials are to be used, adequate information must be provided to show that the substituted materials do not leach contaminants nor cause interferences to the analytical procedures to be used. Acceptance of these materials must be obtained before the sampling event.

If bladder pumps are selected for the collection of VOCs and dissolved gases, the pump setting should be set so that one pulse will deliver a water volume that is sufficient to fill a 40 mL VOC vial. This is not mandatory, but is considered a “best practice”. For the proper operation, the bladder pump will need a minimum amount of water above the pump; consult the manufacturer for the recommended submergence. The pump’s recommended submergence value should be determined during the planning stage, since it may influence well construction and placement of dedicated pumps where water-level fluctuations are significant.

Adjustable rate, peristaltic pumps (suction) are to be used with caution when collecting samples for VOCs and dissolved gases (e.g., methane, carbon dioxide, etc.) analyses. Additional information on the use of peristaltic pumps can be found in Appendix A. If peristaltic pumps are used, the inside diameter of the rotor head tubing needs to match the inside diameter of the tubing installed in the monitoring well.

Inertial pumping devices (motor driven or manual) are not recommended. These devices frequently cause greater disturbance during purging and sampling, and are less easily controlled than submersible pumps (potentially increasing turbidity and sampling variability, etc.). This can lead to sampling results that are adversely affected by purging and sampling operations, and a higher degree of data variability.

D. Tubing

PTFE (Teflon®) or PTFE-lined polyethylene tubing are preferred when sampling is to include VOCs, SVOCs, pesticides, PCBs and inorganics. As discussed in the previous section, PTFE tubing should not be used when sampling for PFAS. In this case, a suitable alternative such as high-density polyethylene tubing should be used.

PVC, polypropylene or polyethylene tubing may be used when collecting samples for metal and other inorganics analyses.

Note: If tubing constructed of other materials is to be used, adequate information must be provided to show that the substituted materials do not leach contaminants nor cause interferences to the analytical procedures to be used. Acceptance of these materials must be obtained before the sampling event.

The use of 1/4 inch or 3/8 inch (inside diameter) tubing is recommended. This will help ensure that the tubing remains liquid filled when operating at very low pumping rates when using centrifugal and peristaltic pumps.

Silastic tubing should be used for the section around the rotor head of a peristaltic pump. It should be less than a foot in length. The inside diameter of the tubing used at the pump rotor head must be the same as the inside diameter of tubing placed in the well. A tubing connector is used to connect the pump rotor head tubing to the well tubing. Alternatively, the two pieces of tubing can be connected to each other by placing the one end of the tubing inside the end of the other tubing. The tubing must not be reused.

E. The water level measuring device

Electronic "tape", pressure transducer, water level sounder/level indicator, etc. should be capable of measuring to 0.01 foot accuracy. Recording pressure transducers, mounted above the pump, are especially helpful in tracking water levels during pumping operations, but their use must include check measurements with a water level "tape" at the start and end of each sampling event.

F. Flow measurement supplies

Graduated cylinder (size according to flow rate) and stopwatch usually will suffice.

Large graduated bucket used to record total water purged from the well.

G. Interface probe

To be used to check on the presence of free phase liquids (LNAPL, or DNAPL) before purging begins (as needed).

H. Power source (generator, nitrogen tank, battery, etc.)

When a gasoline generator is used, locate it downwind and at least 30 feet from the well so that the exhaust fumes do not contaminate samples.

I. Indicator field parameter monitoring instruments

Use of a multi-parameter instrument capable of measuring pH, oxidation/reduction potential (ORP), dissolved oxygen (DO), specific conductance, temperature, and coupled with a flow-through-cell is required when measuring all indicator field parameters, except turbidity. Turbidity is collected using a separate instrument. Record equipment/instrument identification (manufacturer, and model number).

Transparent, small volume flow-through-cells (e.g., 250 mLs or less) are preferred. This allows observation of air bubbles and sediment buildup in the cell, which can interfere with the operation of the monitoring instrument probes, to be easily detected. A small volume

cell facilitates rapid turnover of water in the cell between measurements of the indicator field parameters.

It is recommended to use a flow-through-cell and monitoring probes from the same manufacturer and model to avoid incompatibility between the probes and flow-through-cell.

Turbidity samples are collected before the flow-through-cell. A “T” connector coupled with a valve is connected between the pump’s tubing and flow-through-cell. When a turbidity measurement is required, the valve is opened to allow the groundwater to flow into a container. The valve is closed and the container sample is then placed in the turbidimeter.

Standards are necessary to perform field calibration of instruments. A minimum of two standards are needed to bracket the instrument measurement range for all parameters except ORP which use a Zobell solution as a standard. For dissolved oxygen, a wet sponge used for the 100% saturation and a zero dissolved oxygen solution are used for the calibration.

Barometer (used in the calibration of the Dissolved Oxygen probe) and the conversion formula to convert the barometric pressure into the units of measure used by the Dissolved Oxygen meter are needed.

J. Decontamination supplies

Includes (for example) non-phosphate detergent, distilled/deionized water, isopropyl alcohol, etc.

K. Record keeping supplies

Logbook(s), well purging forms, chain-of-custody forms, field instrument calibration forms, etc.

L. Sample bottles

M. Sample preservation supplies (as required by the analytical methods)

N. Sample tags or labels

O. PID or FID instrument

If appropriate, to detect VOCs for health and safety purposes, and provide qualitative field evaluations.

P. Miscellaneous Equipment

Equipment to keep the sampling apparatus shaded in the summer (e.g., umbrella) and from freezing in the winter. If the pump's tubing is allowed to heat up in the warm weather, the cold groundwater may degas as it is warmed in the tubing.

8.0 EQUIPMENT/INSTRUMENT CALIBRATION

Prior to the sampling event, perform maintenance checks on the equipment and instruments according to the manufacturer's manual and/or applicable SOP. This will ensure that the equipment/instruments are working properly before they are used in the field.

Prior to sampling, the monitoring instruments must be calibrated and the calibration documented. The instruments are calibrated using U.S Environmental Protection Agency Region 1 *Calibration of Field Instruments (temperature, pH, dissolved oxygen, conductivity/specific conductance, oxidation/reduction [ORP], and turbidity)*, March 23, 2017, or latest version or from one of the methods listed in 40CFR136, 40CFR141 and SW-846.

The instruments shall be calibrated at the beginning of each day. If the field measurement falls outside the calibration range, the instrument must be re-calibrated so that all measurements fall within the calibration range. At the end of each day, a calibration check is performed to verify that instruments remained in calibration throughout the day. This check is performed while the instrument is in measurement mode, not calibration mode. If the field instruments are being used to monitor the natural attenuation parameters, then a calibration check at mid-day is highly recommended to ensure that the instruments did not drift out of calibration. Note: during the day if the instrument reads zero or a negative number for dissolved oxygen, pH, specific conductance, or turbidity (negative value only), this indicates that the instrument drifted out of calibration or the instrument is malfunctioning. If this situation occurs the data from this instrument will need to be qualified or rejected.

9.0 PRELIMINARY SITE ACTIVITIES (as applicable)

Check the well for security (damage, evidence of tampering, missing lock, etc.) and record pertinent observations (include photograph as warranted).

If needed, lay out a sheet of clean polyethylene for monitoring and sampling equipment, unless equipment is elevated above the ground (e.g., on a table, etc.).

Remove well cap and if appropriate measure VOCs at the rim of the well with a PID or FID instrument and record reading in field logbook or on the well purge form.

If the well casing does not have an established reference point (usually a V-cut or indelible mark in the well casing), make one. Describe its location and record the date of the mark in the logbook (consider a photographic record as well). All water level measurements must be recorded relative to this reference point (and the altitude of this point should be determined using techniques that are appropriate to site's DQOs).

If water-table or potentiometric surface map(s) are to be constructed for the sampling event, perform synoptic water level measurement round (in the shortest possible time) before any purging and sampling activities begin. If possible, measure water level depth (to 0.01 ft.) and total well depth (to 0.1 ft.) the day before sampling begins, in order to allow for re-settlement of any particulates in the water column. This is especially important for those wells that have not been recently sampled because sediment buildup in the well may require the well to be redeveloped. If measurement of total well depth is not made the day before, it should be measured after sampling of the well is complete. All measurements must be taken from the established referenced point. Care should be taken to minimize water column disturbance.

Check newly constructed wells for the presence of LNAPLs or DNAPLs before the initial sampling round. If none are encountered, subsequent check measurements with an interface probe may not be necessary unless analytical data or field analysis signal a worsening situation. This SOP cannot be used in the presence of LNAPLs or DNAPLs. If NAPLs are present, the project team must decide upon an alternate sampling method. All project modifications must be approved and documented prior to implementation.

If available check intake depth and drawdown information from previous sampling event(s) for each well. Duplicate, to the extent practicable, the intake depth and extraction rate (use final pump dial setting information) from previous event(s). If changes are made in the intake depth or extraction rate(s) used during previous sampling event(s), for either portable or dedicated extraction devices, record new values, and explain reasons for the changes in the field logbook.

10.0 PURGING AND SAMPLING PROCEDURE

Purging and sampling wells in order of increasing chemical concentrations (known or anticipated) are preferred.

The use of dedicated pumps is recommended to minimize artificial mobilization and entrainment of particulates each time the well is sampled. Note that the use of dedicated sampling equipment can also significantly reduce the time needed to complete each sampling event, will promote consistency in the sampling, and may reduce sampling bias by having the pump's intake at a constant depth.

A. Initial Water Level

Measure the water level in the well before installing the pump if a non-dedicated pump is being used. The initial water level is recorded on the purge form or in the field logbook.

B. Install Pump

Lower pump, safety cable, tubing and electrical lines slowly (to minimize disturbance) into the well to the appropriate depth (may not be the mid-point of the screen/open interval). The Sampling and Analysis Plan/Quality Assurance Project Plan should specify the sampling depth (used previously), or provide criteria for selection of intake depth for each new well. If possible keep the pump intake at least two feet above the bottom of the well, to minimize mobilization of particulates present in the bottom of the well.

Pump tubing lengths, above the top of well casing should be kept as short as possible to minimize heating the groundwater in the tubing by exposure to sun light and ambient air temperatures. Heating may cause the groundwater to degas, which is unacceptable for the collection of samples for VOC and dissolved gases analyses.

C. Measure Water Level

Before starting pump, measure water level. Install recording pressure transducer, if used to track drawdowns, to initialize starting condition.

D. Purge Well

From the time the pump starts purging and until the time the samples are collected, the purged water is discharged into a graduated bucket to determine the total volume of groundwater purged. This information is recorded on the purge form or in the field logbook.

Start the pump at low speed and slowly increase the speed until discharge occurs. Check water level. Check equipment for water leaks and if present fix or replace the affected equipment. Try to match pumping rate used during previous sampling event(s). Otherwise, adjust pump speed until there is little or no water level drawdown. If the

minimal drawdown that can be achieved exceeds 0.3 feet, but remains stable, continue purging.

Monitor and record the water level and pumping rate every five minutes (or as appropriate) during purging. Record any pumping rate adjustments (both time and flow rate). Pumping rates should, as needed, be reduced to the minimum capabilities of the pump to ensure stabilization of the water level. Adjustments are best made in the first fifteen minutes of pumping in order to help minimize purging time. During pump start-up, drawdown may exceed the 0.3 feet target and then "recover" somewhat as pump flow adjustments are made. Purge volume calculations should utilize stabilized drawdown value, not the initial drawdown. If the initial water level is above the top of the screen do not allow the water level to fall into the well screen. The final purge volume must be greater than the stabilized drawdown volume plus the pump's tubing volume. If the drawdown has exceeded 0.3 feet and stabilizes, calculate the volume of water between the initial water level and the stabilized water level. Add the volume of the water which occupies the pump's tubing to this calculation. This combined volume of water needs to be purged from the well after the water level has stabilized before samples are collected.

Avoid the use of constriction devices on the tubing to decrease the flow rate because the constrictor will cause a pressure difference in the water column. This will cause the groundwater to degas and result in a loss of VOCs and dissolved gasses in the groundwater samples.

Note: the flow rate used to achieve a stable pumping level should remain constant while monitoring the indicator parameters for stabilization and while collecting the samples.

Wells with low recharge rates may require the use of special pumps capable of attaining very low pumping rates (e.g., bladder, peristaltic), and/or the use of dedicated equipment. For new monitoring wells, or wells where the following situation has not occurred before, if the recovery rate to the well is less than 50 mL/min., or the well is being essentially dewatered during purging, the well should be sampled as soon as the water level has recovered sufficiently to collect the volume needed for all anticipated samples. The project manager or field team leader will need to make the decision when samples should be collected, how the sample is to be collected, and the reasons recorded on the purge form or in the field logbook. A water level measurement needs to be performed and recorded before samples are collected. If the project manager decides to collect the samples using the pump, it is best during this recovery period that the pump intake tubing not be removed, since this will aggravate any turbidity problems. Samples in this specific situation may be collected without stabilization of indicator field parameters. Note that field conditions and efforts to overcome problematic situations must be recorded in order to support field decisions to deviate from normal procedures described in this SOP. If this type of problematic situation persists in a well, then water sample collection should be

changed to a passive or no-purge method, if consistent with the site's DQOs, or have a new well installed.

E. Monitor Indicator Field Parameters

After the water level has stabilized, connect the "T" connector with a valve and the flow-through-cell to monitor the indicator field parameters. If excessive turbidity is anticipated or encountered with the pump startup, the well may be purged for a while without connecting up the flow-through-cell, in order to minimize particulate buildup in the cell (This is a judgment call made by the sampler). Water level drawdown measurements should be made as usual. If possible, the pump may be installed the day before purging to allow particulates that were disturbed during pump insertion to settle.

During well purging, monitor indicator field parameters (turbidity, temperature, specific conductance, pH, ORP, DO) at a frequency of five minute intervals or greater. The pump's flow rate must be able to "turn over" at least one flow-through-cell volume between measurements (for a 250 mL flow-through-cell with a flow rate of 50 mLs/min., the monitoring frequency would be every five minutes; for a 500 mL flow-through-cell it would be every ten minutes). If the cell volume cannot be replaced in the five minute interval, then the time between measurements must be increased accordingly. Note: during the early phase of purging, emphasis should be put on minimizing and stabilizing pumping stress, and recording those adjustments followed by stabilization of indicator parameters. Purging is considered complete and sampling may begin when all the above indicator field parameters have stabilized. Stabilization is considered to be achieved when three consecutive readings are within the following limits:

Turbidity (10% for values greater than 5 NTU; if three Turbidity values are less than 5 NTU, consider the values as stabilized),

Dissolved Oxygen (10% for values greater than 0.5 mg/L, if three Dissolved Oxygen values are less than 0.5 mg/L, consider the values as stabilized),

Specific Conductance (3%),

Temperature (3%),

pH (± 0.1 unit),

Oxidation/Reduction Potential (± 10 millivolts).

All measurements, except turbidity, must be obtained using a flow-through-cell. Samples for turbidity measurements are obtained before water enters the flow-through-cell. Transparent flow-through-cells are preferred, because they allow field personnel to watch for particulate build-up within the cell. This build-up may affect indicator field parameter values measured within the cell. If the cell needs to be cleaned during purging operations, continue pumping and disconnect cell for cleaning, then reconnect after cleaning and

continue monitoring activities. Record start and stop times and give a brief description of cleaning activities.

The flow-through-cell must be designed in a way that prevents gas bubble entrapment in the cell. Placing the flow-through-cell at a 45 degree angle with the port facing upward can help remove bubbles from the flow-through-cell (see Appendix B Low-Flow Setup Diagram). Throughout the measurement process, the flow-through-cell must remain free of any gas bubbles. Otherwise, the monitoring probes may act erratically. When the pump is turned off or cycling on/off (when using a bladder pump), water in the cell must not drain out. Monitoring probes must remain submerged in water at all times.

F. Collect Water Samples

When samples are collected for laboratory analyses, the pump's tubing is disconnected from the "T" connector with a valve and the flow-through-cell. The samples are collected directly from the pump's tubing. Samples must not be collected from the flow-through-cell or from the "T" connector with a valve.

VOC samples are normally collected first and directly into pre-preserved sample containers. However, this may not be the case for all sampling locations; the SAP/QAPP should list the order in which the samples are to be collected based on the project's objective(s). Fill all sample containers by allowing the pump discharge to flow gently down the inside of the container with minimal turbulence.

If the pump's flow rate is too high to collect the VOC/dissolved gases samples, collect the other samples first. Lower the pump's flow rate to a reasonable rate and collect the VOC/dissolved gases samples and record the new flow rate.

During purging and sampling, the centrifugal/peristaltic pump tubing must remain filled with water to avoid aeration of the groundwater. It is recommended that 1/4 inch or 3/8 inch (inside diameter) tubing be used to help ensure that the sample tubing remains water filled. If the pump tubing is not completely filled to the sampling point, use the following procedure to collect samples: collect non-VOC/dissolved gases samples first, then increase flow rate slightly until the water completely fills the tubing, collect the VOC/dissolved gases samples, and record new drawdown depth and flow rate.

For bladder pumps that will be used to collect VOC or dissolved gas samples, it is recommended that the pump be set to deliver long pulses of water so that one pulse will fill a 40 mL VOC vial.

Use pre-preserved sample containers or add preservative, as required by analytical methods, to the samples immediately after they are collected. Check the analytical methods

(e.g. EPA SW-846, 40 CFR 136, water supply, etc.) for additional information on preservation.

If determination of filtered metal concentrations is a sampling objective, collect filtered water samples using the same low flow procedures. The use of an in-line filter (transparent housing preferred) is required, and the filter size (0.45 μm is commonly used) should be based on the sampling objective. Pre-rinse the filter with groundwater prior to sample collection. Make sure the filter is free of air bubbles before samples are collected. Preserve the filtered water sample immediately. Note: filtered water samples are not an acceptable substitute for unfiltered samples when the monitoring objective is to obtain chemical concentrations of total mobile contaminants in groundwater for human health or ecological risk calculations.

Label each sample as collected. Samples requiring cooling will be placed into a cooler with ice or refrigerant for delivery to the laboratory. Metal samples after acidification to a pH less than 2 do not need to be cooled.

G. Post Sampling Activities

If a recording pressure transducer is used to track drawdown, re-measure water level with tape.

After collection of samples, the pump tubing may be dedicated to the well for re-sampling (by hanging the tubing inside the well), decontaminated, or properly discarded.

Before securing the well, measure and record the well depth (to 0.1 ft.), if not measured the day before purging began. Note: measurement of total well depth annually is usually sufficient after the initial low stress sampling event. However, a greater frequency may be needed if the well has a “silting” problem or if confirmation of well identity is needed.

Secure the well.

11.0 DECONTAMINATION

Decontaminate sampling equipment prior to use in the first well, and then following sampling of each subsequent well. Pumps should not be removed between purging and sampling operations. The pump, tubing, support cable and electrical wires which were in contact with the well should be decontaminated by one of the procedures listed below.

The use of dedicated pumps and tubing will reduce the amount of time spent on decontamination of the equipment. If dedicated pumps and tubing are used, only the initial sampling event will require decontamination of the pump and tubing.

Note if the previous equipment blank data showed that contaminant(s) were present after using the below procedure or the one described in the SAP/QAPP, a more vigorous procedure may be needed.

Procedure 1

Decontaminating solutions can be pumped from either buckets or short PVC casing sections through the pump and tubing. The pump may be disassembled and flushed with the decontaminating solutions. It is recommended that detergent and alcohol be used sparingly in the decontamination process and water flushing steps be extended to ensure that any sediment trapped in the pump is removed. The pump exterior and electrical wires must be rinsed with the decontaminating solutions, as well. The procedure is as follows:

Flush the equipment/pump with potable water.

Flush with non-phosphate detergent solution. If the solution is recycled, the solution must be changed periodically.

Flush with potable or distilled/deionized water to remove all of the detergent solution. If the water is recycled, the water must be changed periodically.

Optional - flush with isopropyl alcohol (pesticide grade; must be free of ketones {e.g., acetone}) or with methanol. This step may be required if the well is highly contaminated or if the equipment blank data from the previous sampling event show that the level of contaminants is significant.

Flush with distilled/deionized water. This step must remove all traces of alcohol (if used) from the equipment. The final water rinse must not be recycled.

Procedure 2

Steam clean the outside of the submersible pump.

Pump hot potable water from the steam cleaner through the inside of the pump. This can be accomplished by placing the pump inside a three or four inch diameter PVC pipe with end cap. Hot water from the steam cleaner jet will be directed inside the PVC pipe and the pump exterior will be cleaned. The hot water from the steam cleaner will then be pumped from the PVC pipe through the pump and collected into another container. Note: additives or solutions should not be added to the steam cleaner.

Pump non-phosphate detergent solution through the inside of the pump. If the solution is recycled, the solution must be changed periodically.

Pump potable water through the inside of the pump to remove all of the detergent solution. If the solution is recycled, the solution must be changed periodically.

Pump distilled/deionized water through the pump. The final water rinse must not be recycled.

12.0 FIELD QUALITY CONTROL

Quality control samples are required to verify that the sample collection and handling process has not compromised the quality of the groundwater samples. All field quality control samples must be prepared the same as regular investigation samples with regard to sample volume, containers, and preservation. Quality control samples include field duplicates, equipment blanks, matrix spike/matrix spike duplicates, trip blanks (VOCs), and temperature blanks.

13.0 FIELD LOGBOOK

A field log shall be kept to document all groundwater field monitoring activities (see Appendix C, example table), and record the following for each well:

Site name, municipality, state.

Well identifier, latitude-longitude or state grid coordinates.

Measuring point description (e.g., north side of PVC pipe).

Well depth, and measurement technique.

Well screen length.

Pump depth.

Static water level depth, date, time and measurement technique.

Presence and thickness of immiscible liquid (NAPL) layers and detection method.

Pumping rate, drawdown, indicator parameters values, calculated or measured total volume pumped, and clock time of each set of measurements.

Type of tubing used and its length.

Type of pump used.

Clock time of start and end of purging and sampling activity.

Types of sample bottles used and sample identification numbers.

Preservatives used.

Parameters requested for analyses.

Field observations during sampling event.

Name of sample collector(s).

Weather conditions, including approximate ambient air temperature.

QA/QC data for field instruments.

Any problems encountered should be highlighted.

Description of all sampling/monitoring equipment used, including trade names, model number, instrument identification number, diameters, material composition, etc.

14.0 DATA REPORT

Data reports are to include laboratory analytical results, QA/QC information, field indicator parameters measured during purging, field instrument calibration information, and whatever other field logbook information is needed to allow for a full evaluation of data usability.

Note: the use of trade, product, or firm names in this sampling procedure is for descriptive purposes only and does not constitute endorsement by the U.S. EPA.

15.0 REFERENCES

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Robert W. Puls and Michael J. Barcelona, *Low-Flow (Minimal Drawdown) Ground-Water Sampling Procedures*, April 1996 (EPA/540/S-95/504).

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Vroblesky, Don A., Clifton C. Casey, and Mark A. Lowery, Summer 2007, Influence of Dissolved Oxygen Convection on Well Sampling, *Ground Water Monitoring & Remediation* 27, no. 3: 49-58.

APPENDIX A

PERISTALTIC PUMPS

Before selecting a peristaltic pump to collect groundwater samples for VOCs and/or dissolved gases, (e.g., methane, carbon dioxide, etc.) consideration should be given to the following:

- The decision of whether or not to use a peristaltic pump is dependent on the intended use of the data.
- If the additional sampling error that may be introduced by this device is NOT of concern for the VOC/dissolved gases data's intended use, then this device may be acceptable.
- If minor differences in the groundwater concentrations could affect the decision, such as to continue or terminate groundwater cleanup or whether the cleanup goals have been reached, then this device should NOT be used for VOC/dissolved gases sampling. In these cases, centrifugal or bladder pumps are a better choice for more accurate results.

EPA and USGS have documented their concerns with the use of the peristaltic pumps to collect water sample in the below documents.

- "Suction Pumps are not recommended because they may cause degassing, pH modification, and loss of volatile compounds" *A Compendium of Superfund Field Operations Methods*, EPA/540/P-87/001, December 1987.
- "The agency does not recommend the use of peristaltic pumps to sample ground water particularly for volatile organic analytes" *RCRA Ground-Water Monitoring Draft Technical Guidance*, EPA Office of Solid Waste, November 1992.
- "The peristaltic pump is limited to shallow applications and can cause degassing resulting in alteration of pH, alkalinity, and volatiles loss", *Low-flow (Minimal drawdown) Ground-Water Sampling Procedures*, by Robert Puls & Michael Barcelona, April 1996, EPA/540/S-95/504.
- "Suction-lift pumps, such as peristaltic pumps, can operate at a very low pumping rate; however, using negative pressure to lift the sample can result in the loss of volatile analytes", USGS Book 9 Techniques of Water-Resources Investigation, Chapter A4. (Version 2.0, 9/2006).

APPENDIX B

SUMMARY OF SAMPLING INSTRUCTIONS

These instructions are for using an adjustable rate, submersible pump or a peristaltic pump with the pump's intake placed at the midpoint of a 10 foot or less well screen or an open interval. The water level in the monitoring well is above the top of the well screen or open interval, the ambient temperature is above 32°F, and the equipment is not dedicated. Field instruments are already calibrated. The equipment is setup according to the diagram at the end of these instructions.

1. Review well installation information. Record well depth, length of screen or open interval, and depth to top of the well screen. Determine the pump's intake depth (e.g., mid-point of screen/open interval).
2. On the day of sampling, check security of the well casing, perform any safety checks needed for the site, lay out a sheet of polyethylene around the well (if necessary), and setup the equipment. If necessary a canopy or an equivalent item can be setup to shade the pump's tubing and flow-through-cell from the sun light to prevent the sun light from heating the groundwater.
3. Check well casing for a reference mark. If missing, make a reference mark. Measure the water level (initial) to 0.01 ft. and record this information.
4. Install the pump's intake to the appropriate depth (e.g., midpoint) of the well screen or open interval. Do not turn-on the pump at this time.
5. Measure water level and record this information.
6. Turn-on the pump and discharge the groundwater into a graduated waste bucket. Slowly increase the flow rate until the water level starts to drop. Reduce the flow rate slightly so the water level stabilizes. Record the pump's settings. Calculate the flow rate using a graduated container and a stop watch. Record the flow rate. Do not let the water level drop below the top of the well screen.

If the groundwater is highly turbid or discolored, continue to discharge the water into the bucket until the water clears (visual observation); this usually takes a few minutes. The turbid or discolored water is usually from the well-being disturbed during the pump installation. If the water does not clear, then you need to make a choice whether to continue purging the well (hoping that it will clear after a reasonable time) or continue to

the next step. Note, it is sometimes helpful to install the pump the day before the sampling event so that the disturbed materials in the well can settle out.

If the water level drops to the top of the well screen during the purging of the well, stop purging the well, and do the following:

Wait for the well to recharge to a sufficient volume so samples can be collected. This may take a while (pump may be removed from well, if turbidity is not a problem). The project manager will need to make the decision when samples should be collected and the reasons recorded in the site's log book. A water level measurement needs to be performed and recorded before samples are collected. When samples are being collected, the water level must not drop below the top of the screen or open interval. Collect the samples from the pump's tubing. Always collect the VOCs and dissolved gases samples first. Normally, the samples requiring a small volume are collected before the large volume samples are collected just in case there is not sufficient water in the well to fill all the sample containers. All samples must be collected, preserved, and stored according to the analytical method. Remove the pump from the well and decontaminate the sampling equipment.

If the water level has dropped 0.3 feet or less from the initial water level (water level measure before the pump was installed); proceed to Step 7. If the water level has dropped more than 0.3 feet, calculate the volume of water between the initial water level and the stabilized water level. Add the volume of the water which occupies the pump's tubing to this calculation. This combined volume of water needs to be purged from the well after the water level has stabilized before samples are be collected.

7. Attach the pump's tubing to the "T" connector with a valve (or a three-way stop cock). The pump's tubing from the well casing to the "T" connector must be as short as possible to prevent the groundwater in the tubing from heating up from the sun light or from the ambient air. Attach a short piece of tubing to the other end of the end of the "T" connector to serve as a sampling port for the turbidity samples. Attach the remaining end of the "T" connector to a short piece of tubing and connect the tubing to the flow-through-cell bottom port. To the top port, attach a small piece of tubing to direct the water into a calibrated waste bucket. Fill the cell with the groundwater and remove all gas bubbles from the cell. Position the flow-through-cell in such a way that if gas bubbles enter the cell they can easily exit the cell. If the ports are on the same side of the cell and the cell is cylindrical shape, the cell can be placed at a 45-degree angle with the ports facing upwards; this position should keep any gas bubbles entering the cell away from the monitoring probes and allow the gas bubbles to exit the cell easily (see Low-Flow Setup Diagram). Note:

make sure there are no gas bubbles caught in the probes' protective guard; you may need to shake the cell to remove these bubbles.

8. Turn-on the monitoring probes and turbidity meter.

9. Record the temperature, pH, dissolved oxygen, specific conductance, and oxidation/reduction potential measurements. Open the valve on the "T" connector to collect a sample for the turbidity measurement, close the valve, do the measurement, and record this measurement. Calculate the pump's flow rate from the water exiting the flow-through-cell using a graduated container and a stop watch, and record the measurement. Measure and record the water level. Check flow-through-cell for gas bubbles and sediment; if present, remove them.

10. Repeat Step 9 every 5 minutes or as appropriate until monitoring parameters stabilized. Note: at least one flow-through-cell volume must be exchanged between readings. If not, the time interval between readings will need to be increased. Stabilization is achieved when three consecutive measurements are within the following limits:

Turbidity (10% for values greater than 5 NTUs; if three Turbidity values are less than 5 NTUs, consider the values as stabilized),

Dissolved Oxygen (10% for values greater than 0.5 mg/L, if three Dissolved Oxygen values are less than 0.5 mg/L, consider the values as stabilized),

Specific Conductance (3%),

Temperature (3%),

pH (± 0.1 unit),

Oxidation/Reduction Potential (± 10 millivolts).

If these stabilization requirements do not stabilize in a reasonable time, the probes may have been coated from the materials in the groundwater, from a buildup of sediment in the flow-through-cell, or a gas bubble is lodged in the probe. The cell and the probes will need to be cleaned. Turn-off the probes (not the pump), disconnect the cell from the "T" connector and continue to purge the well. Disassemble the cell, remove the sediment, and clean the probes according to the manufacturer's instructions. Reassemble the cell and connect the cell to the "T" connector. Remove all gas bubbles from the cell, turn-on the probes, and continue the measurements. Record the time the cell was cleaned.

11. When it is time to collect the groundwater samples, turn-off the monitoring probes, and disconnect the pump's tubing from the "T" connector. If you are using a centrifugal or peristaltic pump check the pump's tubing to determine if the tubing is completely filled with water (no air space).

All samples must be collected and preserved according to the analytical method. VOCs and dissolved gases samples are normally collected first and directly into pre-preserved sample containers. However, this may not be the case for all sampling locations; the SAP/QAPP should list the order in which the samples are to be collected based on the project's objective(s). Fill all sample containers by allowing the pump discharge to flow gently down the inside of the container with minimal turbulence.

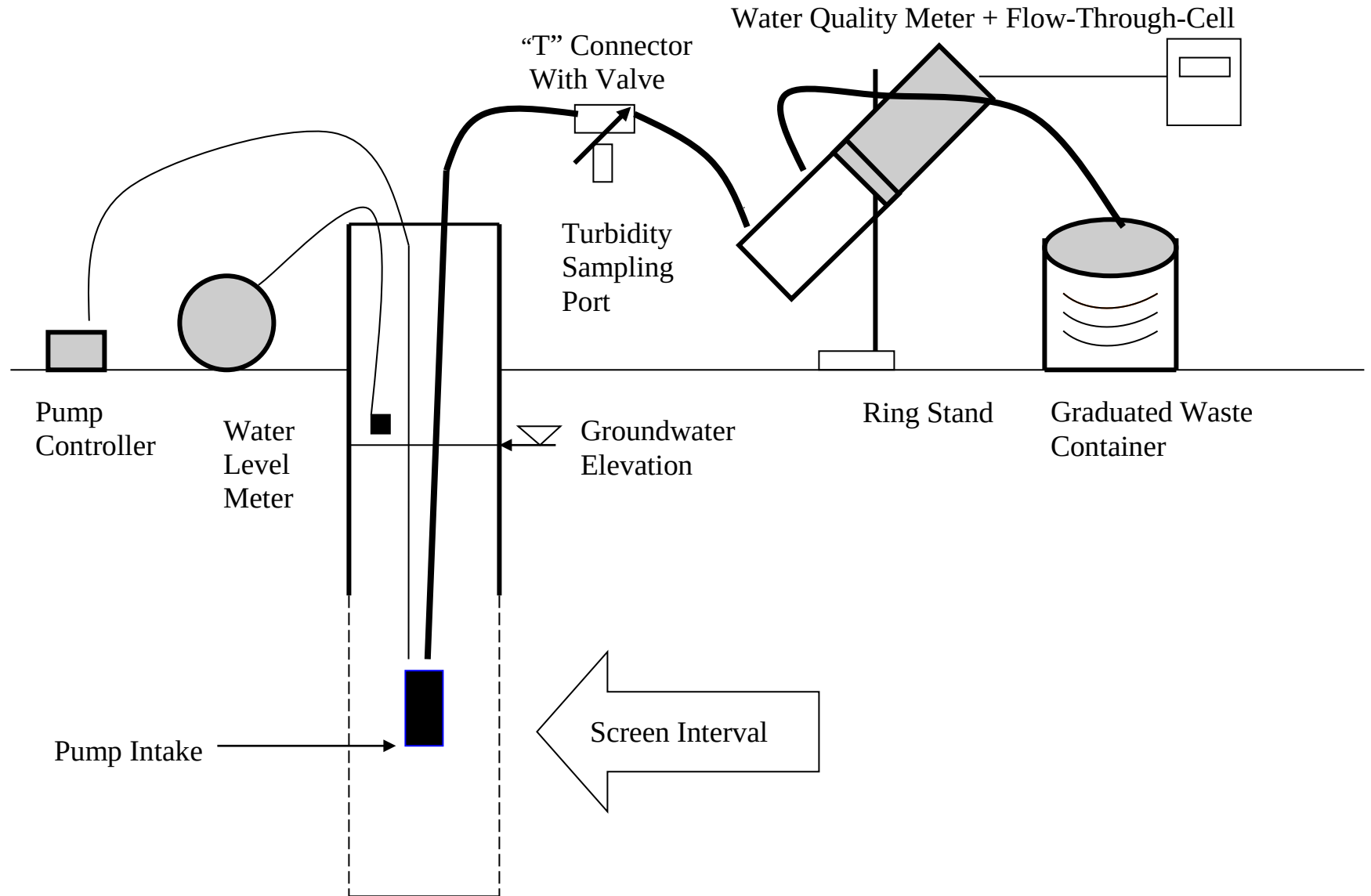
If the pump's tubing is not completely filled with water and the samples are being collected for VOCs and/or dissolved gases analyses using a centrifugal or peristaltic pump, do the following:

All samples must be collected and preserved according to the analytical method. The VOCs and the dissolved gases (e.g., methane, ethane, ethene, and carbon dioxide) samples are collected last. When it becomes time to collect these samples increase the pump's flow rate until the tubing is completely filled. Collect the samples and record the new flow rate.

12. Store the samples according to the analytical method.

13. Record the total purged volume (graduated waste bucket). Remove the pump from the well and decontaminate the sampling equipment.

Low-Flow Setup Diagram



APPENDIX C

EXAMPLE (Minimum Requirements)
WELL PURGING-FIELD WATER QUALITY MEASUREMENTS FORM

Location (Site/Facility Name)_____						Depth to _____/_____ of screen (below MP) top bottom					
Well Number_____ Date_____						Pump Intake at (ft. below MP)_____					
Field Personnel_____						Purging Device; (pump type)_____					
Sampling Organization_____						Total Volume Purged _____					
Identify MP_____											

Clock Time 24 HR	Water Depth below MP ft	Pump Dial ¹	Purge Rate ml/min	Cum. Volume Purged liters	Temp. °C	Spec. Cond. ² µS/cm	pH	ORP ³ mv	DO mg/L	Tur- bidity NTU	Comments

Stabilization Criteria

3%

3%

±0.1

±10 mv

10%

10%

1. Pump dial setting (for example: hertz, cycles/min, etc).

2. µSiemens per cm(same as µmhos/cm)at 25°C.

3. Oxidation reduction potential (ORP)



STANDARD OPERATING PROCEDURE

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Effective Date	04/30/2020	Revision	5
Applicability	EGS Operating Unit		
Prepared	Earth Sciences Department		

Subject EVALUATION OF EXISTING MONITORING WELLS
AND WATER LEVEL MEASUREMENT

Approved
T. Johnston *T.E. Johnston* 04/30/2020

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1.0 PURPOSE

The purpose of this procedure is to provide reference information regarding the proper methods for evaluating the physical condition and project utility of existing monitoring wells and determining water levels.

2.0 SCOPE

The procedures described herein are applicable to all existing monitoring wells and, for the most part, are independent of construction materials and methods.

3.0 GLOSSARY

Hydraulic Head - The height to which water will rise in a well.

Water Table - A surface in an unconfined aquifer where groundwater pressure is equal to atmospheric pressure (i.e., the pressure head is zero).

4.0 RESPONSIBILITIES

Site Geologist/Hydrogeologist - Has overall responsibility for the evaluation of existing wells, obtaining water level measurements and developing groundwater contour maps. The site geologist/hydrogeologist (in concurrence with the Project Manager) shall specify the reference point from which water levels are measured (usually a specific point on the upper edge of the inner well casing and if not otherwise designated, the north edge of the well casing), the number and location of data points which shall be used for constructing a contour map, and how many complete sets of water levels are required to adequately define groundwater flow directions (e.g., if there are seasonal variations).

Field Personnel - Must have a basic familiarity with the equipment and procedures involved in obtaining water levels and must be aware of any project-specific requirements or objectives.

5.0 PROCEDURES

Accurate, valid, and useful groundwater evaluation requires that four important conditions be met:

- Proper characterization of site hydrogeology.
- Proper design of the groundwater monitoring program, including adequate numbers of wells installed at appropriate locations and depths.
- Satisfactory methods of groundwater sampling and analysis to meet the project data quality objectives (DQOs).
- The assurance that specific monitoring well samples are representative of water quality conditions in the monitored interval.

To ensure that these conditions are met, adequate descriptions of subsurface geology, well construction methods and well testing results must be available. The following steps will help to ensure that the required data are available to permit an evaluation of the utility of existing monitoring wells for collecting additional samples.

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5.1 **PRELIMINARY EVALUATION**

A necessary first step in evaluating existing monitoring well data is the study and review of the original work plan for monitoring well installation (if available). This helps to familiarize the site geologist/hydrogeologist with site-specific condition, and will promote an understanding of the original purpose of the monitoring wells.

The next step of the evaluation should involve a review of all available information concerning borehole drilling and well construction. This will allow interpretation of groundwater flow conditions and area geology, and will help to establish consistency between hydraulic properties of the well and physical features of the well or formation. The physical features which should be identified and detailed, if available, include:

- The well identification number, permit number and location by referenced coordinates, the distance from prominent site features, or the location of the well on a map.
- The installation dates, drilling methods, well development methods, past sampling dates, and drilling contractors.
- The depth to bedrock -- where rock cores were not taken, auger refusal, drive casing refusal, or penetration test results (blow counts for split-barrel sampling) may be used to estimate bedrock interface.
- The soil profile and stratigraphy, often referred to as lithology or lithologic description.
- The borehole depth and diameter.
- The elevation of the top of the protective casing, the top of the well riser, and the ground surface.
- The total depth of the well.
- The type of well materials, screen type, slot size, and length, and the elevation/depths of the screen, interval, and/or monitored interval.
- The elevation/depths of the tops and bottom of the filter pack and well seals and the type and size.

5.2 **FIELD INSPECTION**

During the onsite inspection of existing monitoring wells, observe and record the following features and others as appropriate:

- The condition of the protective casing, cap and lock.
- The condition of the cement seal surrounding the protective casing.
- The presence of depressions or standing water around the casing.
- The presence of and condition of dedicated sampling equipment.
- The presence of a survey mark on the inner well casing.

If the protective casing, cap and lock have been damaged or the cement collar appears deteriorated, or if there are any depressions around the well casing capable of holding water, surface water may have infiltrated into the well. This may invalidate previous sampling results unless the time when leakage started can be precisely determined.

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The routine physical inspection must be followed by a more detailed investigation to identify other potential routes of contamination or sampling equipment malfunction. Any of these occurrences may invalidate previously-collected water quality data. If the monitoring well is to be used in the future, considerations shown in the steps described above should be rectified to rehabilitate the well.

After disconnecting any wires, cables or electrical sources, remove the lock and open the cap. Check for the presence of organic vapors with a photoionization detector (PID) or flame-ionization detector (FID) to determine the appropriate worker safety level. The following information should be recorded (at a minimum):

- Cap function.
- Physical characteristics and composition of the inner casing or riser, including inner diameter and annular space.
- Presence of grout between the riser and outer protective casing and the existence of drain holes in the protective casing.
- Presence of a riser cap, method of attachment to casing, and venting of the riser.
- Presence of dedicated sampling equipment; if possible, remove such equipment and inspect size, materials of construction and condition. (A licensed electrician may be required if the pump looks to be wired to an electrical source.)

The final step of the field inspection is to confirm previous hydraulic or physical property data and to obtain data not previously available. This includes the determination of static water levels, total well depth and potential well obstructions. This may be accomplished using a weighted tape measure which can also be used to check for sediment (the weight will advance slowly if sediment is present, and the presence of sediment on the weight upon removal should be noted). If sediment is present and/or the well has not been sampled in 12 or more months, it should be redeveloped before sampling.

Lastly, as a final step, the location, condition and expected water quality of the wells should be reviewed in light of their usefulness for the intended purpose of the investigation.

See Attachment A, Monitoring Well Inspection Sheet.

5.3 WATER LEVEL (HYDRAULIC HEAD) MEASUREMENTS

5.3.1 General

Groundwater level measurements can be made in monitoring wells, private or public water wells, piezometers, open boreholes, or test pits (after stabilization). Groundwater level measurements should generally not be made in boreholes with drilling rods or auger flights present. If groundwater sampling activities are to occur, groundwater level measurements shall take place prior to well purging or sampling.

Measure all groundwater levels to the nearest 0.01 foot, and record the levels in the site geologist/hydrogeologist's field notebook or on the Groundwater Level Measurement Sheet (Attachment B), along with the date and time of the reading with the well identification.

If not already known, measure the total depth of the well and record this depth. Weather changes that occur over the period of time during which water levels are being taken, such as precipitation and barometric pressure changes, should be noted.

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In measuring groundwater levels, ensure that there is a clearly-established reference point of known elevation, which is normally identified by a mark on the upper edge of the inner well casing. To be useful, the reference point should be tied in with an established USGS benchmark or other properly surveyed elevation datum. An arbitrary datum could be used for an isolated group of wells, if necessary.

Cascading water within a borehole or steel well casings can cause false readings with some types of sounding devices (chalked line, electrical). Oil layers may also cause problems in determining the true water level in a well. Special devices (interface probes) are available for measuring the thickness of oil layers and true depth to groundwater, if required.

Water level measurements should be taken regularly, as required by the site geologist/hydrogeologist. Monitoring wells or open-cased boreholes that are subject to tidal fluctuations should be measured in conjunction with a tidal chart (or preferably in conjunction with readings of a tide staff or tide level recorder installed in the adjacent water body); the frequency of such measurements should be established by the site hydrogeologist. All water level measurements at a site used to develop a groundwater contour map should be made in the shortest practical time to minimize affects due to weather changes.

5.3.2 Water Level Measuring Techniques

There are several methods for determining standing or changing water levels in boreholes and monitoring wells. Certain methods have particular advantages and disadvantages depending upon well conditions. A general description of these methods is presented, along with a listing of various advantages and disadvantages of each technique. An effective technique shall be selected for the particular site conditions by the site geologist/hydrogeologist.

In most instances, preparation of accurate potentiometric surface maps require that static water level measurements be obtained to a precision of 0.01 feet. To obtain such measurements in individual accessible wells, electrical water level indicator methods have been found to be best, and thus should be utilized. Other, less precise methods, such as the popper or bell sound, or bailer line methods, should be avoided. When a large number of (or continuous) readings are required, time-consuming individual readings are not usually feasible. In such cases, it is best to use a pressure transducer.

5.3.3 Methods

Water levels may be measured by several different techniques, but the same steps should be followed in each case. The proper sequence is as follows:

1. Check operation of recording equipment above ground. Prior to opening the well, don personal protective equipment, as required. Never remove an air-tight lock (such as a J-plug) with your face over the well. Pressure changes within the well may explosively force the cap off once loosened. Listen closely as the well plug is removed for air exiting or rushing into the well. If pressure or vacuum was evident, then it may take a while for the water level to reach equilibrium to atmospheric pressure for an accurate water level measurement.
2. Record all information specified below in the geologist/hydrogeologist's field notebook or on the Groundwater Level Measurement Sheet (Attachment B):
 - Well number.
 - Water level (to the nearest 0.01 foot). Water levels shall be taken from the surveyed reference mark on the top edge of the inner well casing. If the J-plug was on the well very tightly, it may take several minutes for the water level to stabilize as mentioned above.
 - Time and day of the measurement.
 - Thickness of free product if present.

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Water level measuring devices with permanently marked intervals shall be used. The devices shall be free of kinks or folds which will affect the ability of the equipment to hang straight in the well pipe.

5.3.4 Water Level Measuring Devices

Electric Water Level Indicators

These are the most commonly used devices and consist of a spool of small-diameter cable and a weighted probe attached to the end. When the probe comes in contact with the water, an electrical circuit is closed and a meter, light, and/or buzzer attached to the spool will signal the contact.

There are a number of commercial electric sounders available, none of which is entirely reliable under all conditions likely to occur in a contaminated monitoring well. In conditions where there is oil on the water, groundwater with high specific conductance, water cascading into the well, steel well casing, or a turbulent water surface in the well, measuring with an electric sounder may be difficult.

For accurate readings, lower the probe slowly into the well adjacent to the survey mark on the inner well casing. The electric tape is read (to the nearest 0.01 ft.) at the measuring point and recorded where contact with the water surface was indicated.

Popper or Bell Sounder

A bell- or cup-shaped weight that is hollow on the bottom is attached to a measuring tape and lowered into the well. A "plopping" or "popping" sound is made when the weight strikes the surface of the water. An accurate reading can be determined by lifting and lowering the weight in short strokes, and reading the tape when the weight strikes the water. This method is not sufficiently accurate to obtain water levels to 0.01 feet, and thus is more appropriate for obtaining only approximate water levels quickly.

Pressure Transducer

Pressure transducers can be lowered into a well or borehole to measure the pressure of water and therefore the water elevation above the transducer. If using a hardwired transducer, it will be connected to a recorder at the surface to record changes in water level with time. The recorder digitizes the information and can provide a printout or transfer the information to a computer for evaluation (using a well drawdown/recovery model). The pressure transducer should be initially calibrated with another water level measurement technique to ensure accuracy. This technique is very useful for hydraulic conductivity testing in highly permeable material where repeated, accurate water level measurements are required in a very short period of time. A sensitive transducer element is required to measure water levels to 0.01 foot accuracy.

Borehole Geophysics

Approximate water levels can be determined during geophysical logging of the borehole (although this is not the primary purpose for geophysical logging and such logging is not cost effective if used only for this purpose). Several logging techniques will indicate water level. Commonly used logs which will indicate saturated/unsaturated conditions and include the spontaneous potential (SP) log and the neutron log.

5.3.5 Data Recording

Record water level measurements, time, data, and weather conditions in the geologist/hydrogeologist's field notebook or on the Groundwater Level Measurement Sheet. All water level measurements should be measured from a known reference point (see Section 5.3.1). The reference point is generally a marked point on the upper edge of the inner well casing that has been surveyed for an elevation. The exact

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reference point should be marked with permanent ink on the casing since the top of the casing may not be entirely level. Alternatively, a small notch may be made in the PVC. It is important to note changes in weather conditions because changes in the barometric pressure may affect the water level within a well.

5.3.6 Specific Quality Control Procedures for Water Level Measuring Devices

All groundwater level measurement devices must be cleaned before and after each use to prevent cross contamination of wells. Manufacturer's instructions for cleaning the device should be strictly followed. Some devices used to measure groundwater levels may need to be calibrated. These devices should be calibrated to 0.01 foot accuracy and any adjustments/corrections should be recorded in the field logbook/notebook. After the corrections/adjustments are made to the measuring device and entered in the field logbook/notebook, the corrected readings should be entered onto the Groundwater Level Measurement Sheet (Attachment B). Elevations referenced to a known point will be entered on the sheet when they become available.

5.4 EQUIPMENT DECONTAMINATION

Equipment used for water level measurements provide a mechanism for potentially cross contaminating wells. Therefore, all portions of a device which project down the well casing must be decontaminated prior to advancing to the next well. Decontamination procedures vary based on the project objectives but must be defined prior to conducting any field activities including the collection of water level data. Consult the project planning documents and SA-7.1 Decontamination of Field Equipment.

5.5 HEALTH AND SAFETY CONSIDERATIONS

Groundwater contaminated by volatile organic compounds may release toxic vapors into the air space inside the well pipe. The release of this air when the well is initially opened is a health/safety hazard which must be considered. Under certain conditions, air-tight well caps may explosively fly off the well when the pressure is relieved. Never stand directly over a well when uncapping it.

Perform initial air monitoring of the well headspace and breathing zone concentrations using a PID or FID to determine required levels of protection.

6.0 RECORDS

Record all field procedures, tests and observations in the site logbook or designated field notebook. Entries in the log/notebook should include the individuals participating in the field effort, and the date and time. The use of annotated sketches may help to supplement the evaluation. If one person is measuring the water levels and a second person is recording, the second person should ensure the measurement is correct by repeating the measurement back to the person doing the measurement to confirm.

EVALUATION OF EXISTING MONITORING WELLS AND WATER LEVEL MEASUREMENT

Revision 5

Effective Date
04/30/2020

Monitoring Well Inspection Sheet

Personnel: _____

Well Construction Details (Taken from construction logs)		
Total Depth *	Ground Elev.	Top/Btm Screen *

Riser Pipe Material:
Riser Notched for Surveyors:
Well ID Tag In-place:
Well security:
Photo taken:

Protective Case:
Riser:
Well Pad:
Other:

Standing Water Around Well:	
Existing Sampling Equipment:	
Sediment build-up in Well Btm:	

* = Measurements are from the top of the inner case to the nearest 0.01'



STANDARD OPERATING PROCEDURE (GUIDANCE)

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Effective Date	02/20/2017	Revision	3
Applicability	Tetra Tech, Inc., NUS Operating Unit		
Prepared	Earth Sciences Department		
Approved	T. Johnston	tom.johnston@tetrattech.com	Digitally signed by: tom.johnston@tetrattech.com DN: CN = tom.johnston@tetrattech.com Date: 2017.02.20 14:18:19 -0500

Subject GROUNDWATER CONTOUR MAPS AND FLOW DETERMINATIONS

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1.0 PURPOSE

This guideline describes the process for developing groundwater contour maps and the approaches used to identify and quantify the direction and rate of groundwater flow and contaminant plume movement.

2.0 SCOPE

This document provides only a general overview of the field techniques, mathematical and physical relationships, and data handling procedures used for determining groundwater flow direction and rate. The references identified herein can provide a more complete explanation of particular methods and a more comprehensive discussion on the interpretation of hydrogeologic data.

3.0 GLOSSARY

Aquifer - A geologic formation capable of transmitting usable quantities of groundwater to a well or other discharge point.

Aquitard - A geologic formation that retards the flow of groundwater due to its low permeability.

Confined Aquifer - An aquifer that is overlain and underlain by zones of lower permeability (aquitards). If the aquifer is "artesian," the potentiometric head of the aquifer at a given point is higher than the top of the zone comprising the aquifer at that point.

Equipotential Line - A line connecting points of equal elevation of the water table or potentiometric surface. Equipotential lines are also called water level contours.

Flow Line - A flow line indicates the direction of groundwater movement within the saturated zone. Flow lines are drawn perpendicular to equipotential lines.

Flow Net - A diagram of groundwater flow showing flow lines and equipotential lines. Flow nets can be drawn in plan view or in vertical cross section.

Hydraulic Conductivity (K) - A quantitative measure of the ability of porous media to transmit water. This is defined mathematically as the volume of water that will flow through a unit cross-sectional area of porous media per unit time under a head pressure gradient. Hydraulic conductivity is dependent on properties of the medium and fluid.

Hydraulic Gradient (i) - The rate of change of hydraulic head per unit distance of flow at a given point and in the downgradient direction.

Hydraulic Head - The height to which water will rise inside a well casing; equal to the elevation head plus the pressure head. In a well screened across the water table, hydraulic head equals the elevation head because the pressure head equals zero. In wells screened below the water table in an unconfined aquifer or screened at any interval within a confined aquifer, the head is the sum of the elevation of the aquifer (elevation head) and the fluid pressure of the water confined in the aquifer (pressure head).

North American Vertical Datum of 1988 (NAVD 88) – The current national standard vertical control datum established in 1991, replacing the National Geodetic Vertical Datum of 1929 (NGVD 29).

Potentiometric Surface - A hypothetical surface that coincides with the static level of the water in an aquifer (i.e., the maximum elevation to which water will rise in a well or piezometer penetrating the aquifer). The term "potentiometric surface" is usually applied to confined aquifers, although the water table is the potentiometric surface of an unconfined aquifer.

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Unconfined Aquifer - An aquifer in which the water table forms the upper boundary.

Water Table - The surface in the groundwater system at which the fluid pressure is equal to atmospheric pressure (i.e., the net pressure head is zero) and below which all geologic strata are saturated with water.

4.0 RESPONSIBILITIES AND PERSONNAL REQUIREMENTS

Project Hydrogeologist - The project hydrogeologist has overall responsibility for obtaining water level measurements and developing groundwater contour maps. The hydrogeologist will specify the reference point from which water levels are measured (usually a specific point on the upper edge of the inner well casing), the number of data points needed and which wells will be used for a contour map, and how many complete sets of water levels are required to adequately define groundwater flow directions (e.g., if there are seasonal variations).

Field Personnel - All supporting field personnel must have a basic familiarity with the equipment and procedures involved in obtaining water levels and must be aware of any project-specific requirements.

General personnel qualifications for contour map preparation include the following:

- Technical background in geosciences
- Familiarity with reading and interpreting boring logs and other geologic and hydrogeologic information
- Strong technical understanding of groundwater flow, influences on groundwater flow, and techniques for evaluation of groundwater flow data

5.0 PROCEDURES

5.1 Potentiometric Surface Mapping

Construction of a potentiometric surface map includes the following general steps:

- Selecting wells
- Obtaining water level measurements and calculating groundwater elevations
- Constructing equipotential lines
- Determining groundwater flow direction(s)

Selecting Wells:

- All wells used to construct a single potentiometric map must represent the same hydrogeologic unit. The recorded water levels, monitoring well construction data, site geology, and topographic setting must be reviewed to ascertain that the wells are completed in the same hydrogeologic unit and to determine whether strong vertical hydraulic gradients may be present. Strong vertical hydraulic gradients will be manifested by a pronounced correlation between well depth and water level or by differences in water levels between two wells located near each other but having different screened intervals or screen lengths. Professional judgment of the hydrogeologist is important in this determination. If vertical gradients are significant, the data to be used must be limited vertically, and only wells finished within a chosen vertical zone of the hydrogeologic unit can be used (unless a vertical flow net is being constructed).
- Data from at least three wells are needed to provide an estimation of the direction of groundwater flow, and the three wells should be distributed across the site in a configuration other than a straight line. Information from many more than three wells is generally needed to provide an accurate contour map.

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In general, shallow groundwater flow systems require data from more wells than deep systems for accurate contour mapping.

- Accurate and useful potentiometric surface maps for shallow flow systems require water level measurements from nearby surface water bodies, especially at sites where groundwater/surface water interactions are, or may be, important. Acquisition of these measurements should be incorporated into the field investigation planning process.

Obtaining Water Level Measurements and Calculating Groundwater Elevations

After selecting wells, the next step is to obtain water level measurements from the selected points. In addition, measurements from any other readily available wells/surface water bodies should be collected to ensure that sufficient data are available for interpretation purposes. To calculate groundwater elevations from water level measurements, survey data for all monitoring wells must be available and referenced to the same vertical datum. In addition, elevations of points and areas of groundwater discharge or recharge such as springs, seeps, streams, rivers, and lakes need to be determined, typically through staff gauge measurements.

REMINDER:

All water level measurements to be used to generate potentiometric surface maps must be collected on the same day, preferably within 2 to 3 hours. This is especially important when working in an area where groundwater levels are tidally influenced or influenced by pumping.

Calculate groundwater elevations using water level measurements and elevation (survey) data as follows:

- Compile measurements of depth to water in the wells and surface water points to be used for mapping. Water level measurements in monitoring wells or piezometers are generally taken from the tops of the well casings. Consult Standard Operating Procedure (SOP) GH-1.2. It is important that the measurement points at all wells are documented (e.g., location of notch at top of PVC casing) so that the points at which measurements are taken can be surveyed. The procedure for surface water depth-to-water measurements depends on the location and nature of the point being used. For example, if a surface water measurement is required in an area with an accessible permanent feature that can be used for reference (e.g., a bridge or culvert), a measurement can be taken from a fixed point on the bridge/culvert, and the elevation of that point can later be surveyed.
- Compile surveyed elevations for the wells and surface water points, making sure that all elevations are referenced to the same vertical datum. If the difference in elevations across a site appears to be greater than the total change in elevation across the site, recheck the vertical datum. The North American Vertical Datum of 1988 (NAVD 88) is the current national standard vertical datum, but site-specific datums can also be used.
- Subtract the depth-to-water measurement at each well (generally in feet below top of casing) from the corresponding top-of-casing elevation (generally in feet above mean sea level referenced to NAVD 88) to calculate the groundwater (potentiometric surface) elevation at each well. For surface water elevations, subtract the depth-to-water measurement to the surface water body from the elevation of the reference point at which the measurement was taken.

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Constructing Equipotential Lines:

- Plot water level elevations on a site map. Other hydrogeologic features associated with the zone of interest, such as seeps, wetlands, and surface water bodies, should also be plotted along with their elevations. Topographic contours are also important to consider in interpreting groundwater flow and should be included on the site map.
- Contour the data using mathematically valid and generally accepted techniques. Linear interpolation is the most commonly used technique, however other techniques of trend-surface analysis or data smoothing are also acceptable. Computer-generated contour maps may be useful for rough mapping of large data sets; however, final detailed mapping must ***always*** be performed by hand by an experienced hydrogeologist. Draw all equipotential contours as smooth, continuous lines that never cross one another. Where data is sparse and contouring has significant uncertainty associated with it, dashed contours should be used instead of continuous lines.
- Inspect the contour map, noting known features such as pumping wells, surface water bodies, and site topography, and adjust the contour lines using professional judgment in accordance with these features. Closed contours should be avoided unless a known groundwater sink (i.e., pumping well) or mound exists. Groundwater mounding is common under landfills and lagoons; if the data imply this, the feature must be evident in the contour plot.
- If the hydrogeologic system consists of a water table aquifer and one or more confined aquifers, prepare separate contour maps for each aquifer. Develop water table maps using water level measurements from monitoring wells screened at or near the unsaturated-saturated interface. Contour water level measurements from monitoring wells screened in deeper portions of an unconfined aquifer as a separate potentiometric surface map(s). Tie surface water discharge or recharge features into the water table groundwater flow system.
- If constructing a vertical flow net, use a cross section aligned parallel to the direction of groundwater flow, and plot all water level measurements along this cross section, both deep and shallow, to develop equipotential lines and flow lines.

Determining Groundwater Flow Direction:

- ***Draw flow lines so that they are perpendicular to equipotential lines.*** Flow lines will begin at high head elevations and end at low head elevations. Closed highs (mounding) will be the source of additional flow lines. Closed depressions (i.e., pumping wells) will be the termination of some flow lines. Care must be used in areas with significant vertical gradients to avoid erroneous conclusions concerning gradients and flow directions.
- Include a flow arrow(s) drawn parallel to the direction(s) of flow (and perpendicular to equipotential lines) on each potentiometric surface contour map. The number of arrows required depends on the complexity of the flow system, but only major changes should have separate arrows. Again, professional judgment is required.

REMINDER:

Have the completed cross section reviewed by the project hydrogeologist or another qualified geoscientist with significant groundwater flow contouring experience.

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5.2 Groundwater Flow Considerations

Groundwater movement is an integral part of the hydrologic cycle. Recharge to the shallow groundwater environment generally occurs by infiltration of precipitation through an upper unsaturated soil zone. Movement is downward under the force of gravity until the water reaches the saturated zone of the water table aquifer. After water has become part of the water table aquifer, movement is controlled by differences in hydraulic head, with movement from areas of high head to areas of low head. Areas of low head include natural discharge areas such as springs, lakes, rivers, and, ultimately, oceans. These features can be considered as outcrops of the water table. Points of low head also are created by pumping wells.

Local head differences and consequent vertical flow patterns within an aquifer can be detected by well clusters. A well cluster consists of several adjacent wells, generally installed within a few feet of each other, and screened at different depths. Variations in water levels in these closely spaced wells indicate the vertical component of groundwater flow within an aquifer, provided that the wells are all screened within the same aquifer.

The number, location, and extent of geologic units and their properties with regard to aquifer or aquitard characteristics must be understood to properly interpret water level data gathered from the monitoring system. This firm understanding of the hydrogeologic system must be developed through a program of borings, wells, and interpretation of subsurface geology. The adequacy of the positions and depths of borings/wells used to define relevant subsurface hydrogeologic conditions must also be assessed. The location of surface water discharge or recharge points must be considered. Surface water features influence the system because flow is most likely toward them (if they are discharge points) or away from them (if they are recharge points). Man-made discharge or recharge features such as pumping or injection wells, ditches, and trenches can also affect the flow of groundwater.

5.3 Determination of Flow Rate

Darcy's Law states that the quantity of water flowing through a geologic material is dependent on the permeability of the material, hydraulic gradient, and cross-sectional area through which the water flows. This relation is expressed in the equation:

$$Q = KiA$$

where:

Q	=	volume of water flowing through the cross sectional area of the formation (L^3/T)
K	=	hydraulic conductivity (L/T)
i	=	hydraulic gradient (L/L , i.e., dimensionless)
A	=	cross sectional area of formation being considered (L^2)

The relation is similar to one used in stream flow measurements where:

$$Q = VA$$

where:

Q	=	discharge from the cross sectional area of a stream or pipe (L^3/T)
V	=	average velocity of flowing water (L/T)
A	=	cross sectional area through which water flows (L^2)

To determine the velocity of water movement within a geologic formation, which depends on the specific formation properties and the head differences across the formation, use the following equation:

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$$V = \frac{Ki}{n}$$

where:

V	=	average linear velocity of groundwater through the formation (L/T)
K	=	hydraulic conductivity (L/T)
i	=	hydraulic gradient (dimensionless)
n	=	porosity (expressed as a fraction)

Hydraulic conductivity, K, can be determined using slug tests (see SOP GH-2.4) packer tests (see SOP GH-2.2), pumping tests (see SOP GH-2.3), or geotechnical laboratory analysis of undisturbed soil samples (see SOP SA-1.3). Hydraulic conductivity is related to the permeability of the formation and depends on the size and interconnection of the pore spaces. In isotropic and homogeneous formations, the hydraulic conductivity will be the same vertically and horizontally. In anisotropic formations, horizontal and vertical conductivity can be markedly different, and the vertical hydraulic conductivity can be up to several orders of magnitude lower than the horizontal hydraulic conductivity. Typically, most formations are anisotropic with horizontal hydraulic conductivities at least several times greater than vertical hydraulic conductivities. Generally, hydraulic conductivities are high for sands, gravels, and limestone containing large solution cavities and low for silts, clays, and tightly fractured rock (see Attachment A).

General values of porosity, n, for several geologic materials are provided in Attachment A. More accurate and specific values of porosity can be obtained by laboratory analysis of site-specific formation samples or evaluation of unconfined aquifer pumping test results. Results from field testing usually provide greater (and more representative) hydraulic conductivities than laboratory testing because full-scale field testing includes the effects of the formational macrostructure (i.e., secondary permeability due to jointing or fractures), which is not reflected in the testing of a small sample in the laboratory.

Hydraulic gradient, i, is defined as the rate of change in hydraulic head (dh) per unit distance of flow (dl) determined from field measurements of hydraulic head, as follows:

- Using the potentiometric surface map generated as described above, measure the horizontal distance (generally in feet) between two equipotential lines (dl) that represent a significant portion of the site. If equipotential lines are spaced significantly differently in different areas of the site (e.g., closely spaced in the northern portion and widely separated in the southern portion), consider calculating separate gradients for the different sections of the site in addition to a site-wide gradient. The longer the distance over which the head change is measured, the more representative the gradient is of overall conditions. However, averaging gradients across large areas may not accurately represent the distribution of different gradients across the area.
- Do **not** measure from well to well; measure along a flow line drawn extending (and running perpendicular) from the beginning to ending equipotential lines that are drawn based on well (and other) hydraulic head data.
- Determine the change in head (dh) (generally in feet and in [or converted to] the same units as dl) between the equipotential lines described above (i.e., the difference in elevation between the lines). For example, if lines used in the horizontal distance measurement above represented 100 and 105 feet above mean sea level, the dh would be: 105 feet -100 feet = 5 feet.
- Using the values of dl and dh (in the same units) as determined above, calculate the hydraulic gradient by dividing the change in head by the length of the flow line using the following formula:

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$$i = \frac{dh}{dl}$$

where:

dh = change in head (L)
dl = distance between equipotential lines (L)

- Because the distance and head differences (change in head) are in the same units, hydraulic gradients are dimensionless.

5.4 Contaminant Transport

When chemical solutes are traveling in groundwater, as in cases of groundwater contamination, the calculated groundwater velocity may predict migration rates in excess of those actually observed. The difference in chemical versus water velocities may be due to attenuation or biodegradation of the chemical species in the aquifer. Attenuation is most often caused by adsorption of the chemical contaminant onto the formation grains or matrix. The result is that the chemical does not appear at the downgradient sampling point as quickly as the velocity calculation predicts. An equation to correct for this attenuation is:

$$V_c = V_w / (1 + K_d P_b / n)$$

where:

V_c = velocity of the chemical solute flow (L/T)
V_w = velocity of groundwater flow (L/T)
P_b = formation mass bulk density (M/L³)
n = formation porosity (expressed as a fraction)
K_d = distribution coefficient = (L³/M)

The K_d is equal to the mass of solute per unit mass of solid phase divided by the concentration of solute in a solution that is in equilibrium with the solid phase. The term in the denominator is known as the retardation factor.

Density and/or viscosity differences between water and contaminants can also cause velocity determination errors. Light hydrocarbons such as gasoline are less dense than water and consequently float on the water table. These contaminants can migrate along the water table surface at rates different than the rate of groundwater movement, depending on specific conditions, and may also volatilize into unsaturated soil pore spaces. Oils are more viscous than water and will typically migrate more slowly due to the viscosity difference. Contaminants denser than water such as heavy hydrocarbons (e.g., coal tar) or chlorinated compounds (e.g., TCE, PCE) tend to sink if present in concentrations exceeding their solubility limit (these chemicals are often referred to as dense, nonaqueous phase liquids, or DNAPLs if present as a separate-phase liquid), then pool along the surface of a low permeability layer encountered. Here, the separate-phase contamination may move at faster or slower rates than the overlying groundwater and may actually move in a direction opposite to that of the groundwater, depending on the dip of the surface that the DNAPL pools on.

Other factors involving the physicochemical interaction between the chemical and the groundwater, such as dilution (mixing of contaminated water with uncontaminated or less-contaminated groundwater) and dispersion (molecular diffusion of the chemical throughout the groundwater regime), can also affect the observed rates of contaminant transport in groundwater.

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ATTACHMENT A (EXAMPLE)

GENERALIZED POROSITY AND HYDRAULIC CONDUCTIVITY VALUES FOR GEOLOGIC MATERIALS

Material	Porosity Range (%)	Hydraulic Conductivity Range	
		cm/sec	ft/day
Gravel	30-40	10^{-1} to 10^{-2}	280 to 2.8×10^5
Coarse sand (clean)	30-40	10^{-1} to 1	280 to 2,800
Medium sand (clean)	35-45	10^{-2} to 10^{-1}	28 to 280
Fine sand (clean)	40-50	5×10^{-4} to 10^{-2}	1.4 to 28
Silty sand	25-40	10^{-5} to 10^{-2}	0.03 to 280
Glacial Till	Variable	10^{-10} to 10^{-4}	3×10^{-7} to 0.3
Unweathered Clay/Shale	45-55 (clay)	10^{-7} to 10^{-4}	3×10^{-4} to 0.3 (horizontal)
		10^{-10} to 10^{-6}	3×10^{-7} to 3×10^{-3} (vertical)
Karst Limestone	---	10^{-4} to 10^{-1}	0.3 to 2,800
Fractured Igneous/Metamorphic Rocks	---	10^{-6} to 10^{-1}	3×10^{-3} to 280
Sandstone	5-30	10^{-8} to 10^{-4}	3×10^{-5} to 0.3

Sources: Cedergren, 1977 and Fetter, 1980.

State of Rhode Island
Department of Environmental Management
Division of Site Remediation

Policy Memo 95-01

Guidelines for the Management of Investigation Derived Wastes

1.0 Purpose

The purpose of this policy memo is to provide guidance regarding the management of potentially contaminated materials generated during environmental site investigations, pilot tests, and interim remedial actions (hereafter referred to as investigatory activities) conducted on property in Rhode Island. It is the intent of the Department that the management of investigation derived wastes (IDW) be:

- protective human health and the environment, and accordingly result in no additional site related risks than existed prior to investigation activities;
- cost effective, consider the likely site remedy and consider waste minimization techniques; and
- done in a manner consistent with all applicable or relevant and appropriate requirements to the extent practicable.

The management of IDW should recognize that investigatory activities are not considered comprehensive remedial actions, and therefore final management of most materials encountered during these activities should be postponed to the extent feasible until a final site remedy has been determined.

The potential problems of managing IDW should be a factor in choosing an investigative method. Project managers should strive to minimize the generation of IDW to reduce the need for special storage or disposal requirements that may result in substantial additional costs yet provide little or no reduction in site risks relative to the final remedial actions.

It is important to note that for all investigatory activities in areas which have the potential to encounter a listed hazardous waste stream, all IDW originating from the area of concern must be managed in accordance with the Department's Rules and Regulations for Hazardous Waste Management.

The regulated community may use the management and storage methods outlined in this memo or submit, for review and approval, alternative IDW management proposals on a site-specific basis. The Division reserves the right to require additional IDW handling procedures as it deems necessary.

2.0 Classes of IDW

For the purpose of this policy memo, IDW are separated into the following four classes:

- A. **Solid** – includes unsaturated soils, soil saturated with water, and pre- existing solid wastes;
- B. **Liquid** – includes groundwater, drilling water, and decontamination rinsate;
- C. **Liquid Wastes and Associated Saturated Solids, and Buried Containers** – includes liquid wastes, any solids saturated with liquid wastes (i.e. a matrix containing greater than 1% liquid waste), and buried containers such as drums, electrical transformers, electrical capacitors, unexploded ordnance, and any other type of container which could potentially contain a hazardous substance; and
- D. **Personnel Equipment** – includes equipment and supplies, which are not reusable upon completion of current site activities (such as personal protective equipment and disposable sampling equipment).

3.0 IDW Management Guidelines

The Department recommends the following management guidelines (arranged by IDW class) for initial handling, segregating, storage and disposal of IDW.

3.1 Solid

These guidelines pertain to any investigatory activities, which generate solid IDW, including but not limited to excavations such as boreholes, trenches, and test pits.

- A. All intrusive investigatory activities should be observed for evidence of buried containers or liquid waste saturated solids.
- B. The preferable management alternative for all solids extracted from a site is replacement back into the same excavation from which it originated, however, consideration should be given to the likely site remedy prior to taking this action.
- C. Upon excavation, all solid IDW should be placed on low permeability synthetic sheeting of thickness no less than 10 mils. No material should be stored on synthetic sheeting for a period greater than 48 hours without receiving prior division approval. All solids stored on synthetic sheeting should be covered with similar material during all period when excavation work is not being conducted in that area of concern.
- D. When refilling excavations, the original stratigraphy of the area should be maintained to extent feasible. If the excavation is such that it is not possible to entirely refill the excavation, as in the case of a monitoring well placement, the excess solids should be managed in accordance with section 3.1(C) and (E) of this policy memo unless an alternative management plan has been

approved. Under no circumstances shall solid IDW from one area of concern be disposed of in a different area of concern without prior Division approval.

- E. Long-term storage of solids (typically period greater than 48 hours) should be in secure containers, which are suitable for potential off-site disposal (for example: roll-off dumpsters or 55-gallon drums). The contents of the containers should be characterized to determine the appropriate treatment or disposal method in a manner approved by the Division and consistent with the contaminants of concern at the site. This characterization can utilize either site investigation samples or samples whose sole purpose is to characterize the IDW.

3.2 Liquid

The following guidelines pertain to any investigatory activities, which generate liquid IDW; including, but not limited to, groundwater monitoring well development and sampling, and decontamination procedures. Aquifer pump tests are beyond the scope of this policy memo and require specific Department review and approval.

- A. All liquid IDW, which has been extracted from a site, must be stored in a secure container suitable for off-site disposal, and its contents properly characterized by Division approved laboratory analysis methods for all contaminants of concern at the site to determine the appropriate treatment or disposal method. These analysis methods should be consistent with those proposed for the site investigation work plan. This characterization can utilize either site investigation samples or samples whose sole purpose is to characterize the IDW.
- B. Storage of all liquid IDW should be in a segregated manner (liquid which has been taken from the site, by area of concern, from decontamination liquid).
- C. All non-decontamination liquid IDW that meets the groundwater quality standards for the subject property's groundwater classification, as stated in the Department's Rules and Regulations for the Groundwater Quality may be disposed of on-site. Liquid, which exceeds these criteria, shall be handled on a case-by-case basis. If it is anticipated that there will be exceedances of any groundwater quality standard at the site, the investigation work plan should contain a proposed management plan for this IDW.
- D. The disposal criteria for all non-decontamination liquid IDW containing contaminants for which there are no RIDEM groundwater quality standards must be proposed and approved on a case-by-case basis. Alternative criteria for each contaminant may be proposed as concentrations of individual contaminants or groups of contaminants (i.e., total concentration of volatile organic compounds) below which there will be no demonstrated additional adverse risk to human health or the environment.
- E. The preferred alternative for all liquid IDW generated as a result of decontamination procedures is disposal on site.

- F. Liquid IDW, which, in accordance with this memo, can be disposed of on-site, should be spread uniformly over a relatively level uncontaminated portion of the site. The on-site disposal of liquid IDW may not lead to increased migration of contaminants from the site nor impact a surface water body, wetland, or neighboring property to any degree, and must infiltrate the ground surface. If the volume of liquid IDW generated during a single investigation is expected to exceed 250 gallons, Division approval is required prior to any on-site disposal.

3.3 Liquid Wastes and Associated Saturated Solids, and Buried Containers

The following guidelines pertain to any investigatory activities which could potentially generate liquid wastes and associated saturated solids, or encounter buried containers, including but not limited to excavation such as boreholes, trenches and test pits.

- A. Investigations in areas, which are likely to encounter liquid wastes and associated saturated solids or buried containers, should include a contingency plan for proper handling and disposal of these wastes in the investigation work plan.
- B. The Division of Site Remediation should be contacted immediately upon the discovery of liquid wastes and associated saturated solids or buried containers for all projects which do not have a Division approved contingency plan for addressing this class of IDW.
- C. All liquid wastes and associated saturated solids extracted from a site should be stored in secure containers suitable for potential off-site disposal, and managed in accordance with the Department's Rules and Regulations for Hazardous Waste Management and/or the Rules and Regulations for the Investigation and Remediation of Hazardous Material Releases as appropriate. Liquid wastes and associated saturated solids extracted from a site must be properly characterized by Division approved laboratory analysis methods for all contaminants of concern, and as necessary to determine the proper treatment or disposal method. Under no circumstances shall liquid wastes and associated saturated solids from one area of concern be disposed of in a different area of concern without prior Division approval.
- D. All buried IDW should be extracted from its excavation upon Division approval. This approval may take the form of an approved contingency plan or an incident specific approval by Division personnel. Following extraction, buried containers should be stored in a manner which provided secondary containment for 110% of the container's volume, and managed in accordance with the Department's Rules and Regulations for Solid Waste Management Facilities, and/or the Department's Rules and Regulations for Hazardous Waste Management as appropriate. The contents of any buried containers must be properly characterized by Division approved laboratory analysis methods for all contaminants of concern, and as necessary to determine the proper treatment or disposal method. Under no circumstances should be extracted container not its contents be disposed of on site.

- E. Unexploded ordinance encountered and/or extracted from a site must be managed on a site-specific basis under the direction of personnel from the State Fire Marshal's Office, the appropriate municipal Fire Department, and the Division of Site Remediation.

3.4 Personnel Equipment

All personnel equipment IDW should be ultimately disposed of off-site and must be managed in accordance with the Department's Rules and Regulations for Solid waste management Facilities or the Department's Rules and Regulations for Hazardous Waste Management, as appropriate. Under no circumstances should personnel equipment IDW be disposed of on site.

4.0 IDW Storage Guidelines

The investigation work plan must include provisions for the proper storage and security of IDW in the time period between the generation of the material and the determination of the appropriate treatment or disposal method. The Department recommends the following procedures for the management and storage of IDW.

- A. All IDW, which is determined to be a hazardous waste, must be managed in accordance with the Department's Rules and Regulations for Hazardous Waste Management.
- B. For investigatory activities which are likely to require storage of IDW, the site investigation work plan must include the following:
 - 1. selected containment methods
 - 2. the designated secure storage area
 - 3. a schedule for IDW disposal
 - 4. a point of contact responsible for IDW management
- C. All non-hazardous waste IDW storage containers must be labeled with the following information:
 - 1. class of IDW
 - 2. source area
 - 3. date of generation
 - 4. generator name, address and phone number
- D. The period of storage of non-hazardous waste IDW should logically correspond with ongoing site investigative or remedial work and be completely disposed of within 30 days of the end of that phase of site work.



STANDARD OPERATING PROCEDURE

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Effective Date	7/15/2016	Revision	9
Applicability	Tetra Tech, Inc., NUS Operating Unit		
Prepared	Earth Sciences Department		

Subject
GROUNDWATER SAMPLE ACQUISITION AND
ONSITE WATER QUALITY TESTING

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1.0 PURPOSE

This Standard Operating Procedure (SOP) describes the process to be used for purging groundwater monitoring wells prior to sampling, for collecting groundwater samples, and for measuring groundwater quality parameters.

2.0 SCOPE AND APPLICABILITY

This document provides information on proper sampling equipment, onsite water quality testing, safety measures to ensure the safety of the field technician(s), and techniques for groundwater sampling. All personnel are encouraged to review the information contained herein to facilitate planning of the field sampling effort. The techniques described shall be followed whenever applicable, noting that site-specific conditions or project-specific plans may require modifications to methodology.

WARNING:

- Sample containers can be contaminated during storage if nearby volatile substances infiltrate the surrounding air and migrate into the containers. This can occur even when containers are capped. Therefore, containers that have been stored onsite for longer than 6 months should not be used.

Note: Whereas most air-borne contamination is likely to be associated with organic substances, inorganic substances such as hydrochloric acid (muriatic acid) can emit inorganic vapors that infiltrate sample containers.

3.0 GLOSSARY

Conductivity – Conductivity is a numerical expression of the ability of an aqueous solution to carry an electric current. This ability depends on the presence of ions and their total concentration, mobility, valence, and relative concentrations and on temperature. Conductivity is highly dependent on temperature and should be reported at a particular temperature, e.g., 20.2 milliSiemens per centimeter (mS/cm) at 14 degrees Celsius (°C).

Dissolved Oxygen (DO) – DO concentrations in natural water (including groundwater and surface water) and wastewater depend on the physical, chemical, and biochemical activities in the water sample.

Groundwater Sample – A quantity of water removed from the ground, usually via a monitoring well that may or may not be lined with a well casing, and typically in a manner that is designed to represent the in-situ quality of one or more parameters of interest regarding the water in the ground.

Oxidation-Reduction Potential (ORP) - A measure of the activity ratio of oxidizing and reducing species as determined by the electromotive force developed by a noble metal electrode immersed in water, as referenced against a reference electrode. Also commonly referred to as redox potential or Eh. A reference electrode commonly used in the field is the silver/silver chloride electrode, which has a voltage offset of about 210 millivolts (mV) from the standard hydrogen electrode (SHE). To convert field ORP measurements to equivalent SHE values, approximately 210 mV must be added to the ORP values obtained using the silver/silver chloride electrode. The actual offset depends on the concentration of the potassium chloride (KCl) solution in the field reference electrode and the temperature. Offsets typically range from 199 (saturated KCl) to 205 (3.5 Molar KCl) to 222 mV (1 Molar KCl) at 25°C and are greater at lower temperatures.

pH – A unitless number equal to the negative logarithm (base 10) of the hydrogen ion activity, expressed in moles per liter. The hydrogen ion activity is related to the hydrogen ion concentration, and, in a relatively weak solution, the two are nearly equal. Thus, for all practical purposes, pH is related to the hydrogen ion

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concentration. This is an inverse relationship, meaning that the pH decrease with increasing hydrogen ion concentration.

pH Paper - Indicator paper that turns different colors depending on the pH of the solution to which it is exposed. Comparison with color standards supplied by the manufacturer will then give an indication of the solution's pH.

Representativeness – A qualitative description of the degree to which an individual sample accurately reflects population characteristics or parameter variations at a sampling point. It is therefore an important characteristic not only of assessment and quantification of environmental threats posed by the site, but also for providing information for engineering design and construction. Proper sample location selection and proper sample collection methods are important to ensure that a truly representative sample has been collected.

Salinity – The measurement of dissolved salts in a given mass of solution. Note: most field meters determine salinity automatically from conductivity and temperature. The value will be displayed in either parts per thousand (ppt) or percent (e.g., 35 ppt equals 3.5 percent). The parts per thousand symbol (‰) is not the same as the percent symbol (%).

Specific Conductance – Specific conductance is the electrical conductivity across an electrical cell of specified geometry in an electrolyte solution. The electrical current is carried by charged atoms or molecules in the solution (i.e., ions). Standard measurements are commonly made at, or normalized to, a standard temperature of 25°C. Temperature normalizations commonly are automated by the specific conductance meters used to make these measurements. Temperature compensation, however, adjusts only for changes in instrument response caused by temperature changes. The temperature compensation does not correct for changes in conductance that result from changes in ion mobility. Hence the reason most measurements are made at the standard temperature of 25°C.

Turbidity – Turbidity in water is caused by suspended matter such as clay, silt, and fine organic and inorganic matter. Turbidity is an expression of the optical property that causes light to be scattered and absorbed rather than transmitted in a straight line through the sample.

4.0 RESPONSIBILITIES AND PERSONNEL QUALIFICATIONS

Project Manager - The Project Manager is responsible for determining the sampling objectives, initial sampling locations, and field procedures used in the collection of groundwater samples. Additionally, in consultation with other project personnel (geologist, hydrogeologist, etc.), the Project Manager identifies sampling locations.

Site Safety Officer (SSO) - The SSO (or a qualified designee) is responsible for providing the technical support necessary to implement the project Health and Safety Plan (HASP) or equivalent. This includes but is not be limited to performing air quality monitoring during sampling, boring and excavation activities, and ensuring that workers and offsite (downwind) individuals are not exposed to hazardous levels of airborne contaminants. The SSO or SSO designee may also be required to advise the Field Operations Leader (FOL) on other safety-related matters regarding sampling, such as mitigative measures to address potential hazards from hazardous objects or conditions.

Project Geologist/Sampler - The project geologist/sampler is responsible for the proper acquisition of samples in accordance with this SOP or other project-specific documents. In addition, this individual is responsible for the completion of all required paperwork (e.g., sample log sheets, field notebook, boring logs, container labels, custody seals, and chain-of-custody forms) associated with the collection of those samples.

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Project Hydrogeologist – This individual is responsible for selecting and detailing the specific groundwater sampling techniques, onsite water quality testing (type, frequency, and location), equipment to be used, and providing detailed input in this regard to the project planning documents. The project hydrogeologist is also responsible for properly briefing and overseeing the performance of site sampling personnel.

Field Operations Leader (FOL) – This individual is primarily responsible for the execution of the planning document containing the Sampling and Analysis Plan (SAP). This is accomplished through management of a field sampling team for the proper acquisition of samples. He or she is responsible for the supervision of onsite analyses; ensuring proper instrument calibration, care, and maintenance; sample collection and handling; the completion and accuracy of all field documentation; and making sure that custody of all samples obtained is maintained according to proper procedures. When appropriate and as directed by the FOL, such responsibilities may be performed by other qualified personnel (e.g., field technicians) where credentials and time permit. The FOL is ultimately responsible for adherence to Occupational Safety and Health Administration (OSHA) regulations during these operations through self-acquisition or through the management of a field team of samplers.

General personnel qualifications for groundwater sample collection and onsite water quality testing include the following:

- OSHA 40-hour and applicable refresher training.
- Capability of performing field work under the expected physical and environmental (i.e., weather) conditions.
- Familiarity with appropriate procedures for sample documentation, handling, packaging, and shipping.

5.0 HEALTH AND SAFETY

Specific safety and health precautions are identified throughout this SOP. In addition to those precautions, the following general hazards may be incurred during sampling activities:

- Knee injuries from kneeling on hard surfaces.
- Slips, trips, and falls.
- Cuts and lacerations.
- Traffic hazards associated with sampling in parking areas and roadways and along highways and railways.

Methods of avoiding these hazards are provided below.

Knee injuries – Many monitoring wells are installed as flush mounts. Personnel are required to kneel to open these wells and to take groundwater level measurements, etc. This could result in knee injuries from kneeling on stones/foreign objects and general damage due to stress on the joints. To combat this hazard:

- Clear any foreign objects from the work area.
- Wear hard-sided knee pads.

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Slips, Trips, and Falls – These hazards exist while traversing varying terrains carrying equipment to sample wells. To minimize these hazards:

- Pre-survey well locations. Eliminate, barricade, or otherwise mark physical hazards leading to the locations.
- Carry small loads that do not restrict the field of vision.
- Travel the safest and clearest route (not necessarily the shortest).

Cuts and Lacerations – To prevent cuts and lacerations associated with groundwater sampling, the following provisions are required:

- Always cut away from yourself and others when cutting tubing or rope. This will prevent injury to yourself and others if the knife slips.
- Do not place items to be cut in your hand or on your knee.
- Change blades as necessary to maintain a sharp cutting edge. Many accidents result from struggling with dull cutting attachments.
- Whenever practical, wear cut-resistant gloves (e.g., leather or heavy cotton work gloves) at least on the hand not using the knife.
- Keep cutting surfaces clean and smooth.
- Secure items to be cut -- do not hold them against the opposing hand, a leg, or other body part.
- When transporting glassware, keep it in a hard-sided container such as a cooler so that if there is a fall, you will be less likely to get cut by broken glass.
- DO NOT throw broken glass or glass ampoules into garbage bags. Place broken glass and glass ampoules in hard-sided containers such as a cardboard box or directly into a dumpster. DO NOT reach into garbage bags to retrieve any item accidentally thrown away. Empty the contents onto a flat surface to avoid punctures and lacerations from reaching where you cannot see.

Vehicular and Foot Traffic Hazards – When sampling along the roadway or near traffic patterns, follow the following precautions:

- Motorists may be distracted by onsite activities – ASSUME THEY DO NOT SEE YOU OR MEMBERS OF YOUR FIELD CREW.
- DO NOT place obstructions (such as vehicles) along the sides of the road that may cause site personnel to move into the flow of traffic to avoid your activities or equipment or that will create a blind spot.
- **Provide a required free space of travel.** Maintain at least 6 feet of space between you and moving traffic. Where this is not possible, use flaggers and/or signs to warn oncoming traffic of activities near or within the travel lanes.
- Face Traffic. Whenever feasible, if you must move within the 6 feet of the required free space or into traffic, attempt to face moving traffic at all times. Always leave yourself an escape route.
- Wear high-visibility vests to increase visual recognition by motorists.

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- Do not rely on the vehicle operator's visibility, judgment, or ability. Make eye contact with the driver. Carefully and deliberately use hand signals so they will not startle or confuse motorists or be mistaken for a flagger's direction before moving into traffic.
- Your movements may startle a motorist and cause an accident, so move deliberately. Do not make sudden movements that might confuse a motorist.

6.0 PROCEDURES

6.1 General

For information derived from a groundwater sample to be useful and accurate, the sample must be representative of the particular zone being sampled. The physical, chemical, and bacteriological integrity of the sample must be maintained from the time of sampling to the time of analysis to keep any changes in water quality parameters to a minimum.

CAUTION

A closed well may generate and accumulate gases due to biological degradation, evolution of volatile chemicals from groundwater into the air, or other chemical actions. These gases may also be artificially generated, such as in the case of air sparging or extraction wells, which may take several days to depressurize. See Section 6.6.2 for safety measures to be employed to protect sampling personnel.

Methods for withdrawing samples from completed wells include the use of pumps, compressed air or nitrogen, bailers, and various types of samplers. The primary considerations in obtaining a representative sample of groundwater are to avoid collection of stagnant (standing) water in the well and to avoid physical or chemical alteration of the water sample due to external influences of the sampling technique(s). In a non-pumping well, there will be little or no vertical mixing of water in the well pipe or casing, and stratification will occur. The well water in the screened section will mix with groundwater due to normal flow patterns, but the well water above the screened section will remain isolated and become stagnant. Concentration gradients resulting from mixing and dispersion processes, layers of variable geologic permeability, and the presence of separate-phase product (e.g., floating hydrocarbons) may cause stratification. Excessive pumping or improper sampling methods can dilute or increase contaminant concentrations in the collected sample compared to what is representative of the integrated water column as it naturally occurs at that point, resulting in the collection of a non-representative sample.

To safeguard against collecting non-representative samples, the following approach shall be followed prior to sample acquisition:

CAUTION

Mechanical agitation of well water may cause off-gas generation of volatile contaminants, creating an inhalation exposure to the sampler(s). Where avoiding an inhalation exposure is not possible and mechanical agitation is possible, pump into closed-top containers to control potential air emissions.

1. If possible, position yourself (and the sampling equipment) upwind of the well head.
2. Purge the monitoring well to be sampled prior to obtaining any samples from it. Evacuation of three to five well volumes is recommended prior to sampling, unless low-flow purging and sampling methods are utilized as described in Section 6.7 (Consult the site-specific SAP for exact purging parameters). In a high-yielding groundwater formation and where there is no stagnant water in the well above the

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screened section, extensive evacuation prior to sample withdrawal is not as critical as it is in a low-yielding well or in wells containing stagnant water.

3. For wells with low yields that are purged dry during sampling, evacuate the well and allow it to recover to 75 percent of full capacity prior to sample acquisition. If the recovery rate is fairly rapid [generally 300 milliliters (mL) per minute or greater], attempt to continue evacuation until the number of well volumes specified in the SAP is achieved. If this cannot be accomplished, allow recovery to 75 percent of capacity and begin sampling.

CAUTION

For moderate to high-yielding monitoring wells, an evacuation rate that does not cause excessive turbulence in the well should be selected. There is no absolute safeguard against contaminating the sample with stagnant water; hence, special techniques are required for purging to minimize the potential for sample contamination (see below).

4. For moderate to high-yielding monitoring wells, use one of the following purge techniques:
 - Place a submersible pump or the intake line of a surface pump or bailer just below the water surface when removing the stagnant water.
 - While purging and as the water level decreases, lower the pump or intake line as the water level drops in the well. Three to five volumes of water shall be removed to provide reasonable assurance that all stagnant water has been evacuated. After this is accomplished, a bailer or other approved device may be used to collect the sample for analysis.
 - Unless otherwise directed, place the intake line of the sampling pump (or the submersible pump itself) near the center of the screened section, and pump approximately one casing volume of water from the well at a low purge rate equal to the well's recovery rate (low-flow sampling).

6.2 **Sampling, Monitoring, and Evacuation Equipment**

Sample containers shall conform to the guidelines in SOP SA-6.1.

The following equipment shall be on hand when sampling groundwater wells (reference SOPs SA-6.1 and SA-7.1):

- Sample packaging and shipping equipment – Coolers for sample shipping and cooling, chemical preservatives, appropriate sampling containers and filler materials, ice, labels, and chain-of-custody documents.
- Field tools and instrumentation
 - Multi-parameter water quality meter with an in-line sample chamber capable of measuring ORP, pH, temperature, DO, specific conductance, turbidity, and salinity, or individual meters (as applicable).
 - pH Paper.
 - Camera and film (if appropriate). Approval prior to use may be required (see SOP SA-6.3).
 - Appropriate keys (for locked wells).

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- Water level indicator and/or oil-water interface probe if separate-phase product is expected.

- Pumps

- Shallow-well pumps: Centrifugal, bladder, suction, or peristaltic pumps with drop lines and air-lift apparatus (compressor and tubing) where applicable.
- Deep-well pumps: Submersible pump and electrical power-generating unit, or bladder pumps where applicable.

- Other sampling equipment – Bailers, graduated cylinder, stopwatch, and inert line with tripod-pulley assembly (if necessary).

- Pails – Plastic, graduated.

- Clean paper or cotton towels for cleaning equipment.

- Buckets with lids for collecting purge water.

- Decontamination solutions – Deionized water, potable water, phosphate-free laboratory-grade detergent, and analytical-grade solvent (e.g., pesticide-grade isopropanol), as required.

Ideally, sample withdrawal equipment shall be completely inert, economical, easily cleaned, cleaned prior to use, reusable, able to operate at remote sites in the absence of power sources, and capable of delivering variable rates for well purging and sample collection.

6.3 Calculations of Well Volume

To ensure that the proper volume of water has been removed from the well prior to sampling, it is first necessary to know the volume of standing water in the well pipe (including well screen where applicable). This volume can be easily calculated by the following method. Calculations shall be entered in the site logbook or field notebook or on a sample log sheet form or equivalent electronic form(s) (see SOP SA-6.3):

1. Obtain all available information on well construction (location, casing, screen, etc.).
2. Determine well or inner casing diameter.
3. Measure and record static water level (depth below ground level or top of casing reference point).
4. Determine depth of well by sounding using a clean, decontaminated, weighted tape measure or water level indicator.
5. Calculate number of linear feet of static water (total depth or length of well pipe minus the depth to static water level).
6. Calculate one static well volume in gallons $V = (0.163)(T)(r^2)$

where: V = Static volume of well in gallons.
T = Linear feet of water in the well.
r = Inside radius of well casing in inches.
0.163 = Conversion factor (compensates for conversion of casing radius from inches to feet; cubic feet to gallons; and pi (π)).

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7. Per evacuation volumes discussed above, determine the minimum amount to be evacuated before sampling.

Measuring devices may become contaminated when gathering the above information if they are submerged in contaminated water. Decontamination of the tape or water level indicator must be conducted between measurements in different wells as follows:

1. Saturate a paper towel or clean cotton towel with deionized water.
2. As the measuring device is extracted, wipe the tape, changing the cleaning surface frequently.
3. After it is extracted, rinse the probe or tape using a spray bottle of deionized water over a bucket or similar collection container.

Based on the contaminant (oily, etc.), it may be necessary to use a soap and water wash and rinse to remove contaminants. Isopropanol can be used on the probe/tape. However, it is recommended that the use of solvents on the tape be minimized because they could degrade the protective covering or possibly remove the scale designations. If isopropanol (or some other solvent) is used, assure that the manufacturer/supplier Safety Data Sheet (SDS) is obtained, kept onsite at a readily available location with other SDSs, and reviewed by personnel prior to the first usage of the solvent. Also, add the substance to the site-specific Hazardous Chemical Inventory list (see Section 5 of the Tetra Tech Health and Safety Manual (available on the Health and Safety link of the My Tetra Tech intranet site, Tetra Tech Hazard Communication Program, and OSHA Standard 29 Code of Federal Regulations [CFR] 1910.1200).

6.4 Evacuation of Static Water – Purging

6.4.1 General

The amount to be purged from each well will be determined prior to sample collection. This amount will depend on the intent of the monitoring program and the hydrogeologic conditions. Programs to determine overall quality of water resources may require long pumping periods to obtain a sample that is representative of a large volume of the aquifer. The pumped volume may be specified prior to sampling so that the sample can be a composite of a known volume of the aquifer. Alternately, the well can be pumped until parameters such as temperature, specific conductance, pH, and turbidity (as applicable) have stabilized. Onsite measurements of these parameters shall be recorded in the site logbook or field notebook or on standardized data sheets or an equivalent electronic form(s).

6.4.2 Evacuation Devices

The following discussion is limited to those devices commonly used at hazardous waste sites. Attachment A provides guidance on the proper evacuation device to use for given sampling situations. All of these techniques involve equipment that is portable and readily available.

Bailers

Bailers are the simplest evacuation devices used and have many advantages. They generally consist of a length of tubing equipped with a base plate and ball check-valve at the bottom. Bailers are typically comprised of stainless steel and plastic. They come in a variety of sizes, but the two most often used are 2 inches and 4 inches in diameter. An inert non-absorbent line such as polyethylene rope is used to lower and then raise the bailer to retrieve the sample. As the bailer is lowered into the water column, the ball is

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pushed up allowing the tube to be filled. When the bailer is pulled upward, the ball seats in the base plate preventing water from escaping.

Advantages of bailers include the following:

- There are few limitations on size and materials used.
- No external power source is needed.
- Bailers are inexpensive and can be dedicated and hung in a well to reduce the chances of cross-contamination.
- Bailers are relatively easy to decontaminate.

Limitations on the use of bailers include the following:

- It is time consuming to remove stagnant water using a bailer.
- Splashing the bailer into the water or transfer of sample may cause aeration.
- The use of a bailer does not permit constant in-line monitoring of groundwater parameters.
- Use of bailers is physically demanding, especially in warm temperatures at personal protection equipment (PPE) levels above Level D.

Safety concerns using a bailer include the following:

- Muscle stress and strain, especially when using 4-inch bailers and when pulling from excessively deep wells.
- Entanglement, possible hand/finger injuries, and rope burns during a sudden release of the bailer back down the well.
- Direct contact with contaminants of concern and sample preservatives when discharging the bailer contents because there is not a high level of control during a direct pour, and splashing and indirect contact with contaminants/preservatives could occur.

Control measures for these hazards are provided in Section 6.6.2.

Suction Pumps

There are many different types of inexpensive suction pumps including centrifugal, diaphragm, and peristaltic pumps. Centrifugal and diaphragm pumps can be used for well evacuation at a fast pumping rate and for sampling at a low pumping rate. The peristaltic pump is a low-volume pump that uses rollers to squeeze flexible tubing to create suction. This tubing can be dedicated to a well to prevent cross-contamination from well to well. Suction pumps are all portable, inexpensive, and readily available. However, because they are based on suction, their use is restricted to areas with water levels within 20 to 25 feet of the ground surface. A significant limitation is that the vacuum created by these pumps can cause loss of dissolved gases and volatile organics. Another limitation of these pumps is that they require a secondary energy source to drive them. Electrically driven pumps may require portable generators as energy sources. Air diaphragm pumps require air compressors and/or compressed gas cylinders to drive

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them. The advantage of the peristaltic pump is that it can operate from either a built in power source or a portable battery source. Safety measures associated with these pumps are provided below.

Air-Lift and Gas-Lift Samplers

This group of pump samplers uses gas pressure either in the annulus of the well or in a Venturi to force groundwater up a sampling tube. These pumps are also relatively inexpensive. Air- or gas-lift samplers are more suitable for well development than for sampling because the samples may be aerated as a result of pump action. Aeration can cause pH changes and subsequent trace metal precipitation or loss of volatile organics.

Submersible Pumps

Submersible pumps take in water and push the sample up a sample tube to the surface. The power sources for these samplers may be compressed gas or electricity. Operation principles vary, and displacement of the sample can be by an inflatable bladder, sliding piston, gas bubble, or impeller. Pumps are available for 2-inch-diameter wells and larger. These pumps can lift water from considerable depths (several hundred feet).

Limitations of this class of pumps include the following:

- They may have low delivery rates.
- Many models are expensive.
- Compressed gas or electric power is needed.
- Sediment in water may cause clogging of the valves or eroding of the impellers with some of these pumps.
- Decontamination of internal components can be difficult and time consuming.

Compressed Gases

Safety concerns using compressed gases as an energy source in these pumps are numerous. The nitrogen gas or compressed air is provided in a compressed gas cylinder at a pressure of approximately 2,000 pounds per square inch (psi). If damaged, these cylinders can become dangerous projectiles. Additionally, a sudden release of a cylinder's contents can involve considerable force that could cause significant damage to the eyes and/or skin. Protective measures include the following:

- Always wear safety impact glasses when handling compressed gases.
- Always administer compressed gases through an appropriate pressure-reducing regulator.
- When clearing the cylinder connection port, open the cylinder valve only enough to clear foreign debris. During this process, always position the cylinder valve so that it faces away from you and others.
- If the cylinder is designed to accept a valve protection cap, always keep that protection cap in place, except when the cylinder is connected for use.

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- When using the cylinder, lay the cylinder on its side to avoid the potential of it falling and knocking the valve off (and becoming a missile).
- DO NOT use the compressed nitrogen or air to clean clothing or to spray off the skin. Small cuts in the protective layer of the skin may permit the gas to enter into the bloodstream, presenting the potential danger of an embolism.

See the project-specific HASP (or equivalent) for additional direction concerning cylinder safe handling procedures pertaining to the safe handling, transportation, and storage of compressed gas cylinders.

Electrical Shock

Even in situations where portable batteries are used, the potential for electrical shock exists. This potential risk is increased in groundwater sampling activities because of the presence of groundwater near the batteries. This potential is also increased in (prohibited) situations where jury-rigging of electrical connections is performed. Other potential hazards occur when field samplers open the hood of a running car to access the battery as a power source. To control these hazards:

- If you are unfamiliar with electrical devices, do not experiment, get help, and get the proper equipment necessary to power your device.
- Use the proper portable power inverters for cigarette lighter connections to minimize the need to access the battery under the hood of your vehicle.
- Use of electrical generators may pose a number of hazards including noise, those associated with fueling, and indirect sample influence.

To minimize or eliminate electrical generator hazards:

- Inspect the generator before use. Ensure that the generator and any extension cords are rated for the intended operation and have a Ground Fault Circuit Interrupter (GFCI) in line to control potential electrical shock.
- Fuel the generator before purging and sampling to avoid loss of power during sampling.
- Fuel engines only when they are turned OFF and have cooled sufficiently to prevent a fire hazard.
- Place the generator and any fuel source at least 50 feet from the well to be sampled to avoid indirect influence to the sample from fuel vapors or emission gases.

Lifting Hazards

This hazard may be experienced when moving containers of purge water, equipment, cylinders, etc. To control these potential hazards:

- Do not fill purge buckets to more than 80 percent of their capacity.
- Obtain a gas cylinder of sufficient size to complete the designated task but not too large to handle. K-size cylinders weigh approximately 135 pounds and are difficult to handle. M-size cylinders weigh approximately 50 pounds and are easier to handle and move.

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- When necessary, get help lifting and moving gas cylinders and other heavy objects. Minimize twisting and turning while lifting. If it is necessary to move these cylinders or generators over significant distance, use mechanical means (carts, etc.).
- Use proper lifting techniques as described in Section 4.4 of the Tetra Tech Health and Safety Manual.

6.5 Onsite Water Quality Testing

This section describes the procedures and equipment required to measure the following parameters of an aqueous sample in the field:

- pH
- Specific conductance
- Temperature
- DO
- ORP
- Turbidity
- Salinity

This section is applicable for use in an onsite groundwater quality monitoring program to be conducted at a hazardous or nonhazardous waste site. The procedures and equipment described are applicable to groundwater samples and are not, in general, subject to solution interferences from color, turbidity, or colloidal material or other suspended matter.

This section provides general information for measuring the parameters listed above with instruments and techniques in common use. Because instruments from different manufacturers may vary, review of the manufacturer's literature pertaining to the use of a specific instrument is required before use. Most meters used to measure field parameters require calibration on a daily basis. Refer to SOP SA-6.3 for an example equipment calibration log.

6.5.1 Measurement of pH

6.5.1.1 General

Measurement of pH is one of the most important and frequently used tests in water chemistry. Practically every phase of water supply and wastewater treatment such as acid-base neutralization, water softening, and corrosion control is pH dependent. Likewise, the pH of leachate can be correlated with other chemical analyses to determine the probable source of contamination. It is therefore important that reasonably accurate pH measurements be taken and recorded on the groundwater sample log sheet (Attachment B) or equivalent electronic form.

Two methods are given for pH measurement: the pH meter and pH indicator paper. Indicator paper is used when only an approximation of the pH is required or when pH meter readings need to be verified, and the pH meter is used when a more accurate measurement is needed. The response of a pH meter can be affected by high levels of colloidal or suspended solids, but the effect is generally of little significance. Consequently, specific methods to overcome this interference are not described. The response of pH paper is unaffected by solution interferences from color, turbidity, or colloidal or suspended materials unless extremely high levels capable of coating or masking the paper are encountered. In such cases, use of a pH meter is recommended.

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6.5.1.2 Principles of Equipment Operation

Use of pH papers for pH measurement relies on a chemical reaction caused by the acidity or alkalinity of the solution created by the addition of the water sample reacting with the indicator compound on the paper. Various types of pH papers are available, including litmus (for general acidity or alkalinity determination) and specific, or narrower range, pH range paper.

Use of a pH meter relies on the same principle as other ion-specific electrodes. Measurement relies on establishment of a potential difference across a glass or other type of membrane in response to (in this instance, hydrogen) ion activity (which is usually similar to concentration) across that membrane. The membrane is conductive to ionic species and, in combination with a standard or reference electrode, a potential difference proportional to the ion concentration is generated and measured.

6.5.1.3 Equipment

The following equipment is to be used for obtaining pH measurements:

- A stand-alone portable pH meter or combination meter equipped with an in-line sample chamber (e.g., YSI 600 series and Horiba U-22).
- Combination electrode with polymer body to fit the above meter. Alternately, a pH electrode and a reference electrode can be used if the pH meter is equipped with suitable electrode inputs.
- Buffer solutions, as specified by the manufacturer. If the buffer solutions are considered hazardous per 29 CFR 1910.1200 (Hazard Communication) or the volumes used are greater than consumer commodity levels, the SSO shall obtain SDSs from the manufacturer for the specific buffer solutions (see Section 4 of the Tetra Tech Health and Safety Manual regarding the Hazard Communication Program).
- pH indicator paper to cover the pH range 2 through 12.
- Manufacturer's operation manual. All personnel must be familiar with the equipment operation to ensure that the integrity of samples is preserved and that the equipment is operated safely.

6.5.1.4 Measurement Techniques for Field Determination of pH

pH Meter

The following procedure shall be used for measuring pH with a pH meter (meter standardization is according to manufacturer's instructions):

1. Inspect the instrument and batteries prior to initiation of the field effort.
2. Check that the buffer solutions are used for field calibration prior to their expiration dates. Buffer solutions need to be changed often as a result of degradation upon exposure to the atmosphere.
3. If applicable, make sure all electrolyte solutions within the electrode(s) are at their proper levels and that no air bubbles are present within the electrode(s).

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4. Calibrate the meter and electrode(s) on a daily use basis (or as recommended by manufacturer) following manufacturer's instructions. Record calibration data on a water quality meter calibration log sheet (Attachment C) or equivalent electronic form.
5. Immerse the electrode(s) in the sample. Stabilization may take several seconds to minutes. If the pH continues to drift, the sample temperature may not be stable, a physical reaction (e.g., degassing) may be taking place in the sample, or the meter or electrode may be malfunctioning. The failure of the measurements to stabilize must be clearly noted in the logbook or equivalent electronic form.
6. Read and record the pH of the sample. pH shall be recorded to the nearest 0.01 pH standard unit. Also record the sample temperature (unless otherwise specified in the SAP, record temperatures to the nearest whole degree Fahrenheit or 0.5 degree Celsius).
7. Rinse the electrode(s) with deionized water.
8. Store the electrode(s) in an accordance with manufacturer's instructions when not in use.

Any visual observation of conditions that may interfere with pH measurement, such as oily materials or turbidity, shall be noted and avoided as much as possible.

pH Paper

Use of pH paper is very simple and requires no sample preparation, standardization, etc. pH paper is available in several ranges, including wide-range (indicating approximately pH 1 to 12), mid-range (approximately pH 0 to 6, 6 to 9, or 8 to 14) and narrow-range (many available, with ranges as narrow as 1.5 pH units). The appropriate range of pH paper shall be selected. If the pH is unknown, then the investigation shall start with wide-range paper and proceed with successively narrower range paper until the sample pH is determined. To measure the pH with pH paper:

1. Collect a small portion of sample into a clean container.
2. Dip the pH paper into this small portion of sample.
3. Compare the color of the paper to the color chart that is provided with the pH paper and read the corresponding pH from the chart.
4. Record the pH value from the chart on the sample log sheet.
5. Discard the used pH paper as trash.
6. Discard the small volume of sample that was used for the pH measurement with the other investigative-derived waste.

6.5.2 Measurement of Specific Conductance

6.5.2.1 General

Conductance provides a measure of dissolved ionic species in water and can be used to identify the direction and extent of migration of contaminants in groundwater or surface water. It can also be used as a measure of subsurface biodegradation or to indicate alternate sources of groundwater contamination.

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Conductivity is a numerical expression of the ability of a water sample to carry an electric current. This value depends on the total concentration of ionized substances dissolved in the water and the temperature at which the measurement is made. The mobility of each of the various dissolved ions, their valences, and their actual and relative concentrations affect conductivity.

Specific conductance at 25°C is used as a standard of comparison for different water sources as conductivity changes with temperature. It is important to obtain a specific conductance measurement soon after taking a sample because temperature changes, precipitation reactions, and absorption of carbon dioxide from the air all affect specific conductance. Most conductivity meters in use today display specific conductance in units of mS/cm, which is the conductivity normalized to a temperature of 25°C. These are the required units to be recorded on the groundwater sample log field form or equivalent electronic form. If a sample cannot be cooled or warmed (as appropriate) to 25°C, this should be noted in the field documentation

6.5.2.2 Principles of Equipment Operation

An aqueous system containing ions will conduct an electric current. In a direct-current field, the positive ions migrate toward the negative electrode, and the negatively charged ions migrate toward the positive electrode. Most inorganic acids, bases, and salts such as hydrochloric acid, sodium carbonate, and sodium chloride, respectively, are relatively good conductors. Conversely, organic compounds such as sucrose or benzene, which do not dissociate in aqueous solution, conduct a current very poorly if at all.

A conductance cell and a Wheatstone Bridge (for the measurement of potential difference) may be used for measurement of electrical resistance. The ratio of current applied to voltage across the cell may also be used as a measure of conductance. The core element of the apparatus is the conductivity cell containing the solution of interest. Depending on the ionic strength of the aqueous solution to be tested, a potential difference is developed across the cell, which can be converted directly or indirectly (depending on instrument type) to a measurement of specific conductance.

6.5.2.3 Equipment

The following equipment is needed for taking specific conductance measurements:

- Stand-alone portable conductivity meter or combination meter equipped with an in-line sample chamber (e.g., YSI 600 series and Horiba U-22).
- Calibration solution, as specified by the manufacturer.
- Manufacturer's operation manual.

A variety of conductivity meters are available that may also be used to monitor salinity and temperature. Probe types and cable lengths vary, so equipment must be obtained to meet the specific requirements of the sampling program.

6.5.2.4 Measurement Techniques for Specific Conductance

The steps involved in taking specific conductance measurements are as follows (calibration shall be conducted according to manufacturer's instructions):

1. Check batteries and calibrate instrument before going into the field.

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2. Calibrate on a daily use basis (or as recommended by manufacturer), according to the manufacturer's instructions and record all pertinent information on a water quality meter calibration log sheet or equivalent electronic form. Potassium chloride solutions with a specific conductance closest to the values expected in the field shall be used for calibration.
3. Rinse the cell with one or more portions of the sample to be tested or with deionized water and shake excess water from the cell.
4. Immerse the electrode in the sample and measure the conductivity.
5. Read and record the results in a field logbook or on a sample log sheet or equivalent electronic form.
6. Rinse the electrode with deionized water.

If the specific conductance measurements become erratic, recalibrate the instrument and see the manufacturer's instructions for troubleshooting assistance.

6.5.3 Measurement of Temperature

6.5.3.1 General

In combination with other parameters, temperature can be a useful indicator of the likelihood of biological action in a water sample. It can also be used to trace the flow direction of contaminated groundwater. Temperature measurements shall be taken in-situ, or as quickly as possible in the field because collected water samples may rapidly equilibrate with the temperature of their surroundings.

6.5.3.2 Equipment

Temperature measurements may be taken with alcohol-toluene, mercury-filled, dial-type thermometers or combination meters equipped with an in-line sample chamber (e.g., YSI 600 series and Horiba U-22). In addition, various meters such as specific conductance or DO meters that have temperature measurement capabilities may also be used. Using such instrumentation along with suitable probes and cables, in-situ measurements of temperature at great depths can be performed.

6.5.3.3 Measurement Techniques for Water Temperature

If a thermometer is used to determine the temperature for a water sample, use the following procedure:

1. Immerse the thermometer in the sample until temperature equilibrium is obtained (1 to 3 minutes). To avoid the possibility of cross-contamination, the thermometer shall not be inserted into samples that will undergo subsequent chemical analysis.
2. Record values in a field logbook or on a sample log sheet or equivalent electronic form.

If a temperature meter or probe is used:

1. Calibrate the instrument according to manufacturer's recommendations prior to use.
2. Immerse the meter/probe in the sample until temperature equilibrium is obtained (1 to 3 minutes). To avoid the possibility of cross-contamination, the meter/probe shall not be inserted into samples that will undergo subsequent chemical analysis.

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3. Record values in a field logbook or on a sample log sheet or equivalent electronic form.

6.5.4 Measurement of Dissolved Oxygen

6.5.4.1 General

DO levels in natural water and wastewater depend on the physical, chemical, and biochemical activities in the water body. In addition, the growth of many aquatic organisms and the rate of corrosivity are dependent on DO concentrations. Thus, analysis for DO is a key test in water pollution and waste treatment process control. If at all possible, DO measurements shall be taken in-situ because concentrations may show a large change in a short time if the sample is not adequately preserved.

The monitoring method discussed herein is limited to the use of DO meters. Chemical methods of analysis (i.e., Winkler methods) are available, but require more equipment and greater sample manipulation. Furthermore, DO meters using a membrane electrode are suitable for highly polluted waters because the probe is completely submersible and is not susceptible to interference caused by color, turbidity, or colloidal material or suspended matter.

6.5.4.2 Principles of Equipment Operation

DO probes are normally electrochemical cells that have two solid metal electrodes of different nobility immersed in an electrolyte. The electrolyte is retained by an oxygen-permeable membrane. The metal of highest nobility (the cathode) is positioned at the membrane. When a suitable potential exists between the two metals, reduction of oxygen to hydroxide ion (OH^-) occurs at the cathode surface. An electrical current is developed that is directly proportional to the rate of arrival of oxygen molecules at the cathode. This rate is proportional to the oxygen concentration in the water being measured.

Because the current produced in the probe is directly proportional to the rate of arrival of oxygen at the cathode, it is important that a fresh supply of sample always be in contact with the membrane. Otherwise, the oxygen in the aqueous layer along the membrane is quickly depleted and false low readings are obtained. It is therefore necessary to stir the sample (or the probe) constantly to maintain fresh solution near the membrane interface. Stirring, however, shall not be so vigorous that additional oxygen is introduced through the air-water interface at the sample surface. To avoid this possibility, some probes are equipped with stirrers to agitate the solution near the probe, leaving the surface of the solution undisturbed.

DO probes are relatively unaffected by interferences. Interferences that can occur are reactions with oxidizing gases such as chlorine or with gases such as hydrogen sulfide that are not easily depolarized from the indicating electrode. If a gaseous interference is suspected, it shall be noted in the field logbook and checked if possible. Temperature variations can also cause interference because probes exhibit temperature sensitivity. Automatic temperature compensation is normally provided by the manufacturer. This compensation can counteract some of the temperature effects, but not all of them.

6.5.4.3 Equipment

The following equipment is needed to measure DO concentrations:

- A stand-alone portable DO meter or combination meter equipped with an in-line sample chamber (e.g., YSI 600 series and Horiba U-22).
- Sufficient cable to allow the probe to contact the sample.

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- Manufacturer's operation manual.

6.5.4.4 Measurement Techniques for Dissolved Oxygen Determination

DO probes differ as to instructions for use. Follow the manufacturer's instructions to obtain an accurate reading. The following general steps shall be used to measure DO concentrations:

1. Check the DO meter batteries before going to the field.
2. Condition the probe in a water sample for as long a period as practical before use in the field. Long periods of dry storage followed by short periods of use in the field may result in inaccurate readings.
3. Calibrate the instrument in the field according to manufacturer's recommendations or in a freshly air-saturated water sample of known temperature.
4. Record all pertinent information on a water quality meter calibration log sheet or equivalent electronic form.
5. Rinse the probe with deionized water.
6. Immerse the probe in the sample. Be sure to provide for sufficient flow past the membrane by stirring the sample. Probes without stirrers placed in wells may be moved up and down to achieve the required mixing.
7. Record the DO content and temperature of the sample in a field logbook or on a sample log sheet or equivalent electronic form.
8. Rinse the probe with deionized water.
9. Recalibrate the probe when the membrane is replaced, or as needed. Follow the manufacturer's instructions.

Note that in-situ placement of the probe is preferable because sample handling is not involved. This however may not always be practical.

Special care shall be taken during sample collection to avoid turbulence that can lead to increased oxygen solubilization and positive test interferences.

6.5.5 Measurement of Oxidation-Reduction Potential

6.5.5.1 General

ORP provides a measure of the tendency of organic or inorganic chemicals to exist in an oxidized state. The ORP parameter therefore provides evidence of the likelihood of anaerobic degradation of biodegradable organics or the ratio of activities of reduced to oxidized species in the sample.

6.5.5.2 Principles of Equipment Operation

When an inert metal electrode, such as platinum, is immersed in a solution, a potential is developed at that electrode depending on the ions present in the solution. If a reference electrode is placed in the same solution, an ORP electrode pair is established. This electrode pair allows the potential difference between

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the two electrodes to be measured and is dependent on the concentration of the ions in solution. By this measurement, the ability to oxidize or reduce species in solution may be determined. Supplemental measurements, such as DO, may be correlated with ORP to provide knowledge of the quality of the solution, water, or wastewater.

6.5.5.3 Equipment

The following equipment is needed for measuring the ORP of a solution:

- A combination meter with an in-line sample chamber (e.g., YSI 600 series and Horiba U-22).
- Reference solution as specified by the manufacturer.
- Manufacturer's operation manual.

6.5.5.4 Measurement Techniques for Oxidation-Reduction Potential

The following procedure is used for measuring ORP:

1. Check the equipment using the manufacturer's recommended reference solution and check its batteries before going to the field.
2. Thoroughly rinse the electrode with deionized water.
3. If the probe does not respond properly to the recommended reference solution, verify the sensitivity of the electrodes by noting the change in millivolts when the pH of a test solution is altered. The ORP will increase when the pH of a test solution decreases, and the ORP will decrease when the test solution pH is increased. Place the sample in a clean container and agitate the sample. Insert the electrodes and note that the ORP drops sharply when the caustic is added (i.e., pH increases) thus indicating that the electrodes are sensitive and operating properly. If the ORP increases sharply when the caustic is added, the polarity is reversed and must be corrected in accordance with the manufacturer's instructions or the probe should be replaced.
4. Record all pertinent information on a water quality meter calibration log sheet or equivalent electronic form.

6.5.6 Measurement of Salinity

6.5.6.1 General

Salinity is a unitless property of industrial and natural waters. It is the measurement of dissolved salts in a given mass of solution. Most field meters determine salinity automatically from conductivity and temperature. The displayed value will be displayed in either parts per thousand (ppt) or percent (e.g., 35 ppt equals 3.5 percent).

6.5.6.2 Principles of Equipment Operation

Salinity is determined automatically from the meter's conductivity and temperature readings according to algorithms (such as are found in Standard Methods for the Examination of Water and Wastewater).

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Depending on the meter, the results are displayed in either ppt or percent. The salinity measurements are carried out in reference to the conductivity of standard seawater (corrected to salinity = 35 ppt).

6.5.6.3 Equipment

The following equipment is needed for salinity measurements:

- A multi-parameter water quality meter capable of measuring conductivity and temperature and converting them to salinity (e.g., Horiba U-22 or YSI 600 series).
- Calibration solution as specified by the manufacturer.
- Manufacturer's operation manual.

6.5.6.4 Measurement Techniques for Salinity

The steps involved in taking salinity measurements are as follows (standardization shall be conducted according to manufacturer's instructions):

1. Check the expiration date of the solutions used for field calibration and replace them if they are expired.
2. Check batteries and calibrate the meter before going into the field.
3. Calibrate on a daily use basis, according to the manufacturer's instructions and record all pertinent information on a water quality meter calibration log sheet or equivalent electronic form.
4. Rinse the cell with the sample to be tested. This is typically accomplished as the probe is placed in line during the collection of the purge water up to the time of sample acquisition.
5. Immerse the multi-probe in the sample and measure the salinity. Read and record the results in a field logbook or on a sample log sheet or equivalent electronic form.
6. Rinse the probes with deionized water.

6.5.7 Measurement of Turbidity

6.5.7.1 General

Turbidity is an expression of the optical property that causes light to be scattered and absorbed rather than transmitted in a straight line through the sample. Turbidity in water is caused by suspended matter such as clay, silt, or other finely divided organic and inorganic matter and microscopic organisms including plankton.

It is important to obtain a turbidity reading immediately after taking a sample because irreversible changes in turbidity may occur if the sample is stored too long.

6.5.7.2 Principles of Equipment Operation

Turbidity is measured by the Nephelometric Method, which is based on a comparison of the intensity of light scattered by the sample under defined conditions with the intensity of light scattered by a standard

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reference suspension under the same conditions. The higher the scattered light intensity, the higher the turbidity.

Formazin polymer is used as the reference turbidity standard suspension because of its ease of preparation combined with a higher reproducibility of its light-scattering properties than clay or turbid natural water. The turbidity of a specified concentration of formazin suspension is defined as 40 nephelometric units. This same suspension has an approximate turbidity of 40 Jackson units when measured on the candle turbidimeter. Therefore, nephelometric turbidity units (NTUs) based on the formazin preparation will approximate units derived from the candle turbidimeter, but will not be identical to them.

6.5.7.3 Equipment

The following equipment is needed for turbidity measurements:

- A turbidity meter (preferably a stand-alone meter such as a LaMotte 2020) that calibrates easily using test cells with standards such as 0.0, 1.0, and 10 NTUs, or a combination meter equipped with an in-line sample chamber (e.g., YSI 600 series and Horiba U-22). The range of calibration standards should bracket the measured sample turbidities.
- Calibration solution and sample tubes, as specified by the manufacturer.
- Manufacturer's operation manual.

6.5.7.4 Measurement Techniques for Turbidity

The steps involved in taking turbidity measurements utilizing an electrode (e) or light meter (l) are listed below (standardization shall be done according to manufacturer's instructions):

1. Check the expiration date of the solutions used for field calibration and replace them if they are expired.
2. Check batteries and calibrate the instrument before going into the field.
3. Calibrate on a daily basis according to the manufacturer's instructions, and record all pertinent information on a turbidity meter calibration log sheet (Attachment C) or equivalent electronic form.
4. When using the YSI 600 and/or Horiba U-22, rinse the electrode with one or more portions of the sample to be tested or with deionized water.
5. When using the Lamotte 2020, fill the light meter's glass test cell with approximately 5 mL of sample, screw on the cap, wipe off glass to remove all residue that could intercept the instrument's light beam, place the test cell in the light meter, aligning the arrow on the cap of the test cell to the arrow on the meter, and close the lid.
6. Immerse the electrode in the sample and measure the turbidity.
7. The reading must be taken immediately because suspended solids will settle over time resulting in a lower, inaccurate turbidity reading.
8. Read and record the results in a field logbook or on a sample log sheet or equivalent electronic form. Include a physical description of the sample, including color, qualitative estimate of turbidity, etc.
9. Rinse the electrode or test cell with deionized water.

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6.6 Sampling

6.6.1 Sampling Plan

The sampling approach consisting of the following shall be developed as part of the project planning documents approved prior to beginning work in the field:

- Background and objectives of sampling.
- Brief description of area and waste characterization.
- Identification of sampling locations, with map or sketch, and applicable well construction data (well size, depth, screened interval, reference elevation).
- Intended number, sequence, volumes, and types of samples. If the relative degree of contamination between wells is insignificant, a sampling sequence that facilitates sampling logistics may be followed. Where some wells are known or strongly suspected of being highly contaminated, these shall be sampled last to reduce the risk of cross-contamination between wells. In situations where the well is not well-characterized and the nature or extent of airborne contamination is unknown, it is recommended that head space analysis using a photoionization detector (PID) or flame ionization detector (FID) is performed to rate the wells, sampling from least contaminated to most contaminated. Refer to the project-specific HASP (or equivalent) for appropriate information and direction on air monitoring requirements.
- Sample preservation requirements.
- Work schedule.
- List of team members.
- List of observers and contacts.
- Other information, such as the necessity for a warrant or permission of entry, requirements for split samples, access problems, location of keys, etc.
- The FOL shall ensure that the sampling method(s) to be employed is accurately represented in the HASP (or equivalent), indicating the types of sampling to be employed and the hazards. If the methods are not accurately represented, the FOL should rectify this with the HASP (or equivalent) author.
- The FOL shall ensure that sampling teams understand the sampling approach that they are to follow. Where sampling teams are made up of personnel from multiple locations, personal sampling experiences may vary. Therefore, the FOL shall review project-specific requirements, SOPs, and protocol to be followed. The FOL will conduct periodic surveys to ensure that these methods are being completed per his/her direction.

6.6.2 Sampling Methods as Related to Low-Flow Sampling

The collection of a groundwater sample consists of the following steps:

1. Ensure the safety of the sample location. Take a few minutes to evaluate the area for physical hazards (trip hazards, uneven ground, overhanging branches, etc.) and natural hazards (snakes, bees, spiders, etc.) that may exist in the area or that may have constructed nests in the well head. Snakes often like

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to sun themselves on concrete well pads. Follow provisions in the project-specific HASP (or equivalent) and/or Tetra Tech Health and Safety Manual for addressing natural hazards.

2. As indicated earlier, some monitoring wells have the potential to contain pressurized headspace (e.g., through the generation of gases from contaminated groundwater, due to biological processes, degradation of contaminants, or simply based on location such as near a landfill or in areas that intersect lithological abnormalities) or through intentional artificial means such as those associated with air sparging systems. Injection or extraction wells may be artificially pressurized and may remain so for several days after the system has been turned off. This presents a hazard to people opening these wells. The Field Sampling Technician shall employ the following practices to minimize these hazards:

- Wear safety glasses to protect the eyes. If site-specific observations and conditions indicate that the wells may be pressurized, wear a full-face shield over the safety impact eye protection.
- DO NOT place your face or any other part of your body over the well when opening because this may place you in a strike zone.
- Open the well cover at arms' length, then step away and allow the well to off gas and stabilize.

Follow directions provided in the project-specific HASP (or equivalent), Work Plan, and/or SAP pertaining to the use of volatile chemical detection equipment (PID or FID) within the breathing zone of the sampler during sampling to determine the need to retreat from the work area and/or for the use of respiratory protection (as specified in the HASP or equivalent).

3. When proper respiratory protection has been donned, sound the well for total depth and water level (using clean equipment) and record these data on a groundwater sampling log sheet or equivalent electronic form; then calculate the fluid volume in the well pipe (as previously described in this SOP). It is imperative that downhole equipment be adequately decontaminated between wells to prevent cross-contamination. Just as sampling occurs from the least contaminated to the most contaminated, it is also recommended that groundwater level measurements be taken in this manner.
4. Calculate volume of well water to be removed as described in Section 6.3.
5. Select the appropriate purging equipment (see Attachment A) or as designated within the governing Work Plan/SAP. If an electric submersible pump with packer is chosen, go to Step 10.
6. Lower the purging equipment or intake into the well to a short distance below the water level or mid-screen as indicated in project-specific documentation and begin water removal. Remember that some contaminants are "bottom dwellers," and in these cases, project-specific direction may specify placing the intake just above (1 to 2 feet) the well bottom. Secure the pump intake at the well and secure the effluent at the collection container and begin pumping. The pumping rate will be determined based on the decrease in the water level (see Section 6.7) or as directed in your project-specific documents or this SOP. Purge water is generally collected in a 5-gallon bucket or similar open- or closed-top container. To minimize the potential for spills and back injuries, do not fill 5-gallon buckets beyond approximately 80 percent of their capacity. Dispose of purge water as indicated in the planning document(s). Where necessary, slow the pumping rate or lower the pump intake as required to maintain submergence.
7. Estimate the approximate rate of discharge frequently and record it on the Low Flow Purge Data Sheet (see Attachment D). Estimate flow rate by noting the amount of discharge in a bucket or graduated cylinder per unit time using a watch with a second hand or a stopwatch.

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8. Observe the peristaltic pump tubing intake for degassing "bubbles." If bubbles are abundant and the intake is fully submerged, this pump is not suitable for collecting samples for volatile organics.
9. Purge a minimum of three to five well casing volumes before sampling (or as directed by the site-specific SAP). In low-permeability strata (i.e., if the well is pumped to dryness), one volume will suffice. Allow the well to recover to 75 percent of initial water level before sampling. Do not overfill purge containers because this increases the potential for spills and lifting injuries.
10. If sampling using a submersible pump, lower the pump intake to mid-screen (or the middle of the open section in uncased wells) and collect the sample. If sampling with a bailer, lower the bailer to just below the water surface.
11. For pump and packer assemblies only: Lower the assembly into the well so that the packer is positioned just above the screen or open section. Inflate the packer. Purge a volume equal to at least twice the screened interval (or unscreened open section volume below the packer) before sampling. Packers shall always be tested in a casing section above ground to determine proper inflation pressures for good sealing.
12. If the recovery time of the well is very slow (e.g., 24 hours or greater), sample collection can be delayed until the following day. If the well has been purged early in the morning, sufficient water may be standing in the well by the day's end to permit sample collection. If the well is incapable of producing a sufficient volume of sample at any time, take the largest quantity available and record this occurrence in the site logbook or equivalent electronic form. When this occurs, contact the analytical laboratory to alert them that a reduced sample volume(s) will be submitted for analysis.
13. Fill sample containers and preserve and label them as described in SOP SA-6.1. Many sample bottles will contain preservative when they are shipped to the field. In those cases, do not add preservative.
14. Replace the well cap and lock it as appropriate. Make sure the well is readily identifiable as the source of the sample.
15. Process sample containers as described in SOP SA-6.1.
16. Decontaminate equipment as described in SOP SA-7.1.

6.7 Low-Flow Purging and Sampling

6.7.1 Scope and Application

Low-flow purging and sampling techniques may be required for groundwater sampling activities. The purpose of low-flow purging and sampling is to collect groundwater samples that contain "representative" amounts of mobile organic and inorganic constituents in the vicinity of the selected open well interval, at or near natural flow conditions. This minimum-stress procedure emphasizes negligible water level drawdown and low pumping rates to collect samples with minimal alterations in water chemistry. This procedure is designed primarily to be used in wells with a casing diameter of 1 inch or more and a saturated screen length, or open interval, of 10 feet or less. Samples obtained are suitable for analyses of common types of groundwater contaminants (volatile and semivolatile organic compounds [VOCs and SVOCs], pesticides, polychlorinated biphenyls [PCBs], metals, and other inorganic ions [cyanide, chloride, sulfate, etc.]). This low-flow procedure is not designed for collection of non-aqueous phase liquid samples from wells containing light or dense non-aqueous phase liquids (LNAPLs or DNAPLs).

This procedure is flexible for various well construction types and groundwater yields. The goal of the procedure is to obtain a turbidity level of less than 10 NTUs and to achieve a water level drawdown of less

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than 0.3 foot during purging and sampling. If these goals cannot be achieved, sample collection can take place provided that the remaining criteria in this procedure are met.

6.7.2 Equipment

The following equipment is required (as applicable) for low-flow purging and sampling:

- Adjustable rate submersible pump (e.g., centrifugal or bladder pump constructed of stainless steel or Teflon).
- Disposable clear plastic bottom-filling bailers to be used to check for and obtain samples of LNAPLs or DNAPLs.
- Tubing – Teflon, Teflon-lined polyethylene, polyethylene, polyvinyl chloride (PVC), Tygon, or stainless steel tubing can be used to collect samples for analysis, depending on the analyses to be performed and regulatory requirements.
- Water level measuring device with 0.01-foot accuracy (electronic devices are preferred for tracking water level drawdown during all pumping operations).
- Interface probe.
- Flow measurement supplies.
- Power source (generator, nitrogen tank, etc.). If a gasoline generator is used, it must be located downwind and at a safe distance from the well so that the exhaust fumes do not contaminate the samples.
- Indicator parameter monitoring instruments – pH, turbidity, specific conductance, and temperature. Use of a flow-through cell is recommended. Optional indicators - ORP, salinity, and DO. A flow-through cell (also referred to as an in-line sample chamber) is required.
- Standards to perform field calibration of instruments.
- Decontamination supplies.
- Logbook(s) and other forms (see Attachments B through D) or equivalent electronic form(s).
- Sample bottles.
- Sample preservation supplies (as required by the analytical methods).
- Sample tags and/or labels.
- Well construction data, location map, field data from last sampling event (if available).
- Field Sampling Plan.
- PID or FID instrument for measuring volatile organic vapors per the HASP (or equivalent).

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6.7.3 Purging and Sampling Procedure

1. Open the monitoring well as stated earlier and step away. Prepare sampling equipment while allowing 3 to 5 minutes to allow the water level to reach equilibrium. In situations where VOCs are the primary contaminants of concern, air monitoring of the samplers' breathing zone areas may be required by the HASP or equivalent (typically with a PID or FID).
2. Measure the water level immediately prior to placing the pump in the well and record the water level on the Low-Flow Purge Data Form or equivalent electronic form immediately prior to placing the pump or tubing into the well.
3. Lower the measuring device further into the well to collect the total depth measurement. Again wait 3 to 5 minutes to allow the well to equilibrate to the initial water level prior to placing the pump or pump intake in the well.
4. Record the total well depth on the Low-Flow Purge Data Form or equivalent electronic form immediately prior to placing the pump or tubing into the well.
5. Lower the pump or tubing slowly into the well so that the pump intake is located at the center of the saturated screen length of the well. If possible, keep the pump intake at least 2 feet above the bottom of the well to minimize mobilization of sediment that may be present in the bottom of the well. Collection of turbidity-free water samples may be difficult if there is 3 feet or less of standing water in the well.
6. Start with the initial pump rate set at approximately 0.1 liter per minute. Use a graduated cylinder and stopwatch to measure the pumping rate. Adjust the pumping rate as necessary to prevent drawdown from exceeding 0.3 foot during purging. If no drawdown is noted, the pump rate may be increased (to a maximum of 0.4 liter per minute) to expedite the purging and sampling event. The pump rate will be reduced if turbidity is greater than 10 NTUs after all other field parameters have stabilized. If groundwater is drawn down below the top of the well screen, purging shall cease or the well shall be pumped to dryness and then allowed to recover before purging continues. Well recovery to 75 percent is necessary prior to sampling. Slow-recovering wells should be identified and purged at the beginning of the workday to maximize field work efficiency. If possible, samples should be collected from these wells within the same workday and no later than 24 hours after the end of purging.
7. Measure the water level in the well every 5 to 10 minutes using the water level meter. Record the well water level on the Low Flow Purge Data Form (Attachment D) or equivalent electronic form.
8. Record on the Low Flow Purge Data Form every 5 to 10 minutes the water quality parameters (pH, specific conductance, temperature, turbidity, ORP, DO, and salinity or as specified by the approved site-specific planning document) measured by the water quality meter and turbidity meter. If the cell needs to be cleaned during purging operations, continue pumping (allow the pump to discharge into a container) and disconnect the cell. Rinse the cell with distilled/deionized water. After cleaning is completed, reconnect the flow-through cell and continue purging. Document the cell cleaning on the Low-Flow Purge Data Form or equivalent electronic form.
9. Estimate the flow rate by noting the amount of discharge in a graduated cylinder per unit time using a watch with a second hand. Remeasure the flow rate any time the pump rate is adjusted and periodically during purging. This will determine if a reduction in rate has occurred due to possible battery depletion.
10. During purging, check for the presence of bubbles in the flow-through cell. The presence of bubbles is an indication that connections are not tight. If bubbles are observed, check for loose connections and tighten, repair, or replace them as necessary to achieve a tight connection.

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11. Wait until stabilization is achieved, or a minimum of two saturated screen volumes have been removed and three consecutive readings, taken at 5 to 10 minute intervals, are within the following limits, then begin sampling:

- pH ± 0.2 standard units
- Specific conductance $\pm 10\%$
- Temperature $\pm 10\%$
- Turbidity less than 10 NTUs
- DO $\pm 10\%$

12. If the above conditions have not been met after the well has been purged for 4 hours, purging will be considered complete and sampling can begin. Record the final well stabilization parameters from the Low-Flow Purge Data Form onto the Groundwater Sample Log Form or equivalent electronic form.

NOTE: VOC samples are preferably collected first, directly into pre-preserved sample containers. Fill all sample containers by allowing the pump discharge to flow gently down the inside of the container with minimal turbulence.

13. If the water column in the pump tubing collapses (water does not completely fill the tubing) before exiting the tubing, use one of the following procedures to collect VOC samples:

- Collect samples for non-VOC analyses first, then increase the flow rate incrementally until the water column completely fills the tubing, collect the sample for VOCs, and record the new flow rate.
- Reduce the diameter of the existing tubing until the water column fills the tubing either by adding a connector (Teflon or stainless steel) or clamp, which should reduce the flow rate by constricting the end of the tubing. Proceed with sample collection.
- Insert a narrow-diameter Teflon tube into the pump's tubing so that the end of the tubing is in the water column and the other end of the tubing protrudes beyond the pump's tubing, then collect the sample from the narrow diameter tubing.
- Prepare samples for shipping as per SOP SA-6.1.

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ATTACHMENT A

PURGING EQUIPMENT SELECTION

Diameter Casing		Bailer	Peristaltic Pump	Vacuum Pump	Air-lift	Diaphragm "Trash" Pump	Submersible Diaphragm Pump	Submersible Electric Pump	Submersible Electric Pump w/Packer
1.25-Inch	Water level <25 feet	X	X	X	X	X			
	Water Level >25 feet	X			X				
2-Inch	Water level <25 feet	X	X	X	X	X	X		
	Water Level >25 feet	X			X		X		
4-Inch	Water level <25 feet	X	X	X	X	X	X	X	X
	Water Level >25 feet	X			X		X	X	X
6-Inch	Water level <25 feet				X	X		X	X
	Water Level >25 feet				X			X	X
8-Inch	Water level <25 feet				X	X		X	X
	Water Level >25 feet				X			X	X

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ATTACHMENT A
Purging and Sampling Equipment Selection (Updated July 2016)

Manufacturer	Model Name/ Number	Principle of Operation	IE ¹	Maximum Outer Diameter (OD)/ Length (L) ²	Construction Materials ³ (with Lines and Tubing)	Lift Range (Feet)	Delivery Rates or Volumes ⁴	Main Source(s) of Information ⁵	Comments
Commonly Used Purging/Sampling Equipment									
GeoTech Environmental Equipment, Inc.	Geopump (Series I or II)	Portable, peristaltic vacuum (suction) or pressure - reversible pump	E	3.5" x 8" x 8"	Not applicable (NA)	0 - 27	0 - 1,000 mL/min	Mfg./Pine	Variable speed, operate with 12 volt (V) direct current (DC) or 120V alternating current (AC) power source.
Grundfos Pumps Corporation	Redi-Flo2	Portable, submersible pump	I	1.8" OD x 11" L	SS, PTFE (i.e., Teflon®)	0 - 250	100 mL/min to 9 gpm	Pine/PNNL	Variable speed (adjustable flow rate); ideal for purging or sampling.
GED Environmental Systems	Sample Pro	Portable, bladder pump (positive displacement)	I	1.75" OD x 12" L	SS, PTFE or PE disposable bladder	0 - 250	0 - 800 mL/min	Mfg.	First pump developed specifically to bring the advantages of low-flow sampling to sites with non-dedicated pumps.
GED Environmental Systems	Well Wizard	Dedicated, bladder pump (positive displacement)	I	1.66" OD x 41" L	SS, PTFE bladder, HDPE, PP	0 - 1,000	0 - 2,000 mL/min	Pine/PNNL	Ideal for deep well applications and long-term monitoring. Available in a variety of sizes and materials.
Less Commonly Used Purging/Sampling Equipment									
Bennett Sample Pumps, Inc.	Bennett 1800-8	Portable, piston pump (positive displacement)	I	1.8" OD x 23.625" L	SS, PTFE, PP, Nylon	0 - 1,000	100 mL/min to 2.6 gpm	Mfg./PNNL	Variable speed (adjustable flow rate); ideal for purging or sampling. Discharge rates are difficult to control.
Cole-Parmer Inst. Co.	Masterflex 7571-00	Portable, peristaltic vacuum (suction) or pressure - reversible pump	E	16" x 10.25" x 11"	NA	0 - 27	0.06 - 680 mL/min	Mfg./Pine	Peristaltic pumps are not recommended for sampling volatile organic compounds (VOCs).
Fultz Pumps, Inc.	Fultz SP300	Portable, gear driven pump (positive displacement)	I	1.75" OD x 9.16" L	SS, PTFE gears, Viton®	0 - 200	100 mL/min to 2.4 gpm	Mfg./PNNL	Variable speed (adjustable flow rate); ideal for purging or sampling. Low extraction rates are difficult to control.
GeoTech Environmental Equipment, Inc.	Keck SP12	Portable, progressive cavity pump (positive displacement)	I	1.75" OD x 25" L	SS, EPDM stator, PE	0 - 150	100 mL/min to 1.2 gpm	Mfg./PNNL	Variable speed (adjustable flow rate); ideal for purging or sampling. Low extraction rates are difficult to control.
GeoTech Environmental Equipment, Inc.	SS Geosub	Portable, single speed submersible purge pump	I	1.75" OD x 13.2" L	SS, PTFE, Viton®	0 - 200	low flow to 10 mL/min	Mfg./Pine	Only 12V SS pump with Teflon® cable according to GeoTech brochure.
GeoTech Environmental Equipment, Inc.	NA	Portable, centrifugal purge pump		NA	NA	0 - 55	2.5 gpm max.	Mfg.	Not recommended for sampling.
GeoTech Environmental Equipment, Inc.	Various	Portable, bladder pump (positive displacement)	I	0.675-1.66" OD	SS or PVC; PTFE or PE bladder, FEP or PE tubing	0 - 200 SS; 0 - 250 PVC	NA	Mfg.	Various sizes are available. Pump is also available in PVC instead of SS.
GeoTech Environmental Equipment, Inc.	Bailer	Portable, grab (positive displacement)	I	Various sizes	Various materials - HDPE, PVC, SS, or fluoropolymer	No limit	NA	Mfg.	Low cost, easy operation; potential churning could increase turbidity. Typically a last-resort option.
Grundfos Pumps Corporation	Redi-Flo3	Portable, submersible pump	I	2.9" OD	SS, PVDF, Viton®, Tefzel®	0 - 360	8 gpm max.	Pine/PNNL	Not designed for low flow use.
Grundfos Pumps Corporation	Redi-Flo4	Portable, submersible pump	I	3.8" OD	SS, PTFE	0 - 600	100 mL/min to 10 gpm	Pine/PNNL	Variable speed (adjustable flow rate); ideal for purging or sampling.
KYB	HydroStar™	Portable, single-action piston pump (positive displacement)	I	1.66" OD	SS, PTFE	0 - 400	100 mL/min to 5 gpm	Mfg./PNNL	Variable speed (adjustable flow rate); ideal for purging or sampling. Discharge rates are difficult to control.

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Purging and Sampling Equipment Selection (Updated July 2016)

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Landtec North America	Nu-Matic 4	Dedicated, pneumatic displacement pump	I	3.5" OD x 43.25" L	SS	0 - 275	10 gpm max.	Pine	Float actuated, fully automatic; ideal for the harsh environment of landfill conditions.
ProActive Environmental Products	SS Monsoon (and others)	Portable; Low Flow pumps available in a variety of sizes	I	1.82" OD x 7.5" L	SS, PE	0 - 120	10 mL/min to 3.5 gpm	Pine/GeoTech	12V DC - designed for continuous purging and low flow sampling.
QED Environmental Systems	Eliminator	Portable; bladder pump (positive displacement)	I	3" OD x 40" L	SS; PTFE or elastomer bladder, Q-Tal®, Viton®	0 - 230	6 gpm max.	Pine	High capacity; designed for light non-aqueous phase liquid (LNAPL) and dense non-aqueous phase liquid (DNAPL) removal in 4" wells.
QED Environmental Systems	Pulse Pump	Portable; pneumatic displacement pump	I	1.66" OD x 20" L	SS, PTFE	0 - 230	2 gpm max.	Pine	Especially suited for DNAPL recovery. Requires compressed air, external timer.
Solinst Canada Ltd.	404 Waterra	Portable; inertial reciprocating pump (positive displacement)	I	Various sizes	SS or Delrin® footvalve; HDPE tubing	0 - 200	2 gpm max.	Pine	Low cost, easy operation; potential churning could increase turbidity. Simple alternative to a bailer.
Solinst Canada Ltd.	407 Integra	Portable; bladder pump	I	1" or 1.66" OD	SS or PVC; FEP bladder (disposable PE available)	SS - 0 - 500 PVC - 0 - 100	3.3 L/min at 150"	Pine	Regular or low flow applications.
Solinst Canada Ltd.	408 DVP	Portable; pneumatic drive pump	I	1.66" OD x 24" L	SS or PVC; FEP bladder	SS - 0 - 500 PVC - 0 - 100	3.3 L/min at 150"	Pine	Regular or low flow applications - can deliver high flow rates.
Solinst Canada Ltd.	408M DVP	Portable; micro pneumatic drive pump	I	0.375" OD	FEP tubing	0 - 240	20 - 150 mL/min	Pine	Uses coaxial FEP tubing, easy to transport and install; ideal for low flow and narrow down-hole applications.
Solinst Canada Ltd.	410	Portable; peristaltic (negative pressure) pump	E	13" x 5" x 6"	NA	0 - 27	120 - 3,500 mL/min	Pine	Ideal for vapor sampling, water from shallow wells, or surface water.
TIMCO Mfg. Co., Inc.	Bailer	Portable; grab (positive displacement)	I	Various sizes	PVC, PP, others	No limit	250 mL/foot of bailer	Old SOP	Other sizes, materials, models available; optional bottom-emping device available, no solvents used.
No-Purge Sampling Equipment (Available Options on Some Projects)									
Amplified Geochemical Imaging (AGI), LLC	AGI Universal Sampler (formerly Gore-Sorber™ Module)	Passive container, diffusion-type sampler	I	0.25" OD x 8" L	Waterproof, vapor-permeable GORE-TEX™ polymer membrane of expanded PTFE	0 - 32	NA	ITRC PSC-5	Demonstrated to be applicable to a variety of (but not all) VOCs, semivolatile organic compounds (SVOCs), and polycyclic aromatic hydrocarbons (PAHs); not designed for inorganics.
Geolinsight, Inc.	HydraSleeve™	Passive container, grab-type sampler	I	1.5-2.6" OD x 30" L	PE tubing with PE reed-valve	No limit	650 - 1,250 mL	ITRC PSC-5	Demonstrated to be applicable to a variety of VOCs and inorganics.
Multiple Manufacturers	RPPS	Rigid Porous Polyethylene Sampler (RPPS), diffusion-type sampler	I	1.5" OD x 5" L	Rigid PE tube - thin sheets of foamlike porous PE with a Delrin® plug	No limit	80 - 100 mL	ITRC PSC-5	Developed by Don Vroblesky (USGS). Demonstrated to be applicable to a variety of water-soluble organics and inorganics.
Multiple Manufacturers	PDB	Passive Diffusion Bag (PDB) Sampler, diffusion-type sampler	I	1.22" OD, 18-24" L	LDPE with HDPE cap; some vendors use other materials and sizes	No limit	75 - 350 mL	ITRC PSC-3	Developed by Don Vroblesky (USGS) and Thomas Hyde (GE). Demonstrated to be applicable to a variety of (but not all) VOCs; not for inorganics.

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Purging and Sampling Equipment Selection (Updated July 2016)

Manufacturer	Model Name/ Number	Principle of Operation	IE ¹	Maximum Outer Diameter (OD)/ Length (L) ²	Construction Materials ³ (with Lines and Tubing)	Lift Range (Feet)	Delivery Rates or Volumes ⁴	Main Source(s) of Information ⁵	Comments
Prohydro, Inc.	Snap Sampler™	Pneumatic actuator, grab-type sampler	I	1.7-3.4" OD x 19-22" L	Acetal with PFA-coated SS spring, optional SS weight, 40 mL glass vials or 125-350 mL PP bottles	0 - 2,500	40 - 1,400 mL	ITRC PSC-5	Demonstrated to be applicable to a variety of VOCs and inorganics. Up to 4 bottles may be employed together.
Multi-Level Groundwater Monitoring System (MLS) (Used on Some Projects based on Stakeholder Agreement)									
FLUT, Ltd. Co.	Water FLUTe™	Dedicated, gas drive sampling, liner seals entire borehole wall.	I	Various sizes	NA	NA	NA	Mfg.	Practical to collect large volumes; can use low flow method; minimizes investigation-derived waste (IDW); proven technology.
Solinst Canada Ltd.	Continuous Multichannel Tubing (CMT)	Dedicated; 3 or 7-channel tubing designed to isolate and monitor depth discrete zones.	I	Up to 7 ports	NA	NA	NA	Mfg.	Low cost and easy to install; ideal for shallow applications; single tube for effective sealing.
Solinst Canada Ltd.	Waterloo	Dedicated; double valve or bladder pump sampling, packers separate ports at multiple depths.	I	Various sizes	NA	NA	NA	Mfg.	Practical to collect large volumes; can use low flow method; minimizes IDW; proven technology.
Schlumberger	Westbay	Dedicated; packers separate ports at multiple depths.	I	Various sizes	NA	NA	NA	Mfg.	Transducer and sampler are on a wire line; minimizes IDW; proven technology. Low long-time maintenance costs.

Footnotes:

- 1 - IE - Internal (I) or external (E) to the well.
- 2 - For external pumps, the size is given in three dimensions (i.e. length, width, height); for internal pumps, the size is given in outside diameter (OD) and length (L) (in inches).
- 3 - Construction materials for pumps and associated equipment include: SS = stainless steel; PTFE = polytetrafluoroethylene; HDPE = high density polyethylene; PP = polypropylene; PE = polyethylene; PVC = polyvinyl chloride; FEP = fluorinated ethylene propylene; PMDF = polydimethylsiloxane; EPDM = ethylene propylene diene elastomer; LDPE = low density polyethylene; PFA = perfluoroalkoxy.
- 4 - Rate or volume abbreviations: mL/min = milliliters per minute; gpm = gallons per minute; L/min = liters per minute; max. = maximum; mL = milliliters.
- 5 - Sources of information include specification sheets from the manufacturer (i.e., Mfg.); equipment rental company websites - Pine Environmental (www.pine-environmental.com), GeoTech (www.geotech.com); and guidance documents from Pacific Northwest National Laboratory (PNNL) (i.e., PNNL-13650) and Interstate Technology & Regulatory Council (ITRC) (i.e., PSC-3 and PSC-5 available at www.itrcweb.org).



7/15/2016

SERIAL NUMBER: _____

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ATTACHMENT D (EXAMPLE)
LOW FLOW PURGE DATA SHEET

[illegible]

 TETRA TECH	STANDARD OPERATING PROCEDURE (GUIDANCE)	Number SA-1.6	Page 1 of 21
		Effective Date 02/17/2017	Revision 3
		Applicability Tetra Tech, Inc., NUS Operating Unit	
		Prepared Earth Sciences Department	
Subject NATURAL ATTENUATION PARAMETER COLLECTION	Approved T. Johnston	tom.johnston@tetratech.com	 Digitally signed by: tom.johnston@tetratech.com DN: CN = tom.johnston@tetratech.com Date: 2017.02.20 14:20:19 -05'00'

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1.0 PURPOSE

The purpose of this document is to provide general reference information and guidance regarding natural attenuation parameter and methodology selection, sample collection, and a general understanding of the sample results.

2.0 SCOPE

This document provides information on selection of appropriate groundwater natural attenuation parameters, selection of sampling methods for these parameters, techniques for onsite field analysis of select parameters, and some basic understanding of the field sample results. Review of the information contained herein will facilitate planning of the field sampling effort by describing standard sampling practices and techniques. To a limited extent, it shall also facilitate the understanding and interpretation of the sampling results. It addresses field procedures for collection of data at sites with organic groundwater contaminants (e.g., chlorinated and petroleum hydrocarbons) to the extent practical. The focus of this document is on natural attenuation, not enhanced bioremediation.

The techniques described shall be followed whenever applicable, noting that site-specific conditions, project-specific objectives, local, state, and federal guidelines may be used as a basis for modification of the procedures noted herein. The intent of this document is to supplement the local, state, and federal guidance documents and manufacturer's analytical methods referenced in Section 6.0. It is not intended for this document to supersede this guidance or information. Please note that natural attenuation is a relatively dynamic science with ongoing research in the science and engineering community. It is important that data collectors and interpreters use the most recent regulatory guidance, which may be updated on a periodic basis from that noted in Section 6.

3.0 GLOSSARY

Abiotic: Not associated with or derived from living organisms.

Aerobe: Bacteria that use oxygen as an electron acceptor.

Anaerobe: Organisms that can use electron acceptors other than molecular oxygen to support their metabolism.

Anoxic groundwater: Groundwater that contains oxygen in concentrations less than about 0.5 mg/L. This term is synonymous with the term anaerobic.

Anthropogenic: Man-made.

Biotic: Of, relating to, or resulting from living organisms.

Cometabolism: The process in which a compound is fortuitously degraded by an enzyme or cofactor produced during microbial metabolism of another compound.

Daughter product: A compound that results directly from the biotic or abiotic degradation of another. For example, *cis*-1,2-dichloroethene (*cis*-1,2-DCE) is a common daughter product of trichloroethene (TCE).

Diffusion: The process whereby molecules move from a region of higher concentration to a region of lower concentration as a result of Brownian motion.

Dispersion: The tendency for a solute to spread from the path that it would be expected to follow under advective transport.

Electron acceptor: A compound capable of accepting electrons during oxidation-reduction reactions. Microorganisms obtain energy by transferring electrons from an electron donor such as an organic compound (or sometimes a reduced inorganic compound such as sulfide) to an electron acceptor. Electron acceptors are compounds that are relatively oxidized and include oxygen, nitrate, iron(III), manganese(IV), sulfate, carbon dioxide, or in some cases chlorinated aliphatic hydrocarbons such as tetrachloroethene (PCE), TCE, DCE and vinyl chloride (VC).

Electron donor: A compound capable of supplying (giving up) electrons during oxidation-reduction reactions. Microorganisms obtain energy by transferring electrons from an electron donor such as

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an organic compound (or sometimes a reduced inorganic compound such as sulfide) to an electron acceptor. Electron donors are compounds that are relatively reduced and include fuel hydrocarbons and native organic carbon.

Metabolic byproduct: A product of the reaction between an electron donor and an electron acceptor. Metabolic byproducts include volatile fatty acids, daughter products of chlorinated aliphatic hydrocarbons, methane, and chloride.

Oxic groundwater: Groundwater that contains oxygen in concentrations greater than about 0.5 mg/L.

Oxidation/reduction reaction: A chemical or biological reaction wherein an electron is transferred from an electron donor (donor is oxidized) to an electron acceptor (acceptor is reduced).

Predominant terminal electron-accepting process: The electron-accepting process (oxygen reduction, nitrate reduction, iron(III) reduction, and other chemical species reduction) that sequesters the majority of the electron flow in a given system.

Reductive dechlorination: Reduction of a chlorine-containing organic compound via the replacement of chlorine with hydrogen.

Respiration: The process of coupling the oxidation of organic compounds with the reduction of inorganic compounds such as oxygen, nitrate, iron(III), manganese(IV), and sulfate.

Seepage velocity: The average velocity of groundwater in a porous medium.

Substrate: A compound used by microorganisms to obtain energy for growth. The term can refer to either an electron acceptor or an electron donor.

4.0 RESPONSIBILITIES

Project Manager (PM) - Responsible for ensuring that all field activities are conducted in accordance with this standard operating procedure (SOP).

Project Hydrogeologist or Geochemist - Responsible for selecting and detailing the specific groundwater sampling techniques, onsite water quality testing (type, frequency, and location), and equipment to be used, and providing detailed input in this regard to the project plan documents. The project hydrogeologist or geochemist is also responsible for properly briefing and overseeing the performance of the site sampling personnel.

Field Operations Leader (FOL) - Responsible for the onsite verification that all field activities are performed in compliance with approved SOPs or as otherwise directed by the approved project plan(s).

Project Geologist - is primarily responsible for the proper acquisition of the groundwater samples. He/she is also responsible for the actual analyses of onsite water quality samples, as well as instrument calibration, care, and maintenance. When appropriate, such responsibilities may be performed by other qualified personnel (e.g., field sampling technicians or site personnel).

5.0 PROCEDURES

5.1 General

Natural attenuation includes physical, chemical, and biochemical processes affecting the concentrations of dissolved contaminants in groundwater. These processes may include advection, diffusion, dispersion, volatilization, dilution, sorption to aquifer solids, and/or precipitation or mineralization of compounds. Of greatest importance are those processes that lead to a reduction in contaminant mass (by degrading or destroying contaminants) such as biodegradation. These biochemical processes remove organic contaminants from the aquifer by destruction or decomposition. Depending on the type of contaminant, particularly the organic contaminant (e.g., petroleum hydrocarbons or chlorinated organic solvents), the biochemical environment in the aquifer will vary. The biochemical environment within the aquifer influences and is influenced by the activities of aquifer microbiota. Specific types of microbiota, working singly or in complex consortia, may use organic contaminants as part of their normal cell functions. Natural attenuation

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monitoring is designed to measure indicators of the biochemical environment within the aquifer and, with direct and indirect lines of evidence and associated chemical concentration data, evaluate the likely fate (i.e., transformation, destruction, dilution, attenuation, etc.) of organic contaminants.

5.2 Planning for Natural Attenuation Sampling

5.2.1 Develop a site-specific conceptual model. The first step in developing this model is the analysis and review of available site-specific characterization data. The development and refinement of this model should be supplemented with additional data as needed. The data should include but is not limited to:

- Geologic and hydrogeologic information in three dimensions
- Nature, extent, and magnitude of contamination
- Location and presence of potential receptors to contamination

5.2.2 Develop lines of evidence to establish whether natural attenuation is working. The potential for abiotic and biotic degradation processes should be considered. The most compelling, primary evidence is decreasing groundwater contaminant concentrations over time. Decreasing concentration trends can be demonstrated in several ways including:

- Isoconcentration maps of the dissolved plume over time wherein the extent of the plume is either stable or decreasing.
- Time series plots of contaminant concentrations within a well illustrating a clear downward trend.
- Contaminant concentration profiles in a series of monitoring wells along a groundwater flow path illustrating decreasing concentrations beyond that attributable to dilution and dispersion.

Secondary, or supporting, lines of evidence include:

- Analytical data showing production and subsequent destruction of primary contaminant breakdown products.
- Geochemical data indicating that the biochemical environment is favorable for the appropriate microbiota.
- Geochemical data that indicate the aquifer microbiota are active.

5.2.3 Select the monitoring well location. The number and locations of wells required to monitor natural attenuation will depend on the physical setting at each location. One possible array of monitoring wells is illustrated in Attachment A. In this scenario, one well is used to monitor conditions upgradient of the source, one well is located in the source area, and several wells are used to define and monitor the downgradient and lateral extent of the dissolved plume. At a minimum, there should be at least one upgradient well (ideally with no contamination present), one well in the source area, one well downgradient from the source area in the dissolved plume, and one downgradient well where contaminant concentrations are below regulatory criteria. Note that the number and locations of monitoring wells will vary depending on the site complexity, site objectives, and regulatory requirements.

5.2.4 Select a sampling frequency. Sampling frequency will be dictated by the ultimate use of the data and site-specific characteristics or regulatory requirements and permits. Contaminant concentrations may be used to define statistically meaningful trends in contaminant concentrations. The sampling frequency may be defined by the hydrogeologic and/or geochemical conditions as well as the proposed statistical method for data analysis. For example, groundwater flow and contaminant characteristics (e.g., seepage velocity and contaminant loading) may dictate the sample frequency. Regardless of the factors, sampling

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frequency and duration will need to establish the range of natural chemical variability within the aquifer. After a sufficient amount of data has been collected and the geochemical conditions are understood, the frequency of sampling may be reduced. See Section 5.5 for additional information on sample collection and frequency.

5.3 Selection of Natural Attenuation Parameters

Natural attenuation via biodegradation depends on the nature of the organic contaminants and the oxidation-reduction (redox) environment within the aquifer. Simply stated, if the contaminants are fuels, biodegradation will be most effective if the redox conditions are aerobic or oxidizing. If the contaminants are chlorinated solvents, the biodegradation will be most effective (in the source and near source areas) if redox conditions in the aquifer are anaerobic or reducing.

Several parameters are needed to evaluate whether natural attenuation is taking place and, if so, the rate at which it may be occurring. The primary parameter providing direct evidence of natural attenuation is the aqueous concentrations of parent and daughter volatile organic compounds. More specifically, a decrease in percent products, an increase in daughter products, evidence that the plume is stable or shrinking in size, and overall decline in contaminant concentrations is direct evidence of natural attenuation. Natural attenuation or geochemical parameters that provide information about the redox conditions in the aquifer include:

- Dissolved oxygen
- Nitrate/nitrite
- Dissolved manganese
- Iron
- Sulfate/sulfide
- Methane
- Oxidation-reduction potential (ORP)

Secondary parameters that indicate biological activity in the aquifer and thereby support the natural attenuation evaluation include:

- Dissolved hydrogen
- Alkalinity
- Dissolved carbon dioxide

The concentrations of natural attenuation parameters are used to define the aquifer redox conditions. It is important to record and document the presence or absence (i.e., measurable or not measurable concentration) of certain natural attenuation parameters. The presence or absence of a certain substance may be sufficient to indicate the redox condition within the aquifer. By reference to Attachment B, which illustrates the typical sequence of biologically mediated redox reactions in natural systems, it is apparent that, for example, sulfate reduction (producing dissolved sulfide in groundwater) does not operate in an aerobic environment. Therefore, measurable sulfide should not be present if there is also dissolved oxygen at concentrations indicating an aerobic environment. Attachment B also illustrates the redox potential (measured in millivolts) associated with the redox reactions. ORP readings, also in millivolts, measured during well purging, may be compared with the range of values in Attachment B but with caution. Redox potentials measured with a platinum electrode in natural water samples may be misleading, especially when biologically mediated reactions are important, because many of the critical reactions in Attachment B do not generate a response in the electrode. Dissolved hydrogen concentration ranges associated with important redox reactions are also indicated in Attachment B. Because dissolved hydrogen is actually used by microbiota during redox reactions, its concentration may provide an additional indicator of the overall redox condition in the aquifer.

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Attachments C and D tabulate the natural attenuation parameters for chlorinated volatile organic compound and petroleum hydrocarbon plumes, respectively. The parameters listed in these tables are organized in order of importance. Parameters selected for analysis shall be determined based on site conditions, project-specific plans, and/or other criteria established for the project. Based on these criteria, it is possible that all of the parameters may be selected.

5.4 Selection of Natural Attenuation Analytical Methods and Procedures

There are many analytical methods available to measure concentrations of the natural attenuation parameters discussed in the previous sections. Attachment E summarizes sample methodologies, sampling equipment needed, sample volume, container, preservation, and holding time requirements. This table also summarizes the detection limits and the detection ranges for each method. A number of factors should be considered when selecting the appropriate sample analytical methodology including the required parameters, appropriate detection ranges for each compound, cost, and ease of use in the field. For example, when determining the correct methodology for measuring concentrations of total sulfide, the metabolic byproduct of sulfate reducing conditions, it is important to analyze for each of the forms of sulfide (H_2S , S^{2-} , and HS^-). Also, when the detection limit of the selected method is exceeded, another method may be considered, or the sampler may be able to dilute the sample (per manufacturer's instructions) to quantify it within the detected range. In terms of cost, some parameter analyses are very time consuming when performed in the field. Without sacrificing sample integrity it may be more appropriate to select a methodology performed in a fixed-base laboratory. Finally, in terms of ease of use, certain field methods are generally easier compared to other methods. Using simpler methods may result in better quality sample results and increased sample repeatability without sacrificing sample integrity. For example, in some cases CHEMetrics Titret® Titration Ampule kits may be a good alternative to other hand digital titration methods.

When using field test kits, use the instructions included in the test kit to conduct the analyses. Under certain geochemical conditions, some geochemical parameters may not have positive detections. For example, if dissolved oxygen concentrations indicate aerobic conditions then it is unlikely that dissolved hydrogen is present (see Section 5.10 for additional information). Another example is alkalinity. If the pH of the groundwater sample is less than 4.5, then it is unlikely that alkalinity will be measurable. Despite the potential for non-detect results, in cases such as those described above, all parameters should be collected in the field based upon project plans. The value in collecting the parameters in the future shall be determined by the project hydrogeologist and/or geochemist in accordance with the projects planning documents data quality objectives (DQO) and the items discussed in Section 5.2.

Note: Test kit information provided here is the minimum amount of information required to correctly use the test kits listed in Attachment E. Additional training/instruction may be available from test kit manufacturers in the form of videos posted to the manufacturer's web sites. To access these web sites, conduct an Internet search on the manufacturer's name and view the videos on the web site, if available.

5.5 Procedures for Sample Collection

Groundwater sample collection for natural attenuation sampling should be performed using low flow purging and sampling techniques. These techniques are described in detail in SOP SA-1.1. Low flow purging and sampling procedures should be used to ensure the collection of a sample that is "representative" of the water present in the aquifer formation. Minimizing stress on the aquifer formation during low flow purging and sample collection ensures that there are minimal alternations to the water chemistry of the sample. The criteria used in the purging process should include minimization of drawdown in the well, stabilization of applicable indicator parameters, and evacuation of a sufficient amount of purge volume in accordance with SOP SA-1.1, project plans, and/or applicable regulatory guidance.

Groundwater purging and sampling for natural attenuation should be performed using submersible pumps (e.g., bladder pumps) in accordance with SOP SA-1.1. However, in accordance with project plans and

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applicable regulatory guidance, peristaltic pumps may also be used for this purpose. Limitations of and factors associated with using these devices should be considered (see SOP SA-1.1 for more information). As a result of difficulties in collecting “representative” groundwater samples, bailers should not be used for the collection of natural attenuation samples.

It is critical that disturbance and aeration of samples monitored and collected at the well head are minimized. As a result, a flow-through sampling cell and a direct reading meter shall be used for the measurement of well stabilization indicator parameters (e.g., pH, conductivity, temperature, dissolved oxygen, turbidity, and ORP) at the well head. The pump effluent tubing should be placed at the bottom of the flow-through cell allowing effluent water from the cell to discharge at the top of the meter (above the detector probes) to minimize the agitation of water in the cell.

Documentation of the purging process shall be recorded during and at the completion of purging as discussed in Section 5.8. Immediately following the purging process and before sampling, all applicable indicator parameters must be measured and recorded on the appropriate sample log sheets as discussed in Section 5.8.

After all of the purging requirements have been met, groundwater sampling and natural attenuation data collection can begin. Monitoring wells will be sampled using the same pump and tubing used during well purging.

5.6 Procedures for Field Sample Analysis

Each of the field and fixed-base laboratory sample parameters requires different sampling procedures and holding times. Attachment E presents parameter-specific requirements for sampling, analysis, and storage of all of the parameters and methods sampled as part of natural attenuation analysis.

Due to parameter-specific sampling procedures and holding times, it is important to consider the sequence of sample collection and analysis. Generally speaking, with the exception of volatile organic compounds, analyze field parameters first followed by fixed-base laboratory sample collection. Collect all samples in a sequence and manner that minimizes volatilization, oxidation, and/or chemical transformation of compounds. The following sample and analysis order should be followed:

- | | |
|---|------------------------------------|
| 1. Volatile organic compounds | 8. Nitrate / Nitrite |
| 2. Dissolved oxygen | 9. Dissolved manganese |
| 3. Alkalinity | 10. Semivolatile organic compounds |
| 4. Dissolved carbon dioxide | 11. Other dissolved metals |
| 5. Dissolved ferrous iron | 12. Total metals |
| 6. Dissolved sulfide (hydrogen sulfide and sulfide) | 13. All other constituents |
| 7. Dissolved hydrogen, methane, ethene, and ethane | |

Field-analyzed parameters should be collected and immediately analyzed directly from the pump effluent per the requirements on Attachment E and manufacturer's recommendations. Care should be taken to minimize any unnecessary disturbance, aeration, or agitation of the sample prior to analysis. It is not acceptable to collect and store samples that are to be analyzed immediately at the well head in a temporary holding container (e.g., open topped pitcher) to be analyzed at a later time.

The manufacturer's procedure manual for each of the field-based analyses shall be maintained in the field during the entire sampling program. The procedures give a detailed explanation of how to perform each particular method and include information on sampling, storage, accuracy checks, interferences, reagents, and apparatus needed to perform each analysis.

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5.7 Procedures for Quality Assurance and Quality Control Field Sample Analysis

5.7.1 Prior to analysis, check the expiration dates of reagents. If the reagents have exceeded their expiration date or shelf life, the reagents shall be replaced.

5.7.2 Perform accuracy and precision checks to check the performance of reagents, apparatus, and field analytical procedures per the manufacturer's recommendations. The accuracy checks should include the use of standard solutions (i.e., standard addition), as appropriate. The manufacturer's field test kit manual provides details on how to perform each of the accuracy checks for each parameter where applicable. Refer to Section 6.0 for manufacturer contact information.

Precision checks must include the performance of duplicate analysis. When using a colorimeter, precision checks may also include reagent blank corrections and standard curve adjustments as recommended by the manufacturer. Field duplicate results shall be performed and evaluated for relative percent difference (RPD) at a rate of 1 per 10 samples or as determined by the project plans. The RPD can be calculated as follows:

$$RPD = \left| \frac{\text{First result} - \text{Second result}}{\text{Mean arithmetic (average) of first and second result}} \right| \times 100$$

5.7.3 If the RPD exceeds 50 percent, perform the test again to verify the result. The duplicate results shall be documented in the 'Notes' section for that specific parameter on the appropriate sample logsheet (see Section 5.8).

5.7.4 If a colorimeter (e.g., HACH DR-890 or equivalent) is used for parameter analysis, an instrument performance verification test using absorbance standards may also be performed to ensure the meter is providing accurate measurements.

The following table lists examples of the types and frequencies of accuracy checks required for each parameter. Refer to the manufacturer's instructions for information regarding other analyses.

Parameter	Method	Standard Solution	Field Duplicate	Reagent Blank Correction
Alkalinity	CHEMetrics K-9810, -15, -20	None	1 per 10	None
Carbon dioxide	CHEMetrics K-1910, -20, -25	None	1 per 10	None
Dissolved oxygen	CHEMetrics K-7501, -12	None	1 per 10	None
Ferrous iron	HACH DR-890	None	1 per 10	None
Nitrite	HACH DR-890	1 per round	1 per 10	1 per lot
Nitrate	HACH DR-890	1 per round	1 per 10	1 per lot
Sulfide	HACH DR-890	None	1 per 10	None
Hydrogen sulfide	HACH HS-C	None	1 per 10	None

5.7.4 If deviations from the applicable analytical procedure are identified, correct the deviations and re-analyze the associated samples. If problems are identified with the reagents, apparatus, or procedures, interferences may be present. Interferences may also be due to other factors (e.g., inappropriate pH, presence of other analytes, turbidity, temperature, etc.) that may interfere with the sample result. The

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manufacturer's procedures (e.g., Hach, 1999) should be reviewed prior to analysis to avoid or minimize such interferences. Associated problems or suspected interferences shall be documented in the 'Notes' section of the sample logsheet. Often, interferences cannot be avoided. In these cases, the sampler should be aware of these potential interferences and document them properly.

5.8 Documentation Procedures for Field Sample Analysis

5.8.1 Document field results in the field as noted in SOP SA-6.3. The sample log sheet titled "Field Analytical Log Sheet, Geochemical Parameters" shall be prepared for each sample collected and analyzed in the field. A copy of this form can be found as Attachment F of this SOP. Other field log sheets (e.g., low flow purge log sheet, groundwater sample logsheet, etc.) shall also be completed in accordance with SOP SA-6.3.

5.8.2 Record additional specific information in the project logbook as follows. This information shall include, but is not limited to, the test kit name and model number, lot number and expiration date of the test kit and reagents used, serial number of the instrument (e.g., colorimeter) used for the analysis, and results of the quality assurance and quality control field sample analysis. Because environmental conditions and changes in those conditions may affect the field analytical results, it is important to document the site conditions (weather, temperature, etc.) at the time of sampling in the logbook in accordance with SOP SA-6.3.

5.9 Waste Handling and Disposal

Several of the test kits listed in Attachment E require the use of chemicals and materials that must be properly handled and disposed of in a proper and responsible manner. Refer to specific manufacturer's guidance for handling and disposal practices. See also Section 6.0 for more detailed and complete information. Handling and disposal of these items should be conducted in accordance with all local, state, and federal guidelines.

5.10 Understanding Field Sample Analytical Results

Natural attenuation data interpretation is complicated by the complex inter-relationships of various parameters. The complexity reflects the myriad of biochemical processes. Real-time evaluation of field analytical data can be misleading because a full interpretation often requires combining the field analytical results with fixed-base laboratory results. Regardless, some simple observations and data interpretations in the field may provide insights about the monitoring system or early warnings about sample collection and handling problems.

Data collected from the designated upgradient monitoring well is the baseline from which other interpretations are made. Field analytical data will indicate that the upgradient environment is either oxidizing or reducing. The redox condition within the upgradient area of the aquifer may be natural or impacted by other contaminant source areas (see Section 5.2 for upgradient well selection). Regardless, the redox condition of the upgradient groundwater will influence the source area. Changes in field analytical results from the upgradient well to the source area well will be reflected in samples from monitoring wells further downgradient.

The general characteristics of the two redox environments are summarized in the following table.

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Aerobic/Oxidizing	Anaerobic/Reducing
• Measurable dissolved oxygen (>1 to 2 ppm) • Measurable nitrate • No measurable dissolved manganese • No measurable dissolved ferrous iron • Measurable dissolved sulfate • No measurable dissolved sulfide • No measurable dissolved methane • No measurable dissolved hydrogen	• No measurable dissolved oxygen (<1 ppm) • No measurable nitrate • Measurable dissolved manganese • Measurable dissolved ferrous iron • No measurable dissolved sulfate • Measurable dissolved sulfide • Measurable dissolved methane • Measurable dissolved hydrogen

Transitional environments between these two extremes may have intermediate characteristics and are actually quite common. Because reactions are mediated by biological systems, equilibrium (the basis for the figure in Attachment B) conditions within the aquifer should not be expected. For example, sulfate reduction environments may occur in close proximity to methanogenic environments, and this natural attenuation data may be difficult to interpret. Carefully collected and analyzed field measurements and sample collections for fixed-base laboratory analyses are designed to characterize the aquifer environment along the continuum between strongly aerobic and strongly anaerobic. Because the land surface environment is generally more oxidizing than any groundwater environment, sample handling at the point of collection and analysis is extremely important in preserving the chemical integrity of the groundwater sample.

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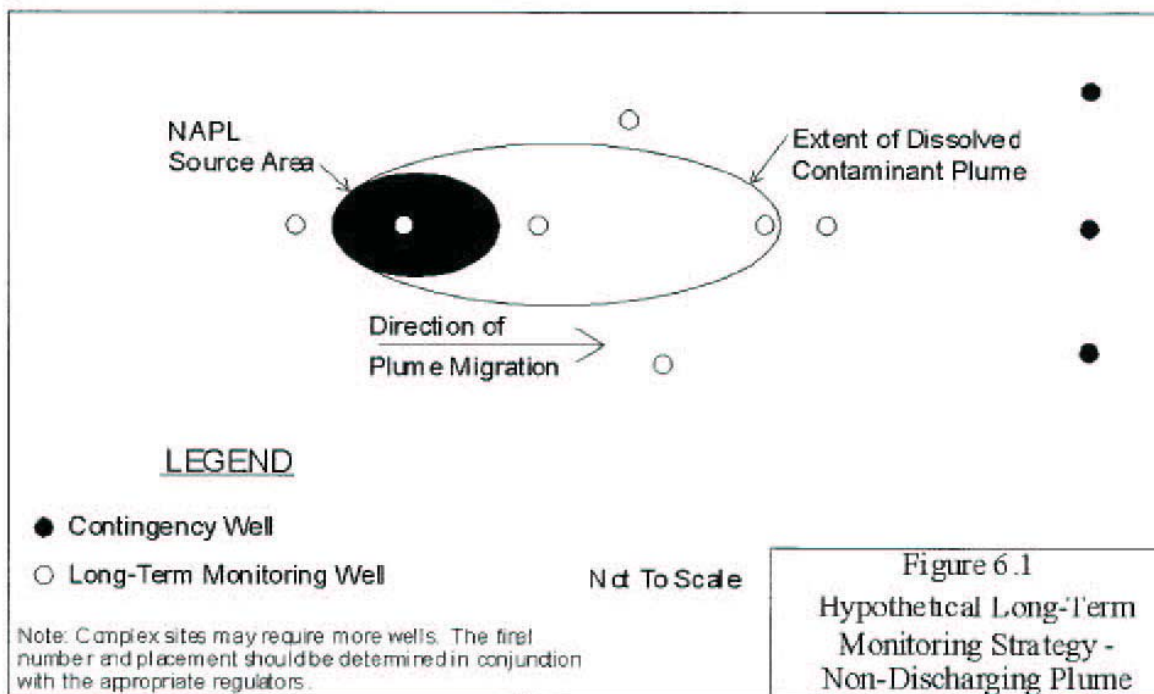
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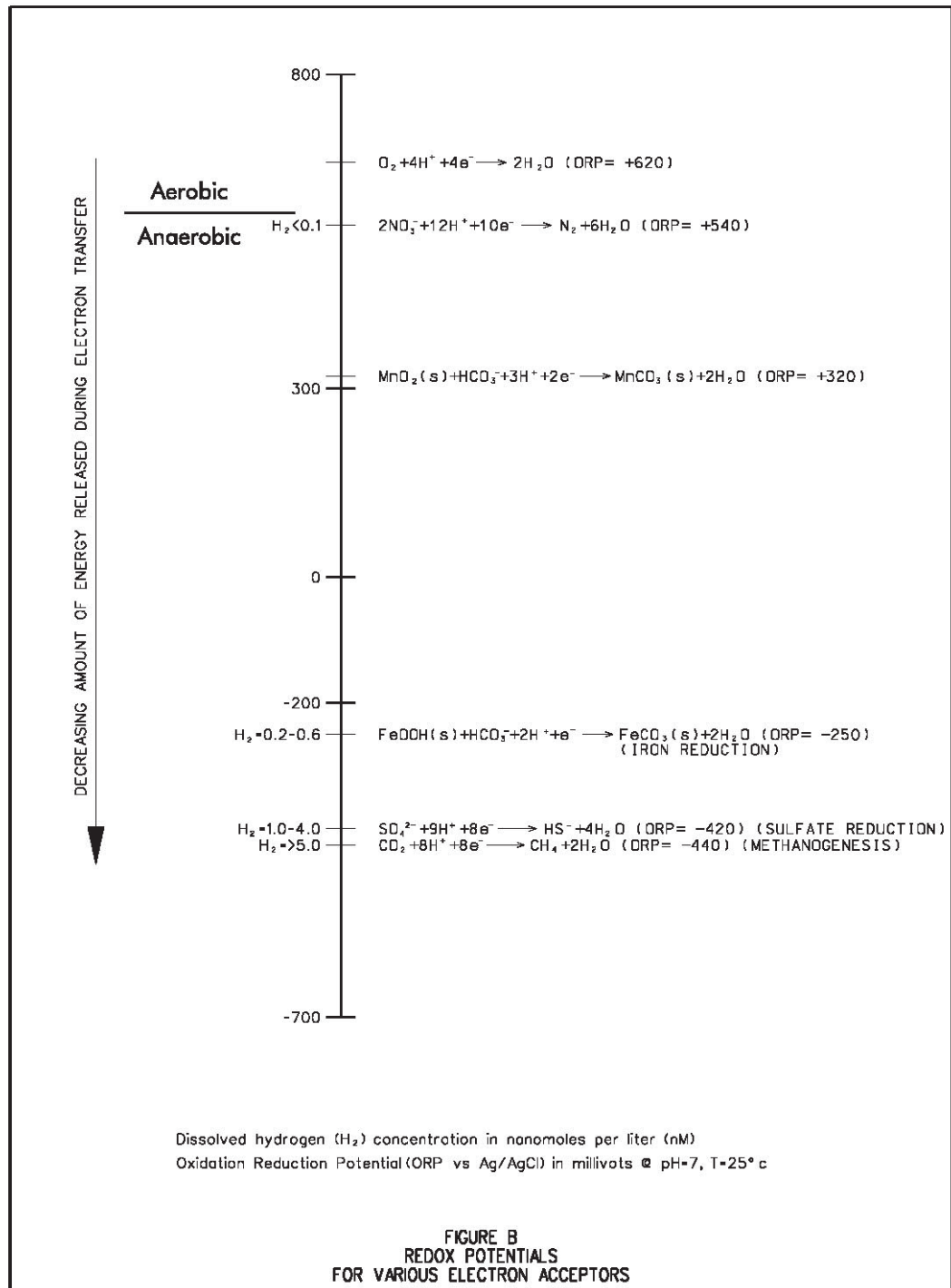
ATTACHMENT A
HYPOTHETICAL LONG-TERM MONITORING STRATEGY



Taken from: Department of the Navy, 1998, Technical Guidelines for Evaluating Monitored Natural Attenuation of Petroleum Hydrocarbons and Chlorinated Solvents in Ground Water at Naval and Marine Corps Facilities, Prepared by Todd Weidemeier and Francis Chappelle.

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ATTACHMENT B
REDOX POTENTIALS FOR VARIOUS ELECTRON ACCEPTORS



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ATTACHMENT C

NATURAL ATTENUATION PARAMETERS FOR CHLORINATED VOLATILE ORGANIC COMPOUND PLUMES SCREENING PROCESS SUMMARY FOR REDUCTIVE (ANAEROBIC) DECHLORINATION

Potential Electron Donors	Electron Acceptors:	Reduced Species:	Related Dechlorination Pathway:
Native total organic carbon (TOC)	Dissolved Oxygen	⇒ Carbon Dioxide (CO ₂)	~ DCE → VC → CO ₂
Anthropogenic carbon (e.g., leachate)	Manganese (Mn ⁴⁺)	⇒ Manganese (Mn ²⁺)	~ DCE → VC
Fuel hydrocarbons (e.g., BTEX)	Nitrate (NO ₃)	⇒ Nitrite (NO ₂)	~ DCE → VC
Lightly chlorinated solvents (DCE/VC)	Ferric Iron (Fe ³⁺)	⇒ Ferrous Iron (Fe ²⁺)	~ DCE → VC → CO ₂
	Sulfate (SO ₄)	⇒ Sulfide (S ²⁻ , HS ⁻ , H ₂ S)	~ TCE → DCE → VC → Ethene
	Carbon Dioxide (CO ₂)	⇒ Methane (CH ₄)	~ PCE → TCE → DCE → VC → Ethene

Geochemical Parameter List:

Parameter	Field or Lab	Rationale	Importance
Volatile organic compounds	L	Source products; daughter products; electron donors (e.g., benzene, toluene, ethylbenzene, and xylene; BTEX)	1
Dissolved oxygen	F	Primary electron acceptor (respiration); an/aerobic indicator	1
Nitrate (and nitrite), dissolved	F or L	Anaerobic electron acceptor (product of nitrate reduction)	1
Manganese, dissolved	F or L	Anaerobic electron acceptor	1
Ferrous Iron (Fe ²⁺)	F	Product of iron reduction	1
Sulfate [and sulfide (S ²⁻)]	F or L	Common anaerobic electron acceptor (product of sulfate reduction)	1
Sulfide (H ₂ S)	F	Common product of sulfate reduction	1
Methane, ethane, ethene	L	Product of methanogenesis; daughter products of reductive dechlorination	1
Chloride	L	Ultimate daughter product of reductive dechlorination	1
TOC - upgradient groundwater	L	Electron donor	1
ORP, pH, specific conductance, temperature, turbidity	F	General water quality determination	1
Carbon dioxide (CO ₂)	F	Anaerobic electron acceptor (methanogenesis); biotic respiration indicator	2
Alkalinity/DIC	F	Buffering capacity; biotic respiration indicator	2
Hydrogen, dissolved	L	Fingerprint for characterizing electron acceptor pathway - indicator of what redox is occurring	2
TOC - upgradient soil	L	Input to analytical NA models; quantifies soil-water distribution coefficient and retardation factor	2
Volatile fatty acids	L	Determination of anthropogenic carbon used as an electron donor	3

Importance: 1=Most important; 3=Least important (depending on DQOs, all may be recommended).
See Attachment E for details regarding analytical methods.

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ATTACHMENT D

NATURAL ATTENUATION PARAMETERS FOR PETROLEUM HYDROCARBON PLUMES SCREENING PROCESS SUMMARY FOR OXIDATIVE (AEROBIC) DEGRADATION

Parameter	Field or Lab	Rationale	Importance
Volatile organic compounds	L	Source products; daughter products; electron donors (BTEX)	1
Dissolved oxygen	F	Primary electron acceptor (respiration); an/aerobic indicator	1
Nitrate (and nitrite), dissolved	F or L	Anaerobic electron acceptor (and product of nitrate reduction)	1
Manganese, dissolved	F or L	Anaerobic electron acceptor	1
Ferrous Iron (Fe ²⁺)	F	Product of iron reduction	1
Sulfate [and Sulfide (S ⁻²)]	F or L	Common anaerobic electron acceptor (product of sulfate reduction)	1
Sulfide (H ₂ S)	F	Common product of sulfate reduction	1
TOC - upgradient groundwater	L	Electron donor	1
ORP, pH, specific conductance temperature, turbidity	F	General water quality determination	1
Dissolved methane (CH ₄)	L	Product of methanogenesis	1
Anions: chloride (Cl), nitrate (NO ₃), nitrite (NO ₂), phosphate (PO ₄), sulfate (SO ₄)	L		1
TOC - Upgradient soil	L	Input to analytical NA models; quantifies soil-water distribution coefficient and retardation factor	2
Biological oxygen demand (BOD)	L	Understanding of aquifer oxygen demand	3
Chemical oxygen demand (COD)	L	Understanding of aquifer oxygen demand	3

Importance: 1=Most important; 3=Least important (depending on DQOs, all may be recommended).

See Attachment E for details regarding analytical methods.

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ATTACHMENT E
GEOCHEMICAL SAMPLING PARAMETERS - METHODS, EQUIPMENT, VOLUMES, CONTAINERS,
PRESERVATION, HOLDING TIMES, AND DETECTION RANGES
PAGE 1 OF 4

Parameter	Method / Reference	Equipment / Method Chemistry	Sample Volume, Container, Preservation, & Holding Time	Test Kit Range (mg/L)	Test Kit Precision (mg/L)	Test Kit Estimated Detection Limit (mg/L)
Alkalinity	CHEMetrics K-9810, K-9815, K-9820 -ASTM D 1067-06 -EPA 310.1 -SM 2320B Fixed-base lab -EPA 310.1, 310.2 -SM 2320B	Titre® Titration Ampoules / Hydrochloric Acid, Phenolphthalein	Avoid agitation and analyze in the field to determine total alkalinity (no holding time). Follow test kit instructions. Filter if turbid (>10 NTU).	10-100 (K-9810) 50-500 (K-9815) 100-1000 (K-9820)	N/A	10 50 100-
Alkalinity		N/A	100 to 250 mL in glass or plastic container. Cool to ≤6 °C. Analyze within 14 days. Filter if turbid.	N/A	N/A	N/A
Alkalinity / Dissolved Inorganic Carbon	HACH AL-DT -HACH 8203 -SM 2320B / EPA 310.2 Fixed-base lab -SW-6010D / 6020A Fixed-base lab -EPA 405.1 -SM 5210B	Digital Titration / Sulfuric Acid, Phenolphthalein (P) and Total (M)	Avoid agitation and analyze samples in the field immediately after collection (no holding time) to determine carbonate, bicarbonate, and hydroxide ions. Follow test kit instructions. Filter if turbid as recommended by manufacturer.	10-4000 (mg/L CaCO3)	N/A	10
Arsenic		N/A	1 liter or 500 mL glass or polyethylene container, HNO3 to pH ≤ 2, analyze within 6 months.	N/A	N/A	N/A
Biochemical Oxygen Demand		N/A	1 liter glass or HDPE. Cool to ≤6 °C. Analyze within 48 hours.	N/A	N/A	N/A
Carbon Dioxide, dissolved	CHEMetrics K-1910, K-1920, K-1925 -ASTM D 513.82 -SM 4500-CO2-C-1997 Fixed-base lab -VOA water sample (VaporTech) Fixed-base lab	Titre® Titration / Sodium Hydroxide, Phenolphthalein Caustic titrant with pH Indicator TCD Detector FID Detector	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions.	10-100 (K-1910) 100-1000 (K-1920) 250-2500 (K-1925)	N/A	10 (K-1910) 100 (K-1920) 250 (K-1925)
Carbon Dioxide, dissolved			40 mL in VOA vial. 2 to 3 vials by (VaporTech).	N/A	N/A	N/A
Carbon Dioxide, dissolved			Field bubble-strip sampling required. Ship in glass septum vial (PACE only).	N/A	N/A	N/A
Carbon Dioxide, dissolved			Do not aerate or agitate and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions.	10-1000 (mg/L CaCO3)	N/A	10
Chemical Oxygen Demand	Fixed-base lab -EPA 410.4 / 5220D Fixed-base lab -EPA 300.0 / 9056 / 4500-CIE	N/A	125 mL HDPE. H2SO4 to pH <2.0. Cool to ≤6 °C. Analyze within 28 days.	N/A	N/A	N/A
Chloride (Cl)		N/A	100 to 250 mL in glass or plastic container. Cool to ≤6 °C. Analyze within 28 days.	N/A	N/A	N/A
Chlorine - Total (Cl)	HACH DPD Total Chlorine Swiftest -HACH 8167 -SM 4500-Cl G	DPD Method	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions.	0.02-2.00	N/A	1
Conductance, Specific	Field Meter -SW-9050 A / 2510 / 120.1	Direct Reading Meter	100 to 250 mL in glass or plastic container. Cool to ≤6 °C. Analyze within 28 days.	N/A	N/A	N/A
Ethane, dissolved	Fixed-base lab -VOA water sample, VaporTech -RSK 175	FID Detector	40 mL in VOA vial. 2 to 3 vials by (VaporTech). Analyze within 14 days.	N/A	N/A	N/A
Ethane, dissolved	Fixed-base lab -water sample, PACE-3810 modified / RSK 175	FID Detector, static headspace technique	3,20 mL headspace vials. If pH <2.0, analyze within 14 days; if unpreserved, analyze within 7 days; cool to 4°C.	N/A	N/A	N/A

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ATTACHMENT E

GEOCHEMICAL SAMPLING PARAMETERS - METHODS, EQUIPMENT, VOLUME, CONTAINER, PRESERVATION, HOLDING TIME, AND DETECTION RANGES PAGE 2 OF 4

Parameter	Method / Reference	Equipment / Method Chemistry	Sample Volume, Container, Preservation, & Holding Time	Test Kit Range (mg/L)	Test Kit Precision (mg/L)	Test Kit Estimated Detection Limit (mg/L)
Ethene, dissolved	Fixed-base lab -VOA water sample, Vaportech -RSK 175	FID Detector	40 mL in VOA vial. 2 to 3 vials by (Vaportech). Analyze within 14 days.	N/A	N/A	N/A
Ethene, dissolved	Fixed-base lab -water sample, PACE Analytical -3810 modified / RSK & 175	FID Detector	320 mL headspace vials, pH <2.0, analyze within 14 days, unpreserved, analyze within 7 days, cool to 4°C.	N/A	N/A	N/A
Fraction Organic Carbon (foc) - Soil	Fixed-base lab -Walkey-Black -SW-846 9060	N/A	200 gram glass jar. Cool to ≤6 °C. Analyze within 14 days.	N/A	N/A	N/A
Hydrogen, dissolved	Fixed-base lab -PACE or Vapor Tech gas stripping cell -RSK 175	RGD Detector	Field bubble-strip sampling required. Ship in glass septum vial.	N/A	N/A	N/A
Iron, ferrous (Fe ²⁺)	HACH DR-800 -HACH 8146	Colorimeter	Field. Follow test kit instructions. Analyze immediately at well head. Filter if turbid (>10 NTU) as recommended by the manufacturer.	0-3.00	±0.017 mg/L with a 2.00 mg/L Fe ²⁺ solution.	0.03
Iron, ferrous (Fe ²⁺)	HACH IR-18C	Color Disc	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions. Filter if turbid (>10 NTU) as recommended by the manufacturer.	0-10	N/A	0.2
Iron, total dissolved (Filtered)	Fixed-base lab -SW-846 6010B	N/A	500 mL in plastic container. Field filter. HNO ₃ to pH <2. Cool to ≤6 °C. Analyze within 6 months.	N/A	N/A	N/A
Manganese (Mn ²⁺)	HACH DR-800 -HACH 8034 -CFR 44(116) 34193	Colorimeter / Cold Periodate Oxidation	Field. Follow test kit instructions. Avoid agitation and analyze at well head. Filter if turbid as recommended by the manufacturer.	0-20.0	± 0.18 mg/L with a 10.00 mg/L Mn solution.	0.2
Manganese (Mn ²⁺)	HACH MN-5 / PAN -CFR 44(116) 34193	Color Disc / Cold Periodate Oxidation	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions. Filter if turbid as recommended by the manufacturer.	0-3	N/A	0.1
Manganese, total dissolved (Filtered)	Fixed-base lab -SW-846 6010B	N/A	500 mL in plastic container. Field filter. HNO ₃ to pH <2. Cool to ≤6 °C. Analyze within 6 months.	N/A	N/A	N/A
Methane, dissolved	Fixed-base lab -VOA water sample, Vaportech -RSK 175	FID/TCD Detector	40 mL in VOA vial. 2 to 3 vials by (Vaportech).	N/A	N/A	N/A
Methane, dissolved	Fixed-base lab -water sample, PACE -3810 modified / RSK 175	FID Detector	320 mL headspace vials, pH <2.0, analyze within 14 days, unpreserved, analyze within 7 days, cool to ≤6 °C.	N/A	N/A	N/A
Nitrate (NO ₃ ⁻)	Fixed-base lab -EPA 300	N/A	100-125 mL plastic or glass container. Cool to ≤6 °C. Analyze within 48 hours.	N/A	N/A	N/A
Nitrate (NO ₃ ⁻)	HACH DR-800 -HACH 8192 -Mod. EPA 353.2 Color Disc, Model NI-12	Color disk / Cadmium Reduction / Diazotization	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions. Pretreatment required if nitrite is present.	0-0.50	± 0.03 mg/L with a 0.25 mg/L of nitrate nitrogen (NO ₃ -N) solution.	0.01
Nitrite (NO ₂ ⁻)	Fixed-base lab -EPA 300 / 9056 / 4500 NO3-F	N/A	125 - 250 mL plastic container. Cool to ≤6 °C. Analyze within 48 hours. Filter if turbid as recommended by the manufacturer.	N/A	N/A	N/A

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ATTACHMENT E

GEOCHEMICAL SAMPLING PARAMETERS - METHODS, EQUIPMENT, VOLUME, CONTAINER, PRESERVATION, HOLDING TIME, AND DETECTION RANGES

PAGE 3 OF 4

Parameter	Method / Reference	Equipment / Method Chemistry	Sample Volume, Container, Preservation, & Holding Time	Test Kit Range (mg/L)	Test Kit Precision (mg/L)	Test Kit Estimated Detection Limit (mg/L)
Nitrite (NO ₂ ⁻)	HACH DR-800 -HACH 8507 -Mod. EPA 354.1 -Mod. SM 419 -CPR 44(85) 25595 Color Disc, Model NI-12	Color disk / Cadmium Reduction / Diazotization	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions. Filter if turbid as recommended by the manufacturer.	0-0.350	± 0.001 mg/L with a 0.250 mg/L nitrite nitrogen solution.	0.005
Nitrogen, dissolved	Fixed-base lab - Vaportech VOA water sample RSK-175	TCD Detector	Ship in VOA vial (Vaportech).	N/A	N/A	N/A
Nitrogen, Total Kjeldahl	Fixed-base lab -EPA 357.4 / 4500NH	N/A	500 mL plastic/glass container. Cool to ≤6 °C. H ₂ SO ₄ to pH ≤ 2. Analyze within 28 days.	N/A	N/A	N/A
Oxidation Reduction Potential	Field Meter - ASTM D-1498	Direct Reading Meter	Avoid aeration or agitation and analyze samples in the field immediately after collection (no holding time). Gently agitate probe using flow over or flow-through method.	N/A	N/A	N/A
Oxygen, dissolved	CHEMetrics (visual kits) K-7501, K-7512 -ASTM D 5543-15 -ASTM D 888-87	CHEMetrics ampoules / Rhodazine D and Indigo Carmine	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions.	0-1 (K-7501) 1-12 (K-7512)	N/A	0.025 1
Oxygen, dissolved	Fixed-base lab -VOA water sample, Vaportech -RSK 175	TCD Detector	40 mL in VOA vial. 2 to 3 vials by (Vaportech).	N/A	N/A	N/A
Oxygen, dissolved	Fixed-base lab -RSK 175	PAES GC /FID Detector	Contact PACE Analytical	N/A	N/A	N/A
Oxygen, dissolved	HACH OX-2P -HACH 8215 -SM 4500-O-G	Digital Titration / Azide Modification of Winkler Digital Titration Method Drop count titration	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions.	1-20	N/A	1
Oxygen, dissolved	HACH HRDO Method (AccuVac Kit)	HRDO -Color Disk	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions.	0-15	N/A	N/A
Oxygen, dissolved	Field Meter	Direct Reading Meter	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Used for well stabilization measurement parameter only.	N/A	N/A	N/A
pH	Field Meter -SW 9040B	Direct Reading Meter	Avoid agitation and analyze samples in the field immediately after collection (no holding time).	N/A	N/A	N/A
Phosphate (ortho)	Fixed-base lab -EPA 300.4500 PE	Ion Chromatography	125 - 250 mL plastic container. Cool to ≤6 °C. Analyze within 48 hours. Filter if turbid as recommended by the manufacturer.	N/A	N/A	N/A
Salinity	Field Meter	Direct Reading Meter	Analyze immediately.	N/A	N/A	N/A
Sulfate (SO ₄ ²⁻)	Fixed-base lab -300.0 / 9056 / ASTM D516-90 (02) / 9038	N/A	125 - 250 mL plastic container. Cool to ≤6 °C. If PO ₄ , NO ₂ , and NO ₃ are not of interest, analyze within 28 days, otherwise 48 hrs. Filter if turbid as recommended by the manufacturer.	N/A	N/A	N/A
Sulfate (SO ₄ ²⁻)	HACH DR-800 -HACH 8051 -EPA 375.4	Colorimeter-based Extinction / Turbimetric	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions. Filter if turbid as recommended by the manufacturer.	0-70	± 0.5 mg/L with a 50 mg/L sulfate solution.	4.9
Sulfide (Hydrogen Sulfide, H ₂ S)	HACH HS-C A	Color Chart / Effervescence of H ₂ S through sulfide reactive paper.	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions.	0-5	N/A	N/A

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GEOCHEMICAL SAMPLING PARAMETERS - METHODS, EQUIPMENT, VOLUME, CONTAINER, PRESERVATION, HOLDING TIME, AND DETECTION RANGES PAGE 4 OF 4

Parameter	Method / Reference	Equipment / Method Chemistry	Sample Volume, Container, Preservation, & Holding Time	Test Kit Range (mg/L)	Test Kit Precision (mg/L)	Test Kit Estimated Detection Limit (mg/L)
Sulfide (S ²⁻)	CHEMetrics K-9510, K-9510A-D -SM 4500-S ²⁻	CHEMetrics® Visual Kits, CHEMetrics Kits and VACUettes® Kits / Methylene Blue	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions.	0-1, 1-10; 0-30, 30-300 0-50, 60-600 0-120, 120-1200 0-1200, 1200-12000	N/A	0.05 5 10 20 200
Sulfide (S ²⁻)	Fixed-base lab -EPA 376.1/376.2	N/A	500 mL in plastic or glass container, no headspace. NaOH to pH >9. Cool to ≤6 °C. Avoid agitation and analyze within 7 days.	N/A	N/A	N/A
Sulfide (S ²⁻)	HACH DR-800 -HACH 8131 -SM 4500-S ²⁻ - D, 376.2	Colorimeter / Methylene Blue	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions. Pretreatment required for turbid samples as recommended by the manufacturer.	0-0.70	± 0.02 mg/L with a 0.73 mg/L sulfide solution.	0.01
Sulfide (S ²⁻)	HACH HS-WR -SM 4500-S ²⁻	Color Disc / Methylene Blue	Avoid agitation and analyze samples in the field immediately after collection (no holding time). Follow test kit instructions. Pretreatment required for turbid samples as recommended by the manufacturer.	0.25-11.25	N/A	0.1-2.5
Temperature	Field Meter / Thermometer - E170.1	Direct Reading Meter / Thermometer	Analyze immediately.	N/A	N/A	N/A
Total Organic Carbon (TOC)-Groundwater	Fixed-base lab -E 415.1	N/A	125 mL HDPE or glass, or 3-40mL VOA vials. H ₂ SO ₄ to pH < 2.0. Cool to ≤6 °C. Analyze within 28 days.	N/A	N/A	N/A
Turbidity	Field Meter - E 180.1	Direct Reading Meter	Analyze immediately.	N/A	N/A	N/A

ASTM = American Society for Testing and Materials
CFR = Code of Federal Regulations
EPA = United States Environmental Protection Agency
FID = Flame Ionization detector
GC = Gas chromatograph
H₂SO₄ = Sulfuric acid
HACH = Hach Chemical Company
HCl = Hydrochloric acid
HDPE = High density polyethylene
HNO₃ = Nitric acid
HRDO = High range dissolved oxygen
L = Liter
Mg = Milligram
mL = Milliliter
Mod. = Modified
N/A = Not applicable
NO₂ = Nitrite
NO₃ = Nitrate
NTU = Nephelometric Turbidity Units
RGD = Reduction gas detector
SM = Standard Methods for the Examination of Water and Wastewater
SO₄ = Sulfate
SW = Test Methods for Evaluating Solid Waste, Physical/Chemical Methods
VOA = Volatile organic analysis

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ATTACHMENT F

FIELD ANALYTICAL LOG SHEET, GEOCHEMICAL PARAMETERS (EXAMPLE) PAGE 1 OF 3

Note: Analyte, method, and/or equipment may be deleted from form if not being performed.



FIELD ANALYTICAL LOG SHEET GEOCHEMICAL PARAMETERS

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Project Site Name: _____		Sample ID No.: _____	
Project No.: _____		Sample Location: _____	
Sampled By: _____		Duplicate: ☐	
Field Analyst: _____		Blank: ☐	
Field Form Checked as per QA/QC Checklist (initials): _____			

SAMPLING DATA:								
Date: _____	Color (Visual)	pH (S U)	S.C. (mS/cm)	Temp. (°C)	Turbidity (NTU)	DO (mg/l)	Salinity (%)	ORP (Eh) (+/- mv)
Time: _____								
Method: _____								

SAMPLE COLLECTION/ANALYSIS INFORMATION:	
ORP (Eh) (+/- mv)	
Electrode Make & Model: _____	
Reference Electrode (circle one): Silver-Silver Chloride / Calomel / Hydrogen	

Dissolved Oxygen:			
Equipment: Chemetrics Test Kit			
Concentration: _____ ppm			
Range Used:	Range	Method	Concentration ppm
☐	0 to 1 ppm	K-7510	
☐	1 to 12 ppm	K-7512	
Analysis Time: _____			
Equipment: HACH Digital Titrator OX-DT			
Range Used:	Range	Sample Vol.	Cartridge Multiplier
☐	1-5 mg/L	200 ml	0.200 N 0.01
☐	2-10 mg/L	100 ml	0.200 N 0.02
Notes: _____			
Titration Count			
Multiplier			
Concentration			
_____ x 0.01 = _____ mg/L			
_____ x 0.02 = _____ mg/L			

Carbon Dioxide:			
Equipment: Chemetrics Test Kit			
Concentration: _____ ppm			
Range Used:	Range	Method	Concentration ppm
☐	10 to 100 ppm	K-1910	
☐	100 to 1000 ppm	K-1920	
☐	250 to 2500 ppm	K-1925	
Analysis Time: _____			
Equipment: HACH Digital Titrator CA-DT			
Range Used:	Range	Sample Vol.	Cartridge Multiplier
☐	10-50 mg/L	200 ml	0.3636 N 0.1
☐	20-100 mg/L	100 ml	0.3636 N 0.2
☐	100-400 mg/L	200 ml	3.636 N 1.0
☐	200-1000 mg/L	100 ml	3.636 N 2.0
Notes: _____			
Titration Count			
Multiplier			
Concentration			
_____ x 0.1 = _____ mg/L			
_____ x 0.2 = _____ mg/L			
_____ x 1.0 = _____ mg/L			
_____ x 2.0 = _____ mg/L			

Standard Additions: ☐ Titrant Molarity: _____ Digits Required: 1st.: _____ 2nd.: _____ 3rd.: _____			
Notes: _____			

Hydrogen, dissolved			
Equipment: Bubble strip sampling field method			
Start stripper at _____ (time)			
End stripper at _____ (time)			
Total stripper time _____			
Pump rate _____ milliliters/minute			

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FIELD ANALYTICAL LOG SHEET, GEOCHEMICAL PARAMETERS PAGE 2 OF 3

Note: Analyte, method, and/or equipment may be deleted from form if not being performed.



FIELD ANALYTICAL LOG SHEET GEOCHEMICAL PARAMETERS

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Project Site Name: _____				Sample ID No.: _____			
Project No.: _____				Sample Location: _____			
Sampled By: _____				Duplicate: ☐			
Field Analyst: _____				Blank: ☐			

Alkalinity:

Equipment: Chemetrics Test Kit

Range Used:	Range	Method	Concentration ppm
☐	10 to 100 ppm	K-9810	
☐	50 to 500 ppm	K-9815	
☐	100 to 1000 ppm	K-9820	

Concentration: _____ ppm

Analysis Time: _____

Filtered: ☐

Equipment: HACH Digital Titrator AL-DT

Range Used:	Range	Sample Vol.	Cartridge	Multiplier	Titration Count	Multiplier	Concentration
☐	10-40 mg/L	100 ml	0.1600 N	0.1	_____ & _____	x 0.1	= _____ mg/L
☐	40-160 mg/L	25 ml	0.1600 N	0.4	_____ & _____	x 0.4	= _____ mg/L
☐	100-400 mg/L	100 ml	1.600 N	1.0	_____ & _____	x 1.0	= _____ mg/L
☐	200-800 mg/L	50 ml	1.600 N	2.0	_____ & _____	x 2.0	= _____ mg/L
☐	500-2000 mg/L	20 ml	1.600 N	5.0	_____ & _____	x 5.0	= _____ mg/L
☐	1000-4000 mg/L	10 ml	1.600 N	10.0	_____ & _____	x 10.0	= _____ mg/L

Parameter:	Hydroxide	Carbonate	Bicarbonate
Relationship:			

Standard Additions: ☐ Titrant Molarity: _____ Digits Required: 1st.: _____ 2nd.: _____ 3rd.: _____

Notes: _____

Ferrous Iron (Fe²⁺):

Equipment: DR-850 DR-8 __ Range: 0 - 3.00 mg/L Concentration: _____ ppm

Program/Module: 500nm 33 Analysis Time: _____

Equipment: IR-18C Color Wheel Range: 0 - 10 mg/L

Notes: _____

Filtered: ☐

Hydrogen Sulfide (H₂S): Range: 0 - 5 mg/L

Equipment: HS-C Other: _____ Concentration: _____ ppm

Exceeded 5.0 mg/L range on color chart: ☐ Analysis Time: _____

Notes: _____

Sulfide (S²⁻):

Equipment: Chemetrics Test Kit Range: 0 - 10 mg/L Concentration: _____ ppm

Range Used:	Range	Method	Concentration ppm
☐	0 to 1 ppm	K-9510	
☐	1 to 10 ppm	K-9510	

Analysis Time: _____

Filtered: ☐

Equipment: DR-850 DR-8 __ Range: 0 - 0.70 mg/L

Program/Module: 610nm 93

Notes: _____

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FIELD ANALYTICAL LOG SHEET, GEOCHEMICAL PARAMETERS PAGE 3 OF 3

Note: Analyte, method, and/or equipment may be deleted from form if not being performed.



FIELD ANALYTICAL LOG SHEET GEOCHEMICAL PARAMETERS

Page of

Project Site Name: _____		Sample ID No.: _____	
Project No.: _____		Sample Location: _____	
Sampled By: _____		Duplicate: ☐	
Field Analyst: _____		Blank: ☐	

Sulfate (SO_4^{2-}):			
Equipment: DR-850	DR-8 <u> </u>	Range: 0 - 70 mg/L	Concentration: _____ ppm
Program/Module: _____	91		Analysis Time: _____
Standard Solution: ☐	Results: _____		Filtered: ☐
Standard Additions: ☐	Digits Required: 0.1ml: _____ 0.2ml: _____ 0.3ml: _____		
Notes: _____			

Nitrate (NO_3^--N):			
Equipment: DR-850	DR-8 <u> </u>	Range: 0 - 0.50 mg/L ⁽¹⁾	Concentration: _____ ppm
Program/Module: _____	55		Analysis Time: _____
Standard Solution: ☐	Results: _____	Nitrite Interference Treatment: ☐	Reagent Blank Correction: ☐
Standard Additions: ☐	Digits Required: 0.1ml: _____ 0.2ml: _____ 0.3ml: _____		
Alternate forms: NO_2 _____ NaNO_2 _____ mg/L			
Notes (1): If results are over limit use dilution method at step 3, 5ml sample 10ml DI result X3, range upto 1.5mg/L			
Notes: _____			

Nitrite (NO_2^--N):			
Equipment: DR-850	DR-8 <u> </u>	Range: 0 - 0.350 mg/L	Concentration: _____ ppm
Program/Module: _____	62		Analysis Time: _____
Standard Solution: ☐	Results: _____	Reagent Blank Correction: ☐	Filtered: ☐
Notes: _____			

Manganese (Mn^{2+}):			
Equipment: DR-850	DR-8 <u> </u>	Range: 0 - 20.0 mg/L	Concentration: _____ ppm
Program/Module: 525nm	41		Analysis Time: _____
Standard Solution: ☐	Results: _____	Digestion: ☐	Reagent Blank Correction: ☐
Standard Additions: ☐	Digits Required: 0.1ml: _____ 0.2ml: _____ 0.3ml: _____		
Notes: _____			

QA/QC Checklist:			
All data fields have been completed as necessary: ☐			
Correct measurement units are cited in the SAMPLING DATA block: ☐			
Values cited in the SAMPLING DATA block are consistent with the Groundwater Sample Log Sheet: ☐			
Multiplication is correct for each Multiplier table: ☐			
Final calculated concentration is within the appropriate Range Used block: ☐			
Alkalinity Relationship is determined appropriately as per manufacturer (HACH) instructions: ☐			
QA/QC sample (e.g., Std. Additions, etc.) frequency is appropriate as per the project planning documents: ☐			
Nitrite Interference treatment was used for Nitrate test if Nitrite was detected: ☐			
Title block on each page of form is initialized by person who performed this QA/QC Checklist: ☐			



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Effective Date

04/30//2020

Revision

6

Applicability

EGS Operating Unit

Prepared

Earth Sciences Department

Subject

NON-RADIOLOGICAL SAMPLE HANDLING

Approved

T. Johnston *T.E. Johnston* 04/30/2020

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1.0 PURPOSE

The purpose of this Standard Operating Procedure (SOP) is to provide information on sample preservation, packaging, and shipping procedures to be used in handling environmental samples submitted for chemical constituent, biological, or geotechnical analysis. Sample chain-of-custody procedures and other aspects of field documentation are addressed in SOP SA-6.3. Sample identification is addressed in SOP CT-04.

2.0 SCOPE

This procedure describes the appropriate containers to be used for samples depending on the analyses to be performed, and the steps necessary to preserve the samples when shipped off site for chemical analysis.

3.0 GLOSSARY

Hazardous Material – A substance or material which has been determined by the Secretary of Transportation to be capable of posing an unreasonable risk to health, safety, and property when transported in commerce, and which has been so designated. Under 49 CFR, the term includes hazardous substances, hazardous wastes, marine pollutants, and elevated temperature materials, as well as materials designated as hazardous under the provisions of §172.101 and §172.102 and materials that meet the defining criteria for hazard classes and divisions in Part 173. With slight modifications, International Air Transport Association (IATA) has adopted Department of Transportation (DOT) "hazardous materials" as IATA "Dangerous Goods."

Hazardous Waste – Any substance listed in 40 CFR, Subpart D (§261.30 et seq.), or otherwise characterized as ignitable, corrosive, reactive, or toxic (as defined by Toxicity Characteristic Leaching Procedure, TCLP, analysis) as specified under 40 CFR, Subpart C (§261.20 et seq.), that would be subject to manifest requirements specified in 40 CFR 262. Such substances are defined and regulated by EPA.

Marking – A descriptive name, identification number, instructions, cautions, weight, specification or UN marks, or combination thereof required on outer packaging of hazardous materials.

n.o.i – Not otherwise indicated (may be used interchangeably with n.o.s.).

n.o.s – Not otherwise specified.

Packaging – A receptacle and any other components or materials necessary for compliance with the minimum packaging requirements of 49 CFR 174, including containers (other than freight containers or overpacks), portable tanks, cargo tanks, tank cars, and multi-unit tank-car tanks to perform a containment function in conformance with the minimum packaging requirements of 49 CFR 173.24(a) & (b).

Placard – Color-coded, pictorial sign which depicts the hazard class symbol and name and which is placed on the side of a vehicle transporting certain hazardous materials.

Common Preservatives:

- Hydrochloric Acid - HCl
- Sulfuric Acid - H₂SO₄
- Nitric Acid - HNO₃
- Sodium Hydroxide - NaOH

Other Preservatives

- Zinc Acetate
- Sodium Thiosulfate - Na₂S₂O₃

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Normality (N) – Concentration of a solution expressed as equivalent per liter, an equivalent being the amount of a substance containing 1 gram-atom of replaceable hydrogen or its equivalent.

Reportable Quantity (RQ) – For the purposes of this SOP, means the quantity specified in column 3 of the Appendix to DOT 49 CFR §172.101 for any material identified in column 1 of the appendix. A spill greater than the amount specified must be reported to the National Response Center.

Sample – A sample is physical evidence collected from a facility or the environment, which is representative of conditions at the location and time of collection.

4.0 RESPONSIBILITIES

Project Manager (PM) – Responsible for all aspects of project implementation and direction. Responsible for ensuring that all field activities are conducted in accordance with approved project plan(s) and requirements and for providing the necessary resources in support of sample handling.

Field Operations Leader – Responsible for implementing field activities as detailed in approved project plans for the specific site. Directly responsible for the bottling, preservation, labeling, packaging, shipping, and custody of samples up to and including release to the shipper.

Field Support Personnel – Responsible for initiating the Chain-of-Custody Record (per SOP SA-6.3), implementing the packaging and shipping requirements, and maintaining custody of samples until they are relinquished to another custodian or to the shipper.

5.0 PROCEDURES

Sample identification, labeling, documentation, and chain-of-custody are addressed by SOP SA-6.3.

5.1 SAMPLE CONTAINERS

Different types of chemicals react differently with sample containers made of various materials. For example, trace metals adsorb more strongly to glass than to plastic, whereas many organic chemicals may dissolve various types of plastic containers. Proper sample containers, sample volumes, preservation methods such as chemical preservatives/cooling requirements, and holding times should be provided in the project-specific work plan. Some of these requirements may be laboratory-specific and should be arranged during the laboratory procurement phase of a project. In general, the sample container shall allow approximately 5-10 percent air space ("ullage") to allow for expansion/vaporization if the sample warms during transport. However, for collection of volatile organic compounds, head space shall be omitted.

The analytical laboratory will generally provide certified-clean containers for samples to be analyzed for chemical constituents. These containers shall be pre-cleaned to meet or exceed requirements of the latest U.S. Environmental Protection Agency (EPA) "Specifications and Guidance for Contaminant-Free Sample Containers." I-Chem 200 and 300 Series containers meet these requirements; other certifications may be available that also meet these requirements. Each case of 300 Series containers comes with a "Certificate of Analysis," and every container bears a similar bar-coded label for traceability. I-Chem Series 200 containers do not come with a Certificate of Analysis, but a certificate may be accessed upon request to the supplier. Shelby tubes or other sample containers are generally provided by the driller for samples requiring geotechnical analysis. Sufficient lead time shall be allowed for a delivery of sample container orders. Therefore, it is critical to use the correct container to maintain the integrity of the sample prior to analysis.

Upon being opened, the container must be used at once for collection/storage of a particular sample. Unused but opened containers are to be considered contaminated and must be discarded. Because of the

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potential for introduction of contamination, they cannot be reclosed and saved for later use. Likewise, any unused containers which appear contaminated upon receipt, or which are found to have loose caps or a missing Teflon liner (if required for the container), shall be discarded.

Sample containers can be contaminated during storage if nearby volatile substances infiltrate the surrounding air and migrate into the containers. This can occur even when containers are capped. Therefore, containers should not be stored on site but if on site storage is necessary, sample containers that have been stored on site for longer than 6 months should not be used.

NOTE: Whereas most air-borne contamination is likely to be associated with organic substances, inorganic substances such as hydrochloric acid (muriatic acid) can emit inorganic vapors that infiltrate sample containers.

5.2 SAMPLE PRESERVATION

Many water and soil samples are unstable and therefore require preservation to prevent changes in either the concentration or the physical condition of the constituent(s) requiring analysis. Although complete and irreversible preservation of samples is not possible, preservation does retard the chemical and biological changes that inevitably take place after the sample is collected. Preservation techniques are usually limited to pH control, chemical addition(s), and refrigeration/ freezing (certain biological samples only).

5.2.1 Overview

Reagents required for sample preservation will either be added to the sample containers by the laboratory prior to their shipment to the field or be added in the field (in a clean environment). Only high purity reagents shall be used for preservation and field preservation of samples is discouraged for reasons of potential cross-contamination of samples and worker safety. Most samples require cooling to <6°C and the project-specific planning documents should be consulted for detailed preservation and storage requirements.

The following subsections describe the procedures for preparing and adding chemical preservatives.

The FOL is responsible for ensuring that an accurate Chemical Inventory is created and maintained for all hazardous chemicals brought to the work site. Furthermore, the FOL must ensure that a corresponding Safety Data Sheet (SDS) is collected for every substance entered on the site Chemical Inventory, and that all persons using/handling/ disposing of these substances review the appropriate SDS for substances they will work with. The Chemical Inventory and the SDSs must be maintained at each work site in a location and manner where they are readily accessible to all personnel.

5.2.2 Preparation and Addition of Reagents

Addition of the following acids or bases may be specified for sample preservation; these reagents shall be analytical reagent (AR) grade or purer and shall be diluted to the required concentration with deionized water before field sampling commences. To avoid uncontrolled reactions, be sure to Add Acid to water (not vice versa). A dilutions guide is provided below.

Acid/Base	Dilution	Concentration	Estimated Amount Required for Preservation
Hydrochloric Acid (HCl)	1 part concentrated HCl: 1 part double-distilled, deionized water	6N	5 to 10 mL

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Acid/Base	Dilution	Concentration	Estimated Amount Required for Preservation
Sulfuric Acid (H ₂ SO ₄)	1 part concentrated H ₂ SO ₄ : 1 part double-distilled, deionized water	18N	2 to 5 mL
Nitric Acid (HNO ₃)	Undiluted concentrated HNO ₃	16N	2 to 5 mL
Sodium Hydroxide (NaOH)	400 grams solid NaOH dissolved in 870 mL double-distilled, deionized water; yields 1 liter of solution	10N	2 mL

NOTE: Pre diluted preservatives are available as ampules or in dropper bottles and are preferred as opposed to handling concentrated acids and bases.

The amounts required for preservation shown in the above table assumes proper preparation of the preservative and addition of the preservative to one liter of aqueous sample. This assumes that the sample is initially at pH 7, is poorly buffered, and does not contain particulate matter; as these conditions vary, more preservative may be required. Consequently, the final sample pH must be checked using narrow-range pH paper, as described in the generalized procedure detailed below:

- Pour off 5 to 10 mL of sample into a dedicated, clean container. Use some of this sample to check the initial sample pH using wide range (0-14) pH paper. Never dip the pH paper into the sample; always apply a drop of sample to the pH paper using a clean stirring rod or pipette.
- Add about one-half of the estimated preservative required to the original sample bottle. Cap and invert gently several times to mix. Check pH (as described above) using medium range pH paper (pH 0-6 or pH 7.5-14, as applicable).
- Cap sample bottle and seal securely.

Additional considerations are discussed below:

- To test if ascorbic acid must be used to remove oxidizing agents (e.g., chlorine, permanganate or persulfate) present in the sample before it can be properly preserved, place a drop of sample on KI-starch paper. A blue color indicates the need for ascorbic acid addition.

If required, add a few crystals of ascorbic acid to the sample and retest with the KI-starch paper. Repeat until a drop of sample produces no color on the KI-starch paper. Then add an additional 0.6 grams of ascorbic acid per each liter of sample volume.

Continue with proper base preservation of the sample as described above.

- Samples for sulfide analysis must be treated by the addition of 4 drops (0.2 mL) of 2N zinc acetate solution per 100 mL of sample.

The 2N zinc acetate solution is made by dissolving 220 grams of zinc acetate in 870 mL of double-distilled, deionized water to make 1 liter of solution.

The sample pH is then raised to 9 using the NaOH preservative.

- Sodium thiosulfate must be added to remove residual chlorine from a sample. To test the sample for residual chlorine, use a field test kit specially made for this purpose.

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If residual chlorine is present (e.g., in municipal water systems), add 0.08 grams of sodium thiosulfate per liter of sample to remove the residual chlorine.

Continue with proper acidification of the sample as described above.

For biological samples, 10% buffered formalin or isopropanol may also be required for preservation. Questions regarding preservation requirements should be resolved through communication with the laboratory before sampling begins.

5.3 **FIELD FILTRATION**

At times, field-filtration may be required to provide for the analysis of dissolved chemical constituents. Field-filtration must be performed prior to the preservation of samples as described above. General procedures for field filtration are described below:

- The sample shall be filtered through a non-metallic, 0.45-micron membrane filter, immediately after collection. The filtration system shall consist of dedicated filter canister, dedicated tubing, and a peristaltic pump with pressure or vacuum pumping squeeze action (since the sample is filtered by mechanical peristalsis, the sample travels only through the tubing) or hand pump.
- To perform filtration, thread the tubing through the peristaltic pump head. Attach the filter canister to the discharge end of the silicone tubing (note flow direction arrow); attach the unfiltered aqueous sample source to the intake end of the silicone tubing. Turn the peristaltic pump on and perform filtration. Run approximately 100 mL of sample through the filter and discard prior to sample collection of the filtered sample.
- Continue by preserving the filtrate, as applicable and generally described above.

5.4 **SAMPLE PACKAGING AND SHIPPING**

The Department of Transportation requires that persons who are required to transport or ship hazardous substances be specially trained. Only employees who have successfully completed the corporate "Shipping Hazardous Materials" training course are authorized to package and ship hazardous substances. These trained individuals are responsible for performing shipping duties in accordance with this training.

Samples collected for shipment from a site shall be classified as either environmental or hazardous material samples. Samples from drums containing materials other than Investigative Derived Waste (IDW) and samples obtained from waste piles or bulk storage tanks are generally shipped as hazardous materials. A distinction must be made between the two types of samples in order to:

- Determine appropriate procedures for transportation of samples (if there is any doubt, a sample shall be considered hazardous and shipped accordingly.)
- Protect the health and safety of transport and laboratory personnel receiving the samples (special precautions are used by the shipper and at laboratories when hazardous materials are received.)

Detailed procedures for packaging environmental samples are outlined in the remainder of this section.

5.4.1 **Environmental Samples**

Environmental samples are packaged as follows:

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- Place properly identified and labeled sample container, with lid securely fastened, in a plastic bag (e.g. Ziploc baggie), and seal the bag.
- Place sample in a cooler constructed of sturdy material which has been lined with a large, plastic bag (e.g. "garbage" bag). Drain plugs on coolers must be taped shut.
- Pack with enough cushioning materials such as bubble wrap (shoulders of bottles must be iced if required) to minimize the possibility of the container breaking. Glass bottles should be individually wrapped in bubble wrap.
- If cooling is required (see Attachments A and B), place ice around sample container shoulders, and on top of packing material (minimum of 8 pounds of ice for a medium-size cooler).
- Seal (i.e., tape or tie top in knot) large liner bag.
- The original (top, signed copy) of the COC form shall be placed inside a large Ziploc-type bag and taped inside the lid of the shipping cooler. If multiple coolers are sent but are included on one COC form, the COC form should be sent with the cooler containing the vials for VOC analysis. The COC form should then state how many coolers are included with that shipment.
- Close and seal outside of cooler as described in SOP SA-6.3. Signed and dated custody seals must be used.

Coolers must be marked as containing "Environmental Samples." The appropriate side of the container must be marked "This End Up" and arrows placed appropriately. No DOT marking or labeling is required; there are no DOT restrictions on mode of transportation.

6.0 REFERENCES

American Public Health Association, 2017 Standard Methods for the Examination of Water and Wastewater, 23rd Edition. APHA, Washington, D.C.

International Air Transport Association (latest issue). Dangerous Goods Regulations, Montreal, Quebec, Canada.

U.S. Department of Transportation (latest issue). Hazardous Materials Regulations, 49 CFR 171-177.

U.S. EPA, 1992. Specifications and Guidance for Contaminant-Free Sample Containers. EPA-540/R-93/051.

US EPA (U.S. Environmental Protection Agency), 2015. Test Methods for Evaluating Solid Waste (SW-846, Third Edition), Physical/Chemical Methods, as amended by Updates I, II, IIA, IIB, III, IIIA, IIIB, IVA, IVB, and VII, August.



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Effective Date

4/30/2020

Revision

7

Applicability

EGS Operating Unit

Prepared

Earth Sciences Department

Subject

FIELD DOCUMENTATION

Approved

T. Johnston *T.E. Johnston* 04/30/2020

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1.0 PURPOSE

The purpose of this Standard Operating Procedure (SOP) is to identify and designate the field data record forms, logs, and reports generally initiated and maintained for documenting EGS Operating Unit field activities.

2.0 SCOPE AND APPLICABILITY

Documents presented within this SOP (and related sampling SOPs) shall be used for all NUS Operating Unit field activities, as applicable. All personnel are encouraged to review the information contained herein to facilitate planning of the field sampling and documentation effort. The techniques described shall be followed whenever applicable, noting that site-specific conditions or project-specific plans may require modifications to methodology. Other or additional documents may be required by specific client contracts or project planning documents. Use of SOPs from certain federal or state environmental agencies may take precedence over these SOPs, depending on the client requirements or jurisdiction of the project.

3.0 GLOSSARY

None.

4.0 RESPONSIBILITIES AND PERSONNEL QUALIFICATIONS

Project Manager (PM) – Responsible for all aspects of project implementation and direction. Responsible for ensuring that all field activities are conducted in accordance with approved project plan(s) and requirements and for providing the necessary resources in support of all air monitoring and sampling applications. The PM is responsible for obtaining hardbound controlled-distribution logbooks (from the appropriate source), as needed. In addition, the PM is responsible for placing all field documentation used in site activities (i.e., records, field reports, sample data sheets, field notebooks, and the site logbook) in the project's central file upon the completion of field work.

Field Operations Leader (FOL) – Responsible for implementing field activities as detailed in approved project plans. The FOL is responsible for ensuring that the site logbook, notebooks, and all appropriate and current forms and field reports included in this SOP (and any additional forms required by the contract) are correctly used, accurately filled out, and completed in the required time frame.

Site Safety Officer (SSO) – The SSO (or a qualified designee) is responsible for providing the technical support necessary to implement the project Health and Safety Plan (HASP) (or equivalent).

Project Geologist/Sampler – The Project Geologist/Sampler is responsible for the proper documentation of field activities in accordance with this SOP and/or other project-specific documents.

General personnel qualifications for field documentation activities include the following:

- Occupational Safety and Health Administration (OSHA) 40-hour and applicable refresher training.
- Capability of performing field work under the expected physical and environmental (i.e., weather) conditions.
- Familiarity with appropriate procedures for documentation, handling, packaging, and shipping.

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5.0 PROCEDURES

5.1 SITE LOGBOOK

5.1.1 General

The site logbook is a hard-bound, paginated, controlled-distribution record book in which all major on-site activities are documented. At a minimum, record or reference the following activities/events (daily) in the site logbook:

- All field personnel present
- Arrival/departure times and names of site visitors
- Times and dates of health and safety training
- Arrival/departure times of equipment
- Times and dates of equipment calibration
- Start and/or completion of borehole, trench, monitoring well installation activities, etc.
- Daily on-site activities
- Sample pickup information
- Health and safety issues (level of protection, personal protective equipment [PPE], etc.)
- Weather conditions

Maintain a site logbook for each project and initiate it at the start of the first on-site activity (e.g., site visit or initial reconnaissance survey). Make entries every day that on-site activities take place involving NUS Operating Unit or subcontractor personnel. Upon completion of the fieldwork, provide the site logbook to the PM (or qualified designee) for inclusion in the project's central file.

Record the following information on the cover of each site logbook:

- Project name
- NUS Operating Unit project number
- Sequential book number
- Start date
- End date

Information recorded daily in the site logbook need not be duplicated in other field notebooks (see Section 5.2) but must summarize the contents of these other notebooks and refer to specific page locations in these notebooks for detailed information (where applicable). An example of a typical site logbook entry is shown in Attachment A. Adding a note to the inside front cover similar to the following is recommended: "If this book is found, please return to:" followed by the office address and project manager's name and phone number.

If measurements are made at any location, either record the measurements and equipment used in the site logbook or reference the field notebook or specific log sheet in which the measurements are recorded (see Attachment A).

Make all logbook, notebook, and log sheet entries in indelible ink (black pen is preferred). No erasures are permitted. If an incorrect entry is made, cross out the entry with a single strike mark, initial, and date it. At the completion of entries by any individual, the logbook pages used must be signed and dated by the person making the entries. Empty spaces generally should not be left on log sheets, forms, or notebooks. Instead, a line should be struck through them accompanied by the date and initials of the person adding the line. This signifies that the empty spaces are not inadvertent and marked them so no entries can be added at a later date. The site logbook must also be signed by the FOL at the end of each day.

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5.1.2 Photographs and Other Images

Permission to take photographs, make movies, or otherwise record visual representations of site features, sampling activities, or facility personnel on certain properties such as military installations is a temporary privilege bestowed upon a person (i.e., the photographer) by the property owner and the particular facility at which the photographs are taken. In the case of U.S. military installations, the U.S. Government and Installation Command bestow this privilege. For the remainder of this section the term “image” is intended to represent all visual forms of documentation of site features, activities, or personnel, such as film and digital still images and movies.

It may be illegal to record images without proper authorization and the photographer may be criminally liable for their actions. This is especially true on or near U.S. Government property. A particular image alone may not constitute an act of espionage or a national security threat; however, nefarious persons and organizations could conceivably combine information from multiple sources to create a national security threat. Therefore, the photographer must comply with any and all restrictions imposed on them by authorized facility personnel.

The instructions below apply to formal and casual photography sessions conducted, for example, during on-site meetings, walkovers, and other visits. Persons recording images should be aware that images recorded on or near military installations, and possibly at other installations, may be subject to censorship and clearance, e.g., through Public Affairs or Public Relations Offices.

Caution

Fieldwork-specific SOPs establish safety precautions for the fieldwork that they govern. When recording images, adhere to all safety precautions associated with fieldwork described in those SOPs. Pay attention to your surroundings. Avoid unsafe locations and positions. Avoid walking or turning your body while looking through the lens or viewfinder of a camera unless the path of movement has been cleared in advance and the movements will be made safely. Be especially alert to trip and fall hazards. Dress appropriately for the weather, topography, and the flora and fauna. Stay hydrated, and be aware of the effects of heat and cold stress on the body. Consult the SSO if in doubt about your safety or the safety of others while recording images.

5.1.2.1 Prior to and during photography sessions, follow these steps:

NOTE: In general, avoidance of specialty photographic equipment reduces the amount of work required of the photographer. The photographer is not required to record aperture, shutter speed, ISO rating, or other camera settings within automatic exposure ranges available on digital cameras; however use of special lenses (e.g., tilt-and-shift), filters (e.g., neutral density, warming, and polarizing filters), and other image enhancement equipment or techniques must be noted in the logbook/notebook. If possible, use of such equipment items or special techniques should be avoided because they can adversely affect the accuracy of recorded images.

- By communicating with the facility point of contact prior to recording images, obtain permission directly from, or on behalf of, the property owner and/or operator to record the images. In the case of U.S. Government military installations, the point of contact typically represents the Installation Command or a higher authority. The actual point of contact is frequently a Public Works or Environmental Department staff member.
- Request from the facility point of contact a written pass that allows permission to record images (hereafter referred to as a photography pass) but do not insist on obtaining one if the facility declines to issue such a pass.

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- If the facility does not issue a written photography pass, either do not record images or verify via written communication with the facility point of contact that image making will be permitted while on the Installation property.

NOTE: The intent of these permissions and communications is to ensure that the photographer clearly communicates his or her intentions to the facility and to avoid disputes regarding appropriateness of images, ownership of images or property depicted in the images, violation of privacy, etc. During communication with the facility point of contact, ensure that they understand the types of scenes, activities, equipment, features, processes, etc., to be recorded visually and under what conditions the images will be made.

- If the facility issues a written photography pass:
 - Prior to recording images, verify that the photography pass is valid for the time period during which images will be made.
 - Carry the written photography pass in your possession when recording images.
 - Be prepared to produce the photography pass as proof of permission to record images.
- Prior to recording images, if using a digital recording medium, enable automatic collection of Exchangeable Image File (EXIF) data. Doing so will provide automatic documentation of the camera settings (ISO rating, shutter speed, aperture, focal length, etc.) used during image capture. Most digital cameras will record this information by default; but the use of special filters or other attachments to the image capture device is not recorded because such equipment is not an integral part of the device.
- While recording images, enter the name of the photographer, date, time, site location, and site description in the logbook/notebook and update this information as necessary to document important changes. Include other pertinent information such as weather conditions, if such information will support interpretation of the images.
- When orientation of the camera is important to interpreting the contents of the images, record the camera angles and positions for each image. This may be done by drawing an arrow on a site figure that points in the same direction the camera was pointed when the image was recorded. Alternatively, some digital cameras may automatically record the geographical coordinates and orientation angle during image capture.
- Sequentially number images of a site or monitoring location to correspond to logbook/notebook entries. Digital still image media typically provide automatic numbering of images. If desired, the image numbers may be altered later (e.g., during download to a computer) but it may be desirable to retain the original file name in the EXIF data for traceability.

5.1.2.2 After recording images follow these steps:

- Verify that the name of the photographer, date, time, site location, site description, and other pertinent notes were correctly entered in the logbook/notebook.
- Compile the images and associated information that will be necessary to interpret them in a manner suitable for the associated project. For example, consecutively number still images and label them according to the logbook/notebook descriptions. If camera angle, camera position, or camera settings are important, record this important information with the images. For moving images, edit and compile the images a manner consistent with the intended use.
- Docket the images and associated film negatives or digital files to the central project file and/or compact disk. If EXIF data were captured and are needed for the project, ensure the EXIF data are transferred

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along with the image to the final storage location. This final storage location may be a project file or a report compiled to document site conditions.

- If multiple versions of image files are generated (e.g., high resolution and low resolution), arrange for proper storage and management of all versions of files for ready retrieval and safe storage.

5.2 FIELD NOTEBOOKS

Key field team personnel may maintain a separate dedicated field notebook to document the pertinent field activities conducted directly under their supervision. For example, on large projects with multiple investigative sites and varying operating conditions, the SSO may elect to maintain a separate field notebook. Where several drill rigs are in operation simultaneously, each Project Geologist/Sampler assigned to oversee a rig must maintain a field notebook.

5.3 FIELD FORMS

Field forms (see list in Section 6.0 of this SOP) can be found on the NUS Operating Unit intranet site under "Field Log Sheets." Forms may be altered or revised for project-specific needs, subject to client approval. Care must be taken to ensure that all essential information can be documented. Guidelines for completing these forms can be found in the related sampling SOPs.

5.3.1 Sample Collection, Labeling, Shipment, Request for Analysis, and Field Test Results

5.3.1.1 Sample Log Sheet

Sample log sheets are used to record specified types of data while sampling. The data recorded on these sheets are useful in describing the sample as well as pointing out any problems, difficulties, or irregularities encountered during sampling. Complete a sample log sheet for each sample obtained, including field quality assurance (QA) samples.

5.3.1.2 Sample Label

A typical sample label is illustrated in Attachment B. Complete the required information on the adhesive labels and apply them to every sample container. Obtain sample labels from the appropriate program/project source, request that they be electronically generated in house, or request them from the laboratory subcontractor. As much as practical, computer generated pre-printed sample labels containing as much specific sample information are preferred for efficiency and legibility.

5.3.1.3 Chain-of-Custody Record

Traditional, hardcopy chain-of-custody records are a multi-part forms that are initiated as samples are acquired and accompany a sample (or group of samples) as they are transferred from person to person. More recently computer generated forms are gaining popularity. Either format may be used. Hardcopy multi-part forms must be used as follows for any samples collected for chemical or geotechnical analysis whether the analyses are performed on-site or off-site:

- Retain one carbonless copy of the completed chain-of-custody form in the field.
- Send one copy to the PM (or qualified designee).
- Send the original (top, signed copy) to the laboratory with the associated samples. Place the original chain-of-custody copy inside a large Ziploc®-type bag taped inside the lid of the shipping cooler. If multiple coolers are sent, but are included on one chain-of-custody form, send the form with the cooler containing vials for volatile organic compounds (VOCs) analysis or the cooler with the air bill attached. Indicate on the air bill how many coolers are included with that shipment.

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If electronic forms are used, they must be used and managed in accordance with the associated instructions. An example of a chain-of-custody form is provided as Attachment C. After the samples are received at the laboratory, the sample cooler and contents are checked, and any problems are noted on the enclosed chain-of-custody form. Any discrepancies between the sample labels and chain-of-custody form and any other problems that are noted are resolved as soon as practical through communication between the laboratory point-of-contact and the PM or qualified designee. The chain-of-custody form is signed by the laboratory sample custodian, and copied. The laboratory will retain the copy, and the original becomes part of the samples' corresponding analytical data package.

5.3.1.4 Custody Seal

Attachment D is an example of a custody seal. The custody seal is an adhesive-backed label that is part of a chain-of-custody process and is used to prevent tampering with samples after they have been collected in the field and sealed in coolers for transport to the laboratory. Sign and date custody seals and affix them across the lid and body of each cooler (front and back of opposite sides) containing environmental samples (see SOP SA-6.1). Obtain custody seals from the laboratory (if available) or purchase them from a supplier.

NOTE: Some projects or contracts may require used of custody seals or wired tags on individual sample bottles.

5.3.1.5 Geochemical Parameters Log Sheets

Complete Field Analytical Log Sheets to record geochemical and/or natural attenuation field test results.

5.3.2 Hydrogeological and Geotechnical Forms

5.3.2.1 Groundwater Level Measurement Sheet

Complete a Groundwater Level Measurement Sheet for each round of water level measurements made at a site.

5.3.2.2 Data Sheet for Pumping Test

During the performance of a pumping test (or an in-situ hydraulic conductivity test), a large amount of data must be recorded, often within a short time period. Use a Pumping Test Data Sheet to facilitate this task by standardizing the data collection format for the pumping well and observation wells, and allowing the time interval for collection to be established in advance.

5.3.2.3 Packer Test Report Form

Complete a Packer Test Report Form for each well at which a packer test is conducted.

5.3.2.4 Boring Log

Complete a Summary Log of Boring, or Boring Log for each soil boring performed to document the materials encountered, operation and driving of casing, and locations/depths of samples collected. In addition, if volatile organic vapors are monitored on cores, samples, cuttings from the borehole, or breathing zone, (using a photoionization detector [PID] or flame ionization detector [FID]), enter these readings on the boring log at the appropriate depth. When they become available, enter the laboratory sample number, concentrations of key contaminants, or other pertinent information in the "Remarks" column. This feature allows direct comparison of contaminant concentrations with soil characteristics. Consult the site-specific sampling plan for level of detail to include, such as soil classification, color, or blow counts for Standard Penetration Test.

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5.3.2.5 Monitoring Well Construction Details Form

Complete a Monitoring Well Construction Details Form for every monitoring well, piezometer, or temporary well point installed. This form contains specific information on length and type of well riser pipe and screen, backfill, filter pack, annular seal and grout characteristics, and surface seal characteristics. This information is important in evaluating the performance of the monitoring well, particularly in areas where water levels show temporal variation or where there are multiple (immiscible) phases of contaminants. Depending on the type of monitoring well (in overburden or bedrock, stick-up or flush mount), different forms are used.

5.3.2.6 Test Pit Log

When a test pit or trench is constructed for investigative or sampling purposes, a Test Pit Log must be filled out by the responsible Project Geologist/Sampler.

5.3.2.7 Miscellaneous Monitoring Well Forms

Miscellaneous monitoring well forms that may be required on a project-specific basis include the Monitoring Well Materials Certificate of Conformance and Monitoring Well Development Record. Use a Monitoring Well Materials Certificate of Conformance to document all materials utilized during each monitoring well installation. Use a Monitoring Well Development Record to document all well development activities.

5.3.2.8 Miscellaneous Field Forms – Quality Assurance and Checklists

Miscellaneous field forms/checklists that may be required on a project-specific basis include the following:

- Container Sample and Inspection Sheet – use this form when a container (drum, tank, etc.) is sampled and/or inspected.
- QA Sample Log Sheet – use this form when a QA sample such as an equipment rinsate blank, source blank, etc. is collected.
- Field Task Modification Request (FTMR) – use this form to document deviations from the project planning document(s). The FOL is responsible for initiating the FTMRs. Maintain copies of all FTMRs with the on-site planning documents, and place originals in the final evidence file.
- Field Project Daily Activities Checklist and Field Project Pre-Mobilization Checklist – use these during both the planning and field effort to ensure that all necessary tasks are planned for and completed. These two forms are not requirements, but are useful tools for most field work.

5.3.3 Equipment Calibration and Maintenance Form

The calibration or standardization of monitoring, measuring, or test equipment is necessary to ensure the proper operation and response of the equipment, to document the accuracy, precision, or sensitivity of the measurements, and to determine if correction should be applied to the readings. Some items of equipment require frequent calibration, others infrequent. Some are calibrated by the manufacturer or the equipment rental vendor, others by the user.

Each instrument requiring calibration has its own Equipment Calibration Log, which documents that the manufacturer's instructions were followed for calibration of the equipment, including frequency and type of standard or calibration device. Maintain an Equipment Calibration Log for each electronic measuring device used in the field; make entries for each day the equipment is used or in accordance with manufacturer recommendations.

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5.4 **FIELD REPORTS**

The primary means of recording on-site activities is the site logbook. Other field notebooks may also be maintained. These logbooks and notebooks (and supporting forms) contain detailed information required for data interpretation or documentation but are not easily used for tracking and reporting of progress. Furthermore, the field logbook/notebooks remain on site for extended periods of time and are thus not accessible for timely review by project management. Other reports useful for tracking and reporting the progress of field activities are described below.

5.4.1 **Daily Activities Report**

To provide timely oversight of on-site contractors, complete and submit Daily Activities Reports (DARs) as described below.

5.4.1.1 Description

The DAR documents the activities and progress for each day's field work. Complete this report on a daily basis whenever there are drilling, test pitting, well construction, or other related activities occurring that involve subcontractor personnel. These sheets summarize the work performed and form the basis of payment to subcontractors. The DAR form can be found on the NUS Operating Unit intranet site.

5.4.1.2 Responsibilities

It is the responsibility of the rig geologist to complete the DAR and obtain the driller's signature acknowledging that the times and quantities of material entered are correct.

5.4.1.3 Submittal and Approval

At the end of the shift, the rig geologist must submit the DAR to the FOL for review and filing. The DAR is not a formal report and thus requires no further approval. The DARs are retained by the FOL for use in preparing the site logbook and in preparing weekly status reports for submission to the PM.

5.4.2 **Weekly Status Reports**

To facilitate timely review by project management, photocopies of logbook/notebook entries may be made for internal use.

In addition to those described herein, other summary reports may also be contractually required. Field forms can be found on the NUS Operating Unit intranet site.

6.0 **LISTING OF FIELD FORMS ON THE NUS OPERATING UNIT INTRANET SITE**

- Boring Log
- Container Sample and Inspection Sheet
- Chain of Custody
- Container Sample and Inspection
- Daily Activities Checklist
- Daily Activities Record
- Daily Activity Report
- Equipment Calibration Log
- Field Task Modification Request Form
- Field Analytical Log Sheet - Geochemical Parameters
- Groundwater Level Measurement Sheet

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- Groundwater Sample Log Sheet
- Hydraulic Conductivity Testing Data Sheet
- Infiltration Test Data Sheet
- Instream Velocity Measurements
- Instrument Cal. Log Sheet (Water Qual. Meter)
- Job Safety Analysis
- Monitoring Well Bedrock Construction (Open Hole)
- Monitoring Well Confining Layer Construction
- Monitoring Well Development Record
- Monitoring Well Inspection Sheet
- Overburden and Bedrock Monitoring Well Construction (Flush Mount)
- Overburden and Bedrock Monitoring Well Construction (Stick Up)
- Packer Test Report Form
- Photo Log
- Pre-Mob ChecklistPrivate Water Supply Sample Log
- Private Well Sampling Log
- Pumping Test Data Sheet
- QA Sample Log Sheet
- Soil & Sediment Sample Log Sheet
- Surface Water Sample Log Sheet
- Tailgate Safety Meeting Attendance Form
- Test Pit Log
- Tetra Tech Certificate of Conformance (Monitoring Well Materials)

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ATTACHMENT A

TYPICAL SITE LOGBOOK ENTRY (EXAMPLE)

START TIME: _____ DATE: _____

SITE LEADER: _____

PERSONNEL: _____

NUS

DRILLER

SITE VISITORS

WEATHER: Clear, 68°F, 2-5 mph wind from SE

ACTIVITIES:

1. Steam Jenney and fire hoses were set up.
2. Drilling activities at well ____ resumes. Rig geologist was _____. See Geologist's Notebook, No. 1, page 29-30, for details of drilling activity. Sample No. 123-21-S4 collected; see sample logbook, page 42. Drilling activities completed at 11:50 and a 4-inch stainless steel well installed. See Geologist's Notebook, No. 1, page 31, and well construction details for well _____.
3. Drilling rig No. 2 steam-cleaned at decontamination pit. Then set up at location of well _____.
4. Well ____ drilled. Rig geologist was _____. See Geologist's Notebook, No. 2, page ____ for details of drilling activities. Sample numbers 123-22-S1, 123-22-S2, and 123-22-S3 collected; see sample logbook, pages 43, 44, and 45.
5. Well ____ was developed. Seven 55-gallon drums were filled in the flushing stage. The well was then pumped using the pitcher pump for 1 hour. At the end of the hour, water pumped from well was "sand free."
6. EPA remedial project manager arrives on site at 14:25 hours.
7. Large dump truck arrives at 14:45 and is steam-cleaned. Backhoe and dump truck set up over test pit _____.
8. Test pit _____ dug with cuttings placed in dump truck. Rig geologist was _____. See Geologist's Notebook, No. 1, page 32, for details of test pit activities. Test pit subsequently filled. No samples taken for chemical analysis. Due to shallow groundwater table, filling in of test pit ____ resulted in a very soft and wet area. A mound was developed and the area roped off.
9. Express carrier picked up samples (see Sample Logbook, pages 42 through 45) at 17:50 hours. Site activities terminated at 18:22 hours. All personnel off site, gate locked.

Field Operations Leader

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ATTACHMENT B
SAMPLE LABEL (EXAMPLE)

	Tetra Tech NUS, Inc. 661 Andersen Drive Pittsburgh, 15220 (412)921-7090		Project: Site: Location:
	Sample No:		Matrix:
Date:	Time:	Preserve:	
Analysis:			
Sampled by:		Laboratory:	

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ATTACHMENT C
CHAIN-OF-CUSTODY FORM (EXAMPLE)

 TETRA TECH NUS, INC.		CHAIN OF CUSTODY		NUMBER 3413	PAGE ____ OF ____
PROJECT NO: _____		FACILITY: _____		LABORATORY NAME AND CONTACT: _____	
SAMPLERS (SIGNATURE) _____		PROJECT MANAGER _____		PHONE NUMBER _____	ADDRESS _____
STANDARD TAT ☐ 24 hr. ☐ 48 hr. ☐ 72 hr. ☐ 7 day ☐ 14 day		FIELD OPERATIONS LEADER _____		PHONE NUMBER _____	CITY, STATE _____
CARRIER/WAYBILL NUMBER _____		CONTAINER TYPE PLASTIC (P) or GLASS (G) _____		PREPRESERVATIVE USED _____	
TOP DEPTH (FT) _____		MATRIX (GW, SO, SW, SD, QC, ETC.) _____		COLLECTION METHOD _____	
BOTTOM DEPTH (FT) _____		NO. OF CONTAINERS _____		TYPE OF ANALYSIS _____	
LOCATION ID _____		SAMPLE ID _____		COMMENTS _____	
DATE _____	TIME _____	1. RELINQUISHED BY _____		DATE _____	TIME _____
2. RELINQUISHED BY _____		DATE _____	1. RECEIVED BY _____	DATE _____	TIME _____
3. RELINQUISHED BY _____		DATE _____	2. RECEIVED BY _____	DATE _____	TIME _____
COMMENTS _____		DATE _____	3. RECEIVED BY _____	DATE _____	TIME _____

4/02R
FORM NO. TNUS-001

PINK (FILE COPY)

YELLOW (FIELD COPY)

WHITE (ACCOMPANIES SAMPLE)

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ATTACHMENT D

CHAIN-OF-CUSTODY SEAL (EXAMPLE)

CUSTODY SEAL Date Signature 		CUSTODY SEAL Date Signature
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STANDARD OPERATING PROCEDURE

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Effective Date 04/30/2020	Revision 9
Applicability Tetra Tech, Inc., EGS Operating Unit	
Prepared Earth Sciences Department	

Subject
DECONTAMINATION OF FIELD EQUIPMENT AND
MANAGEMENT OF INVESTIGATION DERIVED
WASTE

Approved
T. Johnston
T.E. Johnston 04/30/2020

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1.0 PURPOSE

Decontamination is the process of removing, or neutralizing contaminants that have contacted and/or accumulated on equipment and/or personnel. The purpose of this Standard Operating Procedure (SOP) is to protect site personnel, the general public, and the environment while preserving or maintaining sample integrity. It is further intended through this procedure to describe the steps necessary for proper decontamination to prevent either direct or indirect contamination of drilling equipment, earth-moving equipment, chemical sampling equipment, and analytical equipment. It is also intended through the application of these principles to prevent cross-contamination of the environment.

2.0 SCOPE AND APPLICABILITY

This procedure applies to all equipment used to provide access to/acquire environmental samples that may have become contaminated through direct contact with contaminated media including air, water, and soil. This equipment includes drilling and heavy equipment and chemical sampling and field analytical equipment. Where technologically and economically feasible, single-use sealed disposable equipment will be employed to minimize the potential for cross-contamination. This SOP also provides general reference information on the control of contaminated materials.

Decontamination methods and equipment requirements may differ from one project to another. General equipment items are specified in Section 6.0, but project-specific equipment must be obtained to address the project-specific decontamination procedures presented in Section 7.0 and applicable subsections.

3.0 GLOSSARY

Alconox/Liquinox/Luminox – Phosphate-free laboratory-grade detergent products by Alconox, Inc.

Chemical/Hygienic/Sanitizing wipes – These types of devices offer a portable solution when there are no sources of water and soap to allow hands, face, or equipment to be cleaned of gross contamination. Some of these wipes and solution based detergents are chemical specific such as D-Lead or D-Wipes are used to remove lead and other heavy metals from hands and face.

Decontamination Solution – A solution selected/identified in the Health and Safety Plan (HASP), Project-Specific Quality Assurance Project Plan (QAPP), or other governing document. The solution is selected and employed as directed by the project chemist/health and safety professional as it has been shown to be most effective in the removal, neutralization, and/or stabilization of the contaminants in question without creating unwanted byproducts.

Deionized Water (DI) – This is water that has been treated by passing it through a deionizing resin column. This water may also pass through additional filtering media to attain various levels of analyte-free status. The DI water should meet College of American Pathologists (CAP) and National Committee for Clinical Laboratory Standards (NCCLS) specifications for reagent-grade Type I water.

Dry decontamination – This method physically removes contaminants from the surface through the use of High Efficiency Particulate Filter vacuums employed for lead and asbestos operations.

Potable Water – Tap water from any municipal water treatment system. Use of an untreated potable water supply is not an acceptable substitute for tap water.

Pressure Washing – Process employing a high-pressure pump and nozzle configuration to create a high-pressure spray of potable water. High-pressure spray is employed to remove solids from the surface of equipment. This process can be coupled with wash solutions or used solely as a pressure washer to physically remove surface artifacts.

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Solvent – A solvent is a liquid that is able to dissolve other substances, or solutes, (gases, liquids and/or solids) to form a solution. A solution is a homogeneous mixture of solvent and solute. Water is often referred to as a universal solvent because of its ability to dissolve many different types of substances. The category “solvent” includes but is not limited to pesticide-grade isopropanol; methanol; acetone; and hexane. These solvents may be required for particular projects or for a particular purpose (e.g., removal of concentrated waste) and **must be justified** in the project planning documents. **Methanol, acetone, hexane and isopropanol are highly flammable.** For example, it may be necessary to use hexane when analyzing for trace levels of pesticides, polychlorinated biphenyls (PCBs), or fuels. In addition, because many of these solvents are not miscible in water, the equipment should be air dried prior to use. Solvents should not be used on polyvinyl chloride (PVC) equipment or well construction materials. Acidic and caustic materials may also serve as solutes and when mixed with water may be used for the purpose of decontamination. A non-flammable option is available – see “Selection of Detergent” at the end of Section 6.0.

Steam Pressure Washing – A cleaning method employing a high-pressure spray of heated potable water (steam) to remove various organic/inorganic chemicals from equipment. This method is useful for removing organic substances such as pesticides, PCBs, oils, and fuels.

4.0 RESPONSIBILITIES AND PERSONNEL QUALIFICATIONS

Project Manager (PM) – Responsible for all aspects of project implementation and direction. Responsible for ensuring that all field activities are conducted in accordance with approved project plan(s) and requirements.

Decontamination Personnel – Individuals assigned the task of decontamination. It is the responsibility of these individuals to understand the use and application of the decontamination process, including:

- A working knowledge of the use of all equipment involved (pressure washers, steam cleaners, etc.).
- Hazards associated with the solutions used for this purpose.
- The monitoring of the decontamination process to ensure that it is working properly. This is accomplished through:
 - Visual evaluation (i.e., answering the question: “Does it appear to be clean?”),
 - Employing monitoring instruments and scanning of decontaminated items to determine if off-gassing indicates the presence of contaminants , and/or
 - Through the collection of equipment rinsate blanks to verify contaminant removal.
 - Focusing on tools that contact sampled media as well as areas on those tools that trap dirt and contaminated water.
 - Diligent use of PPE and self-decontamination to minimize the potential for cross contamination.
- Replace wash waters often to minimize the collection of contaminants.

Field Operations Leader (FOL) – Responsible for the implementation of project-specific planning documents. This includes on-site verification that all field activities are performed in compliance with approved SOPs or as otherwise dictated by the approved project plan(s). The FOL is also responsible for the completion and accuracy of all field documentation as well as assigning or evaluating the decontamination process. Additionally, through concurrence with the PM, the FOL may:

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- Initiate site investigation/remediation efforts in the area where the least amount of contamination is reported thereby minimizing the potential for cross contamination.
- Use disposable devices to collect sample aliquots to minimize cross contamination that might otherwise occur through the use of reusable items.

Site Safety and Health Officer (SSHO) – Exercises shared responsibility with the FOL concerning decontamination effectiveness. All equipment arriving on site (as part of the equipment inspection), leaving the site, and moving between locations is required to go through a decontamination process and evaluation. This is accomplished through visual examination and/or instrument screening, where contaminants can be detected to determine the effectiveness of the decontamination process. Improper or incomplete decontamination is sufficient to restrict equipment from entering the site, exiting the site, or moving to a new location on the site until the objectives are successfully completed.

General personnel qualifications for decontamination activities include the following:

- Occupational Safety and Health Administration (OSHA) 40-hour General Site Worker and/or applicable 8-Hour General Site Worker Refresher training.
- Capability of performing field work under the expected physical (heavy lifting and moving auger flights and drill rods) and environmental (i.e., weather – heat and/or cold extremes) conditions.
- Familiarity with appropriate decontamination procedures. Throughout this SOP, the procedures remove gross contamination from top down, outside in, then moving to what may be described as “polishing stages” where solvents including laboratory grade isopropanol and deionized water are used to remove residual levels of contaminants.

5.0 HEALTH AND SAFETY

In addition to the health and safety issues and reminders specified in subsections of this SOP, the following considerations and requirements must be observed as SOPs for field equipment decontamination activities:

- If any solvents or hazardous chemicals (e.g., isopropyl alcohol) are to be used in equipment decontamination activities, the FOL must first obtain the manufacturer’s/supplier’s Safety Data Sheet (SDS) and assure that it is reviewed by all users (prior to its use), added to the site Hazardous Chemical Inventory, and maintained on site as part of the project Hazard Communication Program.
- Review and observe specific health and safety requirements (e.g., personal protective equipment [PPE]) specified in the project-specific HASP (or equivalent) for this activity.

6.0 EQUIPMENT LIST

- Wood for decontamination pad construction, when applicable (see Section 7.1).
- Tools for constructing decontamination pad frame, when applicable (see Section 7.1).
- Plastic sheeting or comparable material to cover decontamination pad frame, when applicable (see Section 7.1).
- Wash/drying racks for auger flights and drill/drive rods, when applicable (see Section 7.2).
- PPE as specified in the project HASP (or equivalent).

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- Soap and water for physical washing and rinsing.
- Deionized water for final rinsing.
- Solvents (e.g., pesticide-grade isopropanol) for rinsing (see applicable portions of Section 7.2).
- Tubs, buckets, etc. for containerizing rinse water (see applicable portions of Section 7.2).
- Sample bottles for collecting equipment rinsate blanks (see Section 7.2).
- Calibrated photoionization detector (PID) or flame ionization detector (FID) to monitor decontaminated equipment for organic vapors generated through the existence of residual contamination or the presence of decontamination solvent remaining after the piece was rinsed.
- Aluminum foil or clear clean plastic bag for covering cleaned equipment (see applicable portions of Section 7.2).
- Paper towels or cloths for wiping.
- Brushes, scrapers, or other hand tools useful for removing solid materials from equipment.
- Clear plastic wrap for covering or wrapping large decontaminated equipment items (see Section 7.2.2).
- Drum-moving equipment for moving filled waste drums (optional) (see Section 7.3).
- Drum labels for waste drums (see Attachment A).

Selection of Detergent

A non-phosphate detergent shall be used when a detergent rinse is required. For environmental sampling field decontamination, three options are available and selection depends on the anticipated contaminants. *Alconox* powder is suitable for general purposes. *Liquinox* is a traditional anionic laboratory detergent, also useful for general cleaning and where there is a concern for the stability of the cleaned items in harsher cleaners. *Luminox* is a specialized detergent designed to remove oils and organic contamination, and is best used after an *Alconox* rinse in lieu of a solvent rinse when cleaning equipment for trace contaminant sampling.

If the equipment being cleaned is not highly soiled, and residual trace organics are the primary concern, *Luminox* may be used alone. *Liquinox* detergent is the better choice as a sole cleaning agent for field sampling equipment because it is a powerful emulsifier.

CAUTION

Do not use *Liquinox* when collecting samples for 1,4-dioxane analysis as it may contain trace levels of 1,4-dioxane.

Do not use *Luminox* when collecting samples for PFAS or 1,4-dioxane analysis as it has not been demonstrated to be PFAS- or 1,4-dioxane-free.

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7.0 PROCEDURES

The process of decontamination is accomplished through the removal of contaminants or neutralization of contaminants. To accomplish this activity, preparation is required including site preparation, equipment selection, and evaluation of the decontamination requirements and processes. Site contaminant types, concentrations, and media types are primary drivers in the selection of the types of decontamination and where it will be conducted. For purposes of this SOP, discussion is limited to decontamination procedures for general environmental investigations.

Decontamination processes will be performed at the location(s) specified in project-specific planning documents. Typical decontamination locations include the following:

- Temporary decontamination pads/facilities
- Sample locations
- Centralized decontamination pad/facilities
- Combination of some or all of the above

The following discussion includes general considerations for the decontamination process. Specific construction and implementation procedures will be as specified in the project-specific planning documents and/or may be as dictated by site-specific conditions as long as the intent of the requirements in the planning documents is met. This intent is to contain any residual fluids and solids generated through the decontamination process.

7.1 DECONTAMINATION PAD DESIGN/CONSTRUCTION CONSIDERATIONS

7.1.1 Temporary Decontamination Pads

Temporary decontamination pads may be constructed at satellite locations within the site area in support of temporary work areas. These structures are generally constructed to support the decontamination of heavy equipment such as drill rigs and earth-moving equipment but can be and are generally employed for smaller articles (auger flights, drill rods, split spoons, etc.).

The purpose of the decontamination pad is to contain wash waters and potentially contaminated soil generated during decontamination procedures. Therefore, construction of these pads should take into account the following considerations:

- Site location – Establish the decontamination site far enough from the work site to maximize decontamination effectiveness while minimizing travel distance. The location of the decontamination site shall be selected to provide, in the judgment of the FOL or his/her designee, compliance with as many of the following characteristics as practicable:
 - Away from pedestrian/vehicle thoroughfares.
 - Avoidance of areas where control/custody cannot be maintained.
 - Avoidance of areas where potential releases of contaminated media or decontamination fluids may be compounded through access to storm water transport systems, streams, or other potentially sensitive areas.
 - Avoidance of potentially contaminated areas.
 - Avoidance of areas too close to the ongoing operation, where cross-contamination may occur.

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The selected decontamination site should include the following, where possible:

- Areas where potable water and electricity are provided.

SAFETY REMINDER

Use only electrical power sources and conduits/wires, either hard-wired or portable, that are:

- Routed through a Ground Fault Circuit Interrupter (GFCI).
- In good condition (no physical damage), rated for the intended energy load, and designated for outdoor use.

- Areas where support activities such as removing decontamination waters, soil, and sediment are possible without entering an active exclusion zone.
- Areas that offer sufficient size to carry out the specific decontamination sequence.
- Decontamination pad (decontamination pad) – Construct the decontamination pad to meet the following characteristics:
 - Size – The size of the pad should be sufficient to accept the equipment to be decontaminated as well as permitting free movement around the equipment by the personnel conducting the decontamination. The size of the decontamination pad should permit these movements utilizing pressure/steam washer wands and hoses and minimizing splash due to work in close quarters.
 - Slope – An adequate slope should be constructed to permit the collection of water and potentially contaminated soil within a trough or sump constructed at one end. The collection point for wash waters should be of adequate distance that the decontamination workers do not have to walk through the wash waters while completing their tasks. Because the pad will be sloped, place a light coating of sand over the plastic to minimize potential slips and falls. See the text about liners below. When preparing the pad, remove sticks, roots, stones, and debris that could puncture the liner. The area should be lightly graded and possibly covered with sand to minimize the possibility of puncture.
 - Sidewalls – The sidewalls shall be at least 6 inches in height (or as high as possible if 6 inches is not achievable) to provide adequate containment for wash waters and soil. If splash represents a potential problem, splash guards should be constructed to control overspray. Sidewalls may be constructed of wood, inflatables, sand bags, etc. to permit containment. Splash guards are typically wood frames with plastic sheeting to control overspray.
 - Liner – Depending on the types of equipment and decontamination method to be used, the liner should be of sufficient thickness to provide a puncture-resistant barrier between the decontamination operation and the unprotected environment. Care should be taken to examine the surface area prior to placing the liner to remove sharp articles (sticks, stones, debris) that could puncture the liner. Liners are intended to form an impermeable barrier. The thickness may vary from a minimum recommended thickness of 10 mil to 30 mil. The desired thickness may be achieved through layering materials of lighter construction. It should be noted that various materials (rubber, polyethylene sheeting) become slippery when wet. To minimize this potential hazard associated with a sloped liner, a light coating of sand shall be applied to provide traction as necessary.
 - Wash/drying racks – Auger flights, drill/drive rods, and similar equipment require racks positioned off of the ground to permit these articles to be washed, drained, and dried while secured from falling during this process.

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For decontamination of direct-push technology (DPT) equipment, the pad may be as simple as a mortar tub containing buckets of soapy water for washing and an empty bucket to capture rinse waters. Decontamination may be conducted at the rear of the rig to permit rapid tool exchange, because dual tube or Macro-Core Samplers and drive rods require little space for decontamination

- Maintenance – Maintain the decontamination area by:
 - Periodically clearing the work area of standing water, soil, and debris, and coiling hoses to aid in eliminating slip, trip, and fall hazards. In addition, these articles will reduce potential backsplash and cross-contamination.
 - Regularly changing the decontamination fluids to ensure proper cleaning and prevent cross-contamination.
 - PPE – Periodically evaluate the condition of, and maintain the decontamination equipment, including regular cleaning of face shields and safety glasses. This is critical to ensuring the safety of decontamination personnel and the integrity of the decontamination process, and it will ensure that equipment is functioning properly.

7.1.2 Decontamination Activities at Drill Rigs/DPT Units

7.1.2.1 During subsurface sampling activities including drilling and DPT activities, conduct decontamination of drive rods, Macro Core Samplers, split spoons, etc. at an area adjacent to the operation. Decontamination is generally accomplished using a soap/water wash and rinse utilizing buckets and brushes. This area requires sufficient preparation to accomplish the decontamination objectives.

7.1.2.2 Place buckets within mortar tubs or similar secondary containment tubs to prevent splash and spills from reaching unprotected environmental media.

7.1.2.3 Use drying racks where possible to permit parts to dry and be evaluated prior to use/reuse. Methodology regarding this activity is provided in Section 7.2.

7.1.3 Decontamination Activities at Remote Sample Locations

7.1.3.1 When sampling at remote locations, evacuate sampling equipment such as trowels and pumps/tubing of potentially contaminated media to the extent possible.

7.1.3.2 Wrap this equipment in plastic for transport to the temporary/centralized decontamination location for final cleaning and disposition.

7.1.3.3 Flush and clean single-use equipment such as disposable trowels, tubing, and surgeon's gloves to allow disposal of this equipment after visible soil and water remnants have been removed.

7.2 EQUIPMENT DECONTAMINATION PROCEDURES

The following represents procedures to be employed for the decontamination of equipment that may have contacted and/or accumulated contamination through site investigation activities.

SAFETY REMINDER

Remember when handling powered equipment to disconnect the power source and render the equipment to a zero energy state (both potential and kinetic) before opening valves, disconnecting lines, etc.

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7.2.1 Monitoring Well Sampling Equipment

7.2.1.1 Groundwater sampling equipment – This includes pumps inserted into monitoring wells such as bladder pumps, Whale pumps, and Redi-Flo pumps and reusable bailers, etc.

1. Evacuate to the extent possible, any purge water within the pump/bailer.
2. Scrub the external components using soap and water and/or steam clean* the outside of the pump/bailer and, if applicable, the pump tubing.
3. Insert the pump and tubing/bailer into a clean container of soapy water. Pump/run a sufficient amount of soapy water through the pump/bailer to flush out any residual well water. After the pump is flushed, circulate soapy water through the pump to ensure that the internal components are thoroughly flushed.
4. Remove the pump and tubing/bailer from the container.
5. Rinse external pump components using tap water.
6. Insert the pump and tubing/bailer into a clean container of tap water. Pump/run a sufficient amount of tap water through the pump/bailer to evacuate all of the soapy water (until clear).

CAUTION

Do not rinse polyethylene (PE), PVC, and associated tubing with solvents – These are considered single use items and can be removed and disposed of after use. Use the procedures defined in the project-specific planning documents. If they are not defined, contact the FOL for guidance. The solvent rinse described in Step 7 should be omitted and tubing or plastic components replaced.

7. If groundwater contains or is suspected to contain oil, grease, PAHs, PCBs, or other hard to remove organic materials, rinse the equipment to be cleaned with pesticide-grade isopropanol. If groundwater contains these contaminants employ single use sampling apparatus that can be wash, rinsed, and disposed of after use.
8. Pass deionized water through the hose to flush out the tap water and solvent residue as applicable.
9. Drain residual deionized water to the extent possible.
10. Allow components of the equipment to air dry.
11. For bladder pumps, disassemble the pump and wash the internal components with soap and water, then rinse with tap water, isopropanol, and deionized water and allow to dry. After the parts are dry, conduct a visual inspection and a monitoring instrument scan to ensure that potential contaminants and all decontamination solvent have been removed. Collect an equipment rinsate blank in accordance with the project-specific planning documents to ensure that the decontamination process is functioning as intended. The typical frequency of collection for equipment rinsate blanks is 1 per 20 field samples. In addition, wipe samples or field tests such as ultraviolet light may be used.
12. Wrap pump/bailer in aluminum foil or a clear clean plastic bag for storage.

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7.2.1.2 Electronic Water Level Indicators/Sounders/Tapes

During water level measurements, rinsing the extracted tape and probe with deionized water and wiping the surface of the extracted tape between locations is acceptable. However, periodic full decontamination should be conducted as follows:

1. Wash with soap and water
2. Rinse with tap water
3. Rinse with deionized water

In situations where oil, grease, free product, other hard to remove materials are encountered, probes and exposed tapes should be washed in hot soapy water and wiped with a rag or towel. If probes or tapes cannot be satisfactorily decontaminated (they are still stained, discolored, etc.), they should be removed from service.

7.2.1.3 Miscellaneous Equipment

Miscellaneous equipment including analytical equipment (water quality testing equipment) shall be cleaned per manufacturers' instructions. This generally includes wiping the sensor housing and rinsing with tap and deionized water.

Coolers/shipping containers employed to ship samples are received from the laboratory in a variety of conditions including marginal to extremely poor. Coolers shall be evaluated prior to use for the following:

- Structural integrity – Coolers missing handles or having breaks in the outer housing should be removed and not used. Notify the laboratory that the risk of shipping samples in the cooler(s) provided is too great and request a replacement unit.
- Cleanliness – As per protocol, only volatile organic samples are accompanied by a trip blank. If a cooler's cleanliness is in question (visibly dirty/stained) or if there are noticeable odors, the cooler should be decontaminated prior to use as follows:
 1. Wash with soap and water
 2. Rinse with tap water
 3. Dry

If these measures fail to clean the cooler to an acceptable level, remove the unit from use as a shipping container and ask the cooler provider (e.g., the analytical laboratory) to provide a replacement unit.

7.2.2 **Downhole Drilling Equipment**

This includes any portion of the drill rig that is over the borehole, including auger flights, drill stems, rods, and associated tooling that would extend over the borehole. The following procedure is to be employed prior to initiating the drilling/sampling activity, then between locations:

1. Remove loose soil using shovels, scrapers, etc.
2. Through a combination of scrubbing using soap and water and/or steam cleaning or pressure washing, remove visible dirt/soil from the equipment being decontaminated.

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CAUTION

In Step 3, do not rinse PE, PVC, and associated tubing with solvents. The appropriate procedures should be defined within the project-specific planning documents. If they are not defined, contact the FOL for guidance. The solvent rinse described in Step 4 may be omitted if groundwater does not contain oil, grease, PAHs, PCBs, or other hard to remove organic materials.

3. Rinse the equipment with tap water, where applicable (steam cleaning and pressure washing incorporate rinsing as part of the process).
4. If the equipment has directly or indirectly contacted contaminated sample media and is known or suspected of being contaminated with oil, grease, PAHs, PCBs, or other hard to remove organic materials, rinse equipment with pesticide-grade isopropanol

NOTE: *Steam cleaning does the best job removing contaminants especially the more volatile components. Not only is the water pressure sufficient to remove caked on soils, the heat associated with the steam is very effective in removing contaminants (even those more stubborn ones such as PAHs and PCBs). Steam cleaning units, however, are more dangerous to use due to pressurized steam and water. Care must be taken to avoid thermal burns by ensuring all guards are in place and gloves are employed. When using steam, the items that have been washed dry relatively quickly because heated water and steam are applied.*

5. To the extent possible, allow components to air dry.
6. If the decontaminated equipment is to be used immediately after decontamination, screen it with a calibrated photoionization detector (PID)/flame ionization detector (FID) to ensure that all contaminants and possible decontamination solvents (if they were used) have been adequately removed.
7. Wrap or cover equipment in clear plastic until it is time to be used.

SAFETY REMINDER

Even when equipment is disconnected from power sources, dangers such as the following may persist:

Falls - An auger flight standing on its end may fall and injure someone. Secure all loose articles to prevent heavy articles from falling onto people or equipment.

Burns - Steam cleaner water is heated to more than 212 °F and exhibits thermal energy that can cause burns. Prevent contact of skin with hot water or surfaces.

High water pressure - Pressure washer discharge can have 2,000 to 4,000 psi of water pressure. Water under this amount of pressure can rupture skin and other human tissues. Water at 4,000 psi exiting a 0° tip can be dangerous because of its relatively high cutting power. The exit velocity and cutting power of the water are reduced when exiting a 15-25° fan tip, but damage to soft tissues is still possible.

In general, follow the rules below to avoid injury, equipment damage, or incomplete decontamination:

1. Read the operating manual and follow the manufacturers' recommended safety practices before operating pressure washers and steam cleaners.

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2. Never point the pressure washer or steam cleaner at another person or use to clean your boots or other parts of your body. Water lacerations and burns may appear to be minor at first but can be life threatening. Do not attempt to hold small parts in your hand while washing them with high- temperature or high-pressure water.
3. Always wear PPE as specified in the HASP (or equivalent) such as:
 - Hard hat, safety glasses, splash shield, impermeable apron or splash suit, and hearing protection. Remember that excessive noise is a hazard when operating gas-powered engines and electrically driven pressure washers. PPE will be identified in your project specific planning documents.
4. Inspect each device before use. An inspection checklist will be provided in the project-specific planning documents. If it is a rented device, safety measures are typically provided by the vendor. In all cases, if you are not familiar with the operation of a pressure washer/steam cleaner, do not operate it until you obtain and thoroughly review operating instructions and recommended safety practices.
5. Do not modify equipment unless the manufacturer has approved the modifications.

7.2.3 Soil/Sediment Sampling Equipment

This section applies to soil sampling equipment including but not limited to hand augers, stainless steel trowels/spoons, bowls, dredges, scoops, split spoons, Macro Core samplers, etc.

1. Remove all loose soil from the equipment through manual means.
2. Through a combination of scrubbing using soap and water and/or steam cleaning or pressure washing, remove visible dirt/soil from the equipment.
3. Rinse the equipment with tap water.
4. If the equipment is contaminated or suspected to be contaminated with oil, grease, PAHs, PCBs, or other hard to remove organic materials, rinse the equipment with pesticide-grade isopropanol followed by steam cleaning. Where cleanliness is considered suspect, dispose and replace the suspect item.
5. Rinse the equipment with deionized water.
6. To the extent possible, allow components to air dry.
7. If the equipment is to be used immediately after decontamination, screen it with a calibrated PID/FID to ensure that all solvents (if they were used) and trace contaminants have been adequately removed.
8. After the equipment has dried, wrap it in aluminum foil for storage until use.

Dredges employed in sediment sampling are typically decontaminated as follows:

- Remove the sediment sample from the sampling device. If sufficient surface water exists remove visible sediments by dunking within the water from where the sediment sample was obtained. Because of this always work from downstream to upstream
- Extract the dredge and wash it in soap and water per the project-specific planning documents.

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CAUTION

When handling dredges, the primary safety concern is trapping fingers or extremities in the larger dredge samplers within the jaws or pinch points of the mechanical jaws. Keep hands, fingers, and extremities away from these pinch and compression points. Either handle the device by the rope or preferably lock the jaws in place to control the potential for closing during maintenance and/or cleaning.

NOTE: *In all cases when cleaning stainless steel sampling equipment minimize if not eliminate the use of acids as this will leach certain metals (e.g., chromium, nickel, iron, and antimony) and may result in sample contamination. Use soap and water wash and rinse and, where necessary, steam clean to remove stubborn contaminants.*

7.3 CONTACT WASTE/MATERIALS

During the course of field investigations, disposable/single-use equipment becomes contaminated. These items include tubing, trowels, PPE (gloves, overboots, splash suits, etc.), and broken sample containers.

With the exception of the broken glass, single-use articles should be cleaned (washed and rinsed) of visible materials and disposed as normal refuse. The exception to this rule is that extremely soiled materials that cannot be cleaned shall be containerized for disposal in accordance with project planning documents.

NOTE

Requirements for waste storage may differ from one facility to the next. Facility-specific directions for waste storage areas will be provided in project-specific documents, or separate direction will be provided by the Project Manager.

1. Assume that all investigation-derived waste (IDW) generated from decontamination activities contains the hazardous chemicals associated with the site unless there are analytical or other data to the contrary. Waste solution volumes could vary from a few gallons to several hundred gallons in cases where large equipment required cleaning.
2. Where possible, use filtering systems to extend the use of water within a closed system wash unit to recycle water and to reduce possible waste amounts.

NOTE

Containerized waste rinse solutions are best stored in 55-gallon drums (or equivalent containers); frac tanks, or tankers that can be sealed until ultimate disposal at an approved facility.

3. Properly label waste storage containers (see Attachment A).
4. Ensure that the IDW storage area is configured to meet the following specifications to permit access to the containers and to conduct spill/leak monitoring, sampling, and extraction when the disposal route is determined:
 - Enclose areas accessible by the general public using construction fencing and signs.
 - Store materials in 55-gallon drums on pallets with four (or fewer) drums per pallet.
 - Maintain the retaining bolt and label on the outside of storage containers where readily visible.

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- Provide at least 4 feet of room between each row of pallets to allow access to containers for sampling, drum removal, and spill response.
- As directed in project-specific planning documents, maintain an IDW Inventory List and provide the list to the site Point of Contact at the termination of each shift.
- Maintain spill response equipment at the IDW storage area in case it is required for immediate access.
- Where possible, use equipment for moving containers. Where not possible, obtain help to manipulate containers.

CAUTION

Each container of water or soil can weigh up to 750 pounds, constituting a safety hazard. To minimize hazards such as this as well as spillage and potential expansion due to freezing, limit the filling of drums and temporary containers to 80 percent of capacity. These containers should be managed with appropriate manual equipment like drum dollies or, preferably, powered equipment like drum grapple on forklift truck. See the governing health and safety documents for safe management of IDW.

Airborne Contaminant Exposure

To minimize potential airborne contaminant exposure, close and seal all containers transported inside the sampling support vehicle. Periodically, decontaminate temporary portable containers to minimize off-gassing inside the vehicle. When working at a well or storage container always work from an upwind position.

7.4 DECONTAMINATION EVALUATION

Upon decontamination of equipment, determine the effectiveness of the decontamination process in the following manner:

- Visual evaluation – Visually inspect equipment to verify the removal of particulate matter to ensure that the washing/rinsing process is working as intended. Pay particular attention to points where soils and sediments may become trapped, such as pipe threads, and other small/narrow openings. It is often necessary to combine multiple types of decontamination methods to ensure the decontamination is adequate and will ultimately yield a clean equipment rinsate blank.
- Instrument Screening – Use a properly calibrated PID/FID to evaluate the presence of site contaminants and solvents used in the cleaning process. The air intake of the instrument shall be passed over the article to be evaluated. Avoid placing the instrument probe into residual waters. A PID/FID reading greater than the daily established background level requires a repeat of the decontamination process, followed by rescreening with the PID/FID. This sequence must be repeated until no instrument readings greater than the daily established background level are observed. It should be noted that the instrument scan is only viable if the contaminants are detectable within the instrument's capabilities.

NOTE

When required by project-specific planning documents, collection of equipment rinsate blanks (see next step) shall be completed without exception unless approval to not collect these samples is obtained from the Project Manager.

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- Collection of Equipment Rinsate Blanks – Collect equipment rinsate blanks to evaluate the decontamination procedure representing different equipment types (e.g., pumps versus drilling equipment) and different decontamination processes as follows:
 - Single-use disposable equipment – The number of samples should represent different types of equipment as well as different lot numbers of single-use articles.
 - The collection and the frequency of equipment rinsate blank samples are as follows unless specified differently in the project-specific planning documents:
 - Per decontamination method.
 - Per disposable article/batch number of disposable articles.

NOTE

It is recommended that an initial equipment rinsate blank sample be collected early in the project to ensure that the decontamination process is functioning properly and to avoid using a contaminated batch of single-use articles. It is recommended that a follow-up sample be collected later during the execution of the project to ensure that those conditions do not change.

Equipment rinsate blank sample collection may be driven by types of and/or levels of contaminant. Difficult to remove contaminants, oils/greases, some PAHs/PCBs, etc. may also support the collection of additional equipment rinsate blank samples due to the obvious challenges to the decontamination process. This is a field consideration to be determined by the FOL supporting and/or resigning not to collect.

CAUTION

For equipment rinsate blanks, project-specific applications may require use of water certified to be contaminant free. Consult project planning documents or SOPs devoted to sampling for the particular contaminants (e.g., per- and polyfluoroalkyl substances).



STANDARD OPERATING PROCEDURE

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Applicability	Tetra Tech, Inc., EGS Operating Unit		
Prepared	Earth Sciences Department		

Subject
DECONTAMINATION OF FIELD EQUIPMENT AND
MANAGEMENT OF INVESTIGATION DERIVED
WASTE

Approved
T. Johnston

ATTACHMENT A IDW LABEL (EXAMPLE)

INVESTIGATION DERIVED WASTE

GENERATOR INFORMATION:

SITE _____ JOB NO. _____

LOCATION _____

DATE _____

DRUM# _____

CONTENTS _____

VOLUME _____

CONTACT _____

EMERGENCY PHONE NUMBER _____



STANDARD OPERATING PROCEDURE

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Applicability	Tetra Tech, Inc., EGS Operating Unit		
Prepared	Earth Sciences Department		

Subject
DECONTAMINATION OF FIELD EQUIPMENT AND
MANAGEMENT OF INVESTIGATION DERIVED
WASTE

Approved
T. Johnston

ATTACHMENT B, INVESTIGATIVE DERIVED WASTE DRUM/CONTAINER INVENTORY LOG (EXAMPLE)

This attachment should be completed as Investigative derived waste is accumulated. An updated inventory should be provided to the site Point of Contact at the end of each 10-day shift. Add additional rows on extra pages, as needed.

Drum/ Container Number #	Drum/ Container Type	Media (Contents)	Location (SWMU and Well #, etc.)	Estimated Volume	Date Filled	Comments
1				()-Gallons		
2				()-Gallons		
3				()-Gallons		
4				()-Gallons		

Emergency Point of Contact	Phone Number:
(Name of Facility) Point of Contact:	Phone Number: ()
Information Point of Contact (Name):	Phone Number:
Distribution of Inventory Log: Facility Point of Contact, Phone Number: ()	Phone Number: ()
Fire Dept/Emergency Services Provider, Phone Number: ()	Phone Number: ()
Comments:	

Appendix B

Laboratory Accreditation Certifications

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CERTIFICATE OF ACCREDITATION

The ANSI National Accreditation Board

Hereby attests that

Katahdin Analytical Services, LLC
600 Technology Way
Scarborough, ME 04074

Fulfills the requirements of

ISO/IEC 17025:2017

and

U.S. Department of Defense (DoD) Quality Systems Manual
for Environmental Laboratories (DoD QSM V5.3)

In the field of

TESTING

This certificate is valid only when accompanied by a current scope of accreditation document.
The current scope of accreditation can be verified at www.anab.org.

R. Douglas Leonard Jr., VP, PILR SBU

Expiry Date: 01 February 2022
Certificate Number: L2223



This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2017.
This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory
quality management system (refer to joint ISO-ILAC-IAF Communiqué dated April 2017).

**SCOPE OF ACCREDITATION TO ISO/IEC 17025:2017 AND U.S.
DEPARTMENT OF DEFENSE (DOD) QUALITY SYSTEMS MANUAL
FOR ENVIRONMENTAL LABORATORIES (DOD QSM V 5.3)**

Katahdin Analytical Services, LLC

600 Technology Way
Scarborough, ME 04074
Leslie Dimond
207-874-2400

TESTING

Valid to: **February 1, 2022**

Certificate Number: **L2223**

Environmental

Non-Potable Water		
Technology	Method	Analyte
GC/ECD	EPA 8081B	2, 4`-DDD
GC/ECD	EPA 8081B	2, 4`-DDE
GC/ECD	EPA 8081B	2, 4`-DDT
GC/ECD	EPA 8081B	4, 4`-DDD
GC/ECD	EPA 8081B	4, 4`-DDE
GC/ECD	EPA 8081B	4, 4`-DDT
GC/ECD	EPA 8081B	Aldrin
GC/ECD	EPA 8081B	alpha-BHC (alpha-Hexachlorocyclohexane)
GC/ECD	EPA 8081B	Alpha-Chlordane/cis-chlordane
GC/ECD	EPA 8081B	beta-BHC (beta-Hexachlorocyclohexane)
GC/ECD	EPA 8081B	Cis-Nonaclor
GC/ECD	EPA 8081B	Chlordane (tech.)
GC/ECD	EPA 8081B	delta-BHC
GC/ECD	EPA 8081B	Dieldrin
GC/ECD	EPA 8081B	Endosulfan I

Non-Potable Water		
Technology	Method	Analyte
GC/ECD	EPA 8081B	Endosulfan II
GC/ECD	EPA 8081B	Endosulfan sulfate
GC/ECD	EPA 8081B	Endrin
GC/ECD	EPA 8081B	Endrin aldehyde
GC/ECD	EPA 8081B	Endrin Ketone
GC/ECD	EPA 8081B	gamma-BHC (Lindane gamma-Hexachlorocyclohexane)
GC/ECD	EPA 8081B	gamma-Chlordane/trans-Chlordane
GC/ECD	EPA 8081B	Heptachlor
GC/ECD	EPA 8081B	Heptachlor epoxide
GC/ECD	EPA 8081B	Hexachlorobenzene
GC/ECD	EPA 8081B	Methoxychlor
GC/ECD	EPA 8081B	Mirex
GC/ECD	EPA 8081B	Oxychlordane
GC/ECD	EPA 8081B	Toxaphene (Chlorinated camphene)
GC/ECD	EPA 8081B	trans-Nonachlor
GC/ECD	EPA 8082A	Aroclor-1016 (PCB-1016)
GC/ECD	EPA 8082A	Aroclor-1221 (PCB-1221)
GC/ECD	EPA 8082A	Aroclor-1232 (PCB-1232)
GC/ECD	EPA 8082A	Aroclor-1242 (PCB-1242)
GC/ECD	EPA 8082A	Aroclor-1248 (PCB-1248)
GC/ECD	EPA 8082A	Aroclor-1254 (PCB-1254)
GC/ECD	EPA 8082A	Aroclor-1260 (PCB-1260)
GC/ECD	EPA 8082A MOD	Aroclor-1262 (PCB-1262)
GC/ECD	EPA 8082A MOD	Aroclor-1268 (PCB-1268)
GC/ECD	EPA 8082A	2,2',3,3',4,4',5,5',6-Nonachlorobiphenyl (BZ 206)
GC/ECD	EPA 8082A	2,2',3,3',4,4',5,6-Octachlorobiphenyl (BZ 195)
GC/ECD	EPA 8082A	2,2',3,3',4,4',5-Heptachlorobiphenyl (BZ 170)
GC/ECD	EPA 8082A	2,2',3,3',4,4'-Hexachlorobiphenyl (BZ 128)

Non-Potable Water		
Technology	Method	Analyte
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 5, 5'-Heptachlorobiphenyl (BZ 180)
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 5', 6-Heptachlorobiphenyl (BZ 183)
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 5-Hexachlorobiphenyl (BZ 138)
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 6, 6'-Heptachlorobiphenyl (BZ 184)
GC/ECD	EPA 8082A	2, 2', 3, 4', 5, 5', 6-Heptachlorobiphenyl (BZ 187)
GC/ECD	EPA 8082A	2, 2', 3, 4, 5'-Pentachlorobiphenyl (BZ 87)
GC/ECD	EPA 8082A	2, 2', 3, 5'-Tetrachlorobiphenyl (BZ 44)
GC/ECD	EPA 8082A	2, 2', 4, 4', 5, 5'-Hexachlorobiphenyl (BZ 153)
GC/ECD	EPA 8082A	2, 2', 4, 5, 5'-Pentachlorobiphenyl (BZ 101)
GC/ECD	EPA 8082A	2, 2', 4, 5-Tetrachlorobiphenyl (BZ 48)
GC/ECD	EPA 8082A	2, 2', 4, 5'-Tetrachlorobiphenyl (BZ 49)
GC/ECD	EPA 8082A	2, 2', 5, 5'-Tetrachlorobiphenyl (BZ 52)
GC/ECD	EPA 8082A	2, 2', 5-Trichlorobiphenyl (BZ 18)
GC/ECD	EPA 8082A	2, 3, 3', 4, 4', 5-Hexachlorobiphenyl (BZ 156)
GC/ECD	EPA 8082A	2, 3, 3', 4, 4', 5'-Hexachlorobiphenyl (BZ 157)
GC/ECD	EPA 8082A	2, 3, 3', 4, 4'-Pentachlorobiphenyl (BZ 105)
GC/ECD	EPA 8082A	2, 3, 3', 4, 4', 5, 5'-Heptachlorobiphenyl (BZ 189)
GC/ECD	EPA 8082A	2, 3', 4, 4', 5, 5'-Hexachlorobiphenyl (BZ 167)
GC/ECD	EPA 8082A	2, 3', 4, 4', 5-Pentachlorobiphenyl (BZ 118)
GC/ECD	EPA 8082A	2, 3', 4, 4', 5'-Pentachlorobiphenyl (BZ 123)
GC/ECD	EPA 8082A	2, 3', 4, 4'-Tetrachlorobiphenyl (BZ 66)
GC/ECD	EPA 8082A	2, 3, 4, 4', 5-Pentachlorobiphenyl (BZ 114)
GC/ECD	EPA 8082A	2, 4, 4'-Trichlorobiphenyl (BZ 28)
GC/ECD	EPA 8082A	2, 4'-Dichlorobiphenyl (BZ 8)
GC/ECD	EPA 8082A	3, 3', 4, 4', 5, 5'-Hexachlorobiphenyl (BZ 169)
GC/ECD	EPA 8082A	3, 3', 4, 4', 5-Pentachlorobiphenyl (BZ 126)
GC/ECD	EPA 8082A	3, 3', 4, 4'-Tetrachlorobiphenyl (BZ 77)
GC/ECD	EPA 8082A	3, 4, 4', 5-Tetrachlorobiphenyl (BZ 81)

Non-Potable Water		
Technology	Method	Analyte
GC/ECD	EPA 8082A	Decachlorobiphenyl (BZ 209)
GC/ECD	EPA 8151A	2, 4, 5-T
GC/ECD	EPA 8151A	2, 4-D
GC/ECD	EPA 8151A	2, 4-DB
GC/ECD	EPA 8151A	Dalapon
GC/ECD	EPA 8151A	Dicamba
GC/ECD	EPA 8151A	Dichloroprop
GC/ECD	EPA 8151A	Dinoseb
GC/ECD	EPA 8151A	MCPA
GC/ECD	EPA 8151A	MCP
GC/ECD	EPA 8151A	Pentachlorophenol
GC/ECD	EPA 8151A	Silvex (2, 4, 5-TP)
GC/FID	EPA 8015C/D MOD	Diesel range organics (DRO)
GC/FID	EPA 8015C/D MOD	Total Petroleum Hydrocarbon (TPH)
GC/FID	EPA 8015C/D MOD	Gasoline range organics (GRO)
GC/FID/PID	MA DEP VPH	Volatile Organic Hydrocarbons
GC/FID	MA DEP EPH	Extractable Petroleum Hydrocarbons
GC/FID	CT ETPH	Total Petroleum Hydrocarbons
GC/FID	TNRCC Method 1005	Total Petroleum Hydrocarbons
GC/FID	FL-PRO	Petroleum Range Organics
GC/ECD	EPA 8011; EPA 504	1, 2-Dibromoethane (EDB)
GC/ECD	EPA 8011; EPA 504	1, 2-Dibromo-3-chloropropane
GC/FID	RSK-175	Methane Ethane Ethene
GC/MS	EPA 8260B/C; EPA 524.2	1, 1, 1, 2-Tetrachloroethane
GC/MS	EPA 8260B/C; EPA 524.2	1, 1, 1-Trichloroethane
GC/MS	EPA 8260B/C; EPA 524.2	1, 1, 2, 2-Tetrachloroethane
GC/MS	EPA 8260B/C	1,1,2-Trichloro-1,2,2-trifluoroethane
GC/MS	EPA 8260B/C; EPA 524.2	1, 1, 2-Trichloroethane

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8260B/C; EPA 524.2	1, 1-Dichloroethane
GC/MS	EPA 8260B/C; EPA 524.2	1, 1-Dichloroethene
GC/MS	EPA 8260B/C; EPA 524.2	1, 1-Dichloropropene
GC/MS	EPA 8260B/C; EPA 524.2	1, 2, 3-Trichlorobenzene
GC/MS	EPA 8260B/C; EPA 524.2	1, 2, 3-Trichloropropane
GC/MS	EPA 8260B/C	1,2,3-Trimethylbenzene
GC/MS	EPA 8260B/C; EPA 524.2	1, 2, 4-Trichlorobenzene
GC/MS	EPA 8260B/C; EPA 524.2	1, 2, 4-Trimethylbenzene
GC/MS	EPA 8260B/C; EPA 524.2	1, 2-Dibromo-3-chloropropane
GC/MS	EPA 8260B/C; EPA 524.2	1, 2-Dibromoethane (EDB)
GC/MS	EPA 8260B/C; EPA 524.2	1, 2-Dichlorobenzene
GC/MS	EPA 8260B/C; EPA 524.2	1, 2-Dichloroethane
GC/MS	EPA 8260B/C; EPA 524.2	1, 2-Dichloropropane
GC/MS	EPA 8260B/C	1,3,5-Trichlorobenzene
GC/MS	EPA 8260B/C; EPA 524.2	1, 3, 5-Trimethylbenzene
GC/MS	EPA 8260B/C; EPA 524.2	1, 3-Dichlorobenzene
GC/MS	EPA 8260B/C; EPA 524.2	1, 3-Dichloropropane
GC/MS	EPA 8260B/C; EPA 524.2	1, 4-Dichlorobenzene
GC/MS	EPA 8260B/C	1, 4-Dioxane
GC/MS	EPA 8260B/C; EPA 524.2	2, 2-Dichloropropane
GC/MS	EPA 8260B/C; EPA 524.2	2-Butanone
GC/MS	EPA 8260B/C	2-Chloroethyl vinyl ether
GC/MS	EPA 8260B/C; EPA 524.2	2-Chlorotoluene
GC/MS	EPA 8260B/C; EPA 524.2	2-Hexanone
GC/MS	EPA 8260B/C; EPA 524.2	4-Chlorotoluene
GC/MS	EPA 8260B/C; EPA 524.2	4-Methyl-2-pentanone
GC/MS	EPA 8260B/C; EPA 524.2	Acetone
GC/MS	EPA 8260B/C	Acetonitrile

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8260B/C	Acrolein
GC/MS	EPA 8260B/C; EPA 524.2	Acrylonitrile
GC/MS	EPA 8260B/C; EPA 524.2	Allyl chloride
GC/MS	EPA 8260B/C; EPA 524.2	Benzene
GC/MS	EPA 8260B/C	Benzyl chloride
GC/MS	EPA 8260B/C; EPA 524.2	Bromobenzene
GC/MS	EPA 8260B/C; EPA 524.2	Bromochloromethane
GC/MS	EPA 8260B/C; EPA 524.2	Bromodichloromethane
GC/MS	EPA 8260B/C; EPA 524.2	Bromoform
GC/MS	EPA 8260B/C; EPA 524.2	Carbon disulfide
GC/MS	EPA 8260B/C; EPA 524.2	Carbon tetrachloride
GC/MS	EPA 8260B/C; EPA 524.2	Chlorobenzene
GC/MS	EPA 8260B/C; EPA 524.2	Chloroethane
GC/MS	EPA 8260B/C; EPA 524.2	Chloroform
GC/MS	EPA 8260B/C	Chloroprene
GC/MS	EPA 8260B/C; EPA 524.2	cis-1, 2-Dichloroethene
GC/MS	EPA 8260B/C; EPA 524.2	cis-1, 3-Dichloropropene
GC/MS	EPA 8260B/C	Cis-1,4-Dichloro-2-butene
GC/MS	EPA 8260B/C	Cyclohexane
GC/MS	EPA 8260B/C; EPA 524.2	Dibromochloromethane
GC/MS	EPA 8260B/C; EPA 524.2	Dibromomethane
GC/MS	EPA 8260B/C; EPA 524.2	Dichlorodifluoromethane
GC/MS	EPA 8260B/C; EPA 524.2	Diethyl ether
GC/MS	EPA 8260B/C	Di-isopropylether
GC/MS	EPA 8260B/C; EPA 524.2	Ethyl methacrylate
GC/MS	EPA 8260B/C; EPA 524.2	Ethylbenzene
GC/MS	EPA 8260B/C	Ethyl-t-butylether
GC/MS	EPA 8260B/C; EPA 524.2	Hexachlorobutadiene

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8260B/C	Iodomethane
GC/MS	EPA 8260B/C	Isobutyl alcohol
GC/MS	EPA 8260B/C	Isopropyl alcohol
GC/MS	EPA 8260B/C; EPA 524.2	Isopropyl benzene
GC/MS	EPA 8260B/C; EPA 524.2	m p-xylenes
GC/MS	EPA 8260B/C	Methyl acetate
GC/MS	EPA 8260B/C	Methacrylonitrile
GC/MS	EPA 8260B/C	Methyl bromide (Bromomethane)
GC/MS	EPA 8260B/C; EPA 524.2	Methyl chloride (Chloromethane)
GC/MS	EPA 8260B/C	Methyl methacrylate
GC/MS	EPA 8260B/C; EPA 524.2	Methyl tert-butyl ether
GC/MS	EPA 8260B/C	Methylcyclohexane
GC/MS	EPA 8260B/C; EPA 524.2	Methylene chloride
GC/MS	EPA 8260B/C; EPA 524.2	Naphthalene
GC/MS	EPA 8260B/C; EPA 524.2	n-Butylbenzene
GC/MS	EPA 8260B/C; EPA 524.2	n-Propylbenzene
GC/MS	EPA 8260B/C; EPA 524.2	o-Xylene
GC/MS	EPA 8260B/C	Pentachloroethane
GC/MS	EPA 8260B/C; EPA 524.2	p-Isopropyltoluene
GC/MS	EPA 8260B/C; EPA 524.2	Propionitrile
GC/MS	EPA 8260B/C; EPA 524.2	sec-butylbenzene
GC/MS	EPA 8260B/C; EPA 524.2	Styrene
GC/MS	EPA 8260B/C	t-Amylmethylether
GC/MS	EPA 8260B/C; EPA 524.2	tert-Butyl alcohol
GC/MS	EPA 8260B/C	tert-Butylbenzene
GC/MS	EPA 8260B/C; EPA 524.2	Tetrachloroethene (Perchloroethylene)
GC/MS	EPA 8260B/C; EPA 524.2	Tetrahydrofuran
GC/MS	EPA 8260B/C; EPA 524.2	Toluene

Non-Potable Water

Technology	Method	Analyte
GC/MS	EPA 8260B/C; EPA 524.2	trans-1, 2-Dichloroethylene
GC/MS	EPA 8260B/C; EPA 524.2	trans-1, 3-Dichloropropylene
GC/MS	EPA 8260B/C; EPA 524.2	trans-1, 4-Dichloro-2-butene
GC/MS	EPA 8260B/C; EPA 524.2	Trichloroethene (Trichloroethylene)
GC/MS	EPA 8260B/C; EPA 524.2	Trichlorofluoromethane
GC/MS	EPA 8260B/C	Vinyl acetate
GC/MS	EPA 8260B/C; EPA 524.2	Vinyl chloride
GC/MS	EPA 8260B/C	Xylene
GC/MS	EPA 8260B/C SIM	1,1,1,2-Tetrachloroethane
GC/MS	EPA 8260B/C SIM	1,1,1-Trichloroethane
GC/MS	EPA 8260B/C SIM	1,1,2,2-Tetrachloroethane
GC/MS	EPA 8260B/C SIM	1, 1, 2-Trichloroethane
GC/MS	EPA 8260B/C SIM	1,2,3-Trichlorobenzene
GC/MS	EPA 8260B/C SIM	1,2,3-Trichloropropane
GC/MS	EPA 8260B/C SIM	1,1-Dichloroethane
GC/MS	EPA 8260B/C SIM	1,1-Dichloroethene
GC/MS	EPA 8260B/C SIM	1,2,4-Trichlorobenzene
GC/MS	EPA 8260B/C SIM	1,2,4-Trimethylbenzene
GC/MS	EPA 8260B/C SIM	1,2-Dibromo-3-chloropropane
GC/MS	EPA 8260B/C SIM	1,2-Dibromoethane
GC/MS	EPA 8260B/C SIM	1,2-Dichlorobenzene
GC/MS	EPA 8260B/C SIM	1,2-Dichloroethane
GC/MS	EPA 8260B/C SIM	1,2-Dichloropropane
GC/MS	EPA 8260B/C SIM	1,3-Dichlorobenzene
GC/MS	EPA 8260B/C SIM	1,3-Dichloropropane
GC/MS	EPA 8260B/C SIM	1,4-Dichlorobenzene
GC/MS	EPA 8260B/C SIM	2-Hexanone
GC/MS	EPA 8260B/C SIM	4-Methyl-2-pentanone

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8260B/C SIM	Benzene
GC/MS	EPA 8260B/C SIM	Bromodichloromethane
GC/MS	EPA 8260B/C SIM	Carbon Disulfide
GC/MS	EPA 8260B/C SIM	Carbon Tetrachloride
GC/MS	EPA 8260B/C SIM	Chloroform
GC/MS	EPA 8260B/C SIM	Chloromethane
GC/MS	EPA 8260B/C SIM	cis-1,2-Dichloroethene
GC/MS	EPA 8260B/C SIM	cis-1,3-Dichloropropene
GC/MS	EPA 8260B/C SIM	Dibromochloromethane
GC/MS	EPA 8260B/C SIM	Ethylbenzene
GC/MS	EPA 8260B/C SIM	Isopropylbenzene
GC/MS	EPA 8260B/C SIM	Hexachlorobutadiene
GC/MS	EPA 8260B/C SIM	Methylene chloride
GC/MS	EPA 8260B/C SIM	m,p-Xylene
GC/MS	EPA 8260B/C SIM	o-Xylene
GC/MS	EPA 8260B/C SIM	Tetrachloroethene
GC/MS	EPA 8260B/C SIM	Toluene
GC/MS	EPA 8260B/C SIM	trans-1,2-Dichloroethene
GC/MS	EPA 8260B/C SIM	Trans-1,3-Dichloropropene
GC/MS	EPA 8260B/C SIM	Trichloroethene
GC/MS	EPA 8260B/C SIM	Trichlorofluoromethane
GC/MS	EPA 8260B/C SIM	Vinyl Chloride
GC/MS	EPA 8260B/C SIM	Xylenes (total)
GC/MS	EPA 8270C/D	1, 2, 4, 5-Tetrachlorobenzene
GC/MS	EPA 8270C/D	1, 2, 4-Trichlorobenzene
GC/MS	EPA 8270C/D	1, 2-Dichlorobenzene
GC/MS	EPA 8270C/D	1, 2-Diphenylhydrazine
GC/MS	EPA 8270C/D	1, 3, 5-Trinitrobenzene

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8270C/D	1, 3-Dichlorobenzene
GC/MS	EPA 8270C/D	1, 3-Dinitrobenzene
GC/MS	EPA 8270C/D	1, 4-Dichlorobenzene
GC/MS	EPA 8270C/D	1, 4-Dioxane
GC/MS	EPA 8270C/D	1, 4-Naphthoquinone
GC/MS	EPA 8270C/D	1, 4-Phenylenediamine
GC/MS	EPA 8270C/D	1-Chloronaphthalene
GC/MS	EPA 8270C/D	1-Methylnaphthalene
GC/MS	EPA 8270C/D	1-Naphthylamine
GC/MS	EPA 8270C/D	2, 3, 4, 6-Tetrachlorophenol
GC/MS	EPA 8270C/D	2, 4, 5-Trichlorophenol
GC/MS	EPA 8270C/D	2, 4, 6-Trichlorophenol
GC/MS	EPA 8270C/D	2, 4-Dichlorophenol
GC/MS	EPA 8270C/D	2, 4-Dimethylphenol
GC/MS	EPA 8270C/D	2, 4-Dinitrophenol
GC/MS	EPA 8270C/D	2, 4-Dinitrotoluene (2, 4-DNT)
GC/MS	EPA 8270C/D	2, 6-Dichlorophenol
GC/MS	EPA 8270C/D	2, 6-Dinitrotoluene (2, 6-DNT)
GC/MS	EPA 8270C/D	2-Acetylaminofluorene
GC/MS	EPA 8270C/D	2-Chloronaphthalene
GC/MS	EPA 8270C/D	2-Chlorophenol
GC/MS	EPA 8270C/D	2-Methyl-4 6-dinitrophenol
GC/MS	EPA 8270C/D	2-Methylnaphthalene
GC/MS	EPA 8270C/D	2-Methylphenol
GC/MS	EPA 8270C/D	2-Naphthylamine
GC/MS	EPA 8270C/D	2-Nitroaniline
GC/MS	EPA 8270C/D	2-Nitrophenol
GC/MS	EPA 8270C/D	2-Picoline

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8270C/D	3-Methylcholanthrene
GC/MS	EPA 8270C/D	3-Nitroaniline
GC/MS	EPA 8270C/D	3,4-Dimethylphenol
GC/MS	EPA 8270C/D	4-Aminobiphenyl
GC/MS	EPA 8270C/D	4-Bromophenyl phenyl ether
GC/MS	EPA 8270C/D	4-Chloro-3-methylphenol
GC/MS	EPA 8270C/D	4-Chloroaniline
GC/MS	EPA 8270C/D	4-Chlorophenyl phenylether
GC/MS	EPA 8270C/D	4-Dimethyl aminoazobenzene
GC/MS	EPA 8270C/D	3&4-Methylphenol
GC/MS	EPA 8270C/D	4-Nitroaniline
GC/MS	EPA 8270C/D	4-Nitrophenol
GC/MS	EPA 8270C/D	4-Nitroquinoline-1-oxide
GC/MS	EPA 8270C/D	5-Nitro-o-toluidine
GC/MS	EPA 8270C/D	7, 12-Dimethylbenz(a)anthracene
GC/MS	EPA 8270C/D	a a-Dimethylphenethylamine
GC/MS	EPA 8270C/D	Acenaphthene
GC/MS	EPA 8270C/D	Acenaphthylene
GC/MS	EPA 8270C/D	Acetophenone
GC/MS	EPA 8270C/D	Aniline
GC/MS	EPA 8270C/D	Anthracene
GC/MS	EPA 8270C/D	Aramite
GC/MS	EPA 8270C/D	Atrazine
GC/MS	EPA 8270C/D	Azobenzene
GC/MS	EPA 8270C/D	Benzaldehyde
GC/MS	EPA 8270C/D	Benzidine
GC/MS	EPA 8270C/D	Benzo(a)anthracene
GC/MS	EPA 8270C/D	Benzo(a)pyrene

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8270C/D	Benzo(b)fluoranthene
GC/MS	EPA 8270C/D	Benzo(g h i)perylene
GC/MS	EPA 8270C/D	Benzo(k)fluoranthene
GC/MS	EPA 8270C/D	Benzoic Acid
GC/MS	EPA 8270C/D	Benzyl alcohol
GC/MS	EPA 8270C/D	1,1-Biphenyl
GC/MS	EPA 8270C/D	bis(2-Chloroethoxy)methane
GC/MS	EPA 8270C/D	bis(2-Chloroethyl) ether
GC/MS	EPA 8270C/D	bis(2-Chloroisopropyl) ether (2, 2'-Oxybis(1-chloropropane))
GC/MS	EPA 8270C/D	bis(2-Ethylhexyl)adipate
GC/MS	EPA 8270C/D	bis(2-Ethylhexyl) phthalate (DEHP)
GC/MS	EPA 8270C/D	Butyl benzyl phthalate
GC/MS	EPA 8270C/D	Caprolactam
GC/MS	EPA 8270C/D	Carbazole
GC/MS	EPA 8270C/D	Chlorobenzilate
GC/MS	EPA 8270C/D	Chrysene
GC/MS	EPA 8270C/D	Diallate
GC/MS	EPA 8270C/D	Dibenzo(a,j)acridine
GC/MS	EPA 8270C/D	Dibenzo(a,h)anthracene
GC/MS	EPA 8270C/D	Dibenzofuran
GC/MS	EPA 8270C/D	Diethyladipate
GC/MS	EPA 8270C/D	Diethyl phthalate
GC/MS	EPA 8270C/D	Dimethoate
GC/MS	EPA 8270C/D	Dimethyl phthalate
GC/MS	EPA 8270C/D	Di-n-butyl phthalate
GC/MS	EPA 8270C/D	Di-n-octyl phthalate
GC/MS	EPA 8270C/D	Dinoseb
GC/MS	EPA 8270C/D	Disulfoton

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8270C/D	Ethyl methanesulfonate
GC/MS	EPA 8270C/D	Ethyl parathion
GC/MS	EPA 8270C/D	Ethyl methacrylate
GC/MS	EPA 8270C/D	Famfur
GC/MS	EPA 8270C/D	Fluoranthene
GC/MS	EPA 8270C/D	Fluorene
GC/MS	EPA 8270C/D	Hexachlorobenzene
GC/MS	EPA 8270C/D	Hexachlorobutadiene
GC/MS	EPA 8270C/D	Hexachlorocyclopentadiene
GC/MS	EPA 8270C/D	Hexachloroethane
GC/MS	EPA 8270C/D	Hexachlorophene
GC/MS	EPA 8270C/D	Hexachloropropene
GC/MS	EPA 8270C/D	Indeno(1, 2, 3-cd)pyrene
GC/MS	EPA 8270C/D	Isodrin
GC/MS	EPA 8270C/D	Isophorone
GC/MS	EPA 8270C/D	Isosafrole
GC/MS	EPA 8270C/D	Kepone
GC/MS	EPA 8270C/D	Methapyriline
GC/MS	EPA 8270C/D	Methy methanesulfonate
GC/MS	EPA 8270C/D	Methyl parathion
GC/MS	EPA 8270C/D	Naphthalene
GC/MS	EPA 8270C/D	Nitrobenzene
GC/MS	EPA 8270C/D	Nitroquinoline-1-oxide
GC/MS	EPA 8270C/D	n-Nitrosodiethylamine
GC/MS	EPA 8270C/D	n-Nitrosodimethylamine
GC/MS	EPA 8270C/D	n-Nitroso-di-n-butylamine
GC/MS	EPA 8270C/D	n-Nitrosodi-n-propylamine
GC/MS	EPA 8270C/D	n-Nitrosodiphenylamine

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8270C/D	n-Nitrosomethylethylamine
GC/MS	EPA 8270C/D	n-Nitrosomorpholine
GC/MS	EPA 8270C/D	n-Nitrosopiperidine
GC/MS	EPA 8270C/D	n-Nitrosopyrrolidine
GC/MS	EPA 8270C/D	O,O,O-Triethyl phosphorothioate
GC/MS	EPA 8270C/D	O,O-Diethyl O-2pyrazinyl phosphorothioate
GC/MS	EPA 8270C/D	o-Toluidine
GC/MS	EPA 8270C/D	Pentachlorobenzene
GC/MS	EPA 8270C/D	Pentachloronitrobenzene
GC/MS	EPA 8270C/D	Pentachlorophenol
GC/MS	EPA 8270C/D	Phenacetin
GC/MS	EPA 8270C/D	Phenanthrene
GC/MS	EPA 8270C/D	Phenol
GC/MS	EPA 8270C/D	Phorate
GC/MS	EPA 8270C/D	Pronamide
GC/MS	EPA 8270C/D	Pyrene
GC/MS	EPA 8270C/D	Pyridine
GC/MS	EPA 8270C/D	Safrole
GC/MS	EPA 8270C/D	Sulfotepp
GC/MS	EPA 8270C/D	Thionazin
GC/MS	EPA 8270C/D	3, 3'-Dichlorobenzidine
GC/MS	EPA 8270C/D	3, 3'-Dimethylbenzidine
GC/MS	EPA 8270C/D SIM	1,1'-Biphenyl
GC/MS	EPA 8270C/D SIM	1,2,4,5-Tetrachlorobenzene
GC/MS	EPA 8270C/D SIM	1,4-Dioxane
GC/MS	EPA 8270C/D SIM	1-Methylnaphthalene
GC/MS	EPA 8270C/D SIM	2,2'-Oxybis(1-chloropropane
GC/MS	EPA 8270C/D SIM	2,3,4,6-Tetrachlorophenol

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8270C/D SIM	2,4,5-Trichlorophenol
GC/MS	EPA 8270C/D SIM	2,4,6-Trichlorophenol
GC/MS	EPA 8270C/D SIM	2,4-Dichlorophenol
GC/MS	EPA 8270C/D SIM	2,4-Dimethylphenol
GC/MS	EPA 8270C/D SIM	2,4-Dinitrophenol
GC/MS	EPA 8270C/D SIM	2,4-Dinitrotoluene
GC/MS	EPA 8270C/D SIM	2,6-Dinitrotoluene
GC/MS	EPA 8270C/D SIM	2-Chloronaphthalene
GC/MS	EPA 8270C/D SIM	2-Chlorophenol
GC/MS	EPA 8270C/D SIM	2-Methylnaphthalene
GC/MS	EPA 8270C/D SIM	2-Methylphenol
GC/MS	EPA 8270C/D SIM	2-Nitroaniline
GC/MS	EPA 8270C/D SIM	2-Nitrophenol
GC/MS	EPA 8270C/D SIM	3&4-Methylphenol
GC/MS	EPA 8270C/D SIM	3,3'-Dichlorobenzidine
GC/MS	EPA 8270C/D SIM	3-Nitroaniline
GC/MS	EPA 8270C/D SIM	4,6-Dinitro-2-methylphenol
GC/MS	EPA 8270C/D SIM	4-Bromophenyl-phenylether
GC/MS	EPA 8270C/D SIM	4-Chloro-3-methylphenol
GC/MS	EPA 8270C/D SIM	4-Chloroaniline
GC/MS	EPA 8270C/D SIM	4-Chlorophenyl-phenylether
GC/MS	EPA 8270C/D SIM	4-Nitroaniline
GC/MS	EPA 8270C/D SIM	4-Nitrophenol
GC/MS	EPA 8270C/D SIM	Acenaphthene
GC/MS	EPA 8270C/D SIM	Acenaphthylene
GC/MS	EPA 8270C/D SIM	Acetophenone
GC/MS	EPA 8270C/D SIM	Anthracene
GC/MS	EPA 8270C/D SIM	Atrazine

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8270C/D SIM	Benzaldehyde
GC/MS	EPA 8270C/D SIM	Benzo(a)anthracene
GC/MS	EPA 8270C/D SIM	Benzo(a)pyrene
GC/MS	EPA 8270C/D SIM	Benzo(b)fluoranthene
GC/MS	EPA 8270C/D SIM	Benzo(g,h,i)perylene
GC/MS	EPA 8270C/D SIM	Benzo(k)fluoranthene
GC/MS	EPA 8270C/D SIM	Bis(2-chloroethoxy)methane
GC/MS	EPA 8270C/D SIM	Bis(2-chloroethyl)ether
GC/MS	EPA 8270C/D SIM	Bis(2-ethylhexyl)phthalate
GC/MS	EPA 8270C/D SIM	Butylbenzylphthalate
GC/MS	EPA 8270C/D SIM	Caprolactam
GC/MS	EPA 8270C/D SIM	Carbazole
GC/MS	EPA 8270C/D SIM	Chrysene
GC/MS	EPA 8270C/D SIM	Dibenzo(a,h)anthracene
GC/MS	EPA 8270C/D SIM	Dibenzofuran
GC/MS	EPA 8270C/D SIM	Diethylphthalate
GC/MS	EPA 8270C/D SIM	Dimethyl phthalate
GC/MS	EPA 8270C/D SIM	Di-n-butylphthalate
GC/MS	EPA 8270C/D SIM	Di-n-octylphthalate
GC/MS	EPA 8270C/D SIM	Fluoranthene
GC/MS	EPA 8270C/D SIM	Fluorene
GC/MS	EPA 8270C/D SIM	Hexachlorobenzene
GC/MS	EPA 8270C/D SIM	Hexachlorobutadiene
GC/MS	EPA 8270C/D SIM	Hexachlorocyclopentadiene
GC/MS	EPA 8270C/D SIM	Hexachloroethane
GC/MS	EPA 8270C/D SIM	Indeno(1,2,3-cd)pyrene
GC/MS	EPA 8270C/D SIM	Isophorone
GC/MS	EPA 8270C/D SIM	Naphthalene

Non-Potable Water		
Technology	Method	Analyte
GC/MS	EPA 8270C/D SIM	Nitrobenzene
GC/MS	EPA 8270C/D SIM	n-Nitroso-di-n-propylamine
GC/MS	EPA 8270C/D SIM	n-Nitrosodiphenylamine
GC/MS	EPA 8270C/D SIM	Pentachlorophenol
GC/MS	EPA 8270C/D SIM	Phenanthrene
GC/MS	EPA 8270C/D SIM	Phenol
GC/MS	EPA 8270C/D SIM	Pyrene
HPLC/UV	EPA 8330A/B	1, 3, 5-Trinitrobenzene
HPLC/UV	EPA 8330A/B	1, 3-Dinitrobenzene
HPLC/UV	EPA 8330A/B	2, 4, 6-Trinitrotoluene
HPLC/UV	EPA 8330A/B	2, 4-Dinitrotoluene
HPLC/UV	EPA 8330A/B	2, 6-Dinitrotoluene
HPLC/UV	EPA 8330A/B	2-Amino-4, 6 -Dinitrotoluene
HPLC/UV	EPA 8330A/B	2-Nitrotoluene
HPLC/UV	EPA 8330A/B	3-Nitrotoluene
HPLC/UV	EPA 8330A/B	3,5-Dinitroaniline
HPLC/UV	EPA 8330A/B	4-Amino-2, 6-Dinitrotoluene
HPLC/UV	EPA 8330A/B	4-Nitrotoluene
HPLC/UV	EPA 8330A/B	Ethylene glycol dinitrate (EGDN)
HPLC/UV	EPA 8330A/B	Hexahydro-1, 3, 5-trinitro-1, 3, 5-triazine (RDX)
HPLC/UV	EPA 8330A/B	Nitroguanidine
HPLC/UV	EPA 8330A/B	Nitrobenzene
HPLC/UV	EPA 8330A MOD	Nitroglycerin
HPLC/UV	EPA 8330B	Nitroglycerin
HPLC/UV	EPA 8330A/B	Octahydro-1, 3, 5, 7-tetrazocine (HMX)
HPLC/UV	EPA 8330A/B	Pentaerythritol Tetranitrate (PETN)
HPLC/UV	EPA 8330A/B	Tetryl
CVAA	EPA 245.1; EPA 7470A	Mercury

Non-Potable Water		
Technology	Method	Analyte
CVAF	EPA 1631E	Low Level Mercury
ICP/AES	EPA 200.7; EPA 6010C/D	Aluminum
ICP/AES	EPA 200.7; EPA 6010C/D	Antimony
ICP/AES	EPA 200.7; EPA 6010C/D	Arsenic
ICP/AES	EPA 200.7; EPA 6010C/D	Barium
ICP/AES	EPA 200.7; EPA 6010C/D	Beryllium
ICP/AES	EPA 200.7; EPA 6010C/D	Boron
ICP/AES	EPA 200.7; EPA 6010C/D	Cadmium
ICP/AES	EPA 200.7; EPA 6010C/D	Calcium
ICP/AES	EPA 200.7; EPA 6010C/D	Chromium
ICP/AES	EPA 200.7; EPA 6010C/D	Cobalt
ICP/AES	EPA 200.7; EPA 6010C/D	Copper
ICP/AES	EPA 200.7; EPA 6010C/D	Iron
ICP/AES	EPA 200.7; EPA 6010C/D	Lead
ICP/AES	EPA 200.7; EPA 6010C/D	Magnesium
ICP/AES	EPA 200.7; EPA 6010C/D	Manganese
ICP/AES	EPA 200.7; EPA 6010C/D	Molybdenum
ICP/AES	EPA 200.7; EPA 6010C/D	Nickel
ICP/AES	EPA 200.7; EPA 6010C/D	Potassium
ICP/AES	EPA 200.7; EPA 6010C/D	Selenium
ICP/AES	EPA 200.7; EPA 6010C/D	Silicon
ICP/AES	EPA 200.7; EPA 6010C/D	Silver
ICP/AES	EPA 200.7; EPA 6010C/D	Sodium
ICP/AES	EPA 6010C/D	Strontium
ICP/AES	EPA 200.7; EPA 6010C/D	Thallium
ICP/AES	EPA 200.7; EPA 6010C/D	Tin
ICP/AES	EPA 200.7; EPA 6010C/D	Titanium
ICP/AES	EPA 200.7; EPA 6010C/D	Vanadium

Non-Potable Water		
Technology	Method	Analyte
ICP/AES	EPA 200.7; EPA 6010C/D	Zinc
ICP/MS	EPA 200.8; EPA 6020A/B	Aluminum
ICP/MS	EPA 200.8; EPA 6020A/B	Antimony
ICP/MS	EPA 200.8; EPA 6020A/B	Arsenic
ICP/MS	EPA 200.8; EPA 6020A/B	Barium
ICP/MS	EPA 200.8; EPA 6020A/B	Beryllium
ICP/MS	EPA 200.8; EPA 6020A/B	Boron
ICP/MS	EPA 200.8; EPA 6020A/B	Cadmium
ICP/MS	EPA 200.8; EPA 6020A/B	Calcium
ICP/MS	EPA 200.8; EPA 6020A/B	Chromium
ICP/MS	EPA 200.8; EPA 6020A/B	Cobalt
ICP/MS	EPA 200.8; EPA 6020A/B	Copper
ICP/MS	EPA 200.8; EPA 6020A/B	Iron
ICP/MS	EPA 200.8; EPA 6020A/B	Lead
ICP/MS	EPA 200.8; EPA 6020A/B	Magnesium
ICP/MS	EPA 200.8; EPA 6020A/B	Manganese
ICP/MS	EPA 200.8; EPA 6020A/B	Molybdenum
ICP/MS	EPA 200.8; EPA 6020A/B	Nickel
ICP/MS	EPA 200.8; EPA 6020A/B	Potassium
ICP/MS	EPA 200.8; EPA 6020A/B	Selenium
ICP/MS	EPA 200.8; EPA 6020A/B	Silver
ICP/MS	EPA 200.8; EPA 6020A/B	Sodium
ICP/MS	EPA 6020A/B	Strontium
ICP/MS	EPA 200.8; EPA 6020A/B	Thallium
ICP/MS	EPA 200.8; EPA 6020A/B	Tin
ICP/MS	EPA 200.8; EPA 6020A/B	Tungsten
ICP/MS	EPA 200.8	Uranium
ICP/MS	EPA 200.8; EPA 6020A/B	Vanadium

Non-Potable Water		
Technology	Method	Analyte
ICP/MS	EPA 200.8; EPA 6020A/B	Zinc
IC	EPA 300.0; EPA 9056A	Chloride
IC	EPA 300.0; EPA 9056A	Fluoride
IC	EPA 300.0; EPA 9056A	Nitrate as N
IC	EPA 300.0; EPA 9056A	Nitrite as N
IC	EPA 300.0; EPA 9056A	Nitrate + Nitrite
IC	EPA 300.0; EPA 9056A	Sulfate
IC	SOP CA-776	Lactic Acid
IC	SOP CA-776	Acetic Acid
IC	SOP CA-776	Propionic Acid
IC	SOP CA-776	Formic Acid
IC	SOP CA-776	Butyric Acid
IC	SOP CA-776	Pyruvic Acid
IC	SOP CA-776	i-Pentanoic Acid
IC	SOP CA-776	Pentanoic Acid
IC	SOP CA-776	i-Hexanoic Acid
IC	SOP CA-776	Hexanoic Acid
Titration	EPA 310.1; SM 2320B	Alkalinity
Caculation	SM 2340B	Hardness
Gravimetric	EPA 1664A; EPA 9070A	Oil and Grease, Oil and Grease with SGT
Gravimetric	SM 2540B/C/D	Solids
ISE	EPA 120.1; SM 2510B	Conductivity
ISE	SM 2520B	Practical Salinity
ISE	SM 4500F- C	Fluoride
ISE	SM 4500H+ B	pH
ISE	SM 5210B	TBOD / CBOD
Physical	EPA 1010A	Ignitability
Physical	EPA 9040C	pH

Non-Potable Water		
Technology	Method	Analyte
Titration	SM 2340C	Hardness
Titration	SM 4500SO ₃ B	Sulfite
Titration	EPA 9034; SM 4500-S ²⁻ F	Sulfide
Titration	EPA SW-846 Chapter 7.3.4	Reactive Sulfide
IR	EPA 9060A; SM 5310B	Total organic carbon
Turbidimetric	EPA 180.1; SM 2130B	Turbidity
Turbidimetric	EPA 9038; ASTM 516-02	Sulfate
UV/VIS	EPA 335.4; EPA 9012B; SM 4500-CN G	Amenable cyanide
UV/VIS	EPA 350.1; SM 4500-NH ₃ H	Ammonia as N
UV/VIS	SM 3500Fe D	Ferrous Iron
UV/VIS	EPA 351.2	Kjeldahl nitrogen - total
UV/VIS	EPA 353.2; SM 4500-NO ₃ F	Nitrate + Nitrite
UV/VIS	EPA 353.2; SM 4500-NO ₃ F	Nitrate as N
UV/VIS	EPA 353.2; SM 4500-NO ₃ F	Nitrite as N
UV/VIS	EPA 365.2; SM 4500-P E	Orthophosphate as P
UV/VIS	EPA 365.4	Phosphorus total
UV/VIS	EPA 821/R-91-100	AVS-SEM
UV/VIS	EPA 410.4	COD
UV/VIS	EPA 420.1; EPA 9065	Total Phenolics
UV/VIS	SM 4500-Cl G	Total Residual Chlorine
UV/VIS	SM 5540C	MBAS
UV/VIS	EPA 7196A; SM 3500-Cr D	Chromium VI
UV/VIS	EPA 9012B; EPA 335.4	Total Cyanide
UV/VIS	EPA 9251; SM 4500-Cl E	Chloride
UV/VIS	EPA SW-846 Chapter 7.3.4	Reactive Cyanide

Non-Potable Water		
Preparation	Method	Type
Cleanup Methods	EPA 3640A	Gel Permeation Clean-up
Cleanup Methods	EPA 3630C	Silica Gel
Cleanup Methods	EPA 3660B	Sulfur Clean-Up
Cleanup Methods	EPA 3665A	Sulfuric Acid Clean-Up
Organic Preparation	EPA 3510C	Separatory Funnel Extraction
Organic Preparation	EPA 3520C	Continuous Liquid-Liquid Extraction
Inorganic Preparation	EPA 3010A	Hotblock
Volatile Organic Preparation	EPA 5030C	Purge and Trap

Solid and Chemical Waste		
Technology	Method	Analyte
GC/ECD	EPA 8081B	2,4'-DDD
GC/ECD	EPA 8081B	2,4'-DDE
GC/ECD	EPA 8081B	2,4'-DDT
GC/ECD	EPA 8081B	4, 4'-DDD
GC/ECD	EPA 8081B	4, 4'-DDE
GC/ECD	EPA 8081B	4, 4'-DDT
GC/ECD	EPA 8081B	Aldrin
GC/ECD	EPA 8081B	alpha-BHC (alpha-Hexachlorocyclohexane)
GC/ECD	EPA 8081B	Alpha-Chlordane/cis-chlordane
GC/ECD	EPA 8081B	beta-BHC (beta-Hexachlorocyclohexane)
GC/ECD	EPA 8081B	Chlordane (tech.)
GC/ECD	EPA 8081B	Cis-Nonachlor
GC/ECD	EPA 8081B	delta-BHC
GC/ECD	EPA 8081B	Dieldrin
GC/ECD	EPA 8081B	Endosulfan I
GC/ECD	EPA 8081B	Endosulfan II

Solid and Chemical Waste

Technology	Method	Analyte
GC/ECD	EPA 8081B	Endosulfan sulfate
GC/ECD	EPA 8081B	Endrin
GC/ECD	EPA 8081B	Endrin aldehyde
GC/ECD	EPA 8081B	Endrin Ketone
GC/ECD	EPA 8081B	gamma-BHC (Lindane gamma-Hexachlorocyclohexane)
GC/ECD	EPA 8081B	gamma-Chlordane/trans-Chlordane
GC/ECD	EPA 8081B	Heptachlor
GC/ECD	EPA 8081B	Heptachlor epoxide
GC/ECD	EPA 8081B	Hexachlorobenzene
GC/ECD	EPA 8081B	Methoxychlor
GC/ECD	EPA 8081B	Mirex
GC/ECD	EPA 8081B	Oxychlordane
GC/ECD	EPA 8081B	Toxaphene (Chlorinated camphene)
GC/ECD	EPA 8081B	Trans-Nonachlor
GC/ECD	EPA 8082A	Aroclor-1016 (PCB-1016)
GC/ECD	EPA 8082A	Aroclor-1221 (PCB-1221)
GC/ECD	EPA 8082A	Aroclor-1232 (PCB-1232)
GC/ECD	EPA 8082A	Aroclor-1242 (PCB-1242)
GC/ECD	EPA 8082A	Aroclor-1248 (PCB-1248)
GC/ECD	EPA 8082A	Aroclor-1254 (PCB-1254)
GC/ECD	EPA 8082A	Aroclor-1260 (PCB-1260)
GC/ECD	EPA 8082A MOD	Aroclor-1262 (PCB-1262)
GC/ECD	EPA 8082A MOD	Aroclor-1268 (PCB-1268)
GC/ECD	EPA 8082A	2, 2', 3, 3', 4, 4', 5, 5', 6-Nonachlorobiphenyl (BZ 206)
GC/ECD	EPA 8082A	2, 2', 3, 3', 4, 4', 5, 6-Octachlorobiphenyl (BZ 195)
GC/ECD	EPA 8082A	2, 2', 3, 3', 4, 4', 5-Heptachlorobiphenyl (BZ 170)
GC/ECD	EPA 8082A	2, 2', 3, 3', 4, 4'-Hexachlorobiphenyl (BZ 128)

Solid and Chemical Waste

Technology	Method	Analyte
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 5, 5'-Heptachlorobiphenyl (BZ 180)
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 5', 6-Heptachlorobiphenyl (BZ 183)
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 5-Hexachlorobiphenyl (BZ 138)
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 6, 6'-Heptachlorobiphenyl (BZ 184)
GC/ECD	EPA 8082A	2, 2', 3, 4', 5, 5', 6-Heptachlorobiphenyl (BZ 187)
GC/ECD	EPA 8082A	2, 2', 3, 4, 5'-Pentachlorobiphenyl (BZ 87)
GC/ECD	EPA 8082A	2, 2', 3, 5'-Tetrachlorobiphenyl (BZ 44)
GC/ECD	EPA 8082A	2, 2', 4, 4', 5, 5'-Hexachlorobiphenyl (BZ 153)
GC/ECD	EPA 8082A	2, 2', 4, 5, 5'-Pentachlorobiphenyl (BZ 101)
GC/ECD	EPA 8082A	2, 2', 4, 5-Tetrachlorobiphenyl (BZ 48)
GC/ECD	EPA 8082A	2, 2', 4, 5'-Tetrachlorobiphenyl (BZ 49)
GC/ECD	EPA 8082A	2, 2', 5, 5'-Tetrachlorobiphenyl (BZ 52)
GC/ECD	EPA 8082A	2, 2', 5-Trichlorobiphenyl (BZ 18)
GC/ECD	EPA 8082A	2, 3, 3', 4, 4', 5-Hexachlorobiphenyl (BZ 156)
GC/ECD	EPA 8082A	2, 3, 3', 4, 4', 5'-Hexachlorobiphenyl (BZ 157)
GC/ECD	EPA 8082A	2, 3, 3', 4, 4'-Pentachlorobiphenyl (BZ 105)
GC/ECD	EPA 8082A	2, 3, 3', 4, 4', 5, 5'-Heptachlorobiphenyl (BZ 189)
GC/ECD	EPA 8082A	2, 3', 4, 4', 5, 5'-Hexachlorobiphenyl (BZ 167)
GC/ECD	EPA 8082A	2, 3', 4, 4', 5-Pentachlorobiphenyl (BZ 118)
GC/ECD	EPA 8082A	2, 3', 4, 4', 5'-Pentachlorobiphenyl (BZ 123)
GC/ECD	EPA 8082A	2, 3', 4, 4'-Tetrachlorobiphenyl (BZ 66)
GC/ECD	EPA 8082A	2, 3, 4, 4', 5-Pentachlorobiphenyl (BZ 114)
GC/ECD	EPA 8082A	2, 4, 4'-Trichlorobiphenyl (BZ 28)
GC/ECD	EPA 8082A	2, 4'-Dichlorobiphenyl (BZ 8)
GC/ECD	EPA 8082A	3, 3', 4, 4', 5, 5'-Hexachlorobiphenyl (BZ 169)
GC/ECD	EPA 8082A	3, 3', 4, 4', 5-Pentachlorobiphenyl (BZ 126)
GC/ECD	EPA 8082A	3, 3', 4, 4'-Tetrachlorobiphenyl (BZ 77)
GC/ECD	EPA 8082A	3, 4, 4', 5-Tetrachlorobiphenyl (BZ 81)

Solid and Chemical Waste

Technology	Method	Analyte
GC/ECD	EPA 8082A	Decachlorobiphenyl (BZ 209)
GC/ECD	EPA 8151A	2, 4, 5-T
GC/ECD	EPA 8151A	2, 4-D
GC/ECD	EPA 8151A	2, 4-DB
GC/ECD	EPA 8151A	Dalapon
GC/ECD	EPA 8151A	Dicamba
GC/ECD	EPA 8151A	Dichloroprop
GC/ECD	EPA 8151A	Dinoseb
GC/ECD	EPA 8151A	MCPA
GC/ECD	EPA 8151A	MCPP
GC/ECD	EPA 8151A	Pentachlorophenol
GC/ECD	EPA 8151A	Silvex (2, 4, 5-TP)
GC/FID	EPA 8015C/D MOD	Diesel range organics (DRO)
GC/FID	EPA 8015C/D MOD	Total Petroleum Hydrocarbons (TPH)
GC/FID	EPA 8015C/D MOD	Gasoline range organics (GRO)
GC/FID/PID	MA DEP VPH	Volatile Organic Hydrocarbons
GC/FID	MA DEP EPH	Extractable Petroleum Hydrocarbons
GC/FID	MA DEP EPH EPA 3546	Extractable Petroleum Hydrocarbons Microwave Extraction Preparation
GC/FID	CT-ETPH	Total Petroleum Hydrocarbons
GC/FID	TNRCC Method 1005	Total Petroleum Hydrocarbons
GC/FID	FL-PRO	Petroleum Range Organics
GC/ECD	EPA 8011	1, 2-Dibromoethane (EDB)
GC/ECD	EPA 8011	1, 2-Dibromo-3-chloropropane
GC/MS	EPA 8260B/C	1, 1, 1, 2-Tetrachloroethane
GC/MS	EPA 8260B/C	1,1,2-Trichloro-1,2,2-trifluoroethane
GC/MS	EPA 8260B/C	1, 1, 1-Trichloroethane
GC/MS	EPA 8260B/C	1, 1, 2, 2-Tetrachloroethane
GC/MS	EPA 8260B/C	1, 1, 2-Trichloroethane

Solid and Chemical Waste

Technology	Method	Analyte
GC/MS	EPA 8260B/C	1, 1-Dichloroethane
GC/MS	EPA 8260B/C	1, 1-Dichloroethylene
GC/MS	EPA 8260B/C	1, 1-Dichloropropene
GC/MS	EPA 8260B/C	1, 2, 3-Trichlorobenzene
GC/MS	EPA 8260B/C	1, 2, 3-Trichloropropane
GC/MS	EPA 8260B/C	1,2,3-Trimethylbenzene
GC/MS	EPA 8260B/C	1, 2, 4-Trichlorobenzene
GC/MS	EPA 8260B/C	1, 2, 4-Trimethylbenzene
GC/MS	EPA 8260B/C	1, 2-Dibromo-3-chloropropane
GC/MS	EPA 8260B/C	1, 2-Dibromoethane
GC/MS	EPA 8260B/C	1, 2-Dichlorobenzene
GC/MS	EPA 8260B/C	1, 2-Dichloroethane
GC/MS	EPA 8260B/C	1, 2-Dichloropropane
GC/MS	EPA 8260B/C	1,3,5-Trichlorobenzene
GC/MS	EPA 8260B/C	1, 3, 5-Trimethylbenzene
GC/MS	EPA 8260B/C	1, 3-Dichlorobenzene
GC/MS	EPA 8260B/C	1, 3-Dichloropropane
GC/MS	EPA 8260B/C	1, 4-Dichlorobenzene
GC/MS	EPA 8260B/C	1, 4-Dioxane
GC/MS	EPA 8260B/C	2, 2-Dichloropropane
GC/MS	EPA 8260B/C	2-Butanone
GC/MS	EPA 8260B/C	2-Chloroethyl vinyl ether
GC/MS	EPA 8260B/C	2-Chlorotoluene
GC/MS	EPA 8260B/C	2-Hexanone
GC/MS	EPA 8260B/C	4-Chlorotoluene
GC/MS	EPA 8260B/C	4-Methyl-2-pentanone
GC/MS	EPA 8260B/C	Acetone
GC/MS	EPA 8260B/C	Acetonitrile

Solid and Chemical Waste		
Technology	Method	Analyte
GC/MS	EPA 8260B/C	Acrolein
GC/MS	EPA 8260B/C	Acrylonitrile
GC/MS	EPA 8260B/C	Allyl chloride
GC/MS	EPA 8260B/C	Benzene
GC/MS	EPA 8260B/C	Benzyl chloride
GC/MS	EPA 8260B/C	Bromobenzene
GC/MS	EPA 8260B/C	Bromochloromethane
GC/MS	EPA 8260B/C	Bromodichloromethane
GC/MS	EPA 8260B/C	Bromoform
GC/MS	EPA 8260B/C	Carbon disulfide
GC/MS	EPA 8260B/C	Carbon tetrachloride
GC/MS	EPA 8260B/C	Chlorobenzene
GC/MS	EPA 8260B/C	Chloroethane
GC/MS	EPA 8260B/C	Chloroform
GC/MS	EPA 8260B/C	Chloroprene
GC/MS	EPA 8260B/C	cis-1, 2-Dichloroethene
GC/MS	EPA 8260B/C	cis-1, 3-Dichloropropene
GC/MS	EPA 8260B/C	cis-1,3-Dichloro-2-butene
GC/MS	EPA 8260B/C	Cyclohexane
GC/MS	EPA 8260B/C	Dibromochloromethane
GC/MS	EPA 8260B/C	Dibromomethane
GC/MS	EPA 8260B/C	Dichlorodifluoromethane
GC/MS	EPA 8260B/C	Diethyl ether
GC/MS	EPA 8260B/C	Di-isopropylether
GC/MS	EPA 8260B/C	1,2-Dibromoethane (EDB)
GC/MS	EPA 8260B/C	Ethyl methacrylate
GC/MS	EPA 8260B/C	Ethylbenzene
GC/MS	EPA 8260B/C	Ethyl-t-butylether

Solid and Chemical Waste

Technology	Method	Analyte
GC/MS	EPA 8260B/C	Hexachlorobutadiene
GC/MS	EPA 8260B/C	Iodomethane
GC/MS	EPA 8260B/C	Isobutyl alcohol
GC/MS	EPA 8260B/C	Isopropyl alcohol
GC/MS	EPA 8260B/C	Isopropyl benzene
GC/MS	EPA 8260B/C	m p-xylenes
GC/MS	EPA 8260B/C	Methyl acetate
GC/MS	EPA 8260B/C	Methacrylonitrile
GC/MS	EPA 8260B/C	Methyl bromide (Bromomethane)
GC/MS	EPA 8260B/C	Methyl chloride (Chloromethane)
GC/MS	EPA 8260B/C	Methyl methacrylate
GC/MS	EPA 8260B/C	Methyl tert-butyl ether
GC/MS	EPA 8260B/C	Methylcyclohexane
GC/MS	EPA 8260B/C	Methylene chloride
GC/MS	EPA 8260B/C	Naphthalene
GC/MS	EPA 8260B/C	n-Butylbenzene
GC/MS	EPA 8260B/C	n-propylbenzene
GC/MS	EPA 8260B/C	o-Xylene
GC/MS	EPA 8260B/C	pentachloroethane
GC/MS	EPA 8260B/C	p-Isopropyltoluene
GC/MS	EPA 8260B/C	Propionitrile
GC/MS	EPA 8260B/C	sec-butylbenzene
GC/MS	EPA 8260B/C	Styrene
GC/MS	EPA 8260B/C	t-Amylmethylether
GC/MS	EPA 8260B/C	tert-Butyl alcohol
GC/MS	EPA 8260B/C	tert-Butylbenzene
GC/MS	EPA 8260B/C	Tetrachloroethylene (Perchloroethylene)
GC/MS	EPA 8260B/C	Tetrahydrofuran

Solid and Chemical Waste

Technology	Method	Analyte
GC/MS	EPA 8260B/C	Toluene
GC/MS	EPA 8260B/C	trans-1, 2-Dichloroethylene
GC/MS	EPA 8260B/C	trans-1, 3-Dichloropropylene
GC/MS	EPA 8260B/C	Trans-1, 4-Dichloro-2-butene
GC/MS	EPA 8260B/C	Trichloroethene (Trichloroethylene)
GC/MS	EPA 8260B/C	Trichlorofluoromethane
GC/MS	EPA 8260B/C	Vinyl acetate
GC/MS	EPA 8260B/C	Vinyl chloride
GC/MS	EPA 8260B/C	Xylene
GC/MS	EPA 8270C/D	1, 2, 4, 5-Tetrachlorobenzene
GC/MS	EPA 8270C/D	1, 2, 4-Trichlorobenzene
GC/MS	EPA 8270C/D	1, 2-Dichlorobenzene
GC/MS	EPA 8270C/D	1, 2-Diphenylhydrazine
GC/MS	EPA 8270C/D	1, 3, 5-Trinitrobenzene
GC/MS	EPA 8270C/D	1, 3-Dichlorobenzene
GC/MS	EPA 8270C/D	1, 3-Dinitrobenzene
GC/MS	EPA 8270C/D	1, 4-Dichlorobenzene
GC/MS	EPA 8270C/D	1, 4-Dioxane
GC/MS	EPA 8270C/D	1, 4-Naphthoquinone
GC/MS	EPA 8270C/D	1, 4-Phenylenediamine
GC/MS	EPA 8270C/D	1,1-Biphenyl
GC/MS	EPA 8270C/D	1-Chloronaphthalene
GC/MS	EPA 8270C/D	1-Methylnaphthalene
GC/MS	EPA 8270C/D	1-Naphthylamine
GC/MS	EPA 8270C/D	2, 3, 4, 6-Tetrachlorophenol
GC/MS	EPA 8270C/D	2, 4, 5-Trichlorophenol
GC/MS	EPA 8270C/D	2, 4, 6-Trichlorophenol
GC/MS	EPA 8270C/D	2, 4-Dichlorophenol

Solid and Chemical Waste

Technology	Method	Analyte
GC/MS	EPA 8270C/D	2, 4-Dimethylphenol
GC/MS	EPA 8270C/D	2, 4-Dinitrophenol
GC/MS	EPA 8270C/D	2, 4-Dinitrotoluene (2 4-DNT)
GC/MS	EPA 8270C/D	2, 6-Dichlorophenol
GC/MS	EPA 8270C/D	2, 6-Dinitrotoluene (2 6-DNT)
GC/MS	EPA 8270C/D	2-Acetylaminofluorene
GC/MS	EPA 8270C/D	2-Chloronaphthalene
GC/MS	EPA 8270C/D	2-Chlorophenol
GC/MS	EPA 8270C/D	2-Methyl-4, 6-dinitrophenol
GC/MS	EPA 8270C/D	2-Methylnaphthalene
GC/MS	EPA 8270C/D	2-Methylphenol
GC/MS	EPA 8270C/D	2-Naphthylamine
GC/MS	EPA 8270C/D	2-Nitroaniline
GC/MS	EPA 8270C/D	2-Nitrophenol
GC/MS	EPA 8270C/D	2-Picoline
GC/MS	EPA 8270C/D	3, 3'-Dichlorobenzidine
GC/MS	EPA 8270C/D	3, 3'-Dimethylbenzidine
GC/MS	EPA 8270C/D	3, 4-Dimethylphenol
GC/MS	EPA 8270C/D	3&4-Methylphenol
GC/MS	EPA 8270C/D	3-Methylcholanthrene
GC/MS	EPA 8270C/D	3-Nitroaniline
GC/MS	EPA 8270C/D	4-Aminobiphenyl
GC/MS	EPA 8270C/D	4-Bromophenyl phenyl ether
GC/MS	EPA 8270C/D	4-Chloro-3-methylphenol
GC/MS	EPA 8270C/D	4-Chloroaniline
GC/MS	EPA 8270C/D	4-Chlorophenyl phenylether
GC/MS	EPA 8270C/D	4-Dimethyl aminoazobenzene
GC/MS	EPA 8270C/D	4-Nitroaniline

Solid and Chemical Waste

Technology	Method	Analyte
GC/MS	EPA 8270C/D	4-Nitrophenol
GC/MS	EPA 8270C/D	4-Nitroquinoline-1-oxide
GC/MS	EPA 8270C/D	5-Nitro-o-toluidine
GC/MS	EPA 8270C/D	7,12-Dimethylbenz(a)anthracene
GC/MS	EPA 8270C/D	a a-Dimethylphenethylamine
GC/MS	EPA 8270C/D	Acenaphthene
GC/MS	EPA 8270C/D	Acenaphthylene
GC/MS	EPA 8270C/D	Acetophenone
GC/MS	EPA 8270C/D	Aniline
GC/MS	EPA 8270C/D	Anthracene
GC/MS	EPA 8270C/D	Aramite
GC/MS	EPA 8270C/D	Atrazine
GC/MS	EPA 8270C/D	Azobenzene
GC/MS	EPA 8270C/D	Benzaldehyde
GC/MS	EPA 8270C/D	Benzidine
GC/MS	EPA 8270C/D	Benzo(a)anthracene
GC/MS	EPA 8270C/D	Benzo(a)pyrene
GC/MS	EPA 8270C/D	Benzo(b)fluoranthene
GC/MS	EPA 8270C/D	Benzo(g h i)perylene
GC/MS	EPA 8270C/D	Benzo(k)fluoranthene
GC/MS	EPA 8270C/D	Benzoic Acid
GC/MS	EPA 8270C/D	Benzyl alcohol
GC/MS	EPA 8270C/D	bis(2-Chloroethoxy)methane
GC/MS	EPA 8270C/D	bis(2-Chloroethyl) ether
GC/MS	EPA 8270C/D	bis(2-Chloroisopropyl) ether (2, 2'-Oxybis(1-chloropropane))
GC/MS	EPA 8270C/D	bis(2-Ethylhexyl) phthalate (DEHP)
GC/MS	EPA 8270C/D	Bis(2-Ethylhexyl)adipate
GC/MS	EPA 8270C/D	Butyl benzyl phthalate

Solid and Chemical Waste

Technology	Method	Analyte
GC/MS	EPA 8270C/D	Caprolactam
GC/MS	EPA 8270C/D	Carbazole
GC/MS	EPA 8270C/D	Chlorobenzilate
GC/MS	EPA 8270C/D	Chrysene
GC/MS	EPA 8270C/D	Diallate
GC/MS	EPA 8270C/D	Dibenzo(a,h)anthracene
GC/MS	EPA 8270C/D	Dibenzo(a,j)acridine
GC/MS	EPA 8270C/D	Dibenzofuran
GC/MS	EPA 8270C/D	Diethyl phthalate
GC/MS	EPA 8270C/D	Diethyladipate
GC/MS	EPA 8270C/D	Dimethoate
GC/MS	EPA 8270C/D	Dimethyl phthalate
GC/MS	EPA 8270C/D	Di-n-butyl phthalate
GC/MS	EPA 8270C/D	Di-n-octyl phthalate
GC/MS	EPA 8270C/D	Dinoseb
GC/MS	EPA 8270C/D	Disulfoton
GC/MS	EPA 8270C/D	Ethyl methacrylate
GC/MS	EPA 8270C/D	Ethyl methanesulfonate
GC/MS	EPA 8270C/D	Ethyl parathion
GC/MS	EPA 8270C/D	Famfur
GC/MS	EPA 8270C/D	Fluoranthene
GC/MS	EPA 8270C/D	Fluorene
GC/MS	EPA 8270C/D	Hexachlorobenzene
GC/MS	EPA 8270C/D	Hexachlorobutadiene
GC/MS	EPA 8270C/D	Hexachlorocyclopentadiene
GC/MS	EPA 8270C/D	Hexachloroethane
GC/MS	EPA 8270C/D	Hexachlorophene
GC/MS	EPA 8270C/D	Hexachloropropene

Solid and Chemical Waste

Technology	Method	Analyte
GC/MS	EPA 8270C/D	Indeno(1, 2, 3-cd)pyrene
GC/MS	EPA 8270C/D	Isodrin
GC/MS	EPA 8270C/D	Isophorone
GC/MS	EPA 8270C/D	Isosafrole
GC/MS	EPA 8270C/D	Kepone
GC/MS	EPA 8270C/D	Methapyriline
GC/MS	EPA 8270C/D	Methyl methanesulfonate
GC/MS	EPA 8270C/D	Methyl parathion
GC/MS	EPA 8270C/D	Naphthalene
GC/MS	EPA 8270C/D	Nitrobenzene
GC/MS	EPA 8270C/D	n-Nitrosodiethylamine
GC/MS	EPA 8270C/D	n-Nitrosodimethylamine
GC/MS	EPA 8270C/D	n-Nitroso-di-n-butylamine
GC/MS	EPA 8270C/D	n-Nitrosodi-n-propylamine
GC/MS	EPA 8270C/D	n-Nitrosodiphenylamine
GC/MS	EPA 8270C/D	n-Nitrosomethylethylamine
GC/MS	EPA 8270C/D	n-Nitrosomorpholine
GC/MS	EPA 8270C/D	n-Nitrosopiperidine
GC/MS	EPA 8270C/D	n-Nitrosopyrrolidine
GC/MS	EPA 8270C/D	O, O, O-Triethyl phosphorothioate
GC/MS	EPA 8270C/D	O,O-Diethyl O-2-pyrazinyl phosphorothioate
GC/MS	EPA 8270C/D	o-Toluidine
GC/MS	EPA 8270C/D	Pentachlorobenzene
GC/MS	EPA 8270C/D	Pentachloronitrobenzene
GC/MS	EPA 8270C/D	Pentachlorophenol
GC/MS	EPA 8270C/D	Phenacetin
GC/MS	EPA 8270C/D	Phenanthrene
GC/MS	EPA 8270C/D	Phenol

Solid and Chemical Waste		
Technology	Method	Analyte
GC/MS	EPA 8270C/D	Phorate
GC/MS	EPA 8270C/D	Pronamide
GC/MS	EPA 8270C/D	Pyrene
GC/MS	EPA 8270C/D	Pyridine
GC/MS	EPA 8270C/D	Safrole
GC/MS	EPA 8270C/D	Sulfotepp
GC/MS	EPA 8270C/D	Thionazin
GC/MS	EPA 8270C/D SIM	1,1'-Biphenyl
GC/MS	EPA 8270C/D SIM	1,2,4,5-Tetrachlorobenzene
GC/MS	EPA 8270C/D SIM	1,4-Dioxane
GC/MS	EPA 8270C/D SIM	1-Methylnaphthalene
GC/MS	EPA 8270C/D SIM	2,2'-Oxybis(1-chloropropane
GC/MS	EPA 8270C/D SIM	2,3,4,6-Tetrachlorophenol
GC/MS	EPA 8270C/D SIM	2,4,5-Trichlorophenol
GC/MS	EPA 8270C/D SIM	2,4,6-Trichlorophenol
GC/MS	EPA 8270C/D SIM	2,4-Dichlorophenol
GC/MS	EPA 8270C/D SIM	2,4-Dimethylphenol
GC/MS	EPA 8270C/D SIM	2,4-Dinitrophenol
GC/MS	EPA 8270C/D SIM	2,4-Dinitrotoluene
GC/MS	EPA 8270C/D SIM	2,6-Dinitrotoluene
GC/MS	EPA 8270C/D SIM	2-Chloronaphthalene
GC/MS	EPA 8270C/D SIM	2-Chlorophenol
GC/MS	EPA 8270C/D SIM	2-Methylnaphthalene
GC/MS	EPA 8270C/D SIM	2-Methylphenol
GC/MS	EPA 8270C/D SIM	2-Nitroaniline
GC/MS	EPA 8270C/D SIM	2-Nitrophenol
GC/MS	EPA 8270C/D SIM	3&4-Methylphenol
GC/MS	EPA 8270C/D SIM	3,3'-Dichlorobenzidine

Solid and Chemical Waste

Technology	Method	Analyte
GC/MS	EPA 8270C/D SIM	3,4-Dimethylphenol
GC/MS	EPA 8270C/D SIM	3-Nitroaniline
GC/MS	EPA 8270C/D SIM	4,6-Dinitro-2-methylphenol
GC/MS	EPA 8270C/D SIM	4-Bromophenyl-phenylether
GC/MS	EPA 8270C/D SIM	4-Chloro-3-methylphenol
GC/MS	EPA 8270C/D SIM	4-Chloroaniline
GC/MS	EPA 8270C/D SIM	4-Chlorophenyl-phenylether
GC/MS	EPA 8270C/D SIM	4-Nitroaniline
GC/MS	EPA 8270C/D SIM	4-Nitrophenol
GC/MS	EPA 8270C/D SIM	Acenaphthene
GC/MS	EPA 8270C/D SIM	Acenaphthylene
GC/MS	EPA 8270C/D SIM	Acetophenone
GC/MS	EPA 8270C/D SIM	Anthracene
GC/MS	EPA 8270C/D SIM	Atrazine
GC/MS	EPA 8270C/D SIM	Benzaldehyde
GC/MS	EPA 8270C/D SIM	Benzo(a)anthracene
GC/MS	EPA 8270C/D SIM	Benzo(a)pyrene
GC/MS	EPA 8270C/D SIM	Benzo(b)fluoranthene
GC/MS	EPA 8270C/D SIM	Benzo(g,h,i)perylene
GC/MS	EPA 8270C/D SIM	Benzo(k)fluoranthene
GC/MS	EPA 8270C/D SIM	Bis(2-chloroethoxy)methane
GC/MS	EPA 8270C/D SIM	Bis(2-chloroethyl)ether
GC/MS	EPA 8270C/D SIM	Bis(2-ethylhexyl)phthalate
GC/MS	EPA 8270C/D SIM	Butylbenzylphthalate
GC/MS	EPA 8270C/D SIM	Caprolactam
GC/MS	EPA 8270C/D SIM	Carbazole
GC/MS	EPA 8270C/D SIM	Chrysene
GC/MS	EPA 8270C/D SIM	Dibenzo(a,h)anthracene

Solid and Chemical Waste

Technology	Method	Analyte
GC/MS	EPA 8270C/D SIM	Dibenzofuran
GC/MS	EPA 8270C/D SIM	Diethylphthalate
GC/MS	EPA 8270C/D SIM	Dimethyl phthalate
GC/MS	EPA 8270C/D SIM	Di-n-butylphthalate
GC/MS	EPA 8270C/D SIM	Di-n-octylphthalate
GC/MS	EPA 8270C/D SIM	Fluoranthene
GC/MS	EPA 8270C/D SIM	Fluorene
GC/MS	EPA 8270C/D SIM	Hexachlorobenzene
GC/MS	EPA 8270C/D SIM	Hexachlorobutadiene
GC/MS	EPA 8270C/D SIM	Hexachlorocyclopentadiene
GC/MS	EPA 8270C/D SIM	Hexachloroethane
GC/MS	EPA 8270C/D SIM	Indeno(1,2,3-cd)pyrene
GC/MS	EPA 8270C/D SIM	Isophorone
GC/MS	EPA 8270C/D SIM	Naphthalene
GC/MS	EPA 8270C/D SIM	Nitrobenzene
GC/MS	EPA 8270C/D SIM	n-Nitroso-di-n-propylamine
GC/MS	EPA 8270C/D SIM	n-Nitrosodiphenylamine
GC/MS	EPA 8270C/D SIM	Pentachlorophenol
GC/MS	EPA 8270C/D SIM	Phenanthrene
GC/MS	EPA 8270C/D SIM	Phenol
GC/MS	EPA 8270C/D SIM	Pyrene
HPLC/UV	EPA 8330A	1,3,5-Trinitrobenzene
HPLC/UV	EPA 8330A	1,3-Dinitrobenzene
HPLC/UV	EPA 8330A	2,4,6-Trinitrotoluene
HPLC/UV	EPA 8330A	2,4-Dinitrotoluene
HPLC/UV	EPA 8330A	2,6-Dinitrotoluene
HPLC/UV	EPA 8330A	2-Amino-4,6-dinitrotoluene
HPLC/UV	EPA 8330A	2-Nitrotoluene

Solid and Chemical Waste

Technology	Method	Analyte
HPLC/UV	EPA 8330A	3-Nitrotoluene
HPLC/UV	EPA 8330A	3,5-Dinitroaniline
HPLC/UV	EPA 8330A	4-Amino-2,6-dinitrotoluene
HPLC/UV	EPA 8330A	4-Nitrotoluene
HPLC/UV	EPA 8330A	Ethylene glycol dinitrate (EGDN)
HPLC/UV	EPA 8330A	Hexahydro-1, 3, 5-trinitro-1, 3, 5-triazine (RDX)
HPLC/UV	EPA 8330A	Nitrobenzene
HPLC/UV	EPA 8330A MOD	Nitroglycerin
HPLC/UV	EPA 8330A	Octahydro-1, 3, 5, 7-tetrazocine (HMX)
HPLC/UV	EPA 8330A	Pentaerythritol Tetranitrate (PETN)
HPLC/UV	EPA 8330A	Tetryl
HPLC/UV	EPA 8330A	Nitroguanidine
HPLC/UV	EPA 8330B	1, 3, 5-Trinitrobenzene
HPLC/UV	EPA 8330B	1, 3-Dinitrobenzene
HPLC/UV	EPA 8330B	2, 4, 6-Trinitrotoluene
HPLC/UV	EPA 8330B	2, 4-Dinitrotoluene
HPLC/UV	EPA 8330B	2, 6-Dinitrotoluene
HPLC/UV	EPA 8330B	2-Amino-4, 6 –Dinitrotoluene
HPLC/UV	EPA 8330B	2-Nitrotoluene
HPLC/UV	EPA 8330B	3-Nitrotoluene
HPLC/UV	EPA 8330B	3,5-Dinitroaniline
HPLC/UV	EPA 8330B	4-Amino-2,6,Dinitrotoluene
HPLC/UV	EPA 8330B	4-Nitrotoluene
HPLC/UV	EPA 8330B	Ethylene glycol dinitrate (EGDN)
HPLC/UV	EPA 8330B	Hexahydro-1, 3, 5-trinitro-1, 3, 5-triazine (RDX)
HPLC/UV	EPA 8330B	Nitrobenzene
HPLC/UV	EPA 8330B	Nitroglycerin
HPLC/UV	EPA 8330B	Octahydro-1, 3, 5, 7-tetrazocine (HMX)

Solid and Chemical Waste

Technology	Method	Analyte
HPLC/UV	EPA 8330B	Pentaerythritol Tetranitrate (PETN)
HPLC/UV	EPA 8330B	Tetryl
HPLC/UV	EPA 8330B	Nitroguanidine
CVAA	EPA 7471B	Mercury
CVAF	EPA 1631E	Low Level Mercury
ICP/AES	EPA 6010C/D	Aluminum
ICP/AES	EPA 6010C/D	Antimony
ICP/AES	EPA 6010C/D	Arsenic
ICP/AES	EPA 6010C/D	Barium
ICP/AES	EPA 6010C/D	Beryllium
ICP/AES	EPA 6010C/D	Boron
ICP/AES	EPA 6010C/D	Cadmium
ICP/AES	EPA 6010C/D	Calcium
ICP/AES	EPA 6010C/D	Chromium
ICP/AES	EPA 6010C/D	Cobalt
ICP/AES	EPA 6010C/D	Copper
ICP/AES	EPA 6010C/D	Iron
ICP/AES	EPA 6010C/D	Lead
ICP/AES	EPA 6010C/D	Magnesium
ICP/AES	EPA 6010C/D	Manganese
ICP/AES	EPA 6010C/D	Molybdenum
ICP/AES	EPA 6010C/D	Nickel
ICP/AES	EPA 6010C/D	Potassium
ICP/AES	EPA 6010C/D	Selenium
ICP/AES	EPA 6010C/D	Silicon
ICP/AES	EPA 6010C/D	Silver
ICP/AES	EPA 6010C/D	Sodium
ICP/AES	EPA 6010C/D	Strontium

Solid and Chemical Waste

Technology	Method	Analyte
ICP/AES	EPA 6010C/D	Thallium
ICP/AES	EPA 6010C/D	Tin
ICP/AES	EPA 6010C/D	Titanium
ICP/AES	EPA 6010C/D	Vanadium
ICP/AES	EPA 6010C/D	Zinc
ICP/MS	EPA 6020A/B	Aluminum
ICP/MS	EPA 6020A/B	Antimony
ICP/MS	EPA 6020A/B	Arsenic
ICP/MS	EPA 6020A/B	Barium
ICP/MS	EPA 6020A/B	Beryllium
ICP/MS	EPA 6020A/B	Boron
ICP/MS	EPA 6020A/B	Cadmium
ICP/MS	EPA 6020A/B	Calcium
ICP/MS	EPA 6020A/B	Chromium
ICP/MS	EPA 6020A/B	Cobalt
ICP/MS	EPA 6020A/B	Copper
ICP/MS	EPA 6020A/B	Iron
ICP/MS	EPA 6020A/B	Lead
ICP/MS	EPA 6020A/B	Magnesium
ICP/MS	EPA 6020A/B	Manganese
ICP/MS	EPA 6020A/B	Molybdenum
ICP/MS	EPA 6020A/B	Nickel
ICP/MS	EPA 6020A/B	Potassium
ICP/MS	EPA 6020A/B	Selenium
ICP/MS	EPA 6020A/B	Silver
ICP/MS	EPA 6020A/B	Sodium
ICP/MS	EPA 6020A/B	Strontium
ICP/MS	EPA 6020A/B	Thallium

Solid and Chemical Waste		
Technology	Method	Analyte
ICP/MS	EPA 6020A/B	Tin
ICP/MS	EPA 6020A/B	Tungsten
ICP/MS	EPA 6020A/B	Vanadium
ICP/MS	EPA 6020A/B	Zinc
IC	EPA 9056A	Chloride
IC	EPA 9056A	Fluoride
IC	EPA 9056A	Nitrate as N
IC	EPA 9056A	Nitrite as N
IC	EPA 9056A	Orthophosphate
IC	EPA 9056A	Sulfate
Gravimetric	EPA 9071A/B	Oil and Grease, Oil and Grease with SGT
Physical	EPA 1010A	Ignitability
Physical	EPA 9045D	pH
Titration	EPA SW-846 Chapter 7.3.4	Reactive Sulfide
Titration	Walkley-Black	Total Organic Carbon
IR	Lloyd Kahn	Total organic carbon
Turbidimetric	EPA 9038; ASTM 516-02	Sulfate
UV/VIS	EPA 350.1; SM 4500-NH ₃ H	Ammonia as N
UV/VIS	EPA 9251; SM 4500-Cl E	Chloride
UV/VIS	EPA SW-846 Chapter 7.3.4	Reactive Cyanide
UV/VIS	EPA 821/R-91-100	AVS-SEM
UV/VIS	SM 3500-Fe D	Ferrous Iron
Cleanup Methods	EPA 3630C	Silica Gel
UV/VIS	EPA 7196A	Chromium VI
UV/VIS	EPA 9012B	Total cyanide
Sieves, Hydrometer	ASTM D422	Grain Size

Solid and Chemical Materials		
Preparation	Method	Type
Preparation	EPA 1311	Toxicity Characteristic Leaching Procedure
Preparation	EPA 1312	Synthetic Precipitation Leaching Procedure
Cleanup Methods	EPA 3660B	Sulfur Clean-up
Cleanup Methods	EPA 3620C	Florsil Clean-up
Cleanup Methods	EPA 3630C	Silica Gel Clean-up
Cleanup Methods	EPA 3640A	GPC Clean-up
Organic Preparation	EPA 3540C	Soxhlet Extraction
Organic Preparation	EPA 3545A	Pressurized Fluid Extraction
Organic Preparation	EPA 3546	Microwave Extraction Preparation for EPA 8082A, 8081B and 8270C, D, 8015C/D
Organic Preparation	EPA 3550C	Sonication
Inorganics Preparation	EPA 3050B	Hotblock
Inorganics Preparation	EPA 3060A	Alkaline Digestion
Volatile Organics Preparation	EPA 5035/5035A	Closed System Purge and Trap
Organic Preparation	EPA 8330A/B	ISM

Air		
Technology	Method	Analyte
GC/MS	EPA TO-15	1, 1, 1-Trichloroethane
GC/MS	EPA TO-15	1, 1, 2, 2-Tetrachloroethane
GC/MS	EPA TO-15	1, 1, 2-Trichloroethane
GC/MS	EPA TO-15	1, 1-Dichloroethane
GC/MS	EPA TO-15	1, 1-Dichloroethylene
GC/MS	EPA TO-15	1, 2, 4-Trichlorobenzene
GC/MS	EPA TO-15	1, 2, 4-Trimethylbenzene
GC/MS	EPA TO-15	1, 2-Dibromoethane (EDB)
GC/MS	EPA TO-15	1, 2-Dichlorobenzene

Air		
Technology	Method	Analyte
GC/MS	EPA TO-15	1, 2-Dichloroethane
GC/MS	EPA TO-15	1, 2-Dichloroethenes (Total)
GC/MS	EPA TO-15	1, 2-Dichloropropane
GC/MS	EPA TO-15	1, 3, 5-Trimethylbenzene
GC/MS	EPA TO-15	1, 3-Butadiene
GC/MS	EPA TO-15	1, 3-Dichlorobenzene
GC/MS	EPA TO-15	1, 4-Dichlorobenzene
GC/MS	EPA TO-15	1, 4-Dioxane
GC/MS	EPA TO-15	1,1,2-Trichloro1,2,2-trifluoroethane (Freon 113)
GC/MS	EPA TO-15	1,2-Dichloro-1,1,2,2-tetrafluoroethane (Freon 114)
GC/MS	EPA TO-15	1,4-Difluorobenzene
GC/MS	EPA TO-15	2,2,4-Trimethylpentane
GC/MS	EPA TO-15	2-Butanone
GC/MS	EPA TO-15	2-Chlorotoluene
GC/MS	EPA TO-15	2-Hexanone
GC/MS	EPA TO-15	2-Propanol
GC/MS	EPA TO-15	3-Chloropropene (Allyl chloride)
GC/MS	EPA TO-15	4-Ethyltoluene
GC/MS	EPA TO-15	4-Methyl-2-pentanone
GC/MS	EPA TO-15	Acetone
GC/MS	EPA TO-15	Acrolein
GC/MS	EPA TO-15	Benzene
GC/MS	EPA TO-15	Benzyl chloride
GC/MS	EPA TO-15	Bromochloromethane
GC/MS	EPA TO-15	Bromodichloromethane
GC/MS	EPA TO-15	Bromoform
GC/MS	EPA TO-15	Carbon disulfide
GC/MS	EPA TO-15	Carbon tetrachloride

Air		
Technology	Method	Analyte
GC/MS	EPA TO-15	Chlorobenzene
GC/MS	EPA TO-15	Chloroethane
GC/MS	EPA TO-15	Chloroform
GC/MS	EPA TO-15	Cis-1, 2-Dichloroethene
GC/MS	EPA TO-15	Cis-1, 3-Dichloropropene
GC/MS	EPA TO-15	Cyclohexane
GC/MS	EPA TO-15	Dibromochloromethane
GC/MS	EPA TO-15	Dichlorodifluoromethane (Freon 12)
GC/MS	EPA TO-15	Ethanol
GC/MS	EPA TO-15	Ethyl acetate
GC/MS	EPA TO-15	Ethylbenzene
GC/MS	EPA TO-15	Hexachlorobutadiene
GC/MS	EPA TO-15	Isopropyl alcohol
GC/MS	EPA TO-15	Isopropylbenzene
GC/MS	EPA TO-15	m, p-Xylene
GC/MS	EPA TO-15	Methyl bromide (Bromomethane)
GC/MS	EPA TO-15	Methyl chloride (Chloromethane)
GC/MS	EPA TO-15	Methyl methacrylate
GC/MS	EPA TO-15	Methyl tert-butyl ether
GC/MS	EPA TO-15	Methylene chloride
GC/MS	EPA TO-15	Naphthalene
GC/MS	EPA TO-15	n-Heptane
GC/MS	EPA TO-15	n-Hexane
GC/MS	EPA TO-15	o-Xylene
GC/MS	EPA TO-15	Propene
GC/MS	EPA TO-15	Styrene
GC/MS	EPA TO-15	tert-Butyl alcohol
GC/MS	EPA TO-15	Tetrachloroethylene (Perchloroethylene)

Air		
Technology	Method	Analyte
GC/MS	EPA TO-15	Tetrahydrofuran
GC/MS	EPA TO-15	Toluene
GC/MS	EPA TO-15	trans-1, 2-Dichloroethylene
GC/MS	EPA TO-15	trans-1, 3-Dichloropropylene
GC/MS	EPA TO-15	Trichloroethene (Trichloroethylene)
GC/MS	EPA TO-15	Trichlorofluoromethane (Freon 11)
GC/MS	EPA TO-15	Vinyl acetate
GC/MS	EPA TO-15	Vinyl bromide
GC/MS	EPA TO-15	Vinyl chloride
GC/MS	EPA TO-15	Xylenes (Total)
GC/MS	MA DEP APH	Aliphatic C5-C8 range
GC/MS	MA DEP APH	Aliphatic C9-C12 range
GC/MS	MA DEP APH	Aromatic C9-C10 range
GC/MS	MA DEP APH	1,3-Butadiene
GC/MS	MA DEP APH	Benzene
GC/MS	MA DEP APH	Ethylbenzene
GC/MS	MA DEP APH	m+p-Xylene
GC/MS	MA DEP APH	Methyl tert-butyl ether
GC/MS	MA DEP APH	Naphthalene
GC/MS	MA DEP APH	o-Xylene
GC/MS	MA DEP APH	Toluene

Biological Tissue		
Technology	Method	Analyte
GC/ECD	EPA 8081B	4, 4'-DDD
GC/ECD	EPA 8081B	4, 4'-DDE
GC/ECD	EPA 8081B	4, 4'-DDT
GC/ECD	EPA 8081B	Aldrin

Biological Tissue		
Technology	Method	Analyte
GC/ECD	EPA 8081B	alpha-BHC (alpha-Hexachlorocyclohexane)
GC/ECD	EPA 8081B	Alpha-Chlordane/cis-Chlordane
GC/ECD	EPA 8081B	beta-BHC (beta-Hexachlorocyclohexane)
GC/ECD	EPA 8081B	Cis-Nonaclor
GC/ECD	EPA 8081B	delta-BHC
GC/ECD	EPA 8081B	Dieldrin
GC/ECD	EPA 8081B	Endosulfan I
GC/ECD	EPA 8081B	Endosulfan II
GC/ECD	EPA 8081B	Endosulfan sulfate
GC/ECD	EPA 8081B	Endrin
GC/ECD	EPA 8081B	Endrin aldehyde
GC/ECD	EPA 8081B	Endrin Ketone
GC/ECD	EPA 8081B	gamma-BHC (Lindane gamma-Hexachlorocyclohexane)
GC/ECD	EPA 8081B	gamma-Chlordane/trans-Chlordane
GC/ECD	EPA 8081B	Heptachlor
GC/ECD	EPA 8081B	Heptachlor epoxide
GC/ECD	EPA 8081B	Hexachlorobenzene
GC/ECD	EPA 8081B	Methoxychlor
GC/ECD	EPA 8081B	Oxychlordane
GC/ECD	EPA 8081B	Toxaphene (Chlorinated camphene)
GC/ECD	EPA 8081B	trans-Nonachlor
GC/ECD	EPA 8082A	2,2',3,3',4,4',5,5',6-Nonachlorobiphenyl (BZ 206)
GC/ECD	EPA 8082A	2,2',3,3',4,4',5,6-Octachlorobiphenyl (BZ 195)
GC/ECD	EPA 8082A	2,2',3,3',4,4',5-Heptachlorobiphenyl (BZ 170)
GC/ECD	EPA 8082A	2,2',3,3',4,4'-Hexachlorobiphenyl (BZ 128)
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 5, 5'-Heptachlorobiphenyl (BZ 180)
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 5', 6-Heptachlorobiphenyl (BZ 183)
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 5-Hexachlorobiphenyl (BZ 138)

Biological Tissue		
Technology	Method	Analyte
GC/ECD	EPA 8082A	2, 2', 3, 4, 4', 6, 6'-Heptachlorobiphenyl (BZ 184)
GC/ECD	EPA 8082A	2, 2', 3, 4', 5, 5', 6-Heptachlorobiphenyl (BZ 187)
GC/ECD	EPA 8082A	2, 2', 3, 4, 5'-Pentachlorobiphenyl (BZ 87)
GC/ECD	EPA 8082A	2, 2', 3, 5'-Tetrachlorobiphenyl (BZ 44)
GC/ECD	EPA 8082A	2, 2', 4, 4', 5, 5'-Hexachlorobiphenyl (BZ 153)
GC/ECD	EPA 8082A	2, 2', 4, 5, 5'-Pentachlorobiphenyl (BZ 101)
GC/ECD	EPA 8082A	2, 2', 4, 5'-Tetrachlorobiphenyl (BZ 49)
GC/ECD	EPA 8082A	2, 2', 5, 5'-Tetrachlorobiphenyl (BZ 52)
GC/ECD	EPA 8082A	2, 2', 5-Trichlorobiphenyl (BZ 18)
GC/ECD	EPA 8082A	2, 3, 3', 4, 4'-Pentachlorobiphenyl (BZ 105)
GC/ECD	EPA 8082A	2, 3', 4, 4', 5-Pentachlorobiphenyl (BZ 118)
GC/ECD	EPA 8082A	2, 3', 4, 4'-Tetrachlorobiphenyl (BZ 66)
GC/ECD	EPA 8082A	2, 4, 4'-Trichlorobiphenyl (BZ 28)
GC/ECD	EPA 8082A	2, 4'-Dichlorobiphenyl (BZ 8)
GC/ECD	EPA 8082A	Decachlorobiphenyl (BZ 209)
GC/MS	EPA 8270C/D SIM	1,1'-Biphenyl
GC/MS	EPA 8270C/D SIM	1,2,4,5-Tetrachlorobenzene
GC/MS	EPA 8270C/D SIM	1,2,4-Trichlorobenzene
GC/MS	EPA 8270C/D SIM	1,2-Dichlorobenzene
GC/MS	EPA 8270C/D SIM	1,3-Dichlorobenzene
GC/MS	EPA 8270C/D SIM	1,4-Dichlorobenzene
GC/MS	EPA 8270C/D SIM	1-Methylnaphthalene
GC/MS	EPA 8270C/D SIM	2,2'-Oxybis(1-chloropropane
GC/MS	EPA 8270C/D SIM	2,3,4,6-Tetrachlorophenol
GC/MS	EPA 8270C/D SIM	2,4,5-Trichlorophenol
GC/MS	EPA 8270C/D SIM	2,4,6-Trichlorophenol
GC/MS	EPA 8270C/D SIM	2,4-Dichlorophenol
GC/MS	EPA 8270C/D SIM	2,4-Dimethylphenol

Biological Tissue		
Technology	Method	Analyte
GC/MS	EPA 8270C/D SIM	2,4-Dinitrophenol
GC/MS	EPA 8270C/D SIM	2,4-Dinitrotoluene
GC/MS	EPA 8270C/D SIM	2,6-Dinitrotoluene
GC/MS	EPA 8270C/D SIM	2-Chloronaphthalene
GC/MS	EPA 8270C/D SIM	2-Chlorophenol
GC/MS	EPA 8270C/D SIM	2-Methylnaphthalene
GC/MS	EPA 8270C/D SIM	2-Methylphenol
GC/MS	EPA 8270C/D SIM	2-Nitroaniline
GC/MS	EPA 8270C/D SIM	2-Nitrophenol
GC/MS	EPA 8270C/D SIM	3&4-Methylphenol
GC/MS	EPA 8270C/D SIM	4,6-Dinitro-2-methylphenol
GC/MS	EPA 8270C/D SIM	4-Bromophenyl-phenylether
GC/MS	EPA 8270C/D SIM	4-Chloro-3-methylphenol
GC/MS	EPA 8270C/D SIM	4-Chloroaniline
GC/MS	EPA 8270C/D SIM	4-Chlorophenyl-phenylether
GC/MS	EPA 8270C/D SIM	4-Nitrophenol
GC/MS	EPA 8270C/D SIM	Acenaphthene
GC/MS	EPA 8270C/D SIM	Acenaphthylene
GC/MS	EPA 8270C/D SIM	Acetophenone
GC/MS	EPA 8270C/D SIM	Anthracene
GC/MS	EPA 8270C/D SIM	Atrazine
GC/MS	EPA 8270C/D SIM	Benzo(a)anthracene
GC/MS	EPA 8270C/D SIM	Benzo(a)pyrene
GC/MS	EPA 8270C/D SIM	Benzo(b)fluoranthene
GC/MS	EPA 8270C/D SIM	Benzo(g,h,i)perylene
GC/MS	EPA 8270C/D SIM	Benzo(k)fluoranthene
GC/MS	EPA 8270C/D SIM	Bis(2-chloroethoxy)methane
GC/MS	EPA 8270C/D SIM	Bis(2-chloroethyl)ether

Biological Tissue		
Technology	Method	Analyte
GC/MS	EPA 8270C/D SIM	Butylbenzylphthalate
GC/MS	EPA 8270C/D SIM	Caprolactam
GC/MS	EPA 8270C/D SIM	Carbazole
GC/MS	EPA 8270C/D SIM	Chrysene
GC/MS	EPA 8270C/D SIM	Dibenzo(a,h)anthracene
GC/MS	EPA 8270C/D SIM	Dibenzofuran
GC/MS	EPA 8270C/D SIM	Diethylphthalate
GC/MS	EPA 8270C/D SIM	Dimethyl phthalate
GC/MS	EPA 8270C/D SIM	Di-n-butylphthalate
GC/MS	EPA 8270C/D SIM	Di-n-octylphthalate
GC/MS	EPA 8270C/D SIM	Fluoranthene
GC/MS	EPA 8270C/D SIM	Fluorene
GC/MS	EPA 8270C/D SIM	Hexachlorobenzene
GC/MS	EPA 8270C/D SIM	Hexachlorobutadiene
GC/MS	EPA 8270C/D SIM	Hexachloroethane
GC/MS	EPA 8270C/D SIM	Indeno(1,2,3-cd)pyrene
GC/MS	EPA 8270C/D SIM	Isophorone
GC/MS	EPA 8270C/D SIM	Naphthalene
GC/MS	EPA 8270C/D SIM	n-Nitroso-di-n-propylamine
GC/MS	EPA 8270C/D SIM	n-Nitrosodiphenylamine
GC/MS	EPA 8270C/D SIM	Pentachlorophenol
GC/MS	EPA 8270C/D SIM	Phenanthrene
GC/MS	EPA 8270C/D SIM	Phenol
GC/MS	EPA 8270C/D SIM	Pyrene
ICP/AES	EPA 6010C/D	Aluminum
ICP/AES	EPA 6010C/D	Antimony
ICP/AES	EPA 6010C/D	Arsenic
ICP/AES	EPA 6010C/D	Barium

Biological Tissue		
Technology	Method	Analyte
ICP/AES	EPA 6010C/D	Beryllium
ICP/AES	EPA 6010C/D	Boron
ICP/AES	EPA 6010C/D	Cadmium
ICP/AES	EPA 6010C/D	Calcium
ICP/AES	EPA 6010C/D	Chromium
ICP/AES	EPA 6010C/D	Cobalt
ICP/AES	EPA 6010C/D	Copper
ICP/AES	EPA 6010C/D	Iron
ICP/AES	EPA 6010C/D	Lead
ICP/AES	EPA 6010C/D	Magnesium
ICP/AES	EPA 6010C/D	Manganese
ICP/AES	EPA 6010C/D	Molybdenum
ICP/AES	EPA 6010C/D	Nickel
ICP/AES	EPA 6010C/D	Potassium
ICP/AES	EPA 6010C/D	Selenium
ICP/AES	EPA 6010C/D	Silver
ICP/AES	EPA 6010C/D	Sodium
ICP/AES	EPA 6010C/D	Thallium
ICP/AES	EPA 6010C/D	Tin
ICP/AES	EPA 6010C/D	Vanadium
ICP/AES	EPA 6010C/D	Zinc
ICP/MS	EPA 6020A/B	Aluminum
ICP/MS	EPA 6020A/B	Antimony
ICP/MS	EPA 6020A/B	Arsenic
ICP/MS	EPA 6020A/B	Barium
ICP/MS	EPA 6020A/B	Beryllium
ICP/MS	EPA 6020A/B	Boron
ICP/MS	EPA 6020A/B	Cadmium

Biological Tissue		
Technology	Method	Analyte
ICP/MS	EPA 6020A/B	Calcium
ICP/MS	EPA 6020A/B	Chromium
ICP/MS	EPA 6020A/B	Cobalt
ICP/MS	EPA 6020A/B	Copper
ICP/MS	EPA 6020A/B	Iron
ICP/MS	EPA 6020A/B	Lead
ICP/MS	EPA 6020A/B	Magnesium
ICP/MS	EPA 6020A/B	Manganese
ICP/MS	EPA 6020A/B	Molybdenum
ICP/MS	EPA 6020A/B	Nickel
ICP/MS	EPA 6020A/B	Potassium
ICP/MS	EPA 6020A/B	Selenium
ICP/MS	EPA 6020A/B	Silver
ICP/MS	EPA 6020A/B	Sodium
ICP/MS	EPA 6020A/B	Thallium
ICP/MS	EPA 6020A/B	Tin
ICP/MS	EPA 6020A/B	Vanadium
ICP/MS	EPA 6020A/B	Zinc
CVAA	EPA 7471B	Mercury

Note:

1. This scope is formatted as part of a single document including Certificate of Accreditation No. L2223.



R. Douglas Leonard Jr., VP, PILR SBU



State of Rhode Island and Providence Plantations
DEPARTMENT OF HEALTH
Certifies

LAO00344

KATAHDIN ANALYTICAL SERVICES
600 TECHNOLOGY WAY
SCARBOROUGH ME 04074
Laboratory Director: LESLIE DIMOND

for the analysis of: Potable Water Organic Chemistry - Potable Water Inorganic Chemistry - Non-potable Water Organic Chemistry - Non-potable Water Inorganic Chemistry -

This certificate is issued, pursuant to Rhode Island General Laws 23-16.2 and supersedes all previous Rhode Island certificates issued to this laboratory. Certification is no guarantee of the validity of the laboratory results.

This certificate is valid only when accompanied by the certificate and list of analytes and methods for which certification has been granted based upon the following out of state certification(s):

Certifying Authority
NELAC/NH
MAINE

Certification Number
2001
ME00019

Expiration Date
04/02/2019
06/01/2020

KATAHDIN ANALYTICAL SERVICES is responsible for maintaining each of the certifications listed above. Failure to notify the Laboratory Certification Officer of any change in the status of these certifications may result in the suspension or revocation of certification. Contact the Laboratory Certification Officer to verify the current certification status of this laboratory.

Nicole Alexander-Scott, MD, MPH
Director of Health

Expires: 12/30/2019

THIS CERTIFICATE IS TO BE CONSPICUOUSLY DISPLAYED AT THE LABORATORY IN A LOCATION VISABLE TO THE PUBLIC